

Modeling effect of peaker plant operations on air quality in Pittsfield, MA

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R Markdown

The purpose of this script is to analyze the effect of peaker power plant operations on air quality (PM2.5, PM10, and NO2 concentrations) at two locations: Allendale Elementary School in Pittsfield, MA and the population-weighted centroid in Pittsfield, MA.

Air quality data from BEAT and peaker plant operations data from the EPA CAMPD were downloaded for April 1, 2024 - September 30, 2025. The time period of analysis was Summer 2024 and Summer 2025 (June, July, and August).

```
## Linking to GEOS 3.13.0, GDAL 3.10.1, PROJ 9.5.1; sf_use_s2() is TRUE

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

## Loading required package: spData

## To access larger datasets in this package, install the spDataLarge
## package with: `install.packages('spDataLarge',
## repos='https://nowosad.github.io/drat/', type='source')`

## Loading required package: MASS

##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##   select

## Loading required package: Rcpp

## i Google's Terms of Service: <https://mapsplatform.google.com>
##   Stadia Maps' Terms of Service: <https://stadiamaps.com/terms-of-service>
##   OpenStreetMap's Tile Usage Policy: <https://operations.osmfoundation.org/policies/tiles>
## i Please cite ggmap if you use it! Use `citation("ggmap")` for details.
##
## Attaching package: 'ggmap'
##
```

```

##
## The following object is masked from 'package:cowplot':
##
##   theme_nothing
##
##
## Attaching package: 'data.table'
##
##
## The following objects are masked from 'package:dplyr':
##
##   between, first, last
##
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats   1.0.0      v readr     2.1.5
## v lubridate  1.9.4      v stringr  1.5.1
## v purrr      1.0.2      v tibble   3.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x data.table::between() masks dplyr::between()
## x dplyr::filter()       masks stats::filter()
## x data.table::first()   masks dplyr::first()
## x lubridate::hour()     masks data.table::hour()
## x lubridate::isoweek()  masks data.table::isoweek()
## x dplyr::lag()          masks stats::lag()
## x data.table::last()    masks dplyr::last()
## x lubridate::mday()     masks data.table::mday()
## x lubridate::minute()   masks data.table::minute()
## x lubridate::month()    masks data.table::month()
## x lubridate::quarter()  masks data.table::quarter()
## x lubridate::second()   masks data.table::second()
## x MASS::select()        masks dplyr::select()
## x lubridate::stamp()    masks cowplot::stamp()
## x purrr::transpose()    masks data.table::transpose()
## x lubridate::wday()     masks data.table::wday()
## x lubridate::week()     masks data.table::week()
## x lubridate::yday()     masks data.table::yday()
## x lubridate::year()     masks data.table::year()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
## Loading required package: spatstat.data
##
## Loading required package: spatstat.univar
##
## spatstat.univar 3.1-4
##
## Loading required package: spatstat.geom
##
## spatstat.geom 3.6-0
##
## Attaching package: 'spatstat.geom'
##
##

```

```

## The following object is masked from 'package:data.table':
##
##     shift
##
##
## The following object is masked from 'package:MASS':
##
##     area
##
##
## Loading required package: spatstat.random
##
## spatstat.random 3.4-2
##
## Loading required package: spatstat.explore
##
## Loading required package: nlme
##
##
## Attaching package: 'nlme'
##
##
## The following object is masked from 'package:dplyr':
##
##     collapse
##
##
## spatstat.explore 3.5-3
##
## Loading required package: spatstat.model
##
## Loading required package: rpart
##
## spatstat.model 3.4-2
##
## Loading required package: spatstat.linnet
##
## spatstat.linnet 3.3-2
##
##
## spatstat 3.4-1
## For an introduction to spatstat, type 'beginner'
##
##
## Attaching package: 'gstat'
##
##
## The following object is masked from 'package:spatstat.explore':
##
##     idw
##
##
## Loading required package: abind

```

```
##
## terra 1.8.60
##
##
## Attaching package: 'terra'
##
##
## The following objects are masked from 'package:spatstat.geom':
##
##      area, delaunay, is.empty, rescale, rotate, shift, where.max,
##      where.min
##
##
## The following object is masked from 'package:data.table':
##
##      shift
##
##
## The following object is masked from 'package:ggmap':
##
##      inset
##
##
## The following object is masked from 'package:tidyr':
##
##      extract
##
##
## The following object is masked from 'package:MASS':
##
##      area
```

Loading the input data

```
#time series air quality data from BEAT
sensorsaq_all <- fread("data/input/sensordata_clean.csv")

#weather data (includes hourly temp, wind direction, and windspeed) from National Centers for Environmental Prediction
weather_all <- fread("data/input/WeatherClean.csv")

#sensor location and description data from BEAT
sensorinfo <- fread("data/input/sensors.csv")

#peaker plant operations data from the EPA (cleaned)
peakerops <- fread("data/input/peakerop_clean.csv")
```

#PART 1: JOINING AND VISUALIZING AIR QUALITY DATA

```
#join sensor location/name info to the time series sensor air quality data
sensors_joined_1 <- dplyr::left_join(sensorsaq_all, sensorinfo,
                                   by = c("datasourceId" = "datasourceID"))

head(sensors_joined_1)
```

```
##      datasourceId sourceId    sourceType StartOfPeriod  endOfPeriod
```

```

##          <char>    <char>          <char>          <char>          <char>
## 1:      DBGKL8516 AGK79VTM CLARITY_NODE 2024;04;01;05 2024;04;01;06
## 2:      DDTAU0510 AK639KKR CLARITY_NODE 2024;04;01;05 2024;04;01;06
## 3:      DJFKW1483 ANWQJL6T CLARITY_NODE 2024;04;01;05 2024;04;01;06
## 4:      DKWAZ4010 AKR3XP4V CLARITY_NODE 2024;04;01;05 2024;04;01;06
## 5:      DLWGJ5363 AFVGL7MJ CLARITY_NODE 2024;04;01;05 2024;04;01;06
## 6:      DMTLY7351 AXGPYFJ7 CLARITY_NODE 2024;04;01;05 2024;04;01;06
##      locationLatitude locationLongitude no2Conc1HourMean.qcAssessment
##          <num>          <num>          <char>
## 1:          42.46803          -73.20072          valid
## 2:          42.42506          -73.22515          valid
## 3:          42.42689          -73.28892          valid
## 4:          42.45855          -73.22086          valid
## 5:          42.42172          -73.20349          valid
## 6:          42.45301          -73.26174          valid
##      no2Conc1HourMean.status no2Conc1HourMean.value
##          <char>          <num>
## 1:      calibrated-ready          11.36
## 2:      calibrated-ready          14.80
## 3:      calibrated-ready          12.03
## 4:      calibrated-ready          18.52
## 5:      calibrated-ready          13.81
## 6:      calibrated-ready          21.23
##      pm10ConcMass1HourMean.qcAssessment pm10ConcMass1HourMean.raw
##          <char>          <num>
## 1:          valid          4.74
## 2:          valid          47.01
## 3:          valid          11.82
## 4:          valid          7.43
## 5:          valid          8.90
## 6:          valid          8.58
##      pm2_5ConcMass1HourMean.qcAssessment pm2_5ConcMass1HourMean.status
##          <char>          <char>
## 1:          valid          calibrated-ready
## 2:          valid          calibrated-ready
## 3:          valid          calibrated-ready
## 4:          valid          calibrated-ready
## 5:          valid          calibrated-ready
## 6:          valid          calibrated-ready
##      pm2_5ConcMass1HourMean.value          name
##          <num>          <char>
## 1:          3.09          Flow Station
## 2:          9.19          Pump Station (Sackett Brook)
## 3:          3.37          Pittsfield Municipal Airport
## 4:          3.36          Allendale Elementary School
## 5:          3.25          Residential House (Glory Dr.)
## 6:          3.43          Habitat for Humanity
##
##
##
##          name_full
##          <char>
## 1:          Flow Station @ 694 Merrill Road, Pittsfield MA 01201
## 2:          Pump Station on Sackett Brook @ 233 East New Lenox Road
## 3:          Pittsfield Municipal Airport - National Weather Station @ 832 Tamarack Road, Pittsfield MA 01201
## 4:          Allendale Elementary School @ 180 Connecticut Avenue, Pittsfield MA 01201
## 5:          Residential house @ 75 Glory Drive

```

```
## 6:          Central Berkshire Habitat for Humanity @ 314 Colombus Avenue, Pittsfield MA 01201
##          notes longitude latitude distancepeaker distanceairport
##          <char>      <num>      <num>          <num>          <num>
## 1:          -73.20072 42.46803      1926.7056      8806.0948
## 2: Removed 11/20/24 -73.22515 42.42506      3529.2037      5407.9490
## 3:          -73.28892 42.42689      6685.3583       240.1949
## 4:          -73.22086 42.45855       329.5028      6843.3038
## 5:          -73.20349 42.42172      4035.6887      7201.1790
## 6:          -73.26174 42.45301      3609.3272      3903.1661
## distancemajroads bearing_deg bearing_rad sin_bearing cos_bearing
##          <num>      <num>      <num>          <num>          <num>
## 1:          81.85057   47.88854   0.8358126   0.7418417   0.6705751
## 2:          989.26638  189.46202   3.3067360  -0.1643938  -0.9863948
## 3:          531.66868  240.66254   4.2003538  -0.8717492  -0.4899525
## 4:          200.34768  316.45239   5.5231361  -0.6889571   0.7248021
## 5:          897.84679  162.67021   2.8391307   0.2978713  -0.9546060
## 6:          79.52929  264.03443   4.6082701  -0.9945845  -0.1039309
```

```
#join the peaker operations info to sensors air quality data
sensors_joined_2 <- dplyr::left_join(sensors_joined_1, peakerops,
                                     by = c("StartOfPeriod" = "datetime"))

head(sensors_joined_2)
```

```
##      datasourceId sourceId      sourceType StartOfPeriod  endOfPeriod
##      <char>      <char>      <char>      <char>      <char>
## 1:   DBGKL8516  AGK79VTM  CLARITY_NODE  2024;04;01;05  2024;04;01;06
## 2:   DDTAU0510  AK639KKR  CLARITY_NODE  2024;04;01;05  2024;04;01;06
## 3:   DJFKW1483  ANWQJL6T  CLARITY_NODE  2024;04;01;05  2024;04;01;06
## 4:   DKWAZ4010  AKR3XP4V  CLARITY_NODE  2024;04;01;05  2024;04;01;06
## 5:   DLWGJ5363  AFVGL7MJ  CLARITY_NODE  2024;04;01;05  2024;04;01;06
## 6:   DMTLY7351  AXGPYFJ7  CLARITY_NODE  2024;04;01;05  2024;04;01;06
##      locationLatitude locationLongitude no2Conc1HourMean.qcAssessment
##      <num>          <num>          <char>
## 1:      42.46803      -73.20072      valid
## 2:      42.42506      -73.22515      valid
## 3:      42.42689      -73.28892      valid
## 4:      42.45855      -73.22086      valid
## 5:      42.42172      -73.20349      valid
## 6:      42.45301      -73.26174      valid
##      no2Conc1HourMean.status no2Conc1HourMean.value
##      <char>          <num>
## 1:      calibrated-ready      11.36
## 2:      calibrated-ready      14.80
## 3:      calibrated-ready      12.03
## 4:      calibrated-ready      18.52
## 5:      calibrated-ready      13.81
## 6:      calibrated-ready      21.23
##      pm10ConcMass1HourMean.qcAssessment pm10ConcMass1HourMean.raw
##      <char>          <num>
## 1:      valid      4.74
## 2:      valid      47.01
## 3:      valid      11.82
## 4:      valid      7.43
## 5:      valid      8.90
```

```

## 6: valid 8.58
## pm2_5ConcMass1HourMean.qcAssessment pm2_5ConcMass1HourMean.status
## <char> <char>
## 1: valid calibrated-ready
## 2: valid calibrated-ready
## 3: valid calibrated-ready
## 4: valid calibrated-ready
## 5: valid calibrated-ready
## 6: valid calibrated-ready
## pm2_5ConcMass1HourMean.value name
## <num> <char>
## 1: 3.09 Flow Station
## 2: 9.19 Pump Station (Sackett Brook)
## 3: 3.37 Pittsfield Municipal Airport
## 4: 3.36 Allendale Elementary School
## 5: 3.25 Residential House (Glory Dr.)
## 6: 3.43 Habitat for Humanity
## name_full
## <char>
## 1: Flow Station @ 694 Merrill Road, Pittsfield MA 01201
## 2: Pump Station on Sackett Brook @ 233 East New Lenox Road
## 3: Pittsfield Municipal Airport - National Weather Station @ 832 Tamarack Road, Pittsfield MA 01201
## 4: Allendale Elementary School @ 180 Connecticut Avenue, Pittsfield MA 01201
## 5: Residential house @ 75 Glory Drive
## 6: Central Berkshire Habitat for Humanity @ 314 Colombus Avenue, Pittsfield MA 01201
## notes longitude latitude distancepeak distanceairport
## <char> <num> <num> <num> <num>
## 1: -73.20072 42.46803 1926.7056 8806.0948
## 2: Removed 11/20/24 -73.22515 42.42506 3529.2037 5407.9490
## 3: -73.28892 42.42689 6685.3583 240.1949
## 4: -73.22086 42.45855 329.5028 6843.3038
## 5: -73.20349 42.42172 4035.6887 7201.1790
## 6: -73.26174 42.45301 3609.3272 3903.1661
## distancemajroads bearing_deg bearing_rad sin_bearing cos_bearing state
## <num> <num> <num> <num> <num> <char>
## 1: 81.85057 47.88854 0.8358126 0.7418417 0.6705751 MA
## 2: 989.26638 189.46202 3.3067360 -0.1643938 -0.9863948 MA
## 3: 531.66868 240.66254 4.2003538 -0.8717492 -0.4899525 MA
## 4: 200.34768 316.45239 5.5231361 -0.6889571 0.7248021 MA
## 5: 897.84679 162.67021 2.8391307 0.2978713 -0.9546060 MA
## 6: 79.52929 264.03443 4.6082701 -0.9945845 -0.1039309 MA
## peakername peakerid date hour grossload_mw
## <char> <int> <char> <int> <int>
## 1: Pittsfield Generating 50002 4/1/2024 5 NA
## 2: Pittsfield Generating 50002 4/1/2024 5 NA
## 3: Pittsfield Generating 50002 4/1/2024 5 NA
## 4: Pittsfield Generating 50002 4/1/2024 5 NA
## 5: Pittsfield Generating 50002 4/1/2024 5 NA
## 6: Pittsfield Generating 50002 4/1/2024 5 NA
## steamload_1000lbsperhr so2mass_lbs co2mass_shorttons noxmass_lbs
## <lgcl> <lgcl> <num> <num>
## 1: NA NA NA NA
## 2: NA NA NA NA
## 3: NA NA NA NA

```

```
## 4:      NA      NA      NA      NA
## 5:      NA      NA      NA      NA
## 6:      NA      NA      NA      NA
##   heating_mmbtu operating   year month   day season   timeofday weekday_num
##           <num>      <int> <int> <int> <int> <char>      <char>      <int>
## 1:           NA          0  2024     4     1 Spring Early Morning          1
## 2:           NA          0  2024     4     1 Spring Early Morning          1
## 3:           NA          0  2024     4     1 Spring Early Morning          1
## 4:           NA          0  2024     4     1 Spring Early Morning          1
## 5:           NA          0  2024     4     1 Spring Early Morning          1
## 6:           NA          0  2024     4     1 Spring Early Morning          1
```

*#there are multiple report types, SOM, SOD (summary of month and summary of day)
#with null values. what happens if I remove them?*

```
weather_filtered <- weather_all |> dplyr::filter(REPORT_TYPE == "FM-15")
head(weather_filtered)
```

```
##      STATION      DATE REPORT_TYPE SOURCE HourlyDryBulbTemperature
##      <char>      <char>      <char>  <int>      <int>
## 1: USW00014763 2024;04;01;00      FM-15      343          33
## 2: USW00014763 2024;04;01;01      FM-15      343          33
## 3: USW00014763 2024;04;01;02      FM-15      343          30
## 4: USW00014763 2024;04;01;03      FM-15      343          32
## 5: USW00014763 2024;04;01;04      FM-15      343          32
## 6: USW00014763 2024;04;01;05      FM-15      343          32
##   HourlyRelativeHumidity HourlyWindDirection HourlyWindSpeed
##           <int>           <char>           <char>
## 1:           75             260             6
## 2:           75             0             0
## 3:           82             0             0
## 4:           75             0             0
## 5:           82             0             0
## 6:           82             0             0
```

```
class(weather_filtered$HourlyWindDirection)
```

```
## [1] "character"
```

```
table(weather_filtered$HourlyWindDirection)
```

```
##
##      0   10  100  110  120  130  140  150  160  170  180  190   20  200  210
## 120 3443   56  108  144  113  164  200  103   86  149  180  209   87  215  253
## 220 230  240  250  260  270  280  290   30  300  310  320  330  340  350  360
## 320 397  370  324  390  472  614  738  112  725  403  248  153   90   57   69
##  40   50   60   70   80   90 VRB
## 140 175  253  143  101   73 1041
```

*#There are 1041 values with variable wind direction. Should I force to be NAs?
#decided not to include wind direction as a variable in the model*

#Join the weather data to sensors air quality data

#Got an error that there were multiple weather dates for 2025;03;08;21

```
weather_test <- weather_filtered |> dplyr::filter(
  DATE == "2025;03;08;21")
```

#dropped the second row because both measurements were taken at the same time based on the original wea

```

weather_filtered_unique <- weather_filtered[-8201,]

sensors_joined_3 <- dplyr::left_join(sensors_joined_2, weather_filtered_unique, by = c("StartOfPeriod" =

#sensors_joined_3 is the full dataset of sensor air quality data joined to sensor information
#peaker plant operation information
#and weather information

head(sensors_joined_3)

```

```

##      datasourceId sourceId      sourceType StartOfPeriod  endOfPeriod
##      <char>      <char>      <char>      <char>      <char>
## 1:   DBGKL8516 AGK79VTM CLARITY_NODE 2024;04;01;05 2024;04;01;06
## 2:   DDTAU0510 AK639KKR CLARITY_NODE 2024;04;01;05 2024;04;01;06
## 3:   DJFKW1483 ANWQJL6T CLARITY_NODE 2024;04;01;05 2024;04;01;06
## 4:   DKWAZ4010 AKR3XP4V CLARITY_NODE 2024;04;01;05 2024;04;01;06
## 5:   DLWGJ5363 AFVGL7MJ CLARITY_NODE 2024;04;01;05 2024;04;01;06
## 6:   DMTLY7351 AXGPYFJ7 CLARITY_NODE 2024;04;01;05 2024;04;01;06
##      locationLatitude locationLongitude no2Conc1HourMean.qcAssessment
##      <num>              <num>              <char>
## 1:      42.46803          -73.20072              valid
## 2:      42.42506          -73.22515              valid
## 3:      42.42689          -73.28892              valid
## 4:      42.45855          -73.22086              valid
## 5:      42.42172          -73.20349              valid
## 6:      42.45301          -73.26174              valid
##      no2Conc1HourMean.status no2Conc1HourMean.value
##      <char>              <num>
## 1:      calibrated-ready          11.36
## 2:      calibrated-ready          14.80
## 3:      calibrated-ready          12.03
## 4:      calibrated-ready          18.52
## 5:      calibrated-ready          13.81
## 6:      calibrated-ready          21.23
##      pm10ConcMass1HourMean.qcAssessment pm10ConcMass1HourMean.raw
##      <char>              <num>
## 1:      valid              4.74
## 2:      valid              47.01
## 3:      valid              11.82
## 4:      valid              7.43
## 5:      valid              8.90
## 6:      valid              8.58
##      pm2_5ConcMass1HourMean.qcAssessment pm2_5ConcMass1HourMean.status
##      <char>              <char>
## 1:      valid              calibrated-ready
## 2:      valid              calibrated-ready
## 3:      valid              calibrated-ready
## 4:      valid              calibrated-ready
## 5:      valid              calibrated-ready
## 6:      valid              calibrated-ready
##      pm2_5ConcMass1HourMean.value      name
##      <num>              <char>
## 1:      3.09              Flow Station
## 2:      9.19      Pump Station (Sackett Brook)

```

```

## 3:          3.37  Pittsfield Municipal Airport
## 4:          3.36  Allendale Elementary School
## 5:          3.25  Residential House (Glory Dr.)
## 6:          3.43  Habitat for Humanity
##
##
## 1:          Flow Station @ 694 Merrill Road, Pittsfield MA 01201
## 2:          Pump Station on Sackett Brook @ 233 East New Lenox Road
## 3: Pittsfield Municipal Airport - National Weather Station @ 832 Tamarack Road, Pittsfield MA 01201
## 4:          Allendale Elementary School @ 180 Connecticut Avenue, Pittsfield MA 01201
## 5:          Residential house @ 75 Glory Drive
## 6:          Central Berkshire Habitat for Humanity @ 314 Colombus Avenue, Pittsfield MA 01201
##
##      notes longitude latitude distancepeak distanceairport
##      <char>      <num>      <num>      <num>      <num>
## 1:          -73.20072 42.46803      1926.7056      8806.0948
## 2: Removed 11/20/24 -73.22515 42.42506      3529.2037      5407.9490
## 3:          -73.28892 42.42689      6685.3583      240.1949
## 4:          -73.22086 42.45855      329.5028      6843.3038
## 5:          -73.20349 42.42172      4035.6887      7201.1790
## 6:          -73.26174 42.45301      3609.3272      3903.1661
##
##      distancemajroads bearing_deg bearing_rad sin_bearing cos_bearing state
##      <num>      <num>      <num>      <num>      <num> <char>
## 1:          81.85057      47.88854      0.8358126      0.7418417      0.6705751      MA
## 2:          989.26638      189.46202      3.3067360      -0.1643938      -0.9863948      MA
## 3:          531.66868      240.66254      4.2003538      -0.8717492      -0.4899525      MA
## 4:          200.34768      316.45239      5.5231361      -0.6889571      0.7248021      MA
## 5:          897.84679      162.67021      2.8391307      0.2978713      -0.9546060      MA
## 6:          79.52929      264.03443      4.6082701      -0.9945845      -0.1039309      MA
##
##      peakername peakerid      date      hour grossload_mw
##      <char>      <int>      <char> <int>      <int>
## 1: Pittsfield Generating      50002 4/1/2024      5      NA
## 2: Pittsfield Generating      50002 4/1/2024      5      NA
## 3: Pittsfield Generating      50002 4/1/2024      5      NA
## 4: Pittsfield Generating      50002 4/1/2024      5      NA
## 5: Pittsfield Generating      50002 4/1/2024      5      NA
## 6: Pittsfield Generating      50002 4/1/2024      5      NA
##
##      steamload_1000lbsperhr so2mass_lbs co2mass_shorttons noxmass_lbs
##      <lgcl>      <lgcl>      <num>      <num>
## 1:          NA      NA      NA      NA
## 2:          NA      NA      NA      NA
## 3:          NA      NA      NA      NA
## 4:          NA      NA      NA      NA
## 5:          NA      NA      NA      NA
## 6:          NA      NA      NA      NA
##
##      heatinput_mmbtu operating year month      day season      timeofday weekday_num
##      <num>      <int> <int> <int> <int> <char>      <char>      <int>
## 1:          NA      0 2024      4      1 Spring Early Morning      1
## 2:          NA      0 2024      4      1 Spring Early Morning      1
## 3:          NA      0 2024      4      1 Spring Early Morning      1
## 4:          NA      0 2024      4      1 Spring Early Morning      1
## 5:          NA      0 2024      4      1 Spring Early Morning      1
## 6:          NA      0 2024      4      1 Spring Early Morning      1
##
##      STATION REPORT_TYPE SOURCE HourlyDryBulbTemperature
##      <char>      <char> <int>      <int>

```

```
## 1: USW00014763      FM-15      343      32
## 2: USW00014763      FM-15      343      32
## 3: USW00014763      FM-15      343      32
## 4: USW00014763      FM-15      343      32
## 5: USW00014763      FM-15      343      32
## 6: USW00014763      FM-15      343      32
##      HourlyRelativeHumidity HourlyWindDirection HourlyWindSpeed
##              <int>              <char>              <char>
## 1:              82              0              0
## 2:              82              0              0
## 3:              82              0              0
## 4:              82              0              0
## 5:              82              0              0
## 6:              82              0              0
```

```
write.csv(sensors_joined_3, "data/output/sensors_joined_all.csv")
```

```
#filter the complete dataset to just summer 2024 and summer 2025
sensors_summer_1 <- sensors_joined_3 |> dplyr::filter(season == "Summer")

head(sensors_summer_1)
```

```
##      datasourceId sourceId      sourceType StartOfPeriod  endOfPeriod
##              <char>  <char>          <char>          <char>          <char>
## 1:      DARSM2019 AF64TV9R CLARITY_NODE 2024;06;01;00 2024;06;01;01
## 2:      DBGKL8516 AGK79VTM CLARITY_NODE 2024;06;01;00 2024;06;01;01
## 3:      DDTAU0510 AK639KKR CLARITY_NODE 2024;06;01;00 2024;06;01;01
## 4:      DJFKW1483 ANWQJL6T CLARITY_NODE 2024;06;01;00 2024;06;01;01
## 5:      DKWAZ4010 AKR3XP4V CLARITY_NODE 2024;06;01;00 2024;06;01;01
## 6:      DLWGJ5363 AFVGL7MJ CLARITY_NODE 2024;06;01;00 2024;06;01;01
##      locationLatitude locationLongitude no2Conc1HourMean.qcAssessment
##              <num>              <num>              <char>
## 1:          42.45179          -73.25484              valid
## 2:          42.46803          -73.20072              valid
## 3:          42.42506          -73.22515              valid
## 4:          42.42689          -73.28892              valid
## 5:          42.45855          -73.22086              valid
## 6:          42.42172          -73.20349              valid
##      no2Conc1HourMean.status no2Conc1HourMean.value
##              <char>              <num>
## 1:      calibrated-ready              22.44
## 2:      calibrated-ready              17.25
## 3:      calibrated-ready              11.26
## 4:      calibrated-ready              25.36
## 5:      calibrated-ready              5.01
## 6:      calibrated-ready              19.61
##      pm10ConcMass1HourMean.qcAssessment pm10ConcMass1HourMean.raw
##              <char>              <num>
## 1:              valid              7.06
## 2:              valid              5.60
## 3:              valid              5.03
## 4:              valid              7.79
## 5:              valid              4.46
## 6:              valid              5.31
##      pm2_5ConcMass1HourMean.qcAssessment pm2_5ConcMass1HourMean.status
```

##		<char>		<char>		
## 1:		valid		calibrated-ready		
## 2:		valid		calibrated-ready		
## 3:		valid		calibrated-ready		
## 4:		valid		calibrated-ready		
## 5:		valid		calibrated-ready		
## 6:		valid		calibrated-ready		
##	pm2_5ConcMass1HourMean.value			name		
##		<num>		<char>		
## 1:	5.58		Pittsfield Fire Department			
## 2:	4.71		Flow Station			
## 3:	4.21		Pump Station (Sackett Brook)			
## 4:	4.26		Pittsfield Municipal Airport			
## 5:	4.31		Allendale Elementary School			
## 6:	4.29		Residential House (Glory Dr.)			
##				name_full		
##				<char>		
## 1:			Pittsfield Fire Department @ 74 Columbus Avenue, Pittsfield MA 01201			
## 2:			Flow Station @ 694 Merrill Road, Pittsfield MA 01201			
## 3:			Pump Station on Sackett Brook @ 233 East New Lenox Road			
## 4:			Pittsfield Municipal Airport - National Weather Station @ 832 Tamarack Road, Pittsfield MA 01201			
## 5:			Allendale Elementary School @ 180 Connecticut Avenue, Pittsfield MA 01201			
## 6:			Residential house @ 75 Glory Drive			
##	notes	longitude	latitude	distancepeak	distanceairport	
##	<char>	<num>	<num>	<num>	<num>	
## 1:		-73.25484	42.45179	3065.1987	4178.3037	
## 2:		-73.20072	42.46803	1926.7056	8806.0948	
## 3:	Removed 11/20/24	-73.22515	42.42506	3529.2037	5407.9490	
## 4:		-73.28892	42.42689	6685.3583	240.1949	
## 5:		-73.22086	42.45855	329.5028	6843.3038	
## 6:		-73.20349	42.42172	4035.6887	7201.1790	
##	distancemajroads	bearing_deg	bearing_rad	sin_bearing	cos_bearing	state
##	<num>	<num>	<num>	<num>	<num>	<char>
## 1:	2.628948	260.39541	4.5447573	-0.9859827	-0.1668477	MA
## 2:	81.850571	47.88854	0.8358126	0.7418417	0.6705751	MA
## 3:	989.266375	189.46202	3.3067360	-0.1643938	-0.9863948	MA
## 4:	531.668678	240.66254	4.2003538	-0.8717492	-0.4899525	MA
## 5:	200.347678	316.45239	5.5231361	-0.6889571	0.7248021	MA
## 6:	897.846793	162.67021	2.8391307	0.2978713	-0.9546060	MA
##	peakername	peakrid	date	hour	grossload_mw	
##	<char>	<int>	<char>	<int>	<int>	
## 1:	Pittsfield Generating	50002	6/1/2024	0	NA	
## 2:	Pittsfield Generating	50002	6/1/2024	0	NA	
## 3:	Pittsfield Generating	50002	6/1/2024	0	NA	
## 4:	Pittsfield Generating	50002	6/1/2024	0	NA	
## 5:	Pittsfield Generating	50002	6/1/2024	0	NA	
## 6:	Pittsfield Generating	50002	6/1/2024	0	NA	
##	steamload_1000lbsperhr	so2mass_lbs	co2mass_shorttons	noxmass_lbs		
##	<lgcl>	<lgcl>		<num>	<num>	
## 1:	NA	NA		NA	NA	
## 2:	NA	NA		NA	NA	
## 3:	NA	NA		NA	NA	
## 4:	NA	NA		NA	NA	
## 5:	NA	NA		NA	NA	

```
#add a new column, fireworks, to represent potential July 4th covariate
#dates that fall on or after July 4th on a weekend may have elevated pollutants due to fireworks
#if the date falls on the following dates, assign it a value of 1 for fireworks; otherwise, 0

sensors_summer_2 <- sensors_summer_1

sensors_summer_2$fireworks <- ifelse(sensors_summer_2$date %in% c("7/4/2024", "7/5/2024", "7/6/2024", "7/7/2024"), 1, 0)

#add a new column, smoke, to represent potential covariate from wildfire smoke from Canada
#check for dates that air quality alerts were issued in summer 2024 and summer 2025 due to wildfire smoke
#sources: local news articles, Mass DEP, Berkshire health orgs
#if the date falls on the following dates with an air quality alert due to smoke, assign it a value of 1; otherwise, 0

sensors_summer_2$smoke <- ifelse(sensors_summer_2$date %in% c("8/16/2024", "6/5/2025", "7/26/2025", "7/27/2025"), 1, 0)

head(sensors_summer_2)
```

13

```

## 3:      42.42506      -73.22515      valid
## 4:      42.42689      -73.28892      valid
## 5:      42.45855      -73.22086      valid
## 6:      42.42172      -73.20349      valid
##      no2Conc1HourMean.status no2Conc1HourMean.value
##      <char>                <num>
## 1:      calibrated-ready      22.44
## 2:      calibrated-ready      17.25
## 3:      calibrated-ready      11.26
## 4:      calibrated-ready      25.36
## 5:      calibrated-ready       5.01
## 6:      calibrated-ready      19.61
##      pm10ConcMass1HourMean.qcAssessment pm10ConcMass1HourMean.raw
##      <char>                <num>
## 1:      valid                7.06
## 2:      valid                5.60
## 3:      valid                5.03
## 4:      valid                7.79
## 5:      valid                4.46
## 6:      valid                5.31
##      pm2_5ConcMass1HourMean.qcAssessment pm2_5ConcMass1HourMean.status
##      <char>                <char>
## 1:      valid                calibrated-ready
## 2:      valid                calibrated-ready
## 3:      valid                calibrated-ready
## 4:      valid                calibrated-ready
## 5:      valid                calibrated-ready
## 6:      valid                calibrated-ready
##      pm2_5ConcMass1HourMean.value      name
##      <num>                <char>
## 1:      5.58      Pittsfield Fire Department
## 2:      4.71      Flow Station
## 3:      4.21      Pump Station (Sackett Brook)
## 4:      4.26      Pittsfield Municipal Airport
## 5:      4.31      Allendale Elementary School
## 6:      4.29      Residential House (Glory Dr.)
##
##      name_full
##      <char>
## 1:      Pittsfield Fire Department @ 74 Columbus Avenue, Pittsfield MA 01201
## 2:      Flow Station @ 694 Merrill Road, Pittsfield MA 01201
## 3:      Pump Station on Sackett Brook @ 233 East New Lenox Road
## 4:      Pittsfield Municipal Airport - National Weather Station @ 832 Tamarack Road, Pittsfield MA 01201
## 5:      Allendale Elementary School @ 180 Connecticut Avenue, Pittsfield MA 01201
## 6:      Residential house @ 75 Glory Drive
##
##      notes longitude latitude distancepeaker distanceairport
##      <char>      <num>      <num>      <num>      <num>
## 1:      -73.25484 42.45179      3065.1987      4178.3037
## 2:      -73.20072 42.46803      1926.7056      8806.0948
## 3:      Removed 11/20/24 -73.22515 42.42506      3529.2037      5407.9490
## 4:      -73.28892 42.42689      6685.3583      240.1949
## 5:      -73.22086 42.45855      329.5028      6843.3038
## 6:      -73.20349 42.42172      4035.6887      7201.1790
##      distancemajroads bearing_deg bearing_rad sin_bearing cos_bearing state
##      <num>      <num>      <num>      <num>      <num> <char>

```

```

## 1:      2.628948    260.39541    4.5447573   -0.9859827   -0.1668477    MA
## 2:      81.850571     47.88854    0.8358126    0.7418417    0.6705751    MA
## 3:     989.266375    189.46202    3.3067360   -0.1643938   -0.9863948    MA
## 4:     531.668678    240.66254    4.2003538   -0.8717492   -0.4899525    MA
## 5:     200.347678    316.45239    5.5231361   -0.6889571    0.7248021    MA
## 6:     897.846793    162.67021    2.8391307    0.2978713   -0.9546060    MA
##
##          peakername peakerid      date  hour grossload_mw
##          <char>      <int>    <char> <int>      <int>
## 1: Pittsfield Generating    50002 6/1/2024     0         NA
## 2: Pittsfield Generating    50002 6/1/2024     0         NA
## 3: Pittsfield Generating    50002 6/1/2024     0         NA
## 4: Pittsfield Generating    50002 6/1/2024     0         NA
## 5: Pittsfield Generating    50002 6/1/2024     0         NA
## 6: Pittsfield Generating    50002 6/1/2024     0         NA
##   steamload_1000lbsperhr so2mass_lbs co2mass_shorttons noxmass_lbs
##   <lgcl>      <lgcl>          <num>      <num>
## 1:      NA          NA          NA          NA
## 2:      NA          NA          NA          NA
## 3:      NA          NA          NA          NA
## 4:      NA          NA          NA          NA
## 5:      NA          NA          NA          NA
## 6:      NA          NA          NA          NA
##   heatinput_mmbtu operating  year month  day season  timeofday weekday_num
##   <num>      <int> <int> <int> <int> <char>    <char>      <int>
## 1:      NA          0  2024    6    1 Summer Late Night          0
## 2:      NA          0  2024    6    1 Summer Late Night          0
## 3:      NA          0  2024    6    1 Summer Late Night          0
## 4:      NA          0  2024    6    1 Summer Late Night          0
## 5:      NA          0  2024    6    1 Summer Late Night          0
## 6:      NA          0  2024    6    1 Summer Late Night          0
##
##   STATION REPORT_TYPE SOURCE HourlyDryBulbTemperature
##   <char>      <char> <int>          <int>
## 1: USW00014763    FM-15   343          53
## 2: USW00014763    FM-15   343          53
## 3: USW00014763    FM-15   343          53
## 4: USW00014763    FM-15   343          53
## 5: USW00014763    FM-15   343          53
## 6: USW00014763    FM-15   343          53
##   HourlyRelativeHumidity HourlyWindDirection HourlyWindSpeed fireworks  smoke
##   <int>          <char>          <char>    <char> <char>
## 1:      61          VRB          3          0    0
## 2:      61          VRB          3          0    0
## 3:      61          VRB          3          0    0
## 4:      61          VRB          3          0    0
## 5:      61          VRB          3          0    0
## 6:      61          VRB          3          0    0

```

```
allendale <- sensors_summer_2 |> dplyr::filter(name == "Allendale Elementary School")
```

```

#check for NAs and distribution
table(allendale$no2Conc1HourMean.status)

```

```

##
##          calibrated-ready
##          3          4413

```

```

table(allendale$pm2_5ConcMass1HourMean.status)

##
##           calibrated-ready
##           3           4413
table(allendale$pm10ConcMass1HourMean.qcAssessment)

##
##       valid
##       3 4413
sum(is.na(allendale$no2Conc1HourMean.value))

## [1] 3
sum(is.na(allendale$pm2_5ConcMass1HourMean.value))

## [1] 3
sum(is.na(allendale$pm10ConcMass1HourMean.raw))

## [1] 3
sum(is.na(allendale$HourlyDryBulbTemperature))

## [1] 123
sum(is.na(allendale$HourlyRelativeHumidity))

## [1] 147
sum(is.na(allendale$HourlyWindDirection))

## [1] 60
sum(is.na(allendale$HourlyWindSpeed))

## [1] 60

#all important variables have less than 5% NAs.
#only 0.06% (3/4416) of observations are missing for pollutant concentrations
#2.79% (123/4416) of observations are missing for temperature data
#3.33% (147/4416) of observations are missing for humidity data
#1.36% (60/4416) of observations are missing for hourly wind direction and wind speed

#check for NAs distribution
allendale$date[is.na(allendale$no2Conc1HourMean.value)]

## [1] "6/27/2024" "6/27/2024" "7/28/2025"
allendale$date[is.na(allendale$pm10ConcMass1HourMean.raw)]

## [1] "6/27/2024" "6/27/2024" "7/28/2025"
allendale$date[is.na(allendale$pm2_5ConcMass1HourMean.value)]

## [1] "6/27/2024" "6/27/2024" "7/28/2025"
allendale$date[is.na(allendale$HourlyDryBulbTemperature)]

## [1] "8/6/2024" "8/6/2024" "8/11/2024" "8/11/2024" "8/11/2024" "8/11/2024"
## [7] "8/11/2024" "8/12/2024" "8/12/2024" "8/12/2024" "8/12/2024" "8/12/2024"

```

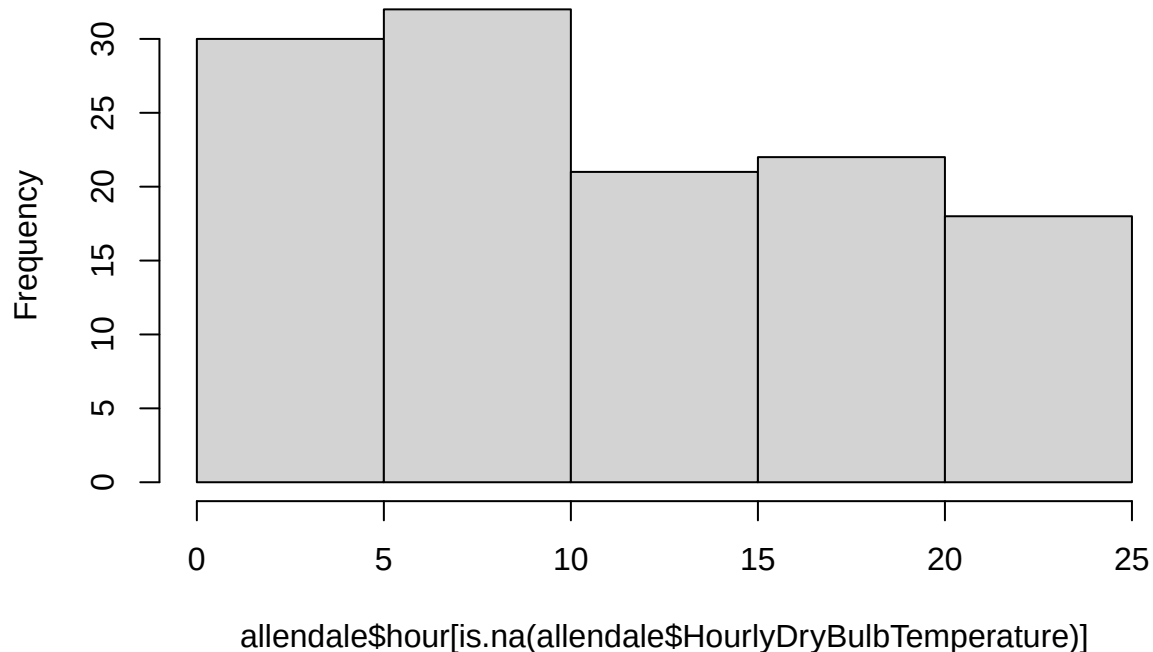
```
## [13] "8/12/2024" "8/12/2024" "8/12/2024" "8/12/2024" "8/12/2024" "8/12/2024"
## [19] "8/12/2024" "8/12/2024" "8/12/2024" "8/12/2024" "8/12/2024" "8/12/2024"
## [25] "8/12/2024" "8/12/2024" "8/12/2024" "8/12/2024" "8/12/2024" "8/12/2024"
## [31] "8/12/2024" "8/13/2024" "8/13/2024" "8/13/2024" "8/13/2024" "8/13/2024"
## [37] "8/13/2024" "8/13/2024" "8/13/2024" "8/19/2024" "8/19/2024" "8/19/2024"
## [43] "8/19/2024" "6/20/2025" "6/20/2025" "6/20/2025" "6/20/2025" "6/20/2025"
## [49] "7/6/2025" "7/8/2025" "7/8/2025" "7/8/2025" "7/8/2025" "7/8/2025"
## [55] "7/9/2025" "7/9/2025" "7/9/2025" "7/9/2025" "7/9/2025" "7/9/2025"
## [61] "7/9/2025" "7/9/2025" "7/9/2025" "7/9/2025" "7/15/2025" "7/15/2025"
## [67] "7/15/2025" "7/15/2025" "7/15/2025" "8/29/2025" "8/29/2025" "8/29/2025"
## [73] "8/29/2025" "8/29/2025" "8/29/2025" "8/30/2025" "8/30/2025" "8/30/2025"
## [79] "8/30/2025" "8/30/2025" "8/30/2025" "8/30/2025" "8/30/2025" "8/30/2025"
## [85] "8/30/2025" "8/30/2025" "8/30/2025" "8/30/2025" "8/30/2025" "8/30/2025"
## [91] "8/30/2025" "8/30/2025" "8/30/2025" "8/30/2025" "8/30/2025" "8/30/2025"
## [97] "8/30/2025" "8/30/2025" "8/30/2025" "8/31/2025" "8/31/2025" "8/31/2025"
## [103] "8/31/2025" "8/31/2025" "8/31/2025" "8/31/2025" "8/31/2025" "8/31/2025"
## [109] "8/31/2025" "8/31/2025" "8/31/2025" "8/31/2025" "8/31/2025" "8/31/2025"
## [115] "8/31/2025" "8/31/2025" "8/31/2025" "8/31/2025" "8/31/2025" "8/31/2025"
## [121] "8/31/2025" "8/31/2025" "8/31/2025"
```

```
allendale$hour[is.na(allendale$HourlyDryBulbTemperature)]
```

```
## [1] 9 10 19 20 21 22 23 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
## [26] 18 19 20 21 22 23 0 1 2 3 4 5 6 7 8 9 10 11 9 10 11 12 13 13 19
## [51] 20 21 22 23 0 1 2 3 4 5 6 7 8 9 7 8 9 10 11 18 19 20 21 22 23
## [76] 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 0
## [101] 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
```

```
hist(allendale$hour[is.na(allendale$HourlyDryBulbTemperature)])
```

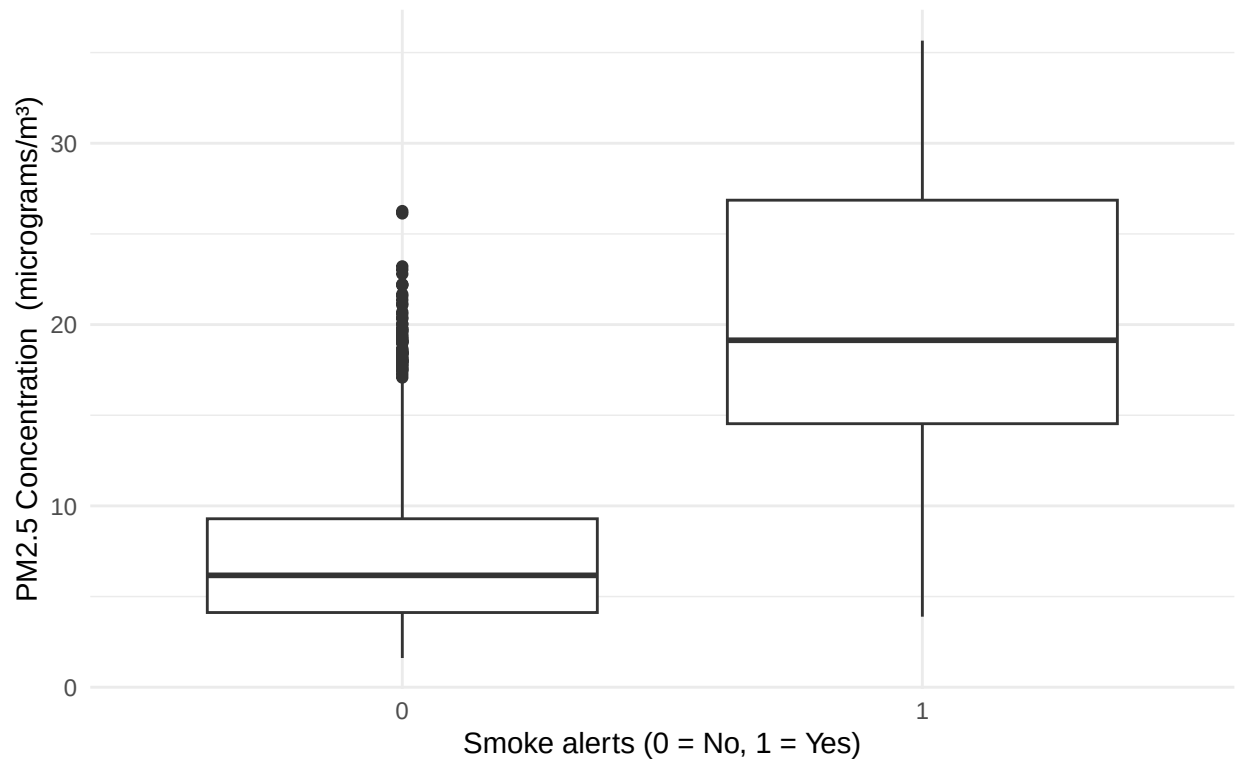
Histogram of allendale\$hour[is.na(allendale\$HourlyDryBulbTemperature)]



```
#boxplot comparing PM2.5 levels when the power plant is operating or not operating
pm25box <- ggplot(data = allendale, aes(x = factor(operating), y = pm2_5ConcMass1HourMean.value)) +
  geom_boxplot(na.rm = TRUE) +
  labs(title = "Mean hourly PM2.5 levels (µg/m3)",
        x = "Operating (0 = No, 1 = Yes)",
        y = "Concentration (µg/m3)") +
  theme_minimal() +
  theme(plot.title = element_text(size = 40),
        axis.title.x = element_text(size = 30),
        axis.title.y = element_text(size = 30),
        axis.text = element_text(size = 25))

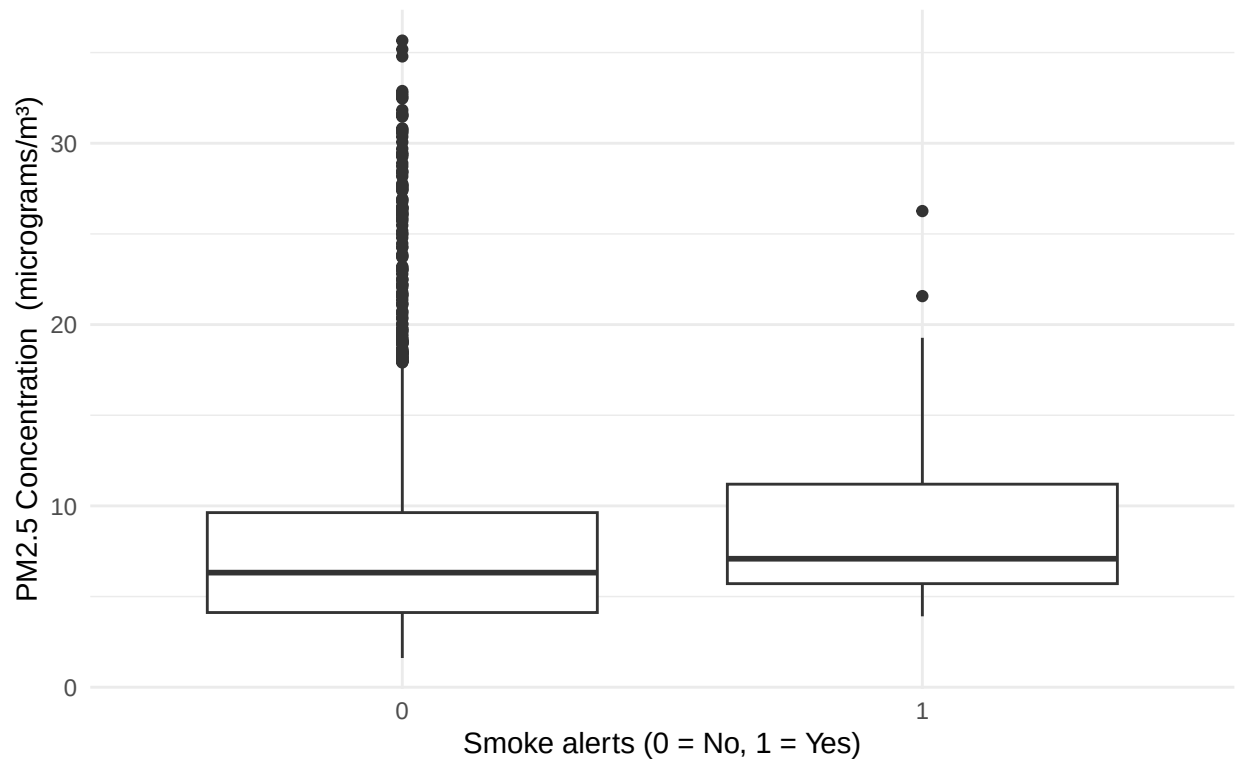
#factor by smoke
ggplot(data = allendale, aes(x = factor(smoke), y = pm2_5ConcMass1HourMean.value)) +
  geom_boxplot(na.rm = TRUE) +
  labs(title = "Mean hourly PM2.5 levels at Allendale Elementary School \nSummer 2024 and 2025",
        x = "Smoke alerts (0 = No, 1 = Yes)",
        y = "PM2.5 Concentration (micrograms/m³)") +
  theme_minimal()
```

Mean hourly PM2.5 levels at Allendale Elementary School Summer 2024 and 2025



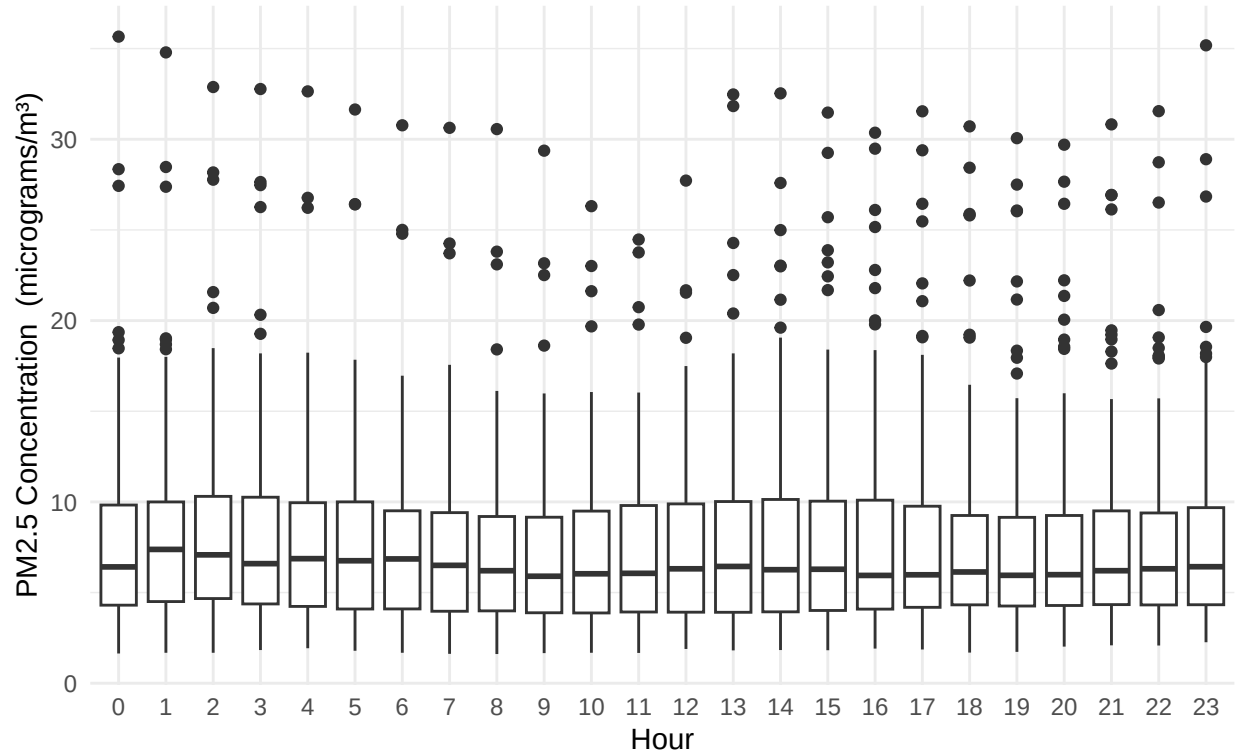
```
#factor by fireworks
ggplot(data = allendale, aes(x = factor(fireworks), y = pm2_5ConcMass1HourMean.value)) +
  geom_boxplot(na.rm = TRUE) +
  labs(title = "Mean hourly PM2.5 levels at Allendale Elementary School \nSummer 2024 and 2025",
        x = "Smoke alerts (0 = No, 1 = Yes)",
        y = "PM2.5 Concentration (micrograms/m³)") +
  theme_minimal()
```

Mean hourly PM2.5 levels at Allendale Elementary School
Summer 2024 and 2025



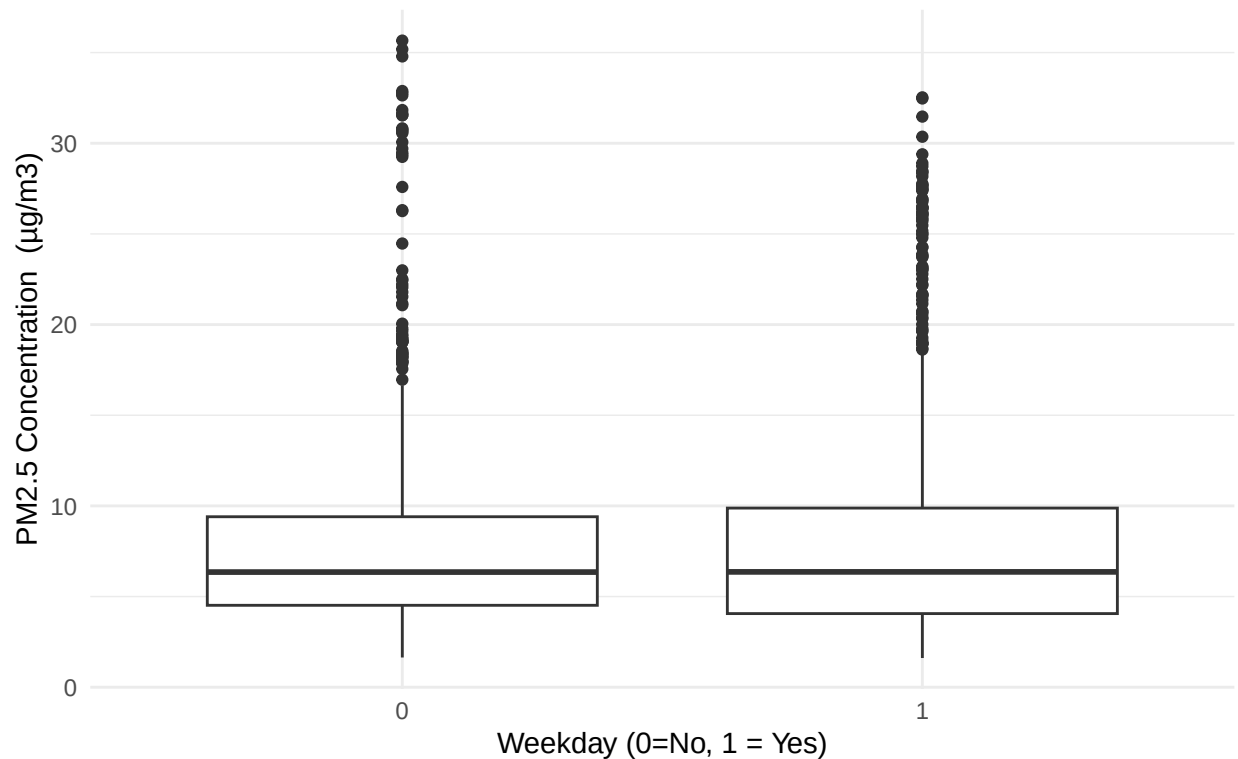
```
#factor by hour
ggplot(data = allendale, aes(x = factor(hour), y = pm2_5ConcMass1HourMean.value)) +
  geom_boxplot(na.rm = TRUE) +
  labs(title = "Mean hourly PM2.5 levels at Allendale Elementary School \nSummer 2024 and 2025",
        x = "Hour",
        y = "PM2.5 Concentration (micrograms/m³)") +
  theme_minimal()
```

Mean hourly PM2.5 levels at Allendale Elementary School Summer 2024 and 2025



```
#factor by weekday
ggplot(data = allendale, aes(x = factor(weekday_num), y = pm2_5ConcMass1HourMean.value)) +
  geom_boxplot(na.rm = TRUE) +
  labs(title = "Mean hourly PM2.5 levels at Allendale Elementary School \nSummer 2024 and 2025",
        x = "Weekday (0=No, 1 = Yes)",
        y = "PM2.5 Concentration (µg/m3)") +
  theme_minimal()
```

Mean hourly PM2.5 levels at Allendale Elementary School Summer 2024 and 2025



```
#scatterplot of PM2.5 over time
class(allendale$date)
```

```
## [1] "character"
```

```
allendale$date <- as.Date(allendale$date, format = "%m/%d/%Y")
class(allendale$date)
```

```
## [1] "Date"
```

```
allendale$year <- format(allendale$date, "%Y")
```

```
allendale$operating_f <- factor(allendale$operating,
                                levels = c(0, 1),
                                labels = c("Not operating", "Operating"))
```

```
#plot scatterplot of data faceted by year and colored by operational status
```

```
pm25date<- ggplot(data = allendale, aes(x = date,
                                          y = pm2_5ConcMass1HourMean.value,
                                          color = operating_f)) +
  geom_point(alpha = 0.4, size = 1) +
  facet_wrap(~year, scales = "free_x") +
  labs(title = "Mean hourly PM2.5 levels at Allendale Elementary",
       x = "Date",
       y = "Concentration ( $\mu\text{g}/\text{m}^3$ )",
       color = "Power plant status") +
  theme_minimal(base_size = 16)
```

```
ggsave("plots/pm25date.png", pm25date, width = 8, height = 6, units = "in")
```

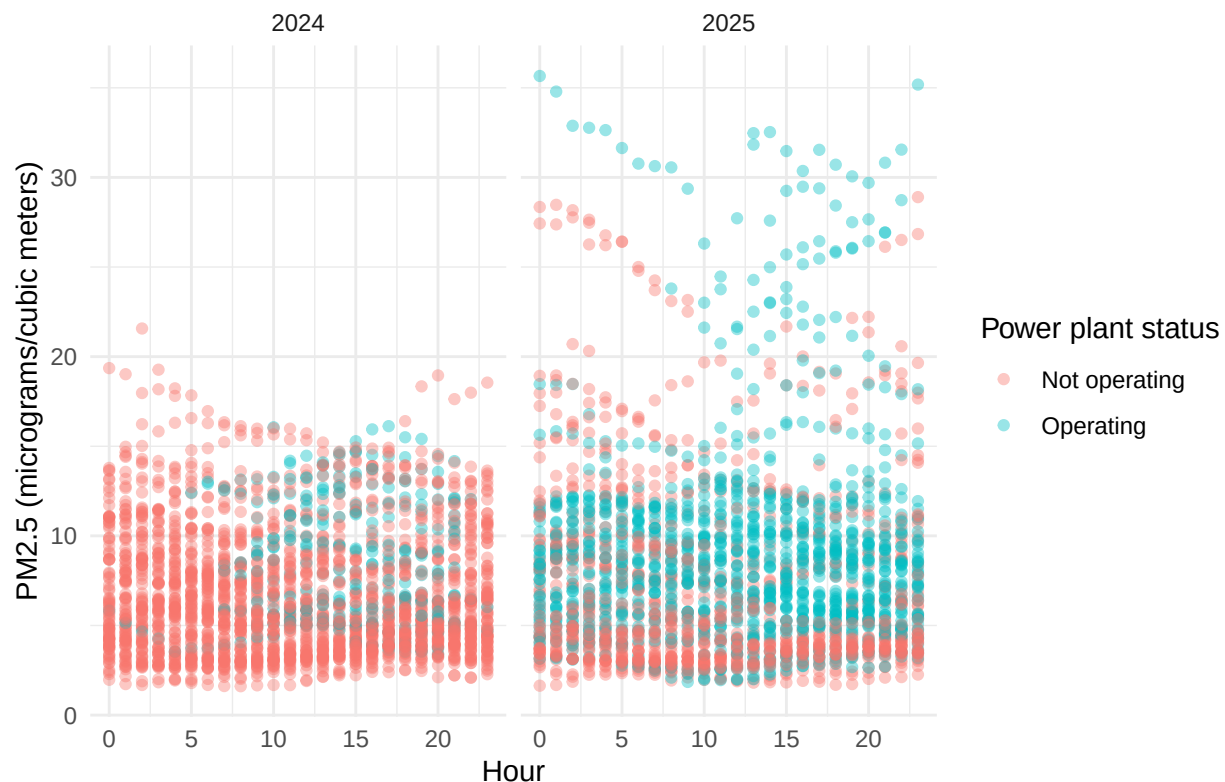
```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

```
#plot scatterplot of data faceted by hour and colored by operational status
#don't use, not very helpful
```

```
ggplot(data = allendale, aes(x = hour,
                             y = pm2_5ConcMass1HourMean.value,
                             color = operating_f)) +
  geom_point(alpha = 0.4) +
  facet_wrap(~year, scales = "free_x") +
  labs(title = "Hourly PM2.5 Levels at Allendale Elementary School",
       x = "Hour",
       y = "PM2.5 (micrograms/cubic meters)",
       color = "Power plant status") +
  theme_minimal()
```

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

Hourly PM2.5 Levels at Allendale Elementary School

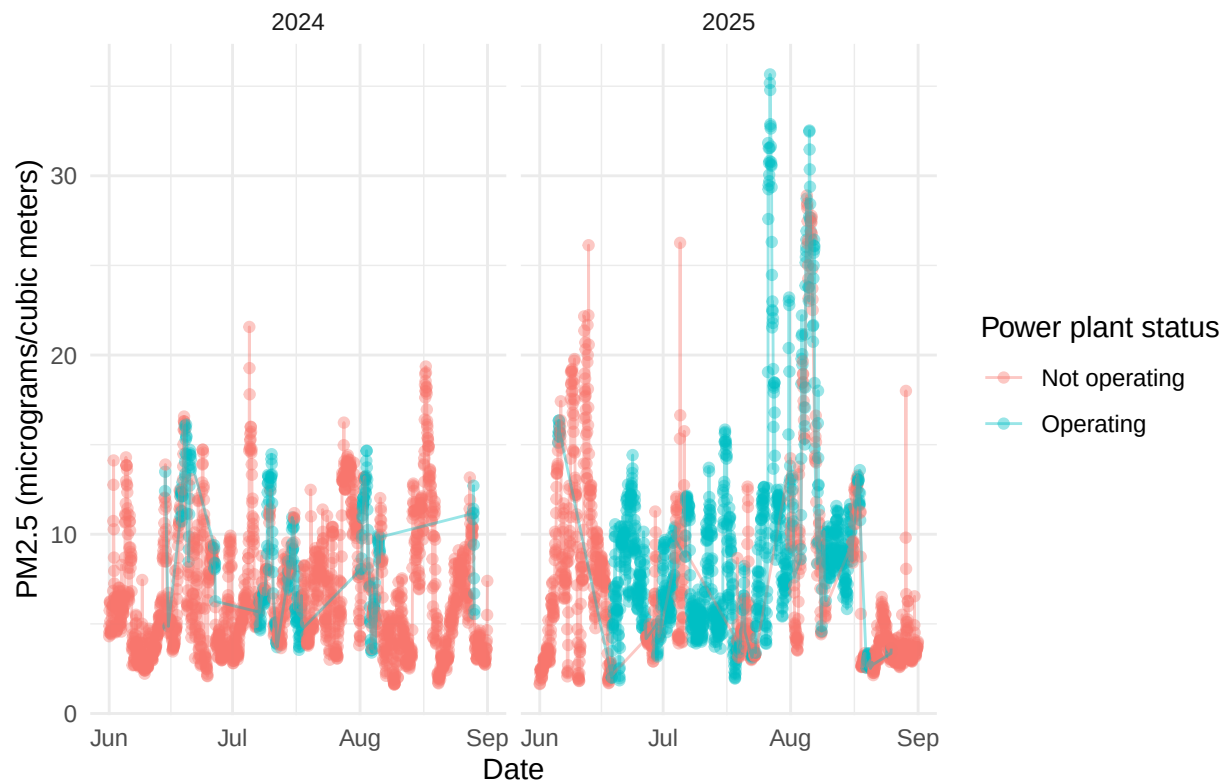


```
#plot scatterplot over datetime
allendale$datetime <- as.POSIXct(
  paste(allendale$date, allendale$hour),
  format = "%Y-%m-%d %H")
```

```
ggplot(data = allendale, aes(x = datetime,
                             y = pm2_5ConcMass1HourMean.value,
                             color = operating_f)) +
  geom_point(alpha = 0.4) +
  geom_line(alpha = 0.4) +
  facet_wrap(~year, scales = "free_x") +
  labs(title = "Hourly PM2.5 Levels at Allendale Elementary School",
       x = "Date",
       y = "PM2.5 (micrograms/cubic meters)",
       color = "Power plant status") +
  theme_minimal()
```

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

Hourly PM2.5 Levels at Allendale Elementary School



```
summary(allendale$pm2_5ConcMass1HourMean.value)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##  1.610   4.190   6.360   7.555   9.700  35.660         3
```

```
#allendale /> dplyr::group_by(operating)
```

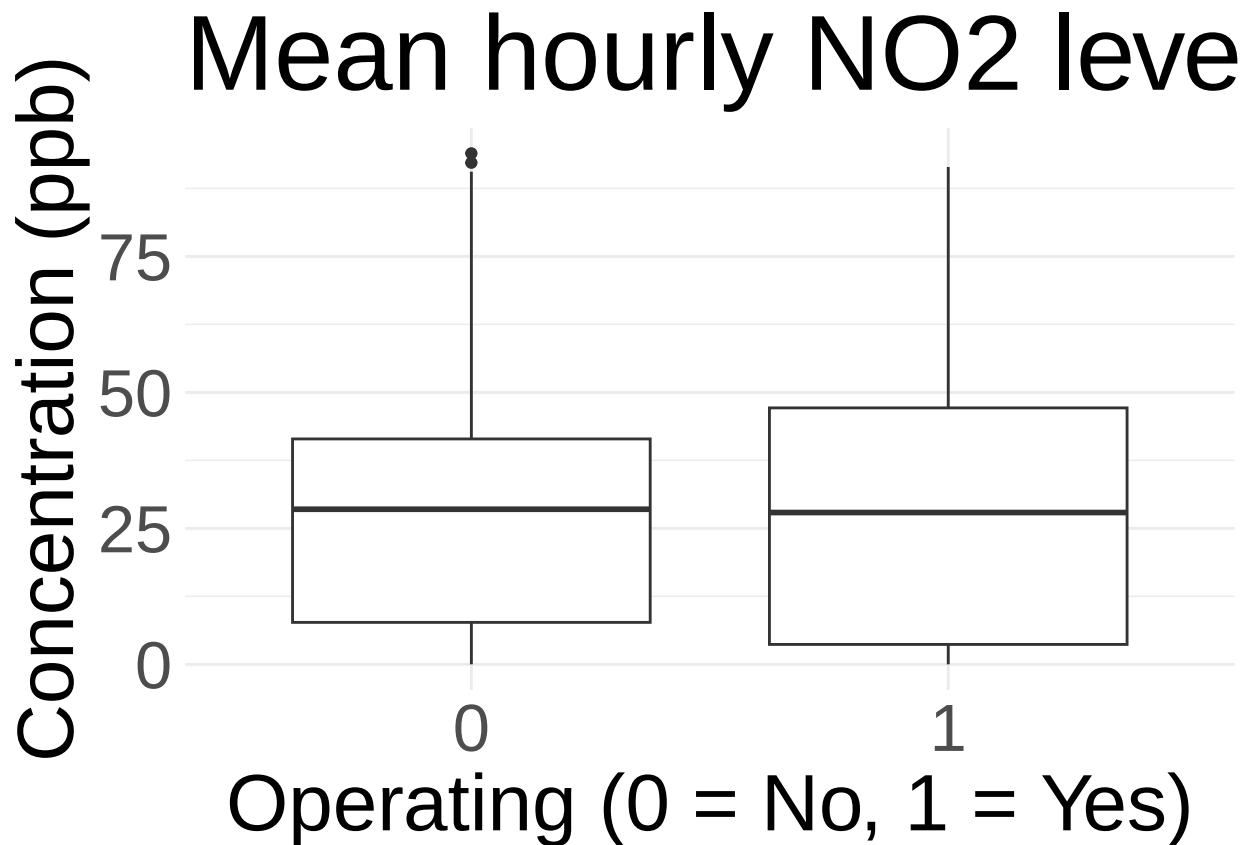
```
#boxplot comparing NOX levels when the power plant is operating or not operating
no2box <- ggplot(data = allendale, aes(x = factor(operating), y = no2Conc1HourMean.value)) +
  geom_boxplot(na.rm = TRUE) +
  labs(title = "Mean hourly N02 levels (ppb)",
       x = "Operating (0 = No, 1 = Yes)",
```

```

y = "Concentration (ppb)" +
theme_minimal() +
theme(plot.title = element_text(size = 40),
      axis.title.x = element_text(size = 30),
      axis.title.y = element_text(size = 30),
      axis.text = element_text(size = 25))

```

no2box



```

no2date<- ggplot(data = allendale, aes(x = date,
                                       y = no2Conc1HourMean.value,
                                       color = operating_f)) +
  geom_point(alpha = 0.4, size = 1) +
  facet_wrap(~year, scales = "free_x") +
  labs(title = "Mean hourly NO2 levels at Allendale Elementary",
       x = "Date",
       y = "Concentration (ppb)",
       color = "Power plant status") +
  theme_minimal(base_size = 16)

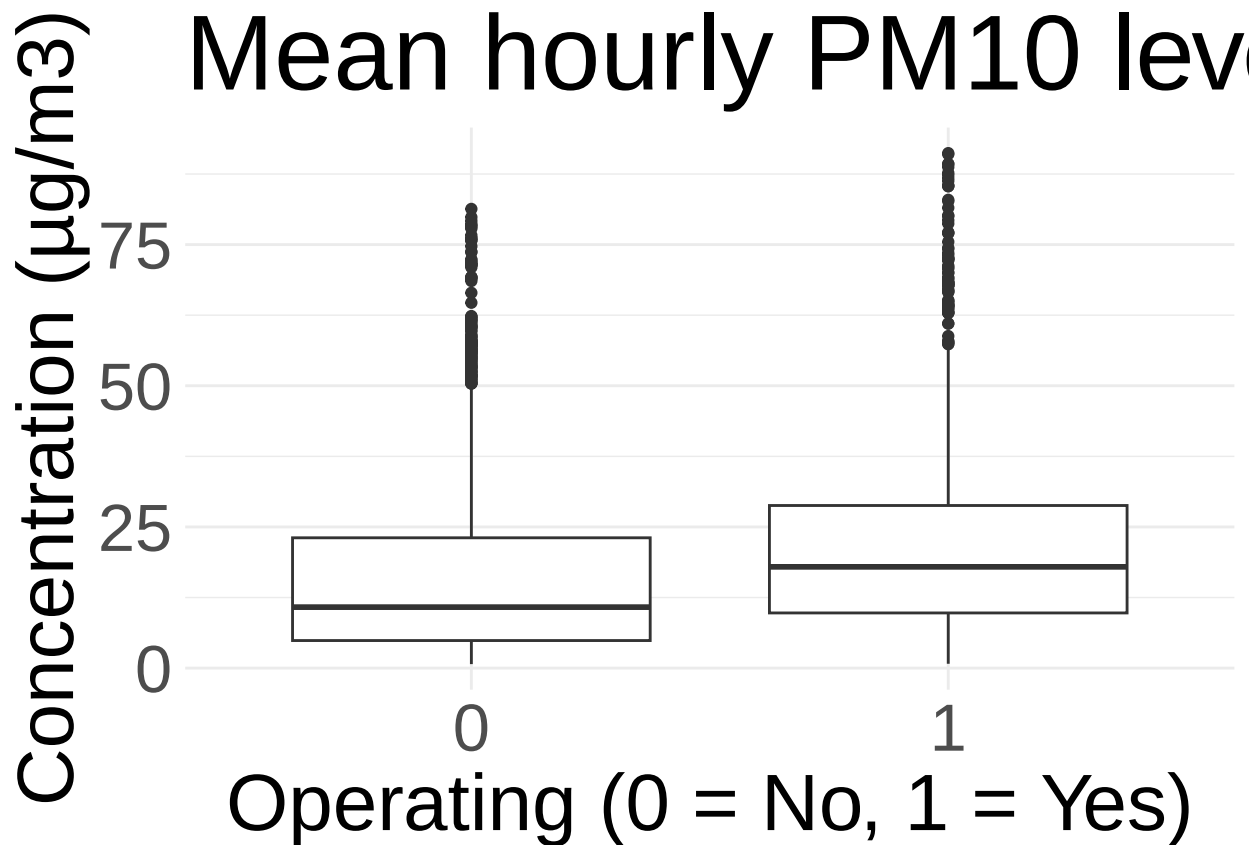
ggsave("plots/no2date.png", no2date, width = 8, height = 6, units = "in")

```

Warning: Removed 3 rows containing missing values or values outside the scale range

```
## (`geom_point()`).
#boxplot comparing NOX levels when the power plant is operating or not operating
pm10box <- ggplot(data = allendale, aes(x = factor(operating), y = allendale$pm10ConcMass1HourMean.raw)) +
  geom_boxplot(na.rm = TRUE) +
  labs(title = "Mean hourly PM10 levels (µg/m3)",
       x = "Operating (0 = No, 1 = Yes)",
       y = "Concentration (µg/m3)") +
  theme_minimal() +
  theme(plot.title = element_text(size = 40),
        axis.title.x = element_text(size = 30),
        axis.title.y = element_text(size = 30),
        axis.text = element_text(size = 25))
```

pm10box



```
operatingbox <- cowplot::plot_grid(pm25box, pm10box, no2box)
cowplot::save_plot("plots/operatingboxplot.png", operatingbox, base_height = 21, ncol = 1)

pm10date <- ggplot(data = allendale, aes(x = date,
                                         y = allendale$pm10ConcMass1HourMean.raw,
                                         color = operating_f)) +
  geom_point(alpha = 0.4, size = 1) +
  facet_wrap(~year, scales = "free_x") +
  labs(title = "Mean hourly PM10 levels at Allendale Elementary",
       x = "Date",
```

```

    y = "Concentration (µg/m3)",
    color = "Power plant status") +
  theme_minimal(base_size = 16)

ggsave("plots/pm10date.png", pm10date, width = 8, height = 6, units = "in")

## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).
#pm2.5 statistics at allendale
mean(allendale$pm2_5ConcMass1HourMean.value, na.rm = TRUE)

## [1] 7.555377
median(allendale$pm2_5ConcMass1HourMean.value, na.rm = TRUE)

## [1] 6.36
quantile(allendale$pm2_5ConcMass1HourMean.value, na.rm = TRUE)

##      0%    25%    50%    75%   100%
##  1.61  4.19  6.36  9.70 35.66
sd(allendale$pm2_5ConcMass1HourMean.value, na.rm = TRUE)

## [1] 4.653725
#pm10 statistics at allendale
mean(allendale$pm10ConcMass1HourMean.raw, na.rm = TRUE)

## [1] 17.88262
median(allendale$pm10ConcMass1HourMean.raw, na.rm = TRUE)

## [1] 13.08
quantile(allendale$pm10ConcMass1HourMean.raw, na.rm = TRUE)

##      0%    25%    50%    75%   100%
##  0.71  5.90 13.08 25.02 91.20
sd(allendale$pm10ConcMass1HourMean.raw, na.rm = TRUE)

## [1] 15.66642
#no2 statistics at allendale
mean(allendale$no2Conc1HourMean.value, na.rm = TRUE)

## [1] 27.35955
median(allendale$no2Conc1HourMean.value, na.rm = TRUE)

## [1] 28.39
quantile(allendale$no2Conc1HourMean.value, na.rm = TRUE)

##      0%    25%    50%    75%   100%
##  0.00  6.01 28.39 43.11 93.94
sd(allendale$no2Conc1HourMean.value, na.rm = TRUE)

## [1] 20.79638

```

```

#filter peaker power plant data when power plant is on vs. off
allendaleon <- allendale |>filter(operating == 1)
allendaleoff <- allendale |> filter(operating == 0)

#statistics for pm2.5 at allendale when peaker is on
mean(allendaleon$pm2_5ConcMass1HourMean.value, na.rm = TRUE)

## [1] 9.128465

median(allendaleon$pm2_5ConcMass1HourMean.value, na.rm = TRUE)

## [1] 8.31

quantile(allendaleon$pm2_5ConcMass1HourMean.value, na.rm = TRUE)

##      0%      25%      50%      75%     100%
## 1.8500  5.6500  8.3100 11.0475 35.6600

sd(allendaleon$pm2_5ConcMass1HourMean.value, na.rm = TRUE)

## [1] 5.294341

#statistics for pm10 at allendale when peaker is on
mean(allendaleon$pm10ConcMass1HourMean.raw, na.rm = TRUE)

## [1] 21.67515

median(allendaleon$pm10ConcMass1HourMean.raw, na.rm = TRUE)

## [1] 17.935

quantile(allendaleon$pm10ConcMass1HourMean.raw, na.rm = TRUE)

##      0%      25%      50%      75%     100%
## 0.7700  9.7725 17.9350 28.7975 91.2000

sd(allendaleon$pm10ConcMass1HourMean.raw, na.rm = TRUE)

## [1] 16.64125

#statistics for no2 at allendale when peaker is on
mean(allendaleon$no2Conc1HourMean.value, na.rm = TRUE)

## [1] 27.59556

median(allendaleon$no2Conc1HourMean.value, na.rm = TRUE)

## [1] 27.91

quantile(allendaleon$no2Conc1HourMean.value, na.rm = TRUE)

##      0%      25%      50%      75%     100%
## 0.0000  3.6625 27.9100 47.1450 91.4300

sd(allendaleon$no2Conc1HourMean.value, na.rm = TRUE)

## [1] 22.60325

#statistics for pm2.5 at allendale when peaker is off
mean(allendaleoff$pm2_5ConcMass1HourMean.value, na.rm = TRUE)

## [1] 6.829016

```

```

median(allendaleoff$pm2_5ConcMass1HourMean.value, na.rm = TRUE)

## [1] 5.56
quantile(allendaleoff$pm2_5ConcMass1HourMean.value, na.rm = TRUE)

##      0%    25%    50%    75%   100%
##  1.61  3.80  5.56  8.73 28.90
sd(allendaleoff$pm2_5ConcMass1HourMean.value, na.rm = TRUE)

## [1] 4.129414
#statistics for pm10 at allendale when peaker is off
mean(allendaleoff$pm10ConcMass1HourMean.raw, na.rm = TRUE)

## [1] 16.13145
median(allendaleoff$pm10ConcMass1HourMean.raw, na.rm = TRUE)

## [1] 10.8
quantile(allendaleoff$pm10ConcMass1HourMean.raw, na.rm = TRUE)

##      0%    25%    50%    75%   100%
##  0.710  4.885 10.800 23.080 81.320
sd(allendaleoff$pm10ConcMass1HourMean.raw, na.rm = TRUE)

## [1] 14.87515
#statistics for no2 at allendale when peaker is off
mean(allendaleoff$no2Conc1HourMean.value, na.rm = TRUE)

## [1] 27.25057
median(allendaleoff$no2Conc1HourMean.value, na.rm = TRUE)

## [1] 28.51
quantile(allendaleoff$no2Conc1HourMean.value, na.rm = TRUE)

##      0%    25%    50%    75%   100%
##  0.000  7.715 28.510 41.440 93.940
sd(allendaleoff$no2Conc1HourMean.value, na.rm = TRUE)

## [1] 19.9098
write.csv(allendale, "data/output/allendale_aq.csv")

#PART 2: RUNNING SIMPLE LINEAR MODELS FOR PM2.5, NO2, AND PM10
#check variable classes
allendale$hour <- as.factor(allendale$hour)

#run first linear model
pm25lm1 <- lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f +
              hour + weekday_num, data = allendale)

summary(pm25lm1)

##

```

```
## Call:
## lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f + hour +
##     weekday_num, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.944 -3.087 -1.162  1.902 25.738
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      7.46987    0.35035  21.321 < 2e-16 ***
## operating_f0operating  2.45185    0.15010  16.335 < 2e-16 ***
## hour1           0.34892    0.47156   0.740  0.45938
## hour2           0.47033    0.47220   0.996  0.31929
## hour3           0.32822    0.47156   0.696  0.48645
## hour4           0.13071    0.47220   0.277  0.78193
## hour5          -0.09359    0.47156  -0.198  0.84269
## hour6          -0.30892    0.47156  -0.655  0.51243
## hour7          -0.49808    0.47157  -1.056  0.29092
## hour8          -0.59246    0.47159  -1.256  0.20907
## hour9          -0.76185    0.47237  -1.613  0.10685
## hour10         -0.82103    0.47187  -1.740  0.08194 .
## hour11         -0.76593    0.47211  -1.622  0.10480
## hour12         -0.67555    0.47232  -1.430  0.15271
## hour13         -0.48687    0.47242  -1.031  0.30279
## hour14         -0.44804    0.47252  -0.948  0.34309
## hour15         -0.44141    0.47252  -0.934  0.35028
## hour16         -0.42140    0.47247  -0.892  0.37250
## hour17         -0.41623    0.47247  -0.881  0.37838
## hour18         -0.53754    0.47247  -1.138  0.25530
## hour19         -0.60851    0.47247  -1.288  0.19783
## hour20         -0.58622    0.47242  -1.241  0.21471
## hour21         -0.52725    0.47215  -1.117  0.26418
## hour22         -0.32482    0.47161  -0.689  0.49102
## hour23         -0.16644    0.47156  -0.353  0.72413
## weekday_num     -0.49123    0.15055  -3.263  0.00111 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.523 on 4387 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.06074,    Adjusted R-squared:  0.05539
## F-statistic: 11.35 on 25 and 4387 DF,  p-value: < 2.2e-16
```

#add in more variables

```
pm25lm2 <- lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f +
              hour + weekday_num + fireworks + smoke, data = allendale)

summary(pm25lm2)
```

```
##
## Call:
## lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f + hour +
##     weekday_num + fireworks + smoke, data = allendale)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18.3525  -2.7212  -0.9121   2.0892  19.9883
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.92775    0.29382   23.578 < 2e-16 ***
## operating_f0operating  1.88525    0.12638   14.917 < 2e-16 ***
## hour1            0.34584    0.39402    0.878  0.38013
## hour2            0.46505    0.39455    1.179  0.23859
## hour3            0.32514    0.39402    0.825  0.40931
## hour4            0.12544    0.39455    0.318  0.75056
## hour5           -0.09359    0.39401   -0.238  0.81226
## hour6           -0.30584    0.39402   -0.776  0.43766
## hour7           -0.48577    0.39402   -1.233  0.21770
## hour8           -0.57091    0.39404   -1.449  0.14745
## hour9           -0.71759    0.39469   -1.818  0.06912 .
## hour10          -0.75636    0.39428   -1.918  0.05513 .
## hour11          -0.67971    0.39448   -1.723  0.08495 .
## hour12          -0.57393    0.39467   -1.454  0.14596
## hour13          -0.37910    0.39475   -0.960  0.33693
## hour14          -0.33410    0.39483   -0.846  0.39750
## hour15          -0.32747    0.39483   -0.829  0.40693
## hour16          -0.31054    0.39479   -0.787  0.43156
## hour17          -0.30538    0.39479   -0.774  0.43926
## hour18          -0.42668    0.39479   -1.081  0.27986
## hour19          -0.49766    0.39479   -1.261  0.20753
## hour20          -0.47845    0.39475   -1.212  0.22557
## hour21          -0.43795    0.39452   -1.110  0.26702
## hour22          -0.29710    0.39406   -0.754  0.45092
## hour23          -0.15721    0.39402   -0.399  0.68993
## weekday_num      -0.34808    0.12663   -2.749  0.00601 **
## fireworks1       1.86464    0.30015    6.212  5.7e-10 ***
## smoke1          12.96440    0.29933   43.311 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.779 on 4385 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.3445, Adjusted R-squared:  0.3405
## F-statistic: 85.37 on 27 and 4385 DF, p-value: < 2.2e-16

#add in more variables

pm25lm3 <- lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f +
              hour + year + weekday_num + fireworks + smoke, data = allendale)

summary(pm25lm3)

##
## Call:
## lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f + hour +
##     year + weekday_num + fireworks + smoke, data = allendale)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18.3739  -2.7198  -0.9201   2.0838  20.0898
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.98978    0.29881  23.392 < 2e-16 ***
## operating_f0operating  1.95309    0.13970  13.981 < 2e-16 ***
## hour1            0.34621    0.39400   0.879  0.3796
## hour2            0.46572    0.39454   1.180  0.2379
## hour3            0.32551    0.39400   0.826  0.4088
## hour4            0.12610    0.39454   0.320  0.7493
## hour5           -0.09359    0.39400  -0.238  0.8123
## hour6           -0.30621    0.39400  -0.777  0.4371
## hour7           -0.48724    0.39401  -1.237  0.2163
## hour8           -0.57349    0.39404  -1.455  0.1456
## hour9           -0.72365    0.39472  -1.833  0.0668 .
## hour10          -0.76410    0.39432  -1.938  0.0527 .
## hour11          -0.69004    0.39457  -1.749  0.0804 .
## hour12          -0.58610    0.39480  -1.485  0.1377
## hour13          -0.39200    0.39490  -0.993  0.3209
## hour14          -0.34774    0.39500  -0.880  0.3787
## hour15          -0.34111    0.39500  -0.864  0.3879
## hour16          -0.32381    0.39495  -0.820  0.4123
## hour17          -0.31865    0.39495  -0.807  0.4198
## hour18          -0.43995    0.39495  -1.114  0.2654
## hour19          -0.51093    0.39495  -1.294  0.1958
## hour20          -0.49135    0.39490  -1.244  0.2135
## hour21          -0.44864    0.39462  -1.137  0.2556
## hour22          -0.30042    0.39406  -0.762  0.4459
## hour23          -0.15831    0.39401  -0.402  0.6879
## year2025        -0.14483    0.12709  -1.140  0.2545
## weekday_num     -0.35615    0.12682  -2.808  0.0050 **
## fireworks1       1.86431    0.30014   6.211 5.74e-10 ***
## smoke1          13.00018    0.30096  43.195 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.779 on 4384 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.3447, Adjusted R-squared:  0.3405
## F-statistic: 82.37 on 28 and 4384 DF, p-value: < 2.2e-16

#add in more variables
allendale$HourlyWindSpeed <- as.integer(allendale$HourlyWindSpeed)

pm25lm4 <- lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f +
              hour + year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature,
              data = allendale)

summary(pm25lm4)

##
## Call:
## lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f + hour +
```

```
##      year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature,
##      data = allendale)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -17.8082  -2.5017  -0.8639   1.8978  19.6957
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -2.106351   0.650584  -3.238  0.001215 **
## operating_f0operating  0.606888   0.160061   3.792  0.000152 ***
## hour1          0.418292   0.389687   1.073  0.283150
## hour2          0.614602   0.390305   1.575  0.115407
## hour3          0.564976   0.389954   1.449  0.147459
## hour4          0.418496   0.390631   1.071  0.284080
## hour5         -0.004988   0.389699  -0.013  0.989788
## hour6         -0.579786   0.390026  -1.487  0.137213
## hour7         -1.146988   0.392476  -2.922  0.003491 **
## hour8         -1.692257   0.396318  -4.270  2.00e-05 ***
## hour9         -2.094734   0.401630  -5.216  1.92e-07 ***
## hour10        -2.369284   0.403975  -5.865  4.83e-09 ***
## hour11        -2.441525   0.406265  -6.010  2.01e-09 ***
## hour12        -2.482422   0.407489  -6.092  1.21e-09 ***
## hour13        -2.355561   0.409016  -5.759  9.05e-09 ***
## hour14        -2.263663   0.407181  -5.559  2.87e-08 ***
## hour15        -2.188626   0.405889  -5.392  7.34e-08 ***
## hour16        -2.045221   0.403684  -5.066  4.23e-07 ***
## hour17        -1.798923   0.399964  -4.498  7.05e-06 ***
## hour18        -1.516913   0.395686  -3.834  0.000128 ***
## hour19        -1.169162   0.393391  -2.972  0.002975 **
## hour20        -0.884039   0.391954  -2.255  0.024155 *
## hour21        -0.653647   0.391088  -1.671  0.094725 .
## hour22        -0.528053   0.390538  -1.352  0.176410
## hour23        -0.274691   0.390292  -0.704  0.481591
## year2025       0.568879   0.131870   4.314  1.64e-05 ***
## weekday_num   -0.400881   0.125760  -3.188  0.001445 **
## fireworks1     0.967048   0.297992   3.245  0.001183 **
## smoke1        12.839870   0.293906  43.687  < 2e-16 ***
## HourlyDryBulbTemperature 0.147973   0.009312  15.891  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.686 on 4260 degrees of freedom
## (126 observations deleted due to missingness)
## Multiple R-squared:  0.3817, Adjusted R-squared:  0.3775
## F-statistic: 90.69 on 29 and 4260 DF, p-value: < 2.2e-16
```

```
#lm5
pm25lm5 <- lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f +
              hour + year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
              HourlyRelativeHumidity, data = allendale)

summary(pm25lm5)
```

```
##
```

```
## Call:
## lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f + hour +
##      year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
##      HourlyRelativeHumidity, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.997  -2.407  -0.724   1.872  21.228
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -8.423449   0.876135  -9.614 < 2e-16 ***
## operating_f0operating    0.256435   0.161609   1.587 0.112641
## hour1          0.418776   0.385888   1.085 0.277882
## hour2          0.622278   0.386502   1.610 0.107467
## hour3          0.518994   0.386172   1.344 0.179037
## hour4          0.353832   0.386864   0.915 0.360445
## hour5         -0.054871   0.385921  -0.142 0.886942
## hour6         -0.469577   0.386377  -1.215 0.224307
## hour7         -0.805953   0.389955  -2.067 0.038815 *
## hour8         -1.053522   0.397344  -2.651 0.008046 **
## hour9         -1.211046   0.406446  -2.980 0.002903 **
## hour10        -1.354027   0.410118  -3.302 0.000969 ***
## hour11        -1.304126   0.414936  -3.143 0.001684 **
## hour12        -1.247418   0.418515  -2.981 0.002893 **
## hour13        -1.086537   0.420730  -2.583 0.009842 **
## hour14        -1.020003   0.418404  -2.438 0.014816 *
## hour15        -1.011968   0.415930  -2.433 0.015014 *
## hour16        -0.925753   0.412632  -2.244 0.024914 *
## hour17        -0.818623   0.406053  -2.016 0.043858 *
## hour18        -0.766616   0.397902  -1.927 0.054091 .
## hour19        -0.656078   0.392506  -1.672 0.094694 .
## hour20        -0.531154   0.389575  -1.363 0.172823
## hour21        -0.402225   0.387948  -1.037 0.299888
## hour22        -0.379538   0.386968  -0.981 0.326748
## hour23        -0.214126   0.386531  -0.554 0.579630
## year2025       0.733490   0.131874   5.562 2.83e-08 ***
## weekday_num   -0.356226   0.124361  -2.864 0.004198 **
## fireworks1     0.846690   0.294481   2.875 0.004058 **
## smoke1        12.600318   0.291035  43.295 < 2e-16 ***
## HourlyDryBulbTemperature 0.176721   0.009664  18.286 < 2e-16 ***
## HourlyRelativeHumidity 0.051408   0.004704  10.929 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.64 on 4235 degrees of freedom
## (150 observations deleted due to missingness)
## Multiple R-squared:  0.3971, Adjusted R-squared:  0.3928
## F-statistic: 92.99 on 30 and 4235 DF, p-value: < 2.2e-16
```

```
#lm6
```

```
pm25lm6 <- lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f +
              hour + year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
```

```

HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)

summary(pm25lm6)

```

```

##
## Call:
## lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f + hour +
##      year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
##      HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.9875  -2.3744  -0.7638   1.8782  21.2951
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -7.878289   0.882086  -8.931 < 2e-16 ***
## operating_f0operating    0.262824   0.162071   1.622  0.10495
## hour1           0.415510   0.384768   1.080  0.28025
## hour2           0.634428   0.385390   1.646  0.09980 .
## hour3           0.531984   0.385063   1.382  0.16718
## hour4           0.370032   0.385760   0.959  0.33750
## hour5          -0.006939   0.385369  -0.018  0.98563
## hour6          -0.492843   0.386357  -1.276  0.20216
## hour7          -0.750878   0.389533  -1.928  0.05397 .
## hour8          -0.963557   0.396885  -2.428  0.01523 *
## hour9          -1.102745   0.407562  -2.706  0.00684 **
## hour10         -1.257238   0.413165  -3.043  0.00236 **
## hour11         -1.183126   0.418925  -2.824  0.00476 **
## hour12         -1.170330   0.423896  -2.761  0.00579 **
## hour13         -0.846271   0.425392  -1.989  0.04672 *
## hour14         -0.838259   0.420370  -1.994  0.04621 *
## hour15         -0.866922   0.416594  -2.081  0.03750 *
## hour16         -0.805551   0.413960  -1.946  0.05173 .
## hour17         -0.733887   0.406922  -1.804  0.07138 .
## hour18         -0.722617   0.397078  -1.820  0.06885 .
## hour19         -0.621766   0.392034  -1.586  0.11281
## hour20         -0.507782   0.389057  -1.305  0.19191
## hour21         -0.359531   0.387413  -0.928  0.35344
## hour22         -0.368193   0.385862  -0.954  0.34003
## hour23         -0.206079   0.385415  -0.535  0.59289
## year2025        0.724358   0.132106   5.483 4.42e-08 ***
## weekday_num    -0.360055   0.124658  -2.888  0.00389 **
## fireworks1      0.811250   0.294965   2.750  0.00598 **
## smoke1        12.455983   0.292749  42.548 < 2e-16 ***
## HourlyDryBulbTemperature  0.178498   0.009682  18.435 < 2e-16 ***
## HourlyRelativeHumidity   0.046113   0.004830   9.547 < 2e-16 ***
## HourlyWindSpeed  -0.069381   0.015504  -4.475 7.84e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.63 on 4196 degrees of freedom
## (188 observations deleted due to missingness)
## Multiple R-squared:  0.4013, Adjusted R-squared:  0.3969

```

```
## F-statistic: 90.72 on 31 and 4196 DF, p-value: < 2.2e-16
#run first linear model
no2lm1 <- lm(formula = no2Conc1HourMean.value ~ operating_f +
             hour + weekday_num, data = allendale)

summary(no2lm1)

##
## Call:
## lm(formula = no2Conc1HourMean.value ~ operating_f + hour + weekday_num,
##     data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -44.477  -9.328  -2.632   7.564  76.408
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      34.6682     1.1477  30.207 < 2e-16 ***
## operating_f0operating    5.3787     0.4917  10.939 < 2e-16 ***
## hour1              6.3312     1.5448   4.098 4.23e-05 ***
## hour2              8.1084     1.5469   5.242 1.66e-07 ***
## hour3              8.4521     1.5448   5.471 4.71e-08 ***
## hour4              7.5500     1.5469   4.881 1.09e-06 ***
## hour5              6.9169     1.5448   4.478 7.74e-06 ***
## hour6              5.2428     1.5448   3.394 0.000695 ***
## hour7              3.7054     1.5448   2.399 0.016497 *
## hour8              3.2171     1.5449   2.082 0.037359 *
## hour9              3.0185     1.5474   1.951 0.051159 .
## hour10             4.3289     1.5458   2.800 0.005125 **
## hour11             1.6108     1.5466   1.042 0.297689
## hour12            -6.0573     1.5473  -3.915 9.18e-05 ***
## hour13           -16.6410     1.5476 -10.753 < 2e-16 ***
## hour14           -24.9630     1.5479 -16.127 < 2e-16 ***
## hour15           -28.8327     1.5479 -18.627 < 2e-16 ***
## hour16           -29.6825     1.5477 -19.178 < 2e-16 ***
## hour17           -29.9124     1.5477 -19.326 < 2e-16 ***
## hour18           -30.1492     1.5477 -19.479 < 2e-16 ***
## hour19           -28.6032     1.5477 -18.481 < 2e-16 ***
## hour20           -25.0028     1.5476 -16.156 < 2e-16 ***
## hour21           -21.6746     1.5467 -14.013 < 2e-16 ***
## hour22           -16.3548     1.5449 -10.586 < 2e-16 ***
## hour23            -8.2083     1.5448  -5.314 1.13e-07 ***
## weekday_num        -0.4921     0.4932  -0.998 0.318373
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.82 on 4387 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.4953, Adjusted R-squared:  0.4924
## F-statistic: 172.2 on 25 and 4387 DF, p-value: < 2.2e-16
#add in more variables
```

```
no2lm2 <- lm(formula = no2Conc1HourMean.value ~ operating_f +
             hour + weekday_num + fireworks + smoke, data = allendale)

summary(no2lm2)

##
## Call:
## lm(formula = no2Conc1HourMean.value ~ operating_f + hour + weekday_num +
##     fireworks + smoke, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -44.185  -9.118  -2.655   7.594  76.696
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    34.0849    1.1476  29.702 < 2e-16 ***
## operating_f0operating    5.3596    0.4936  10.858 < 2e-16 ***
## hour1          6.3311    1.5389   4.114 3.96e-05 ***
## hour2          8.1068    1.5410   5.261 1.50e-07 ***
## hour3          8.4520    1.5389   5.492 4.19e-08 ***
## hour4          7.5484    1.5410   4.898 1.00e-06 ***
## hour5          6.9169    1.5389   4.495 7.14e-06 ***
## hour6          5.2429    1.5389   3.407 0.000663 ***
## hour7          3.7058    1.5389   2.408 0.016078 *
## hour8          3.2178    1.5390   2.091 0.036597 *
## hour9          3.0186    1.5415   1.958 0.050273 .
## hour10         4.3311    1.5399   2.813 0.004937 **
## hour11         1.6137    1.5407   1.047 0.294985
## hour12        -6.0538    1.5414  -3.927 8.72e-05 ***
## hour13       -16.6374    1.5417 -10.791 < 2e-16 ***
## hour14       -24.9592    1.5421 -16.185 < 2e-16 ***
## hour15       -28.8289    1.5421 -18.695 < 2e-16 ***
## hour16       -29.6787    1.5419 -19.248 < 2e-16 ***
## hour17       -29.9087    1.5419 -19.397 < 2e-16 ***
## hour18       -30.1455    1.5419 -19.551 < 2e-16 ***
## hour19       -28.5994    1.5419 -18.548 < 2e-16 ***
## hour20       -24.9992    1.5417 -16.215 < 2e-16 ***
## hour21       -21.6716    1.5408 -14.065 < 2e-16 ***
## hour22       -16.3539    1.5391 -10.626 < 2e-16 ***
## hour23        -8.2080    1.5389  -5.334 1.01e-07 ***
## weekday_num    -0.1810    0.4946  -0.366 0.714327
## fireworks1      6.1591    1.1723   5.254 1.56e-07 ***
## smoke1         3.5063    1.1691   2.999 0.002722 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.76 on 4385 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.4993, Adjusted R-squared:  0.4962
## F-statistic: 162 on 27 and 4385 DF, p-value: < 2.2e-16

#add in more variables
```

```
no2lm3 <- lm(formula = no2Conc1HourMean.value ~ operating_f +
             hour + year + weekday_num + fireworks + smoke, data = allendale)

summary(no2lm3)
```

```
##
## Call:
## lm(formula = no2Conc1HourMean.value ~ operating_f + hour + year +
##     weekday_num + fireworks + smoke, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -43.701  -9.184  -2.143   7.790  77.463
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    35.4095     1.1620  30.472 < 2e-16 ***
## operating_f0operating    6.8084     0.5433  12.532 < 2e-16 ***
## hour1           6.3390     1.5322   4.137 3.58e-05 ***
## hour2           8.1210     1.5343   5.293 1.26e-07 ***
## hour3           8.4599     1.5322   5.521 3.56e-08 ***
## hour4           7.5626     1.5343   4.929 8.57e-07 ***
## hour5           6.9169     1.5322   4.514 6.52e-06 ***
## hour6           5.2350     1.5322   3.417 0.000640 ***
## hour7           3.6743     1.5323   2.398 0.016528 *
## hour8           3.1627     1.5324   2.064 0.039081 *
## hour9           2.8892     1.5350   1.882 0.059872 .
## hour10          4.1657     1.5335   2.717 0.006623 **
## hour11          1.3932     1.5345   0.908 0.363946
## hour12         -6.3137     1.5353  -4.112 3.99e-05 ***
## hour13        -16.9130     1.5357 -11.013 < 2e-16 ***
## hour14        -25.2505     1.5361 -16.438 < 2e-16 ***
## hour15        -29.1202     1.5361 -18.957 < 2e-16 ***
## hour16        -29.9622     1.5359 -19.508 < 2e-16 ***
## hour17        -30.1921     1.5359 -19.658 < 2e-16 ***
## hour18        -30.4289     1.5359 -19.812 < 2e-16 ***
## hour19        -28.8829     1.5359 -18.805 < 2e-16 ***
## hour20        -25.2748     1.5357 -16.458 < 2e-16 ***
## hour21        -21.9000     1.5346 -14.271 < 2e-16 ***
## hour22        -16.4248     1.5325 -10.718 < 2e-16 ***
## hour23         -8.2316     1.5322  -5.372 8.18e-08 ***
## year2025        -3.0931     0.4942  -6.258 4.26e-10 ***
## weekday_num     -0.3535     0.4932  -0.717 0.473613
## fireworks1       6.1521     1.1672   5.271 1.42e-07 ***
## smoke1          4.2705     1.1704   3.649 0.000267 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.7 on 4384 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.5038, Adjusted R-squared:  0.5006
## F-statistic: 158.9 on 28 and 4384 DF, p-value: < 2.2e-16
```

```
#add in more variables
```

```
no2lm4 <- lm(formula = no2Conc1HourMean.value ~ operating_f +  
             hour + year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature,  
             data = allendale)
```

```
summary(no2lm4)
```

```
##
```

```
## Call:
```

```
## lm(formula = no2Conc1HourMean.value ~ operating_f + hour + year +  
##     weekday_num + fireworks + smoke + HourlyDryBulbTemperature,  
##     data = allendale)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max  
## -44.250  -9.250  -2.199   7.200  77.584
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)  
## (Intercept)      4.23796    2.55272   1.660 0.096953 .  
## operating_fOperating      2.26029    0.62804   3.599 0.000323 ***  
## hour1              6.52357    1.52903   4.266 2.03e-05 ***  
## hour2              8.50277    1.53145   5.552 2.99e-08 ***  
## hour3              9.04402    1.53007   5.911 3.67e-09 ***  
## hour4              8.37944    1.53273   5.467 4.84e-08 ***  
## hour5              7.10609    1.52907   4.647 3.46e-06 ***  
## hour6              4.05781    1.53036   2.652 0.008042 **  
## hour7              1.08185    1.53997   0.703 0.482399  
## hour8             -0.90866    1.55505  -0.584 0.559031  
## hour9             -2.06830    1.57589  -1.312 0.189434  
## hour10            -1.32688    1.58509  -0.837 0.402583  
## hour11            -4.82535    1.59407  -3.027 0.002484 **  
## hour12           -13.04783    1.59888  -8.161 4.35e-16 ***  
## hour13           -23.70140    1.60487 -14.768 < 2e-16 ***  
## hour14           -31.93488    1.59767 -19.988 < 2e-16 ***  
## hour15           -35.58127    1.59260 -22.342 < 2e-16 ***  
## hour16           -36.04860    1.58395 -22.759 < 2e-16 ***  
## hour17           -35.52332    1.56935 -22.636 < 2e-16 ***  
## hour18           -34.40417    1.55257 -22.160 < 2e-16 ***  
## hour19           -31.36931    1.54356 -20.323 < 2e-16 ***  
## hour20           -26.82627    1.53792 -17.443 < 2e-16 ***  
## hour21           -22.79483    1.53452 -14.855 < 2e-16 ***  
## hour22           -17.39614    1.53237 -11.352 < 2e-16 ***  
## hour23            -8.71068    1.53140  -5.688 1.37e-08 ***  
## year2025          -0.92826    0.51742  -1.794 0.072882 .  
## weekday_num       -0.41934    0.49345  -0.850 0.395474  
## fireworks1         3.30119    1.16924   2.823 0.004774 **  
## smoke1             3.97355    1.15321   3.446 0.000575 ***  
## HourlyDryBulbTemperature  0.50799    0.03654  13.904 < 2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Residual standard error: 14.46 on 4260 degrees of freedom
```

```

## (126 observations deleted due to missingness)
## Multiple R-squared: 0.5239, Adjusted R-squared: 0.5206
## F-statistic: 161.6 on 29 and 4260 DF, p-value: < 2.2e-16

#lm5
no2lm5 <- lm(formula = no2Conc1HourMean.value ~ operating_f +
             hour + year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
             HourlyRelativeHumidity, data = allendale)

summary(no2lm5)

##
## Call:
## lm(formula = no2Conc1HourMean.value ~ operating_f + hour + year +
##     weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
##     HourlyRelativeHumidity, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -38.343  -8.605  -1.103   6.563  75.081
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -58.26644     3.16091  -18.433 < 2e-16 ***
## operating_fOperating    -1.16499     0.58305   -1.998 0.045770 *
## hour1           6.52405     1.39220    4.686 2.87e-06 ***
## hour2           8.57837     1.39442    6.152 8.36e-10 ***
## hour3           8.69278     1.39323    6.239 4.83e-10 ***
## hour4           7.89088     1.39572    5.654 1.67e-08 ***
## hour5           6.76235     1.39232    4.857 1.24e-06 ***
## hour6           5.27110     1.39397    3.781 0.000158 ***
## hour7           4.54796     1.40688    3.233 0.001236 **
## hour8           5.36077     1.43353    3.740 0.000187 ***
## hour9           6.65488     1.46637    4.538 5.83e-06 ***
## hour10          8.66837     1.47962    5.859 5.02e-09 ***
## hour11          6.35483     1.49700    4.245 2.23e-05 ***
## hour12          -1.04167     1.50991   -0.690 0.490301
## hour13          -11.41857     1.51791   -7.523 6.52e-14 ***
## hour14          -19.83301     1.50951  -13.139 < 2e-16 ***
## hour15          -24.05665     1.50059  -16.031 < 2e-16 ***
## hour16          -25.05493     1.48869  -16.830 < 2e-16 ***
## hour17          -25.88696     1.46495  -17.671 < 2e-16 ***
## hour18          -26.97637     1.43555  -18.792 < 2e-16 ***
## hour19          -26.23091     1.41608  -18.524 < 2e-16 ***
## hour20          -23.28376     1.40551  -16.566 < 2e-16 ***
## hour21          -20.61839     1.39963  -14.731 < 2e-16 ***
## hour22          -16.22806     1.39610  -11.624 < 2e-16 ***
## hour23          -8.21448     1.39452   -5.891 4.15e-09 ***
## year2025         0.87244     0.47577    1.834 0.066765 .
## weekday_num     -0.11824     0.44867   -0.264 0.792142
## fireworks1       2.16230     1.06243    2.035 0.041889 *
## smoke1          1.64788     1.04999    1.569 0.116623
## HourlyDryBulbTemperature 0.79759     0.03487   22.876 < 2e-16 ***
## HourlyRelativeHumidity 0.50327     0.01697   29.655 < 2e-16 ***
## ---

```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.13 on 4235 degrees of freedom
## (150 observations deleted due to missingness)
## Multiple R-squared:  0.6079, Adjusted R-squared:  0.6052
## F-statistic: 218.9 on 30 and 4235 DF,  p-value: < 2.2e-16

#lm6

no2lm6 <- lm(formula = no2Conc1HourMean.value ~ operating_f +
             hour + year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
             HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)

summary(no2lm6)

##
## Call:
## lm(formula = no2Conc1HourMean.value ~ operating_f + hour + year +
##     weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
##     HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -37.620  -8.608  -1.092   6.492  76.228
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -57.41480     3.18550  -18.024  < 2e-16 ***
## operating_f0operating    -1.22690     0.58529   -2.096  0.036123 *
## hour1           6.51165     1.38952    4.686  2.87e-06 ***
## hour2           8.60509     1.39177    6.183  6.90e-10 ***
## hour3           8.70986     1.39059    6.263  4.14e-10 ***
## hour4           7.91191     1.39311    5.679  1.44e-08 ***
## hour5           6.78341     1.39169    4.874  1.13e-06 ***
## hour6           5.44492     1.39526    3.902  9.67e-05 ***
## hour7           4.83356     1.40673    3.436  0.000596 ***
## hour8           5.74401     1.43328    4.008  6.24e-05 ***
## hour9           6.95752     1.47184    4.727  2.35e-06 ***
## hour10          9.21512     1.49208    6.176  7.20e-10 ***
## hour11          6.63205     1.51288    4.384  1.20e-05 ***
## hour12          -0.24207     1.53083   -0.158  0.874360
## hour13         -10.77876     1.53623   -7.016  2.64e-12 ***
## hour14         -19.00516     1.51809  -12.519  < 2e-16 ***
## hour15         -23.40283     1.50446  -15.556  < 2e-16 ***
## hour16         -24.34399     1.49495  -16.284  < 2e-16 ***
## hour17         -25.33048     1.46953  -17.237  < 2e-16 ***
## hour18         -26.68493     1.43398  -18.609  < 2e-16 ***
## hour19         -26.05948     1.41576  -18.407  < 2e-16 ***
## hour20         -23.21769     1.40501  -16.525  < 2e-16 ***
## hour21         -20.49208     1.39908  -14.647  < 2e-16 ***
## hour22         -16.16197     1.39348  -11.598  < 2e-16 ***
## hour23          -8.17722     1.39186   -5.875  4.56e-09 ***
## year2025         0.90584     0.47708    1.899  0.057669 .
## weekday_num     -0.08964     0.45018   -0.199  0.842174
## fireworks1       2.00321     1.06522    1.881  0.060100 .
```

```

## smoke1          1.27254    1.05721    1.204 0.228783
## HourlyDryBulbTemperature 0.80045    0.03497   22.892 < 2e-16 ***
## HourlyRelativeHumidity   0.49686    0.01744   28.485 < 2e-16 ***
## HourlyWindSpeed         -0.18411    0.05599   -3.288 0.001016 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.11 on 4196 degrees of freedom
## (188 observations deleted due to missingness)
## Multiple R-squared:  0.61, Adjusted R-squared:  0.6072
## F-statistic: 211.7 on 31 and 4196 DF, p-value: < 2.2e-16

#run first linear model
pm10lm1 <- lm(formula = pm10ConcMass1HourMean.raw ~ operating_f +
              hour + weekday_num, data = allendale)

summary(pm10lm1)

##
## Call:
## lm(formula = pm10ConcMass1HourMean.raw ~ operating_f + hour +
##     weekday_num, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.656 -10.647  -4.831   6.318  65.812
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      17.0746     1.1854  14.404 < 2e-16 ***
## operating_f0operating    6.3836     0.5079  12.569 < 2e-16 ***
## hour1              1.8047     1.5955   1.131 0.258077
## hour2              2.8859     1.5977   1.806 0.070943 .
## hour3              2.9351     1.5955   1.840 0.065900 .
## hour4              2.9465     1.5977   1.844 0.065220 .
## hour5              2.8040     1.5955   1.757 0.078921 .
## hour6              2.3561     1.5955   1.477 0.139827
## hour7              2.0413     1.5956   1.279 0.200831
## hour8              2.0450     1.5956   1.282 0.200040
## hour9              1.8850     1.5983   1.179 0.238305
## hour10             1.8229     1.5966   1.142 0.253619
## hour11             1.9256     1.5974   1.205 0.228080
## hour12             1.3384     1.5981   0.837 0.402378
## hour13             0.5286     1.5984   0.331 0.740889
## hour14            -0.5866     1.5988  -0.367 0.713710
## hour15            -1.5159     1.5988  -0.948 0.343089
## hour16            -2.0258     1.5986  -1.267 0.205140
## hour17            -2.5589     1.5986  -1.601 0.109514
## hour18            -3.1632     1.5986  -1.979 0.047911 *
## hour19            -3.5642     1.5986  -2.230 0.025826 *
## hour20            -3.4470     1.5984  -2.156 0.031099 *
## hour21            -3.1426     1.5975  -1.967 0.049225 *
## hour22            -2.2413     1.5957  -1.405 0.160217
## hour23            -1.1951     1.5955  -0.749 0.453879
## weekday_num       -1.9374     0.5094  -3.803 0.000145 ***

```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.3 on 4387 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.05117,    Adjusted R-squared:  0.04576
## F-statistic: 9.463 on 25 and 4387 DF,  p-value: < 2.2e-16

#add in more variables

pm10lm2 <- lm(formula = pm10ConcMass1HourMean.raw ~ operating_f +
              hour + weekday_num + fireworks + smoke, data = allendale)

summary(pm10lm2)

##
## Call:
## lm(formula = pm10ConcMass1HourMean.raw ~ operating_f + hour +
##     weekday_num + fireworks + smoke, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -59.173  -9.542  -3.810   7.098  61.700
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      15.5104     1.0255   15.125 < 2e-16 ***
## operating_f0operating    4.5655     0.4411   10.350 < 2e-16 ***
## hour1              1.7948     1.3752    1.305 0.191904
## hour2              2.8694     1.3770    2.084 0.037239 *
## hour3              2.9252     1.3752    2.127 0.033462 *
## hour4              2.9300     1.3770    2.128 0.033412 *
## hour5              2.8040     1.3752    2.039 0.041508 *
## hour6              2.3660     1.3752    1.721 0.085406 .
## hour7              2.0809     1.3752    1.513 0.130315
## hour8              2.1142     1.3753    1.537 0.124290
## hour9              2.0275     1.3775    1.472 0.141143
## hour10             2.0304     1.3761    1.476 0.140149
## hour11             2.2023     1.3768    1.600 0.109760
## hour12             1.6644     1.3774    1.208 0.226969
## hour13             0.8744     1.3777    0.635 0.525654
## hour14            -0.2210     1.3780   -0.160 0.872593
## hour15            -1.1503     1.3780   -0.835 0.403883
## hour16            -1.6701     1.3779   -1.212 0.225541
## hour17            -2.2032     1.3779   -1.599 0.109893
## hour18            -2.8075     1.3779   -2.038 0.041653 *
## hour19            -3.2085     1.3779   -2.329 0.019924 *
## hour20            -3.1012     1.3777   -2.251 0.024436 *
## hour21            -2.8561     1.3769   -2.074 0.038112 *
## hour22            -2.1524     1.3753   -1.565 0.117654
## hour23            -1.1655     1.3752   -0.848 0.396758
## weekday_num       -1.5731     0.4419   -3.559 0.000376 ***
## fireworks1         4.0916     1.0476    3.906 9.53e-05 ***
## smoke1            40.6481     1.0447   38.909 < 2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.19 on 4385 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.2955, Adjusted R-squared:  0.2912
## F-statistic: 68.12 on 27 and 4385 DF,  p-value: < 2.2e-16
```

```
#add in more variables
```

```
pm10lm3 <- lm(formula = pm10ConcMass1HourMean.raw ~ operating_f +
               hour + weekday_num + fireworks + smoke + year, data = allendale)
```

```
summary(pm10lm3)
```

```
##
## Call:
## lm(formula = pm10ConcMass1HourMean.raw ~ operating_f + hour +
##     weekday_num + fireworks + smoke + year, data = allendale)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-59.019	-9.634	-3.713	6.971	60.973

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	15.0644	1.0424	14.452	< 2e-16 ***
operating_f0operating	4.0777	0.4873	8.367	< 2e-16 ***
hour1	1.7922	1.3744	1.304	0.192332
hour2	2.8646	1.3763	2.081	0.037459 *
hour3	2.9225	1.3744	2.126	0.033531 *
hour4	2.9252	1.3763	2.125	0.033610 *
hour5	2.8040	1.3744	2.040	0.041403 *
hour6	2.3687	1.3744	1.723	0.084896 .
hour7	2.0915	1.3745	1.522	0.128174
hour8	2.1328	1.3746	1.552	0.120836
hour9	2.0710	1.3769	1.504	0.132635
hour10	2.0861	1.3756	1.517	0.129460
hour11	2.2765	1.3764	1.654	0.098213 .
hour12	1.7519	1.3772	1.272	0.203412
hour13	0.9672	1.3776	0.702	0.482634
hour14	-0.1229	1.3779	-0.089	0.928932
hour15	-1.0523	1.3779	-0.764	0.445121
hour16	-1.5747	1.3778	-1.143	0.253135
hour17	-2.1078	1.3778	-1.530	0.126126
hour18	-2.7120	1.3778	-1.968	0.049078 *
hour19	-3.1131	1.3778	-2.260	0.023899 *
hour20	-3.0084	1.3776	-2.184	0.029025 *
hour21	-2.7792	1.3766	-2.019	0.043558 *
hour22	-2.1285	1.3747	-1.548	0.121596
hour23	-1.1575	1.3745	-0.842	0.399747
weekday_num	-1.5151	0.4424	-3.425	0.000622 ***
fireworks1	4.0939	1.0470	3.910	9.37e-05 ***
smoke1	40.3908	1.0499	38.472	< 2e-16 ***
year2025	1.0415	0.4433	2.349	0.018859 *

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.18 on 4384 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.2964, Adjusted R-squared:  0.2919
## F-statistic: 65.95 on 28 and 4384 DF,  p-value: < 2.2e-16

#add in more variables

pm10lm4 <- lm(formula = pm10ConcMass1HourMean.raw ~ operating_f +
              hour + year + weekday_num + fireworks + smoke + year + HourlyDryBulbTemperature,
              data = allendale)

summary(pm10lm4)

##
## Call:
## lm(formula = pm10ConcMass1HourMean.raw ~ operating_f + hour +
##     year + weekday_num + fireworks + smoke + year + HourlyDryBulbTemperature,
##     data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -57.541  -9.166  -3.494   6.494  63.105
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -9.44793     2.29488  -4.117 3.91e-05 ***
## operating_f0operating    0.34624     0.56460   0.613 0.539749
## hour1           1.99611     1.37459   1.452 0.146532
## hour2           3.28236     1.37677   2.384 0.017164 *
## hour3           3.59042     1.37553   2.610 0.009080 **
## hour4           3.75400     1.37791   2.724 0.006468 **
## hour5           3.08169     1.37463   2.242 0.025023 *
## hour6           1.66537     1.37578   1.210 0.226159
## hour7           0.35045     1.38442   0.253 0.800172
## hour8          -0.89222     1.39798  -0.638 0.523364
## hour9          -1.57094     1.41671  -1.109 0.267551
## hour10         -2.18050     1.42499  -1.530 0.126045
## hour11         -2.39741     1.43306  -1.673 0.094415 .
## hour12         -3.34626     1.43738  -2.328 0.019958 *
## hour13         -4.32860     1.44277  -3.000 0.002714 **
## hour14         -5.30166     1.43629  -3.691 0.000226 ***
## hour15         -6.04859     1.43174  -4.225 2.44e-05 ***
## hour16         -6.22952     1.42396  -4.375 1.24e-05 ***
## hour17         -6.11583     1.41084  -4.335 1.49e-05 ***
## hour18         -5.60990     1.39575  -4.019 5.94e-05 ***
## hour19         -4.86118     1.38765  -3.503 0.000464 ***
## hour20         -4.03166     1.38258  -2.916 0.003564 **
## hour21         -3.30082     1.37953  -2.393 0.016767 *
## hour22         -2.72221     1.37759  -1.976 0.048212 *
## hour23         -1.44884     1.37672  -1.052 0.292682
## year2025        3.10873     0.46516   6.683 2.64e-11 ***
## weekday_num    -1.75334     0.44361  -3.952 7.86e-05 ***
## fireworks1      1.53976     1.05114   1.465 0.143035
```

```

## smoke1          39.84733      1.03673  38.436 < 2e-16 ***
## HourlyDryBulbTemperature 0.40040      0.03285  12.190 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13 on 4260 degrees of freedom
## (126 observations deleted due to missingness)
## Multiple R-squared:  0.3216, Adjusted R-squared:  0.317
## F-statistic: 69.65 on 29 and 4260 DF, p-value: < 2.2e-16

#lm5
pm10lm5 <- lm(formula = pm10ConcMass1HourMean.raw ~ operating_f +
               hour + year + weekday_num + fireworks + smoke + year + HourlyDryBulbTemperature +
               HourlyRelativeHumidity, data = allendale)

summary(pm10lm5)

##
## Call:
## lm(formula = pm10ConcMass1HourMean.raw ~ operating_f + hour +
##     year + weekday_num + fireworks + smoke + year + HourlyDryBulbTemperature +
##     HourlyRelativeHumidity, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -58.362  -8.632  -3.112   6.158  62.246
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -37.14223     3.06867  -12.104 < 2e-16 ***
## operating_f0operating    -1.18587     0.56604   -2.095 0.036227 *
## hour1           1.99803     1.35158    1.478 0.139404
## hour2           3.31758     1.35373    2.451 0.014298 *
## hour3           3.40127     1.35257    2.515 0.011951 *
## hour4           3.49034     1.35500    2.576 0.010032 *
## hour5           2.88690     1.35169    2.136 0.032756 *
## hour6           2.17197     1.35329    1.605 0.108578
## hour7           1.86607     1.36582    1.366 0.171929
## hour8           1.91192     1.39170    1.374 0.169577
## hour9           2.31003     1.42358    1.623 0.104730
## hour10          2.26391     1.43644    1.576 0.115088
## hour11          2.57737     1.45332    1.773 0.076229 .
## hour12          2.04040     1.46586    1.392 0.164009
## hour13          1.19200     1.47361    0.809 0.418621
## hour14          0.11165     1.46546    0.076 0.939272
## hour15         -0.91072     1.45680   -0.625 0.531905
## hour16         -1.33577     1.44525   -0.924 0.355410
## hour17         -1.83006     1.42220   -1.287 0.198242
## hour18         -2.31559     1.39366   -1.662 0.096682 .
## hour19         -2.59575     1.37476   -1.888 0.059074 .
## hour20         -2.46625     1.36449   -1.807 0.070763 .
## hour21         -2.21128     1.35879   -1.627 0.103730
## hour22         -2.08054     1.35536   -1.535 0.124846
## hour23         -1.18618     1.35383   -0.876 0.380990
## year2025         3.86298     0.46189    8.363 < 2e-16 ***

```

```
## weekday_num          -1.58306    0.43557   -3.634 0.000282 ***
## fireworks1           1.02016    1.03143    0.989 0.322680
## smoke1               38.80645    1.01935   38.070 < 2e-16 ***
## HourlyDryBulbTemperature 0.52754    0.03385   15.585 < 2e-16 ***
## HourlyRelativeHumidity 0.22415    0.01648   13.605 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.75 on 4235 degrees of freedom
## (150 observations deleted due to missingness)
## Multiple R-squared:  0.3485, Adjusted R-squared:  0.3439
## F-statistic: 75.53 on 30 and 4235 DF, p-value: < 2.2e-16
```

```
#lm6
```

```
pm10lm6 <- lm(formula = pm10ConcMass1HourMean.raw ~ operating_f +
              hour + year + weekday_num + fireworks + smoke + year + HourlyDryBulbTemperature +
              HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)
```

```
summary(pm10lm6)
```

```
##
## Call:
## lm(formula = pm10ConcMass1HourMean.raw ~ operating_f + hour +
##     year + weekday_num + fireworks + smoke + year + HourlyDryBulbTemperature +
##     HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -58.333  -8.514  -3.076   6.238  62.493
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -35.29485     3.08209  -11.452 < 2e-16 ***
## operating_f0operating    -1.14222     0.56629   -2.017  0.04376 *
## hour1           1.98742     1.34442    1.478  0.13941
## hour2           3.36050     1.34659    2.496  0.01261 *
## hour3           3.44794     1.34545    2.563  0.01042 *
## hour4           3.54853     1.34788    2.633  0.00850 **
## hour5           3.05280     1.34652    2.267  0.02343 *
## hour6           2.04610     1.34997    1.516  0.12968
## hour7           2.05384     1.36107    1.509  0.13138
## hour8           2.20998     1.38676    1.594  0.11109
## hour9           2.67005     1.42406    1.875  0.06087 .
## hour10          2.41487     1.44364    1.673  0.09445 .
## hour11          2.94127     1.46376    2.009  0.04456 *
## hour12          2.20434     1.48113    1.488  0.13675
## hour13          1.95552     1.48636    1.316  0.18837
## hour14          0.73091     1.46881    0.498  0.61878
## hour15         -0.43537     1.45562   -0.299  0.76488
## hour16         -0.96101     1.44642   -0.664  0.50647
## hour17         -1.56060     1.42182   -1.098  0.27244
## hour18         -2.18088     1.38743   -1.572  0.11605
## hour19         -2.49973     1.36980   -1.825  0.06809 .
## hour20         -2.40353     1.35940   -1.768  0.07712 .
```

```

## hour21                -2.07472    1.35366   -1.533   0.12543
## hour22                -2.04536    1.34824   -1.517   0.12933
## hour23                -1.16025    1.34668   -0.862   0.38897
## year2025              3.80594     0.46159    8.245 < 2e-16 ***
## weekday_num          -1.58382     0.43557   -3.636   0.00028 ***
## fireworks1            0.90116     1.03064    0.874   0.38196
## smoke1               38.31972     1.02289   37.462 < 2e-16 ***
## HourlyDryBulbTemperature 0.53448     0.03383   15.798 < 2e-16 ***
## HourlyRelativeHumidity 0.20566     0.01688   12.186 < 2e-16 ***
## HourlyWindSpeed       -0.23881     0.05417   -4.408 1.07e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.68 on 4196 degrees of freedom
## (188 observations deleted due to missingness)
## Multiple R-squared:  0.3531, Adjusted R-squared:  0.3484
## F-statistic: 73.9 on 31 and 4196 DF, p-value: < 2.2e-16

#compare linear model results without weather data (lm3) and with weather data (lm6)
summary(pm25lm3)

##
## Call:
## lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f + hour +
##      year + weekday_num + fireworks + smoke, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18.3739  -2.7198  -0.9201   2.0838  20.0898
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.98978    0.29881  23.392 < 2e-16 ***
## operating_f0operating 1.95309    0.13970  13.981 < 2e-16 ***
## hour1            0.34621    0.39400   0.879  0.3796
## hour2            0.46572    0.39454   1.180  0.2379
## hour3            0.32551    0.39400   0.826  0.4088
## hour4            0.12610    0.39454   0.320  0.7493
## hour5           -0.09359    0.39400  -0.238  0.8123
## hour6           -0.30621    0.39400  -0.777  0.4371
## hour7           -0.48724    0.39401  -1.237  0.2163
## hour8           -0.57349    0.39404  -1.455  0.1456
## hour9           -0.72365    0.39472  -1.833  0.0668 .
## hour10          -0.76410    0.39432  -1.938  0.0527 .
## hour11          -0.69004    0.39457  -1.749  0.0804 .
## hour12          -0.58610    0.39480  -1.485  0.1377
## hour13          -0.39200    0.39490  -0.993  0.3209
## hour14          -0.34774    0.39500  -0.880  0.3787
## hour15          -0.34111    0.39500  -0.864  0.3879
## hour16          -0.32381    0.39495  -0.820  0.4123
## hour17          -0.31865    0.39495  -0.807  0.4198
## hour18          -0.43995    0.39495  -1.114  0.2654
## hour19          -0.51093    0.39495  -1.294  0.1958
## hour20          -0.49135    0.39490  -1.244  0.2135
## hour21          -0.44864    0.39462  -1.137  0.2556

```

```
## hour22          -0.30042    0.39406  -0.762    0.4459
## hour23          -0.15831    0.39401  -0.402    0.6879
## year2025        -0.14483    0.12709  -1.140    0.2545
## weekday_num     -0.35615    0.12682  -2.808    0.0050 **
## fireworks1       1.86431    0.30014    6.211 5.74e-10 ***
## smoke1          13.00018    0.30096   43.195 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.779 on 4384 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.3447, Adjusted R-squared:  0.3405
## F-statistic: 82.37 on 28 and 4384 DF, p-value: < 2.2e-16
```

```
summary(pm25lm6)
```

```
##
## Call:
## lm(formula = pm2_5ConcMass1HourMean.value ~ operating_f + hour +
##      year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
##      HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.9875  -2.3744  -0.7638   1.8782  21.2951
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -7.878289   0.882086  -8.931 < 2e-16 ***
## operating_f0operating    0.262824   0.162071   1.622  0.10495
## hour1           0.415510   0.384768   1.080  0.28025
## hour2           0.634428   0.385390   1.646  0.09980 .
## hour3           0.531984   0.385063   1.382  0.16718
## hour4           0.370032   0.385760   0.959  0.33750
## hour5          -0.006939   0.385369  -0.018  0.98563
## hour6          -0.492843   0.386357  -1.276  0.20216
## hour7          -0.750878   0.389533  -1.928  0.05397 .
## hour8          -0.963557   0.396885  -2.428  0.01523 *
## hour9          -1.102745   0.407562  -2.706  0.00684 **
## hour10         -1.257238   0.413165  -3.043  0.00236 **
## hour11         -1.183126   0.418925  -2.824  0.00476 **
## hour12         -1.170330   0.423896  -2.761  0.00579 **
## hour13         -0.846271   0.425392  -1.989  0.04672 *
## hour14         -0.838259   0.420370  -1.994  0.04621 *
## hour15         -0.866922   0.416594  -2.081  0.03750 *
## hour16         -0.805551   0.413960  -1.946  0.05173 .
## hour17         -0.733887   0.406922  -1.804  0.07138 .
## hour18         -0.722617   0.397078  -1.820  0.06885 .
## hour19         -0.621766   0.392034  -1.586  0.11281
## hour20         -0.507782   0.389057  -1.305  0.19191
## hour21         -0.359531   0.387413  -0.928  0.35344
## hour22         -0.368193   0.385862  -0.954  0.34003
## hour23         -0.206079   0.385415  -0.535  0.59289
## year2025        0.724358   0.132106   5.483 4.42e-08 ***
## weekday_num    -0.360055   0.124658  -2.888  0.00389 **
```

```
## fireworks1          0.811250    0.294965    2.750  0.00598 **
## smoke1              12.455983    0.292749   42.548  < 2e-16 ***
## HourlyDryBulbTemperature 0.178498    0.009682   18.435  < 2e-16 ***
## HourlyRelativeHumidity  0.046113    0.004830    9.547  < 2e-16 ***
## HourlyWindSpeed       -0.069381    0.015504   -4.475  7.84e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.63 on 4196 degrees of freedom
## (188 observations deleted due to missingness)
## Multiple R-squared:  0.4013, Adjusted R-squared:  0.3969
## F-statistic: 90.72 on 31 and 4196 DF, p-value: < 2.2e-16
```

```
summary(pm10lm3)
```

```
##
## Call:
## lm(formula = pm10ConcMass1HourMean.raw ~ operating_f + hour +
##     weekday_num + fireworks + smoke + year, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -59.019  -9.634  -3.713   6.971  60.973
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      15.0644      1.0424  14.452  < 2e-16 ***
## operating_fOperating  4.0777      0.4873   8.367  < 2e-16 ***
## hour1              1.7922      1.3744   1.304  0.192332
## hour2              2.8646      1.3763   2.081  0.037459 *
## hour3              2.9225      1.3744   2.126  0.033531 *
## hour4              2.9252      1.3763   2.125  0.033610 *
## hour5              2.8040      1.3744   2.040  0.041403 *
## hour6              2.3687      1.3744   1.723  0.084896 .
## hour7              2.0915      1.3745   1.522  0.128174
## hour8              2.1328      1.3746   1.552  0.120836
## hour9              2.0710      1.3769   1.504  0.132635
## hour10             2.0861      1.3756   1.517  0.129460
## hour11             2.2765      1.3764   1.654  0.098213 .
## hour12             1.7519      1.3772   1.272  0.203412
## hour13             0.9672      1.3776   0.702  0.482634
## hour14            -0.1229      1.3779  -0.089  0.928932
## hour15            -1.0523      1.3779  -0.764  0.445121
## hour16            -1.5747      1.3778  -1.143  0.253135
## hour17            -2.1078      1.3778  -1.530  0.126126
## hour18            -2.7120      1.3778  -1.968  0.049078 *
## hour19            -3.1131      1.3778  -2.260  0.023899 *
## hour20            -3.0084      1.3776  -2.184  0.029025 *
## hour21            -2.7792      1.3766  -2.019  0.043558 *
## hour22            -2.1285      1.3747  -1.548  0.121596
## hour23            -1.1575      1.3745  -0.842  0.399747
## weekday_num       -1.5151      0.4424  -3.425  0.000622 ***
## fireworks1         4.0939      1.0470   3.910  9.37e-05 ***
## smoke1            40.3908      1.0499  38.472  < 2e-16 ***
## year2025           1.0415      0.4433   2.349  0.018859 *
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.18 on 4384 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.2964, Adjusted R-squared:  0.2919
## F-statistic: 65.95 on 28 and 4384 DF,  p-value: < 2.2e-16
```

```
summary(pm10lm6)
```

```
##
## Call:
## lm(formula = pm10ConcMass1HourMean.raw ~ operating_f + hour +
##      year + weekday_num + fireworks + smoke + year + HourlyDryBulbTemperature +
##      HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -58.333  -8.514  -3.076   6.238  62.493
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -35.29485     3.08209  -11.452 < 2e-16 ***
## operating_fOperating    -1.14222     0.56629   -2.017  0.04376 *
## hour1           1.98742     1.34442    1.478  0.13941
## hour2           3.36050     1.34659    2.496  0.01261 *
## hour3           3.44794     1.34545    2.563  0.01042 *
## hour4           3.54853     1.34788    2.633  0.00850 **
## hour5           3.05280     1.34652    2.267  0.02343 *
## hour6           2.04610     1.34997    1.516  0.12968
## hour7           2.05384     1.36107    1.509  0.13138
## hour8           2.20998     1.38676    1.594  0.11109
## hour9           2.67005     1.42406    1.875  0.06087 .
## hour10          2.41487     1.44364    1.673  0.09445 .
## hour11          2.94127     1.46376    2.009  0.04456 *
## hour12          2.20434     1.48113    1.488  0.13675
## hour13          1.95552     1.48636    1.316  0.18837
## hour14          0.73091     1.46881    0.498  0.61878
## hour15         -0.43537     1.45562   -0.299  0.76488
## hour16         -0.96101     1.44642   -0.664  0.50647
## hour17         -1.56060     1.42182   -1.098  0.27244
## hour18         -2.18088     1.38743   -1.572  0.11605
## hour19         -2.49973     1.36980   -1.825  0.06809 .
## hour20         -2.40353     1.35940   -1.768  0.07712 .
## hour21         -2.07472     1.35366   -1.533  0.12543
## hour22         -2.04536     1.34824   -1.517  0.12933
## hour23         -1.16025     1.34668   -0.862  0.38897
## year2025         3.80594     0.46159    8.245 < 2e-16 ***
## weekday_num     -1.58382     0.43557   -3.636  0.00028 ***
## fireworks1       0.90116     1.03064    0.874  0.38196
## smoke1          38.31972     1.02289   37.462 < 2e-16 ***
## HourlyDryBulbTemperature  0.53448     0.03383   15.798 < 2e-16 ***
## HourlyRelativeHumidity   0.20566     0.01688   12.186 < 2e-16 ***
## HourlyWindSpeed    -0.23881     0.05417   -4.408 1.07e-05 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.68 on 4196 degrees of freedom
## (188 observations deleted due to missingness)
## Multiple R-squared:  0.3531, Adjusted R-squared:  0.3484
## F-statistic: 73.9 on 31 and 4196 DF, p-value: < 2.2e-16
```

```
summary(no2lm3)
```

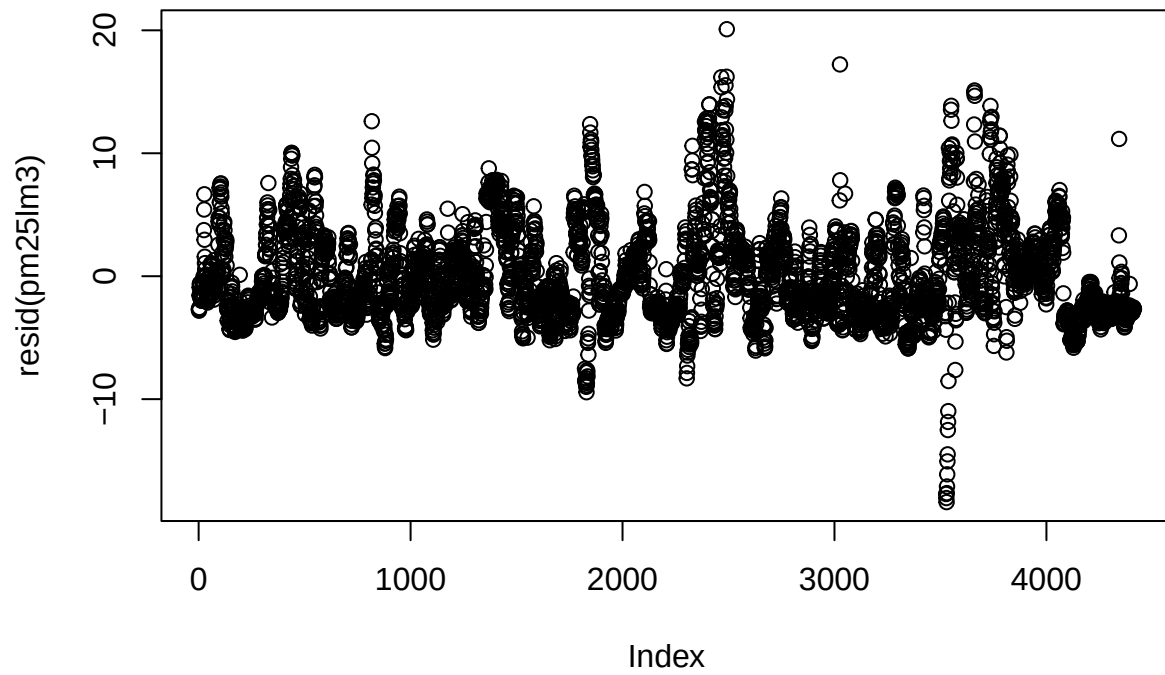
```
##
## Call:
## lm(formula = no2Conc1HourMean.value ~ operating_f + hour + year +
##     weekday_num + fireworks + smoke, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -43.701  -9.184  -2.143   7.790  77.463
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      35.4095     1.1620  30.472 < 2e-16 ***
## operating_fOperating   6.8084     0.5433  12.532 < 2e-16 ***
## hour1             6.3390     1.5322   4.137 3.58e-05 ***
## hour2             8.1210     1.5343   5.293 1.26e-07 ***
## hour3             8.4599     1.5322   5.521 3.56e-08 ***
## hour4             7.5626     1.5343   4.929 8.57e-07 ***
## hour5             6.9169     1.5322   4.514 6.52e-06 ***
## hour6             5.2350     1.5322   3.417 0.000640 ***
## hour7             3.6743     1.5323   2.398 0.016528 *
## hour8             3.1627     1.5324   2.064 0.039081 *
## hour9             2.8892     1.5350   1.882 0.059872 .
## hour10            4.1657     1.5335   2.717 0.006623 **
## hour11            1.3932     1.5345   0.908 0.363946
## hour12           -6.3137     1.5353  -4.112 3.99e-05 ***
## hour13          -16.9130     1.5357 -11.013 < 2e-16 ***
## hour14          -25.2505     1.5361 -16.438 < 2e-16 ***
## hour15          -29.1202     1.5361 -18.957 < 2e-16 ***
## hour16          -29.9622     1.5359 -19.508 < 2e-16 ***
## hour17          -30.1921     1.5359 -19.658 < 2e-16 ***
## hour18          -30.4289     1.5359 -19.812 < 2e-16 ***
## hour19          -28.8829     1.5359 -18.805 < 2e-16 ***
## hour20          -25.2748     1.5357 -16.458 < 2e-16 ***
## hour21          -21.9000     1.5346 -14.271 < 2e-16 ***
## hour22          -16.4248     1.5325 -10.718 < 2e-16 ***
## hour23           -8.2316     1.5322  -5.372 8.18e-08 ***
## year2025         -3.0931     0.4942  -6.258 4.26e-10 ***
## weekday_num      -0.3535     0.4932  -0.717 0.473613
## fireworks1        6.1521     1.1672   5.271 1.42e-07 ***
## smoke1           4.2705     1.1704   3.649 0.000267 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.7 on 4384 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.5038, Adjusted R-squared:  0.5006
```

```
## F-statistic: 158.9 on 28 and 4384 DF, p-value: < 2.2e-16
```

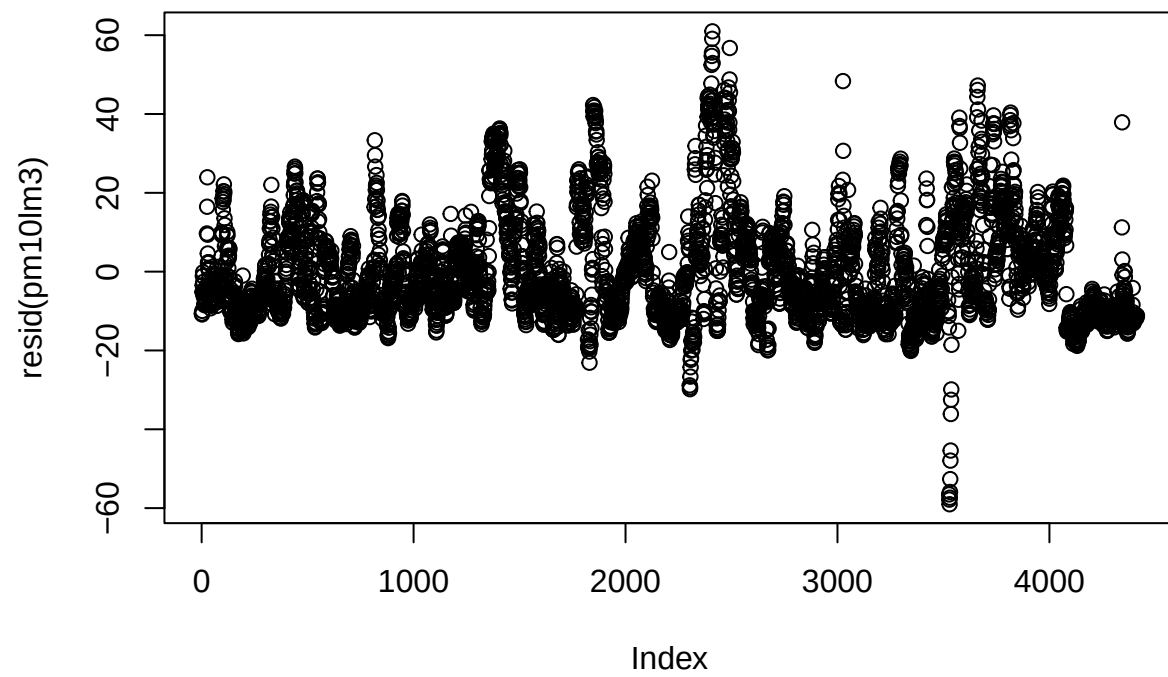
```
summary(no2lm6)
```

```
##
## Call:
## lm(formula = no2Conc1HourMean.value ~ operating_f + hour + year +
##     weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
##     HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -37.620  -8.608  -1.092   6.492  76.228
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -57.41480     3.18550  -18.024 < 2e-16 ***
## operating_f0operating    -1.22690     0.58529   -2.096 0.036123 *
## hour1           6.51165     1.38952    4.686 2.87e-06 ***
## hour2           8.60509     1.39177    6.183 6.90e-10 ***
## hour3           8.70986     1.39059    6.263 4.14e-10 ***
## hour4           7.91191     1.39311    5.679 1.44e-08 ***
## hour5           6.78341     1.39169    4.874 1.13e-06 ***
## hour6           5.44492     1.39526    3.902 9.67e-05 ***
## hour7           4.83356     1.40673    3.436 0.000596 ***
## hour8           5.74401     1.43328    4.008 6.24e-05 ***
## hour9           6.95752     1.47184    4.727 2.35e-06 ***
## hour10          9.21512     1.49208    6.176 7.20e-10 ***
## hour11          6.63205     1.51288    4.384 1.20e-05 ***
## hour12         -0.24207     1.53083   -0.158 0.874360
## hour13        -10.77876     1.53623   -7.016 2.64e-12 ***
## hour14        -19.00516     1.51809  -12.519 < 2e-16 ***
## hour15        -23.40283     1.50446  -15.556 < 2e-16 ***
## hour16        -24.34399     1.49495  -16.284 < 2e-16 ***
## hour17        -25.33048     1.46953  -17.237 < 2e-16 ***
## hour18        -26.68493     1.43398  -18.609 < 2e-16 ***
## hour19        -26.05948     1.41576  -18.407 < 2e-16 ***
## hour20        -23.21769     1.40501  -16.525 < 2e-16 ***
## hour21        -20.49208     1.39908  -14.647 < 2e-16 ***
## hour22        -16.16197     1.39348  -11.598 < 2e-16 ***
## hour23         -8.17722     1.39186   -5.875 4.56e-09 ***
## year2025         0.90584     0.47708    1.899 0.057669 .
## weekday_num     -0.08964     0.45018   -0.199 0.842174
## fireworks1       2.00321     1.06522    1.881 0.060100 .
## smoke1          1.27254     1.05721    1.204 0.228783
## HourlyDryBulbTemperature  0.80045     0.03497  22.892 < 2e-16 ***
## HourlyRelativeHumidity   0.49686     0.01744  28.485 < 2e-16 ***
## HourlyWindSpeed    -0.18411     0.05599   -3.288 0.001016 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.11 on 4196 degrees of freedom
## (188 observations deleted due to missingness)
## Multiple R-squared:  0.61, Adjusted R-squared:  0.6072
## F-statistic: 211.7 on 31 and 4196 DF, p-value: < 2.2e-16
```

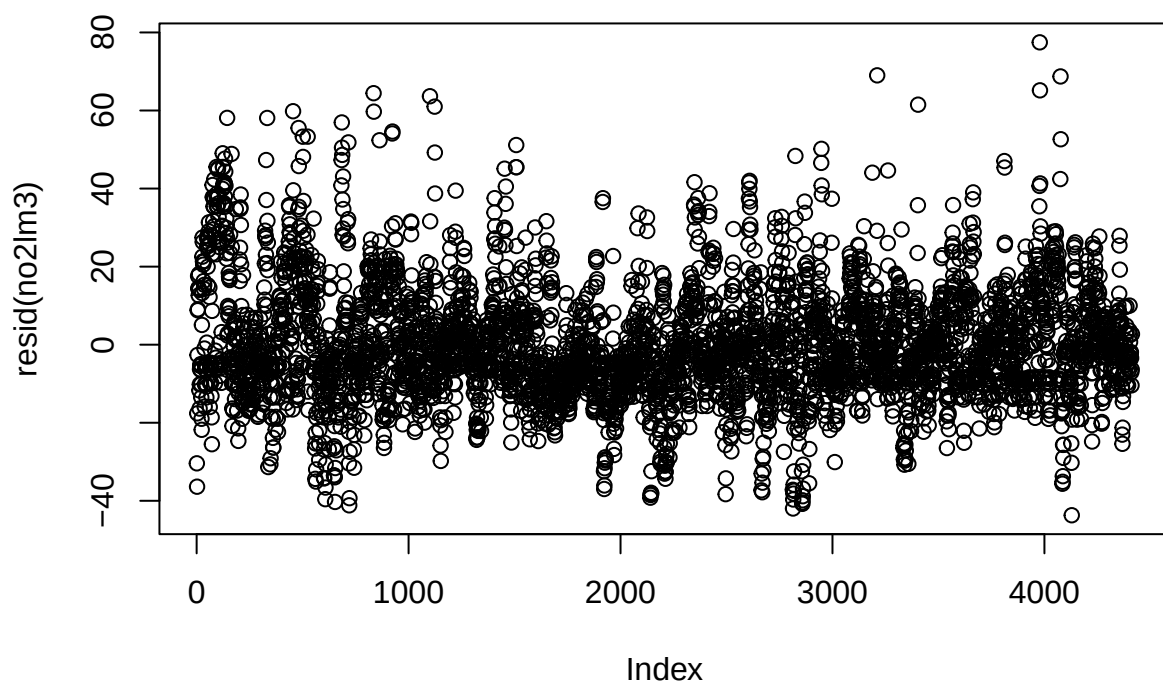
```
#plot model residuals  
plot(resid(pm25lm3))
```



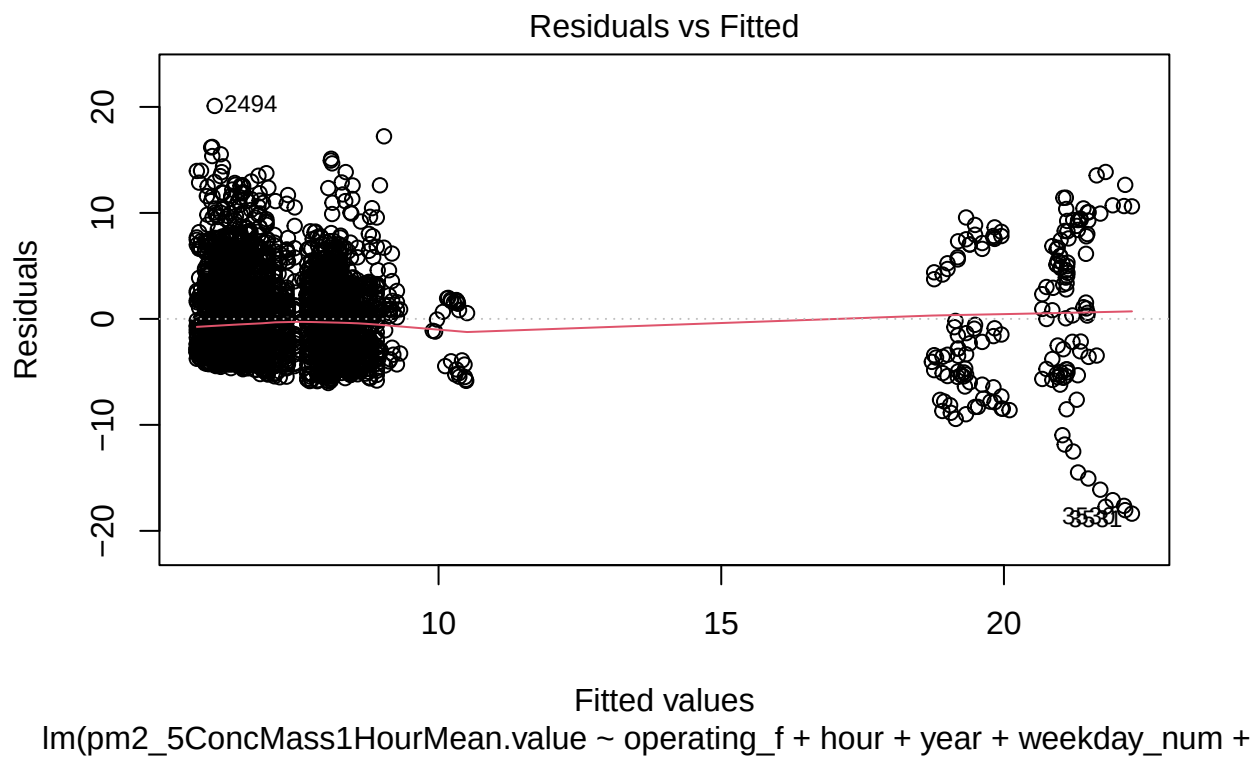
```
plot(resid(pm10lm3))
```

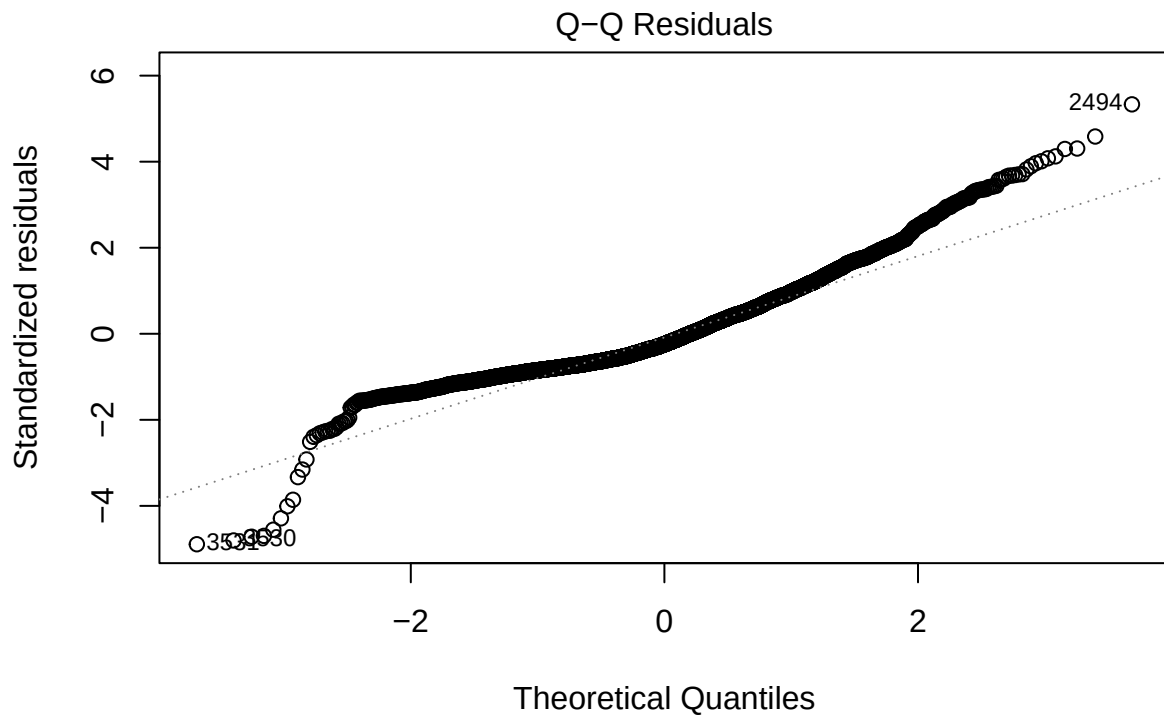


```
plot(resid(no2lm3))
```

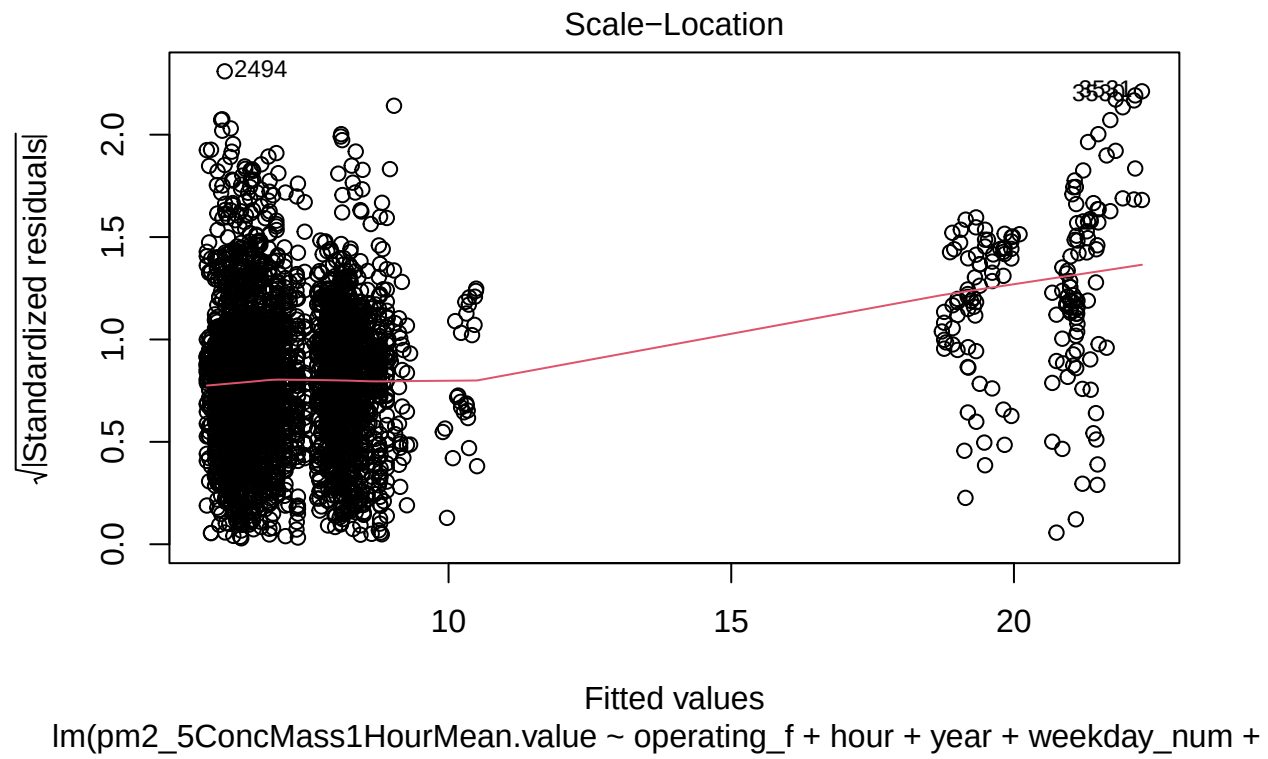


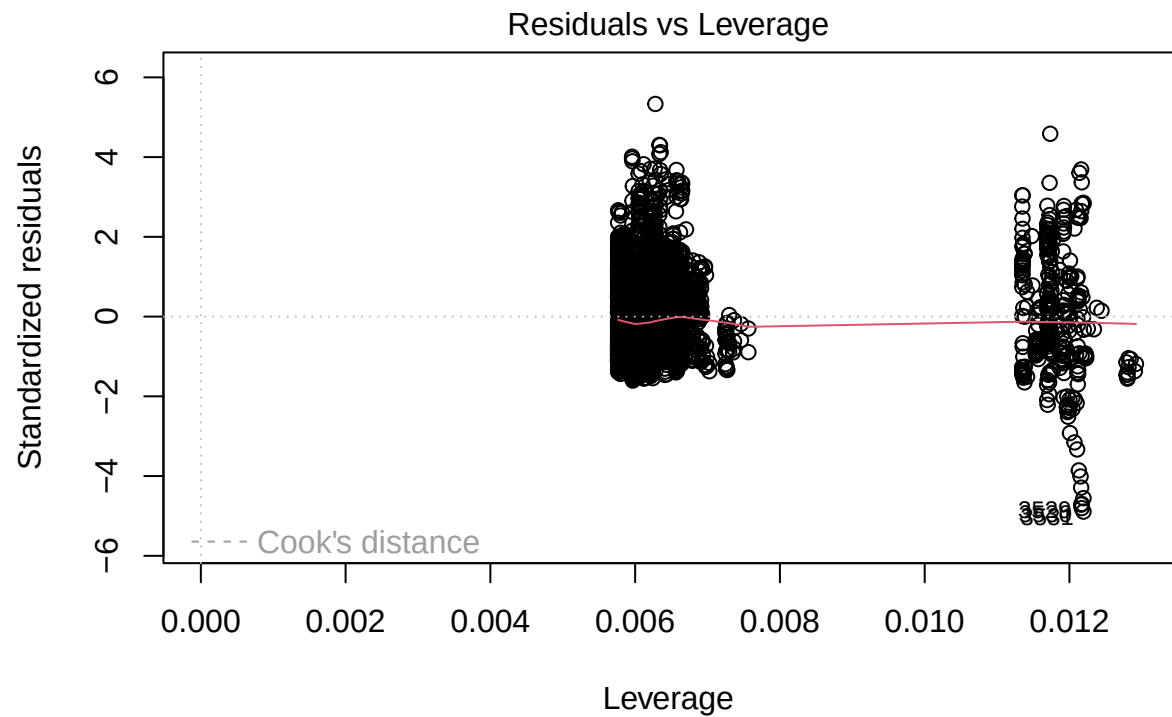
```
plot(pm25lm3)
```





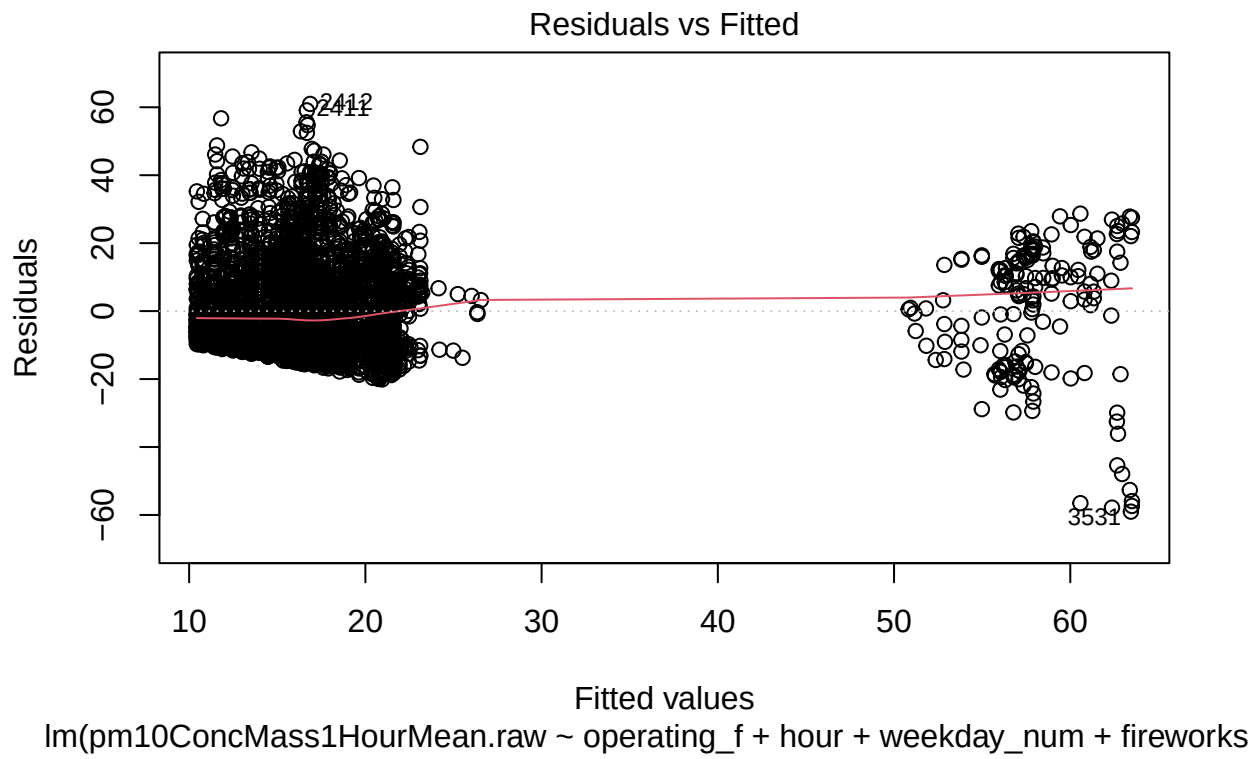
lm(pm2_5ConcMass1HourMean.value ~ operating_f + hour + year + weekday_num +

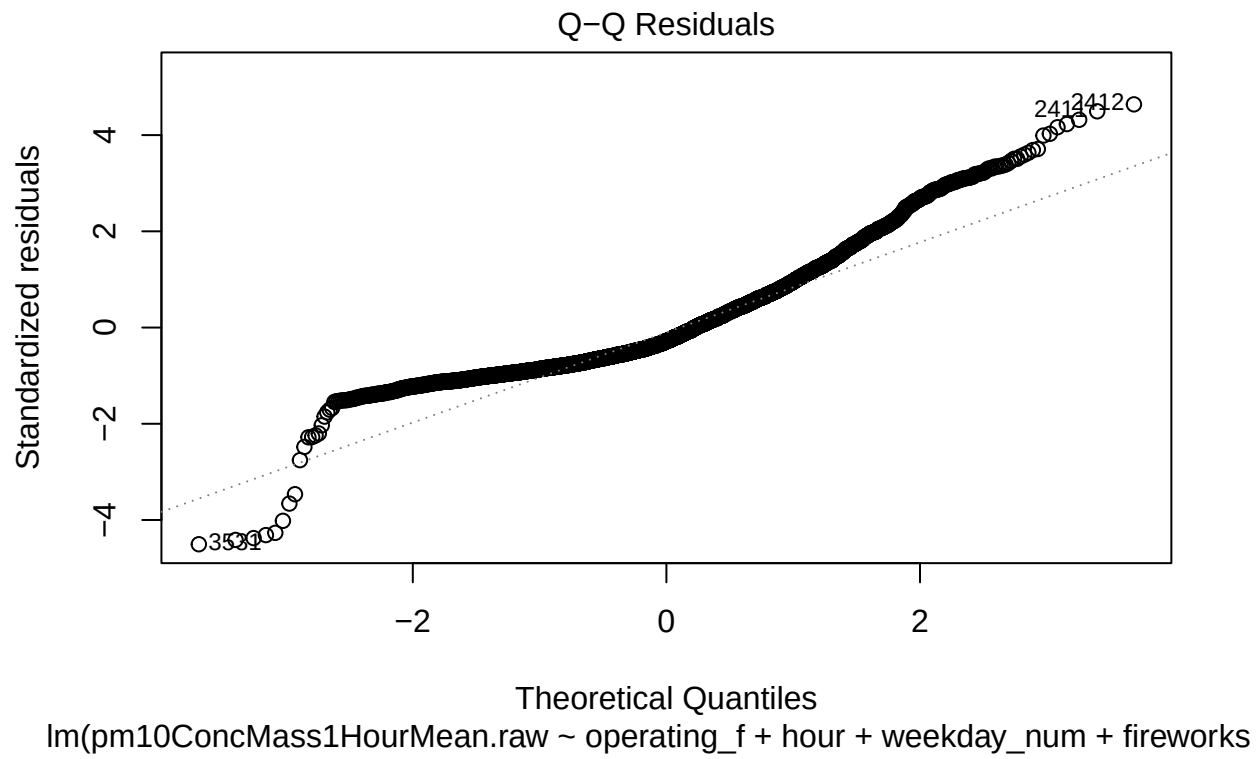


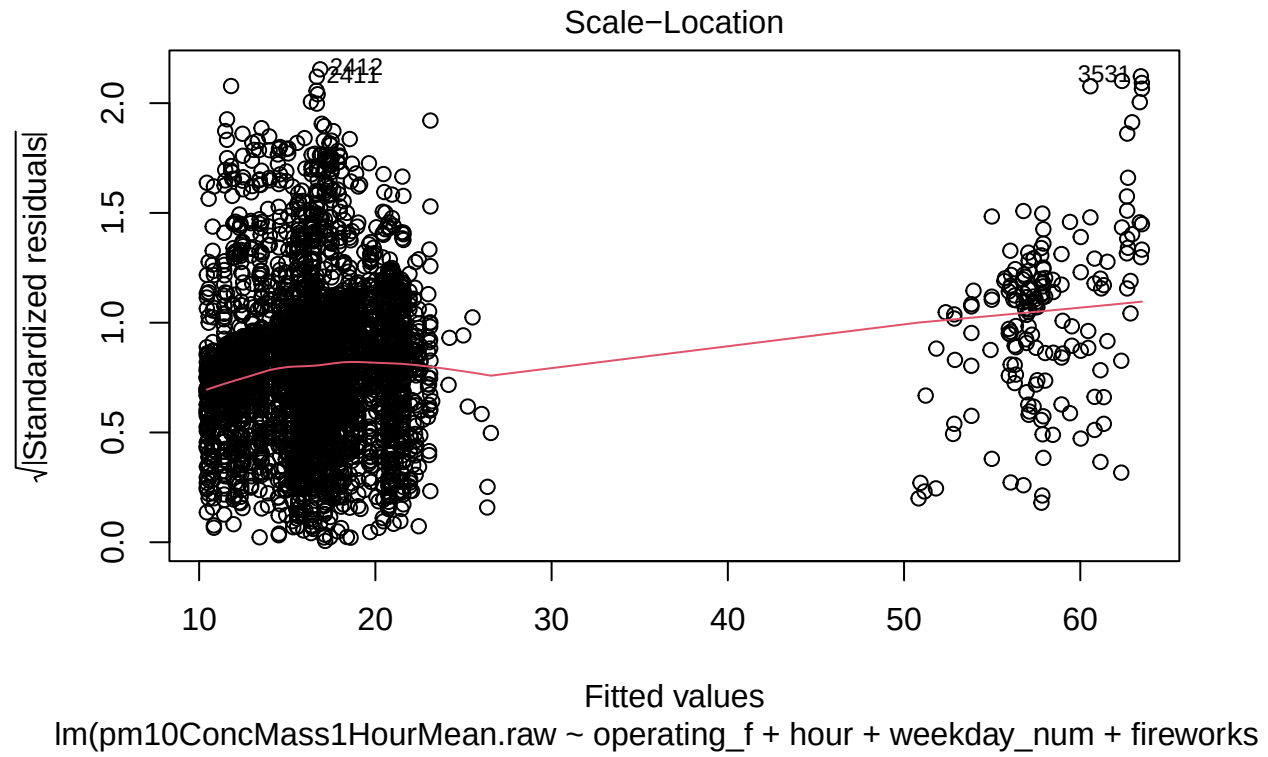


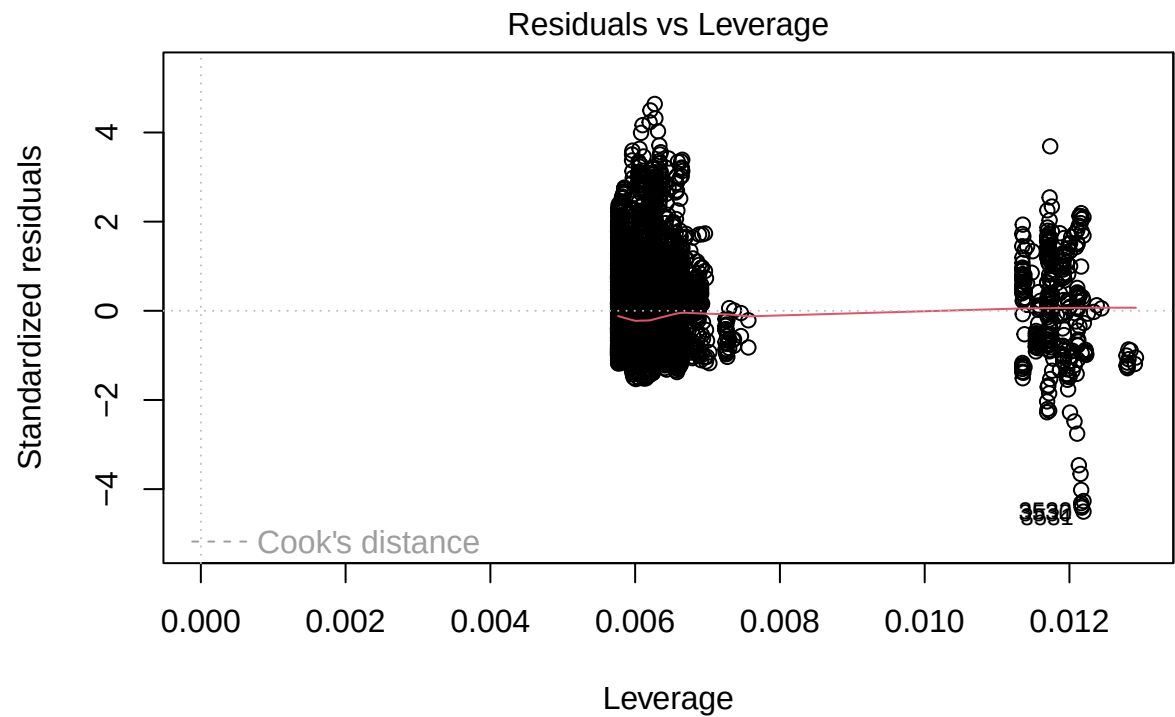
`lm(pm2_5ConcMass1HourMean.value ~ operating_f + hour + year + weekday_num +`

`plot(pm10lm3)`



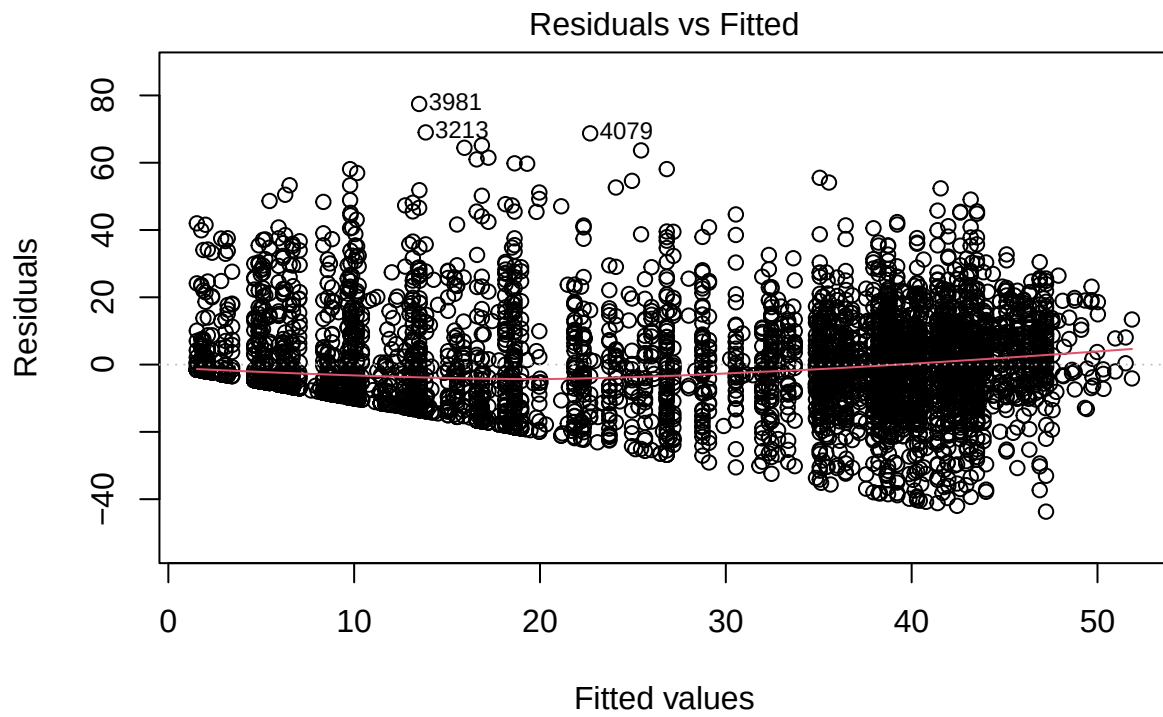




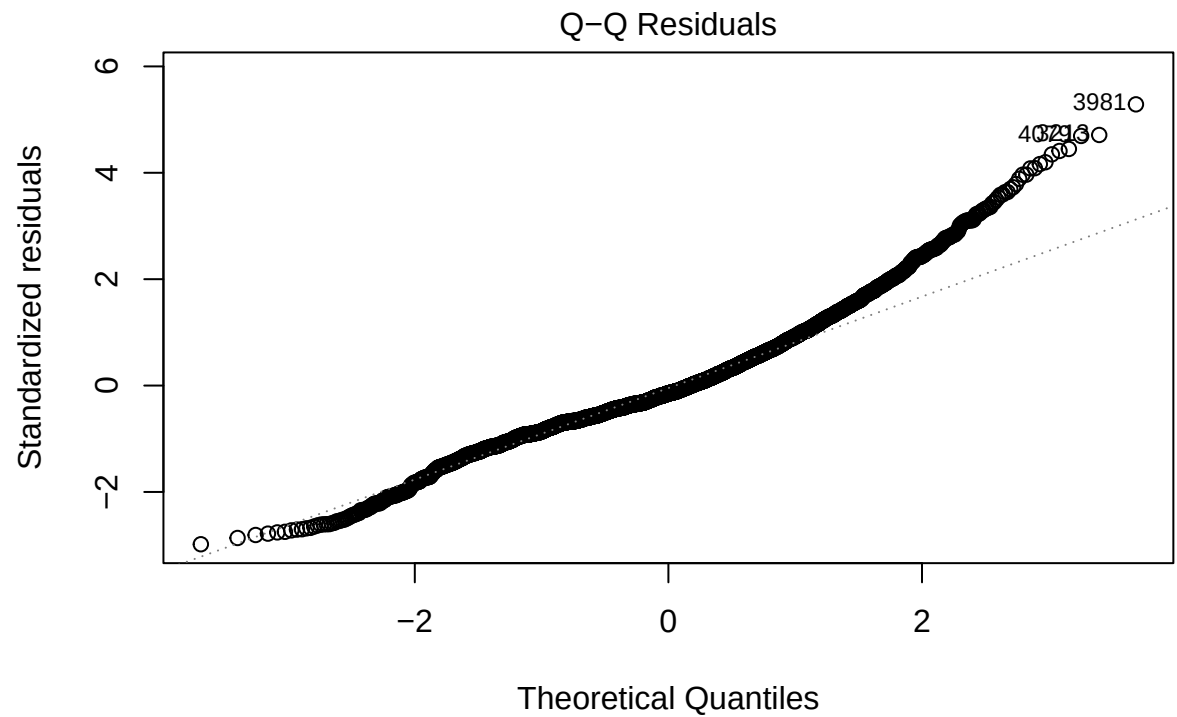


$\text{lm}(\text{pm10ConcMass1HourMean.raw} \sim \text{operating_f} + \text{hour} + \text{weekday_num} + \text{fireworks})$

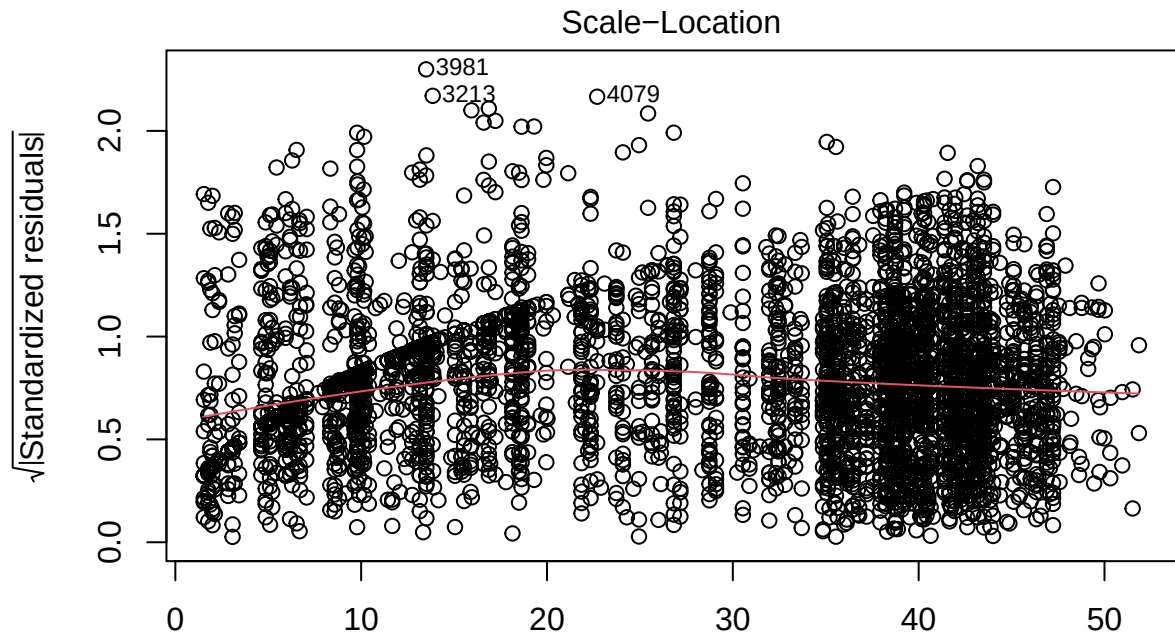
```
plot(no2lm3)
```



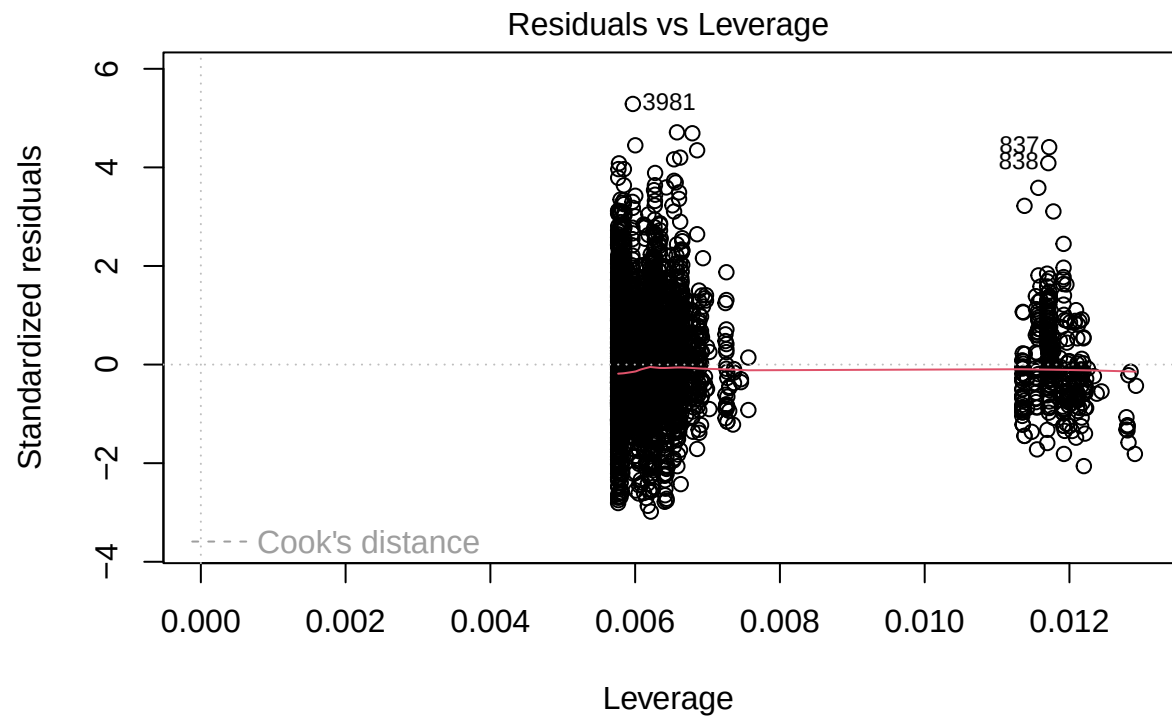
lm(no2Conc1HourMean.value ~ operating_f + hour + year + weekday_num + firew ..



lm(no2Conc1HourMean.value ~ operating_f + hour + year + weekday_num + firew ..

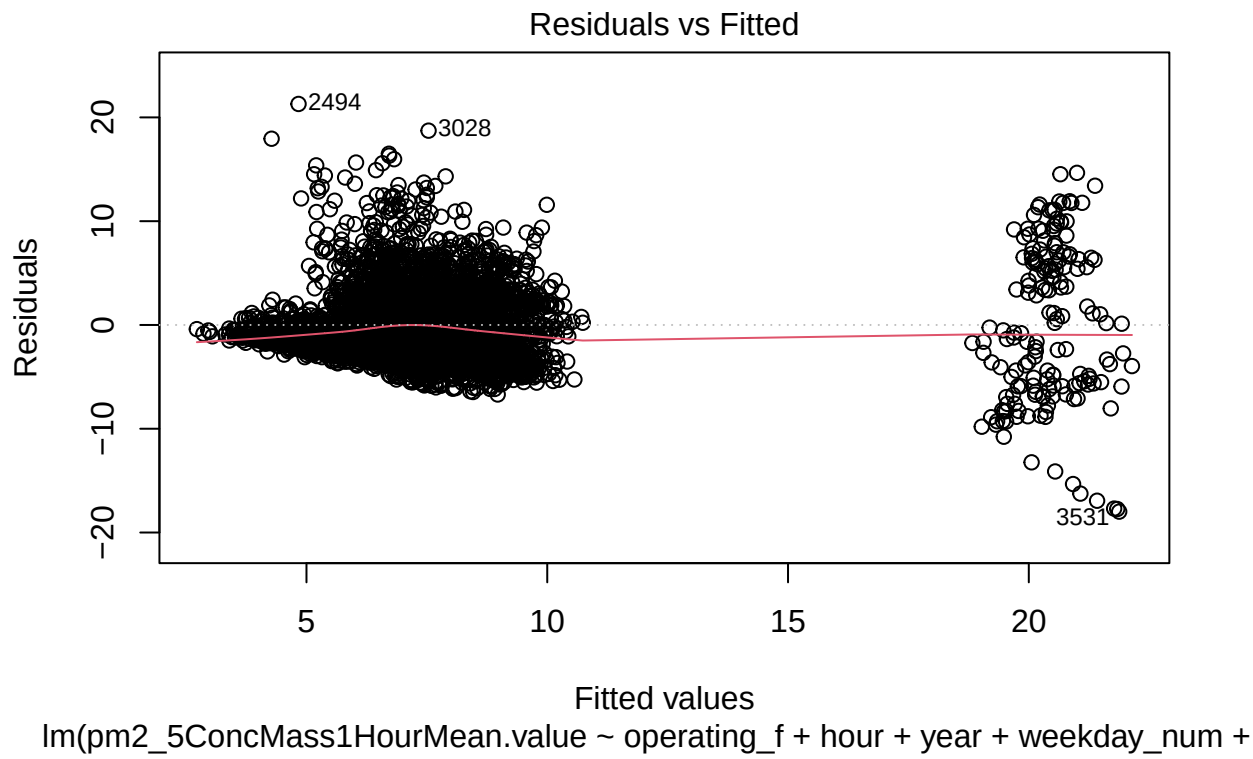


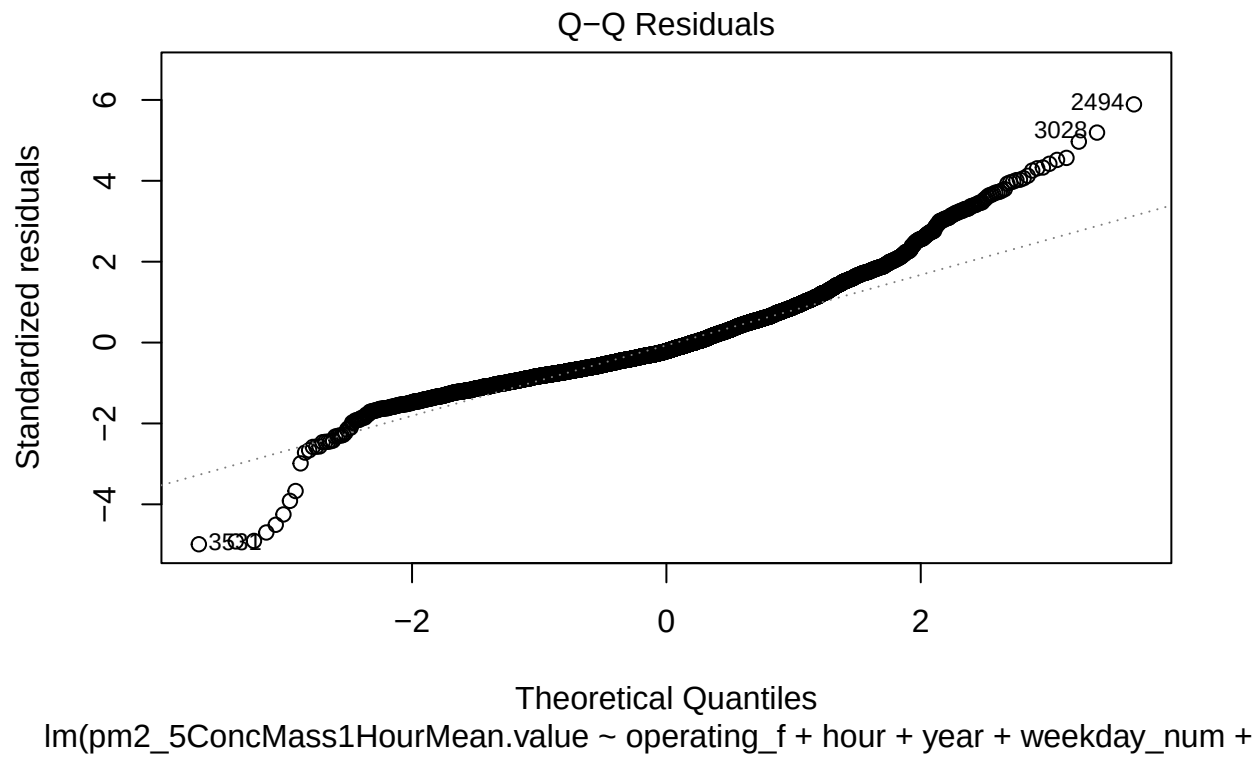
Fitted values
 $\text{lm}(\text{no2Conc1HourMean.value} \sim \text{operating_f} + \text{hour} + \text{year} + \text{weekday_num} + \text{firew} \dots)$

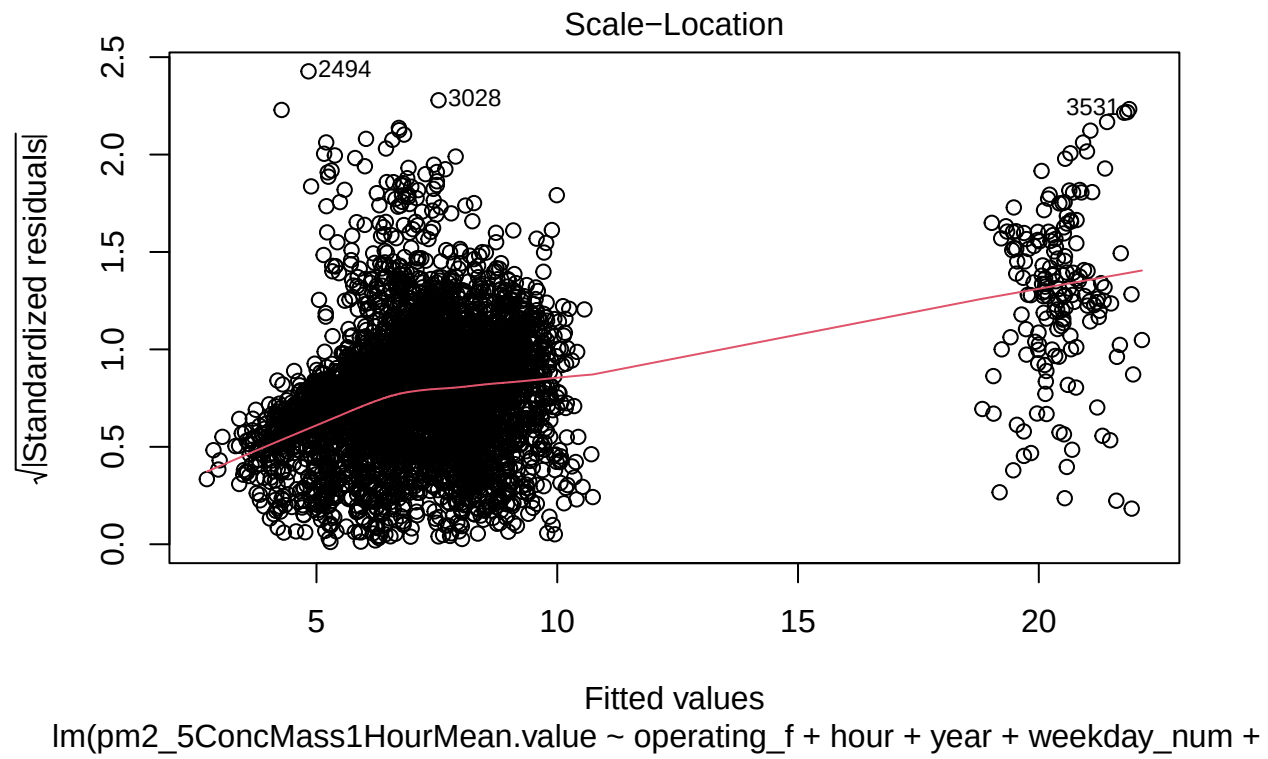


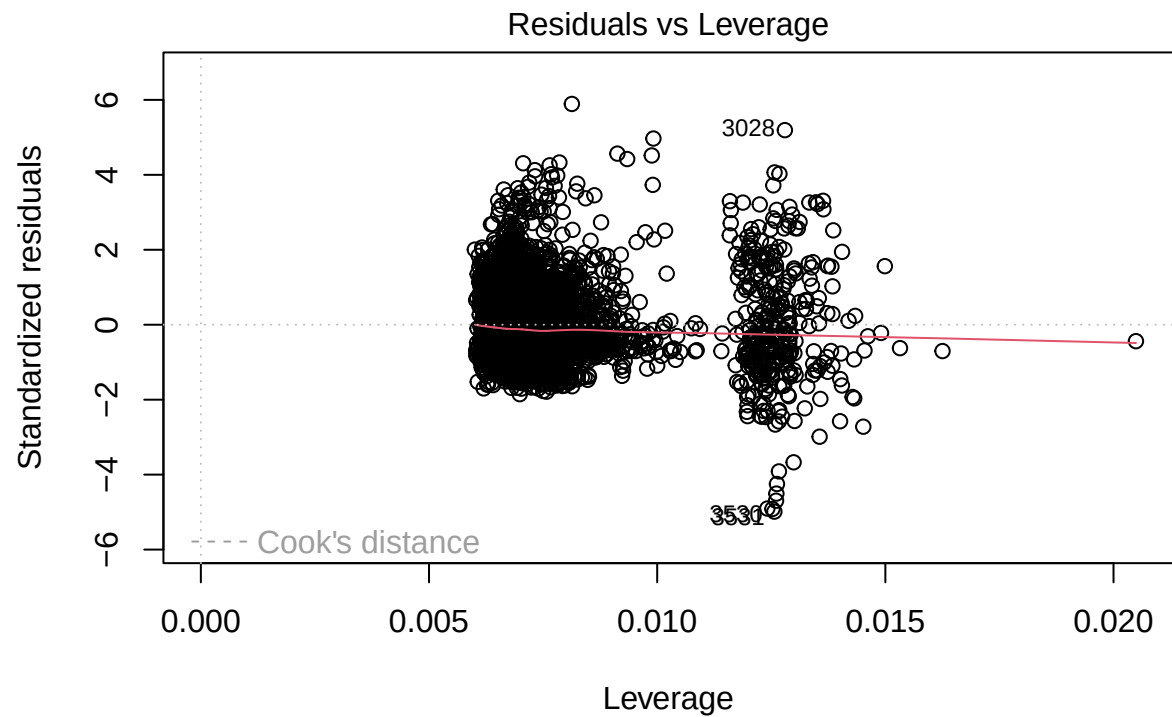
$\text{lm}(\text{no2Conc1HourMean.value} \sim \text{operating_f} + \text{hour} + \text{year} + \text{weekday_num} + \text{firew} \dots)$

```
plot(pm25lm6)
```



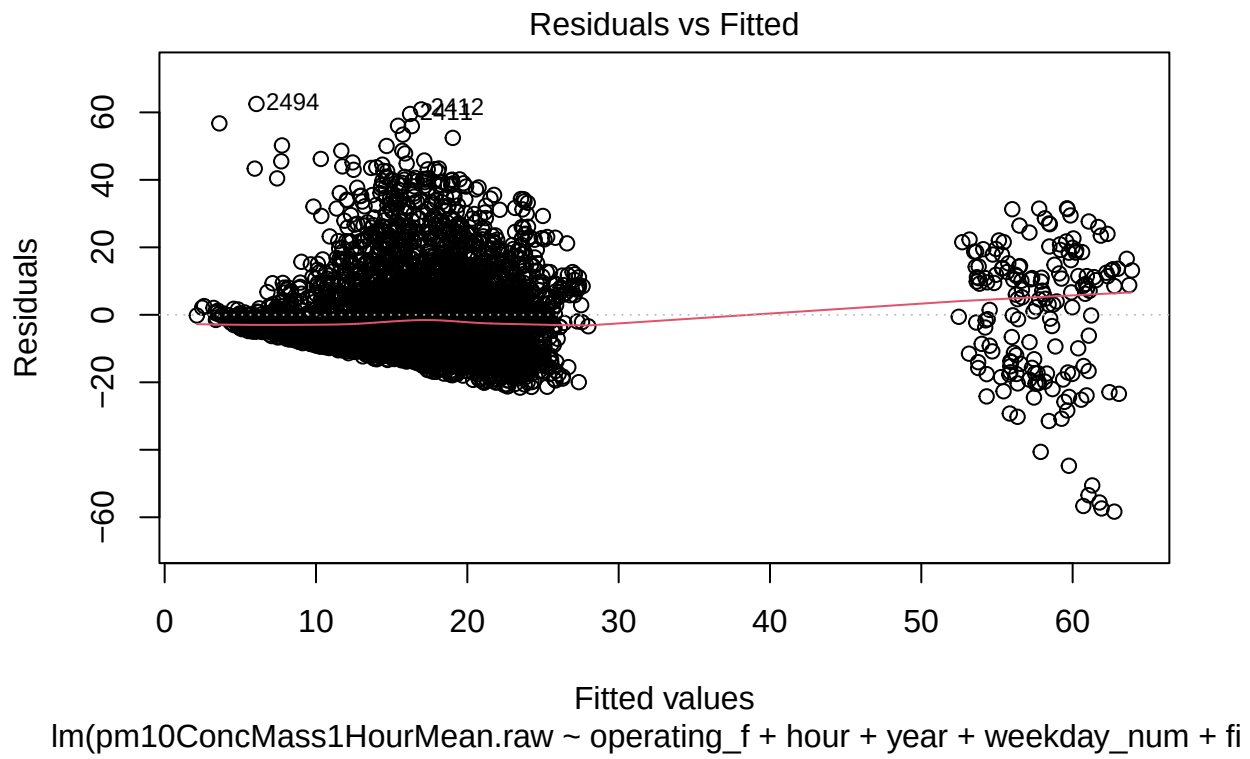


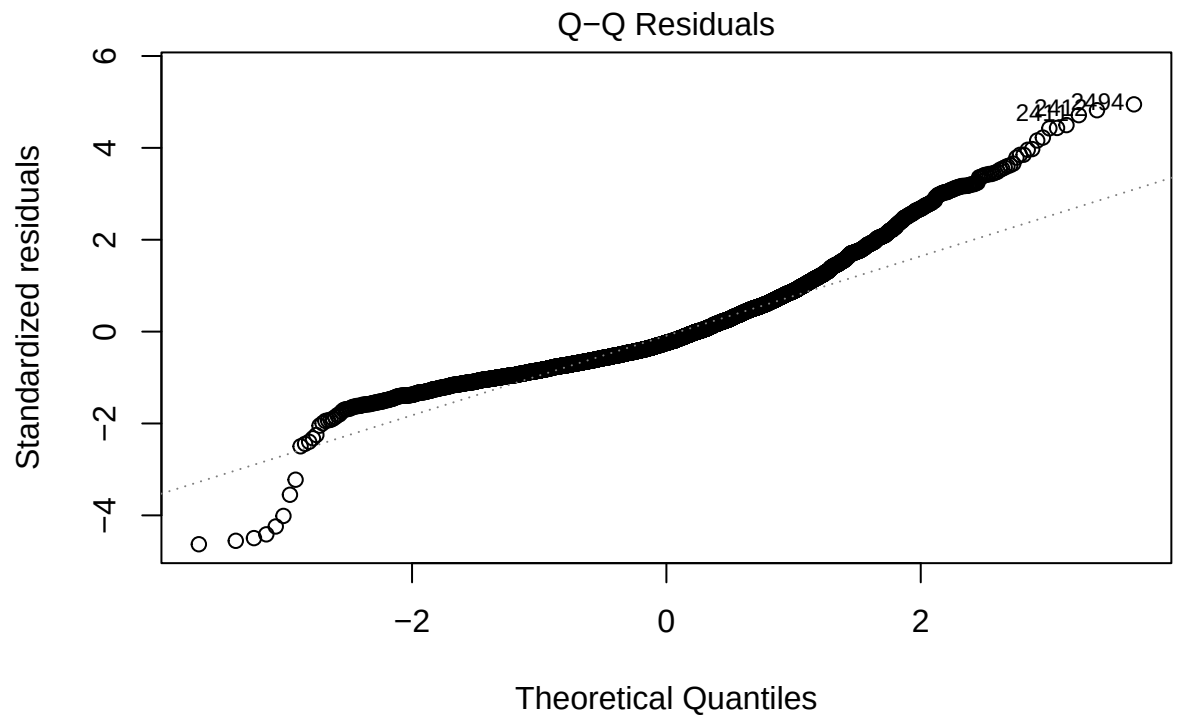




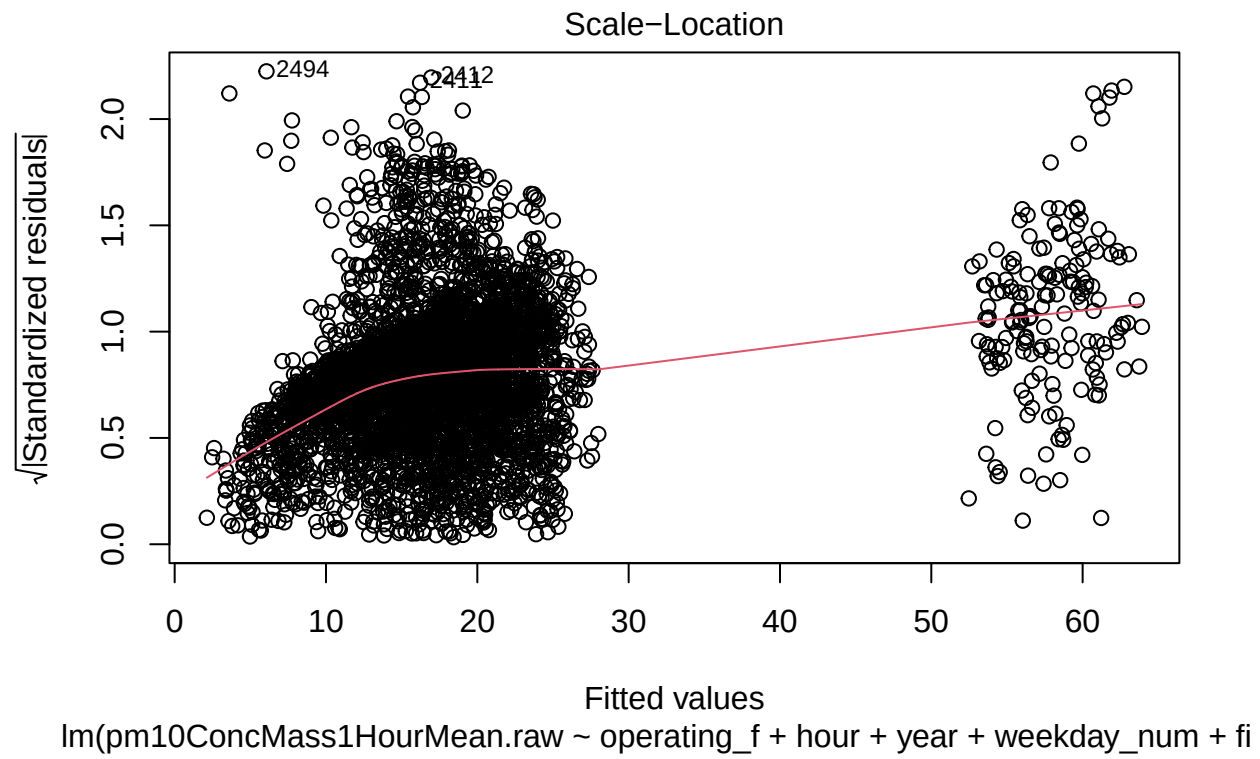
$\text{lm}(\text{pm2_5ConcMass1HourMean.value} \sim \text{operating_f} + \text{hour} + \text{year} + \text{weekday_num} +$

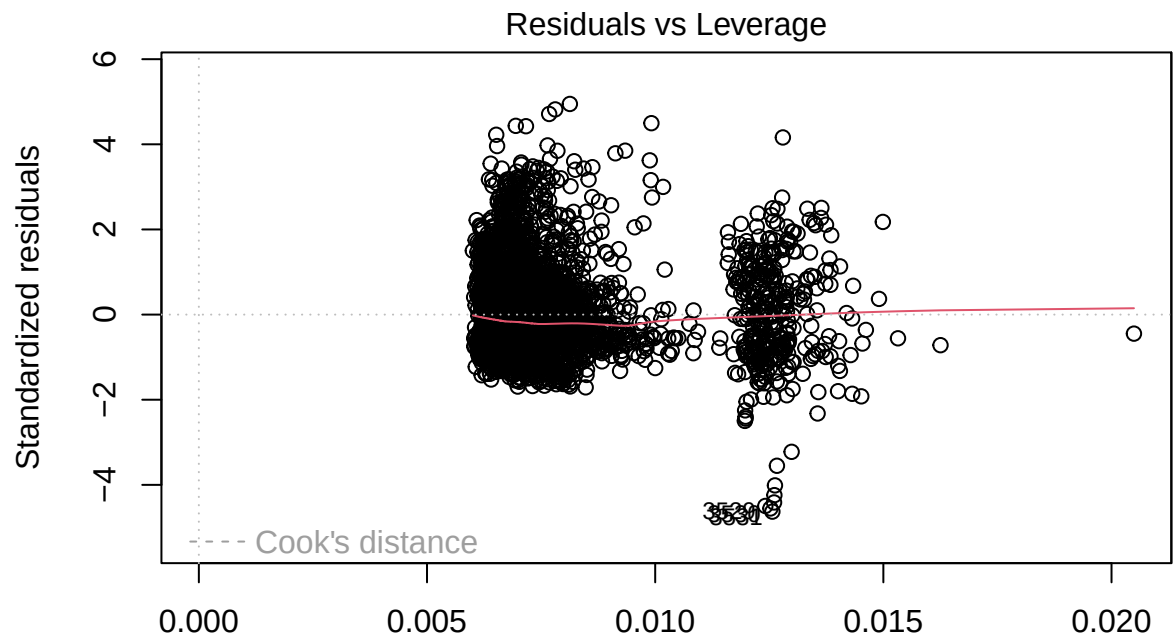
`plot(pm10lm6)`





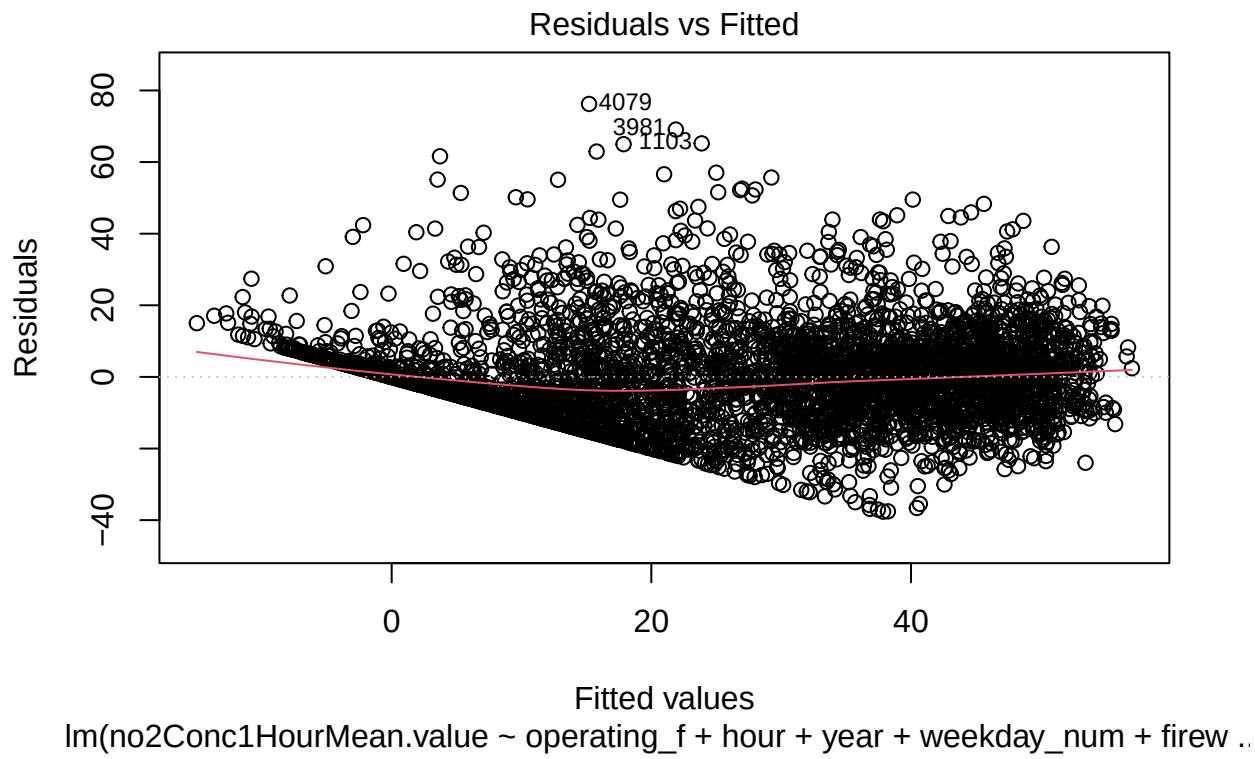
$\text{lm}(\text{pm10ConcMass1HourMean.raw} \sim \text{operating_f} + \text{hour} + \text{year} + \text{weekday_num} + \text{fi})$

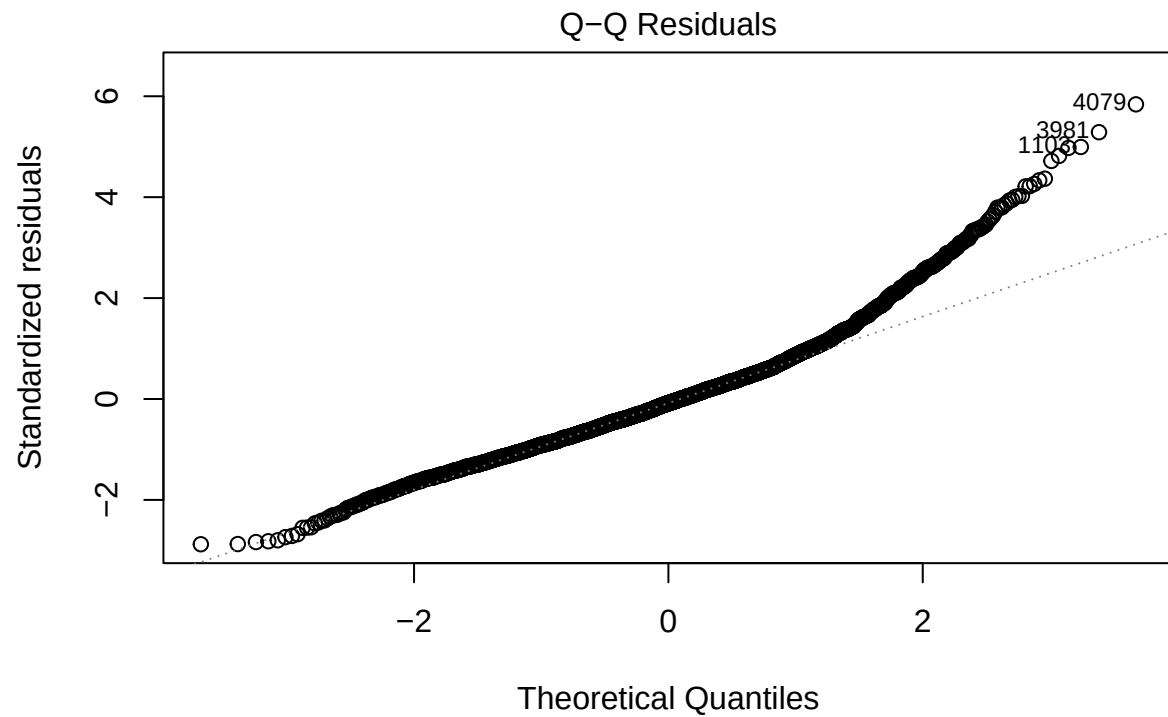




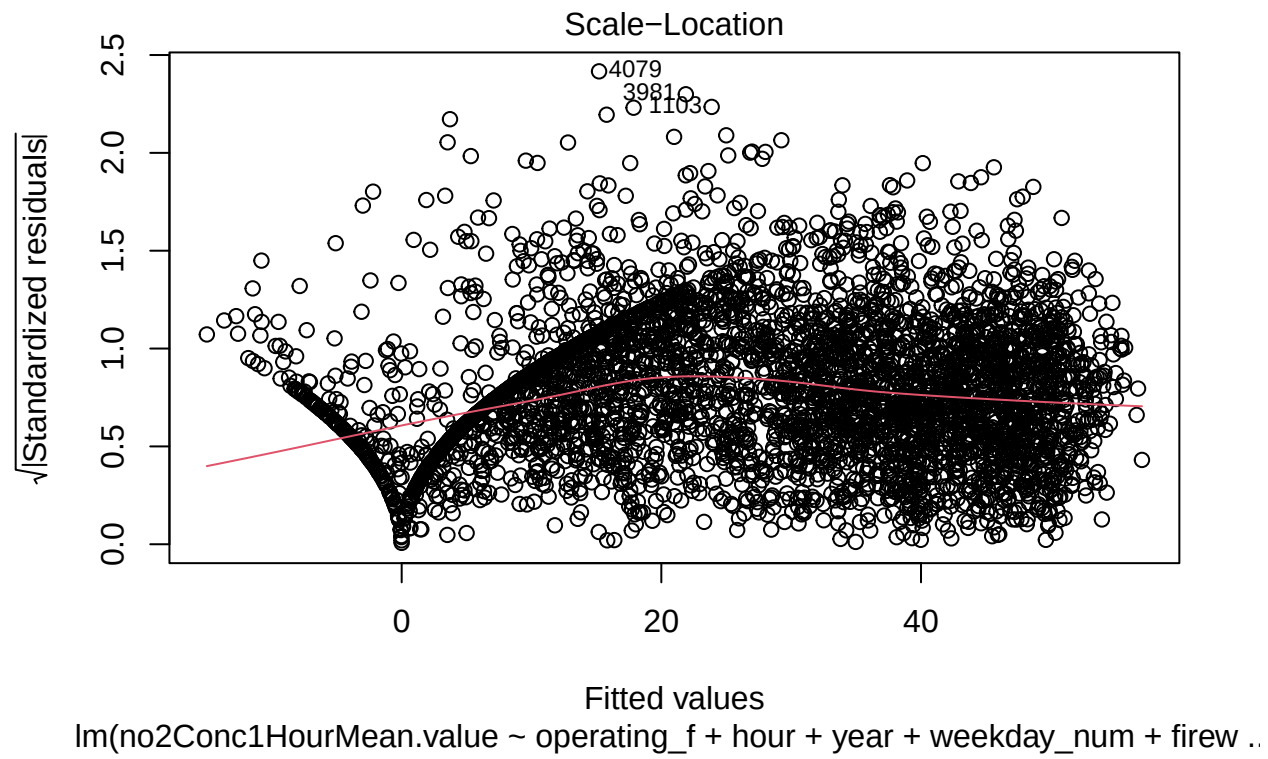
lm(pm10ConcMass1HourMean.raw ~ operating_f + hour + year + weekday_num + fi

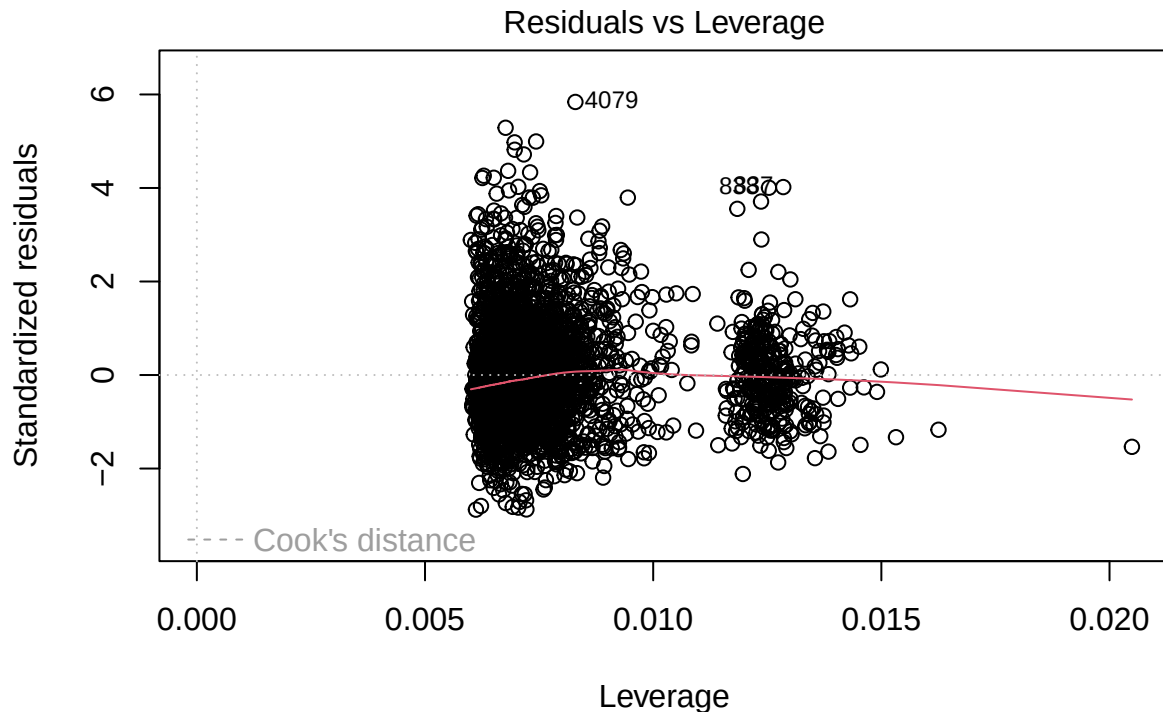
```
plot(no2lm6)
```





lm(no2Conc1HourMean.value ~ operating_f + hour + year + weekday_num + firew ..





`lm(no2Conc1HourMean.value ~ operating_f + hour + year + weekday_num + firew ..`

#they don't look the most normal...

#does log transforming the models change the model significance we see?

#it does not

#pm2.5

```
pm25lm3log <- lm(formula = log(pm2_5ConcMass1HourMean.value) ~ operating_f +
                  hour + year + weekday_num + fireworks + smoke, data = allendale)
summary(pm25lm3log)
```

##

Call:

```
## lm(formula = log(pm2_5ConcMass1HourMean.value) ~ operating_f +
##     hour + year + weekday_num + fireworks + smoke, data = allendale)
```

##

Residuals:

```
##      Min       1Q   Median       3Q      Max
## -1.84700 -0.36753 -0.01628  0.37216  1.62264
```

##

Coefficients:

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.841230   0.039105  47.084 < 2e-16 ***
## operating_f0operating 0.324931   0.018282  17.773 < 2e-16 ***
## hour1          0.041626   0.051563   0.807 0.419545
## hour2          0.051719   0.051634   1.002 0.316569
## hour3          0.029478   0.051563   0.572 0.567565
## hour4          0.009071   0.051634   0.176 0.860549
```

```

## hour5          -0.016339   0.051563  -0.317  0.751351
## hour6          -0.043194   0.051563  -0.838  0.402253
## hour7          -0.069850   0.051565  -1.355  0.175611
## hour8          -0.085807   0.051568  -1.664  0.096190 .
## hour9          -0.108237   0.051657  -2.095  0.036201 *
## hour10         -0.116404   0.051605  -2.256  0.024142 *
## hour11         -0.107827   0.051638  -2.088  0.036844 *
## hour12         -0.098461   0.051667  -1.906  0.056759 .
## hour13         -0.090478   0.051680  -1.751  0.080065 .
## hour14         -0.087152   0.051694  -1.686  0.091883 .
## hour15         -0.083612   0.051694  -1.617  0.105853
## hour16         -0.076811   0.051687  -1.486  0.137333
## hour17         -0.069098   0.051687  -1.337  0.181343
## hour18         -0.073084   0.051687  -1.414  0.157441
## hour19         -0.081855   0.051687  -1.584  0.113343
## hour20         -0.075522   0.051680  -1.461  0.143995
## hour21         -0.069656   0.051644  -1.349  0.177474
## hour22         -0.037628   0.051571  -0.730  0.465649
## hour23         -0.016728   0.051564  -0.324  0.745648
## year2025       -0.070575   0.016632  -4.243  2.25e-05 ***
## weekday_num    -0.060553   0.016597  -3.648  0.000267 ***
## fireworks1      0.284072   0.039280   7.232  5.59e-13 ***
## smoke1         1.058108   0.039387  26.864  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4946 on 4384 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.2154, Adjusted R-squared:  0.2104
## F-statistic: 42.98 on 28 and 4384 DF, p-value: < 2.2e-16

pm25lm6log <- lm(formula = log(pm2_5ConcMass1HourMean.value) ~ operating_f +
                  hour + year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature + HourlyRel.
summary(pm25lm6log)

##
## Call:
## lm(formula = log(pm2_5ConcMass1HourMean.value) ~ operating_f +
##     hour + year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
##     HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.76373 -0.32452 -0.02508  0.31547  1.78991
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.6197306  0.1093294  -5.668 1.54e-08 ***
## operating_f0operating  0.0253250  0.0200878   1.261 0.207482
## hour1         0.0541602  0.0476897   1.136 0.256156
## hour2         0.0832809  0.0477668   1.743 0.081322 .
## hour3         0.0699289  0.0477263   1.465 0.142939
## hour4         0.0573592  0.0478127   1.200 0.230338
## hour5         0.0030119  0.0477642   0.063 0.949723
## hour6        -0.0794141  0.0478866  -1.658 0.097316 .

```

```
## hour7          -0.1309975  0.0482804 -2.713 0.006690 **
## hour8          -0.1793463  0.0491916 -3.646 0.000270 ***
## hour9          -0.2057960  0.0505150 -4.074 4.71e-05 ***
## hour10         -0.2361174  0.0512094 -4.611 4.13e-06 ***
## hour11         -0.2358357  0.0519233 -4.542 5.73e-06 ***
## hour12         -0.2421417  0.0525394 -4.609 4.17e-06 ***
## hour13         -0.2203197  0.0527249 -4.179 2.99e-05 ***
## hour14         -0.2208218  0.0521024 -4.238 2.30e-05 ***
## hour15         -0.2238002  0.0516344 -4.334 1.50e-05 ***
## hour16         -0.2034740  0.0513080 -3.966 7.44e-05 ***
## hour17         -0.1820643  0.0504356 -3.610 0.000310 ***
## hour18         -0.1563562  0.0492154 -3.177 0.001499 **
## hour19         -0.1265085  0.0485903 -2.604 0.009258 **
## hour20         -0.0961839  0.0482213 -1.995 0.046147 *
## hour21         -0.0646541  0.0480176 -1.346 0.178224
## hour22         -0.0567510  0.0478254 -1.187 0.235441
## hour23         -0.0288982  0.0477699 -0.605 0.545247
## year2025        0.0777154  0.0163738  4.746 2.14e-06 ***
## weekday_num    -0.0574546  0.0154506 -3.719 0.000203 ***
## fireworks1      0.0921509  0.0365592  2.521 0.011752 *
## smoke1         0.9643107  0.0362846 26.576 < 2e-16 ***
## HourlyDryBulbTemperature 0.0319133  0.0012001 26.593 < 2e-16 ***
## HourlyRelativeHumidity  0.0061261  0.0005987 10.233 < 2e-16 ***
## HourlyWindSpeed   -0.0165313  0.0019216 -8.603 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4499 on 4196 degrees of freedom
## (188 observations deleted due to missingness)
## Multiple R-squared:  0.34, Adjusted R-squared:  0.3351
## F-statistic: 69.73 on 31 and 4196 DF, p-value: < 2.2e-16
```

```
#PM10
```

```
pm10lm3log <- lm(formula = log(pm10ConcMass1HourMean.raw) ~ operating_f +
                  hour + year + weekday_num + fireworks + smoke, data = allendale)
summary(pm10lm3log)
```

```
##
## Call:
## lm(formula = log(pm10ConcMass1HourMean.raw) ~ operating_f + hour +
##     year + weekday_num + fireworks + smoke, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.05524 -0.57262  0.05147  0.64089  2.31596
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.42971    0.06931  35.057 < 2e-16 ***
## operating_f0operating  0.48004    0.03240  14.815 < 2e-16 ***
## hour1          0.12933    0.09139   1.415  0.15709
## hour2          0.19766    0.09151   2.160  0.03083 *
## hour3          0.19567    0.09139   2.141  0.03232 *
## hour4          0.19665    0.09151   2.149  0.03170 *
## hour5          0.19081    0.09139   2.088  0.03686 *
```

```
## hour6          0.16933    0.09139    1.853    0.06397 .
## hour7          0.14325    0.09139    1.567    0.11708
## hour8          0.14083    0.09140    1.541    0.12343
## hour9          0.12654    0.09155    1.382    0.16699
## hour10         0.11647    0.09146    1.273    0.20293
## hour11         0.11804    0.09152    1.290    0.19719
## hour12         0.06615    0.09157    0.722    0.47010
## hour13        -0.01781    0.09159   -0.194    0.84583
## hour14        -0.11092    0.09162   -1.211    0.22610
## hour15        -0.17820    0.09162   -1.945    0.05183 .
## hour16        -0.21885    0.09161   -2.389    0.01693 *
## hour17        -0.24882    0.09161   -2.716    0.00663 **
## hour18        -0.27519    0.09161   -3.004    0.00268 **
## hour19        -0.30039    0.09161   -3.279    0.00105 **
## hour20        -0.29908    0.09159   -3.265    0.00110 **
## hour21        -0.27555    0.09153   -3.010    0.00262 **
## hour22        -0.19292    0.09140   -2.111    0.03485 *
## hour23        -0.09486    0.09139   -1.038    0.29932
## year2025       -0.10071    0.02948   -3.416    0.00064 ***
## weekday_num    -0.14171    0.02942   -4.817    1.50e-06 ***
## fireworks1     0.37108    0.06962    5.330    1.03e-07 ***
## smoke1         1.46073    0.06981   20.925    < 2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.8766 on 4384 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.1703, Adjusted R-squared:  0.165
## F-statistic: 32.13 on 28 and 4384 DF, p-value: < 2.2e-16
```

```
pm10lm6log <- lm(formula = log(pm10ConcMass1HourMean.raw) ~ operating_f +
                  hour + year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
                  HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)
summary(pm10lm6log)
```

```
##
## Call:
## lm(formula = log(pm10ConcMass1HourMean.raw) ~ operating_f + hour +
##     year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
##     HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.86791 -0.51680  0.01067  0.53500  2.84699
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -2.353215   0.191299  -12.301   < 2e-16 ***
## operating_f0operating  -0.064038   0.035149   -1.822  0.068537 .
## hour1          0.149391   0.083445    1.790  0.073480 .
## hour2          0.249600   0.083580    2.986  0.002840 **
## hour3          0.258141   0.083509    3.091  0.002007 **
## hour4          0.270007   0.083660    3.227  0.001259 **
## hour5          0.216424   0.083576    2.590  0.009643 **
## hour6          0.114935   0.083790    1.372  0.170229
```

```
## hour7          0.068920    0.084479    0.816 0.414644
## hour8          0.036839    0.086073    0.428 0.668676
## hour9          0.048890    0.088389    0.553 0.580206
## hour10         0.010852    0.089604    0.121 0.903607
## hour11         0.008483    0.090853    0.093 0.925610
## hour12        -0.059300    0.091931   -0.645 0.518928
## hour13        -0.120176    0.092256   -1.303 0.192769
## hour14        -0.217217    0.091166   -2.383 0.017233 *
## hour15        -0.299942    0.090348   -3.320 0.000908 ***
## hour16        -0.321142    0.089776   -3.577 0.000351 ***
## hour17        -0.340572    0.088250   -3.859 0.000115 ***
## hour18        -0.334353    0.086115   -3.883 0.000105 ***
## hour19        -0.316038    0.085021   -3.717 0.000204 ***
## hour20        -0.291506    0.084375   -3.455 0.000556 ***
## hour21        -0.232970    0.084019   -2.773 0.005582 **
## hour22        -0.205076    0.083683   -2.451 0.014301 *
## hour23        -0.107453    0.083586   -1.286 0.198672
## year2025       0.173371    0.028650    6.051 1.56e-09 ***
## weekday_num   -0.139777    0.027035   -5.170 2.45e-07 ***
## fireworks1     0.031470    0.063970    0.492 0.622782
## smoke1        1.278481    0.063489   20.137 < 2e-16 ***
## HourlyDryBulbTemperature 0.056749    0.002100   27.026 < 2e-16 ***
## HourlyRelativeHumidity 0.015485    0.001048   14.783 < 2e-16 ***
## HourlyWindSpeed -0.027647    0.003362   -8.222 2.63e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7872 on 4196 degrees of freedom
## (188 observations deleted due to missingness)
## Multiple R-squared:  0.314, Adjusted R-squared:  0.3089
## F-statistic: 61.95 on 31 and 4196 DF, p-value: < 2.2e-16
```

```
#no2
no2lm3log <- lm(formula = log(allendale$no2Conc1HourMean.value + 1) ~ operating_f +
               hour + year + weekday_num + fireworks + smoke, data = allendale)
summary(no2lm3log)
```

```
##
## Call:
## lm(formula = log(allendale$no2Conc1HourMean.value + 1) ~ operating_f +
##     hour + year + weekday_num + fireworks + smoke, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.6397 -0.4606  0.0889  0.4082  2.9962
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.46080    0.07666  45.144 < 2e-16 ***
## operating_f0operating 0.25696    0.03584   7.170 8.79e-13 ***
## hour1          0.17559    0.10108   1.737 0.082449 .
## hour2          0.24060    0.10122   2.377 0.017500 *
## hour3          0.25805    0.10108   2.553 0.010720 *
## hour4          0.24892    0.10122   2.459 0.013964 *
## hour5          0.23096    0.10108   2.285 0.022367 *
```

```

## hour6          0.15001    0.10108    1.484 0.137875
## hour7          0.10222    0.10109    1.011 0.311961
## hour8          0.11414    0.10109    1.129 0.258947
## hour9          0.11576    0.10127    1.143 0.253056
## hour10         0.15097    0.10117    1.492 0.135683
## hour11         0.05781    0.10123    0.571 0.567958
## hour12        -0.25452    0.10129   -2.513 0.012013 *
## hour13        -0.91149    0.10131   -8.997 < 2e-16 ***
## hour14        -1.76055    0.10134  -17.373 < 2e-16 ***
## hour15        -2.27778    0.10134  -22.477 < 2e-16 ***
## hour16        -2.44818    0.10133  -24.161 < 2e-16 ***
## hour17        -2.38491    0.10133  -23.537 < 2e-16 ***
## hour18        -2.45692    0.10133  -24.248 < 2e-16 ***
## hour19        -2.22073    0.10133  -21.917 < 2e-16 ***
## hour20        -2.05160    0.10131  -20.250 < 2e-16 ***
## hour21        -1.62064    0.10124  -16.008 < 2e-16 ***
## hour22        -1.03265    0.10110  -10.214 < 2e-16 ***
## hour23        -0.38473    0.10109   -3.806 0.000143 ***
## year2025       -0.07914    0.03261   -2.427 0.015254 *
## weekday_num    -0.04310    0.03254   -1.325 0.185322
## fireworks1     0.25721    0.07700    3.340 0.000844 ***
## smoke1         0.16718    0.07721    2.165 0.030427 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9696 on 4384 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.5394, Adjusted R-squared:  0.5364
## F-statistic: 183.3 on 28 and 4384 DF, p-value: < 2.2e-16

no2lm6log <- lm(formula = log(no2Conc1HourMean.value + 1) ~ operating_f +
                hour + year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
                HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)
summary(no2lm6log)

##
## Call:
## lm(formula = log(no2Conc1HourMean.value + 1) ~ operating_f +
##     hour + year + weekday_num + fireworks + smoke + HourlyDryBulbTemperature +
##     HourlyRelativeHumidity + HourlyWindSpeed, data = allendale)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.3571 -0.4181  0.0202  0.4490  3.1166
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -2.425376   0.211398  -11.473 < 2e-16 ***
## operating_f0operating -0.168977   0.038841   -4.350 1.39e-05 ***
## hour1          0.183737   0.092212    1.993 0.046376 *
## hour2          0.257627   0.092361    2.789 0.005305 **
## hour3          0.256141   0.092283    2.776 0.005534 **
## hour4          0.245568   0.092450    2.656 0.007932 **
## hour5          0.209486   0.092356    2.268 0.023365 *
## hour6          0.178194   0.092593    1.924 0.054361 .

```

```
## hour7          0.219508    0.093354    2.351 0.018752 *
## hour8          0.342394    0.095116    3.600 0.000322 ***
## hour9          0.451875    0.097675    4.626 3.83e-06 ***
## hour10         0.533809    0.099018    5.391 7.39e-08 ***
## hour11         0.484294    0.100398    4.824 1.46e-06 ***
## hour12         0.220782    0.101589    2.173 0.029815 *
## hour13        -0.435443    0.101948   -4.271 1.99e-05 ***
## hour14        -1.259572    0.100744  -12.503 < 2e-16 ***
## hour15        -1.785400    0.099840  -17.883 < 2e-16 ***
## hour16        -1.984315    0.099208  -20.001 < 2e-16 ***
## hour17        -1.985192    0.097521  -20.356 < 2e-16 ***
## hour18        -2.118478    0.095162  -22.262 < 2e-16 ***
## hour19        -1.964371    0.093953  -20.908 < 2e-16 ***
## hour20        -1.887628    0.093240  -20.245 < 2e-16 ***
## hour21        -1.503460    0.092846  -16.193 < 2e-16 ***
## hour22        -0.995501    0.092474  -10.765 < 2e-16 ***
## hour23        -0.370682    0.092367   -4.013 6.10e-05 ***
## year2025       0.143650    0.031660    4.537 5.86e-06 ***
## weekday_num   -0.038493    0.029875   -1.288 0.197655
## fireworks1     0.075338    0.070690    1.066 0.286600
## smoke1        0.051424    0.070159    0.733 0.463627
## HourlyDryBulbTemperature 0.041175    0.002320   17.745 < 2e-16 ***
## HourlyRelativeHumidity  0.036974    0.001158   31.942 < 2e-16 ***
## HourlyWindSpeed  0.026649    0.003716    7.172 8.68e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8699 on 4196 degrees of freedom
## (188 observations deleted due to missingness)
## Multiple R-squared:  0.6335, Adjusted R-squared:  0.6308
## F-statistic: 234 on 31 and 4196 DF, p-value: < 2.2e-16
```

#PART 3: PREPARING DATA FOR INTERPOLATION

```
#convert classes for all sensors not just allendale
sensors_summer_2$hour <- as.factor(sensors_summer_2$hour)
sensors_summer_2$HourlyWindSpeed <- as.integer(sensors_summer_2$HourlyWindSpeed)
sensors_summer_2$date <- as.Date(sensors_summer_2$date, format = "%m/%d/%Y")
sensors_summer_2$year <- format(sensors_summer_2$date, "%Y")
sensors_summer_2$operating_f <- factor(sensors_summer_2$operating,
                                       levels = c(0, 1),
                                       labels = c("Not operating", "Operating"))
sensors_summer_2$datetime <- as.POSIXct(
  paste(sensors_summer_2$date, sensors_summer_2$hour),
  format = "%Y-%m-%d %H")

#write to a csv. updated December 13.
write.csv(sensors_summer_2, "data/output/summersensors.csv")

#prepare data for interpolation
#create a sequential week column of every 7 days
sensors_summer_2$date <- as.Date(sensors_summer_2$date, format = "%m/%d/%Y")

start <- min(sensors_summer_2$date)
```

```

sensors_summer_2$week <- factor(
  floor(as.numeric(sensors_summer_2$date - start) / 7) + 1
)

table(sensors_summer_2$week)

##
##      1      2      3      4      5      6      7      8      9     10     11     12     13     14     53     54
## 1512 1512 1512 1512 1512 1512 1512 1512 1512 1512 1512 1512 1512 216 1296 1512
##      55     56     57     58     59     60     61     62     63     64     65     66
## 1512 1512 1512 1512 1512 1512 1512 1512 1512 1512 1512 432
#i see low numbers of observations in week 14 and week 66 because they fall at the tail end.
#recode sensor weeks 14 to be part of 13 and week 66 to be part of 65
sensors_summer_2$week <- recode(sensors_summer_2$week,
                                "14" = "13",
                                "66" = "65")

table(sensors_summer_2$week)

##
##      1      2      3      4      5      6      7      8      9     10     11     12     13     53     54     55
## 1512 1512 1512 1512 1512 1512 1512 1512 1512 1512 1512 1512 1728 1296 1512 1512
##      56     57     58     59     60     61     62     63     64     65
## 1512 1512 1512 1512 1512 1512 1512 1512 1512 1944

#write to a csv.
write.csv(sensors_summer_2, "data/output/summersensors.csv")

#calculate the average hourly no2 per week for each sensor when the power plant is on
#keep the longitude and latitude columns
#if any of those weeks contains a value of 1 for fireworks
#(indicating on or around july 4)
#put a value of 1 for fireworks2

#if any of those weeks contains a value of 1 for smoke
#(indicating air quality alert due to wildfire smoke
#put a value of 1 for smoke2)

no2_weekly_on <- sensors_summer_2 |> dplyr::filter(operating == 1,
                                                  !is.na(no2Conc1HourMean.value)) |>

  group_by(week, datasourceId) |>
  summarise(
    lon = first(longitude),
    lat = first(latitude),
    no2mean = mean(no2Conc1HourMean.value, na.rm = TRUE),
    fireworks2 = ifelse(any(fireworks == 1), 1, 0),
    smoke2 = ifelse(any(smoke == 1), 1, 0),
    temp = mean(HourlyDryBulbTemperature),
    humidity = mean(HourlyRelativeHumidity),
    windspeed = mean(HourlyWindSpeed),
    n_obs = n(),
    .groups = "drop"
  )

```

```
table(no2_weekly_on$week)
```

```
##
##  1  2  3  4  5  6  7  8  9 10 11 12 13 53 54 55 56 57 58 59 60 61 62 63 64 65
##  0  9  9  9  0  9  9  0  9  9  0  0  9  9  0  9  9  9  9  8  8  8  8  8  8  8
```

```
no2_weekly_on$operating = "Operating"
```

```
no2_weekly_on$year = ifelse(no2_weekly_on$week %in% c("1", "2", "3", "4", "5", "6", "7", "8", "9", "10"
```

```
#calculate the average hourly no2 per week for each sensor when the power plant is OFF
#keep the longitude and latitude columns
```

```
#if any of those weeks contains a value of 1 for fireworks
```

```
 #(indicating on or around july 4)
```

```
#put a value of 1 for fireworks2
```

```
#if any of those weeks contains a value of 1 for smoke
```

```
 #(indicating air quality alert due to wildfire smoke
```

```
#put a value of 1 for smoke2)
```

```
no2_weekly_off <- sensors_summer_2 |> dplyr::filter(operating == 0,
                                                    !is.na(no2Conc1HourMean.value)) |>
```

```
  group_by(week, datasourceId) |>
  summarise(
    lon = first(longitude),
    lat = first(latitude),
    no2mean = mean(no2Conc1HourMean.value, na.rm = TRUE),
    fireworks2 = ifelse(any(fireworks == 1), 1, 0),
    smoke2 = ifelse(any(smoke == 1), 1, 0),
    temp = mean(HourlyDryBulbTemperature),
    humidity = mean(HourlyRelativeHumidity),
    windspeed = mean(HourlyWindSpeed),
    n_obs = n(),
    .groups = "drop"
  )
```

```
table(no2_weekly_off$week)
```

```
##
##  1  2  3  4  5  6  7  8  9 10 11 12 13 53 54 55 56 57 58 59 60 61 62 63 64 65
##  9  9  9  9  9  9  9  9  9  9  9  9  9  9  9  9  9  9  9  0  8  8  8  8  8  8
```

```
no2_weekly_off$operating = "Not operating"
```

```
no2_weekly_off$year = ifelse(no2_weekly_off$week %in% c("1", "2", "3", "4", "5", "6", "7", "8", "9", "10"
```

```
#bind rows
```

```
no2 <- bind_rows(no2_weekly_on, no2_weekly_off)
```

```
write.csv(no2, "data/output/no2.csv")
```

```
#PART 4: INTERPOLATION
```

```
#no2 is the data that we want to use for interpolation
```

```
#convert no2 to a spatial object
```

```
no2_sf <- st_as_sf(no2, coords = c("lon", "lat"), crs=4326)
```

```
no2_26986 <- st_transform(no2_sf, crs = 26986)
```

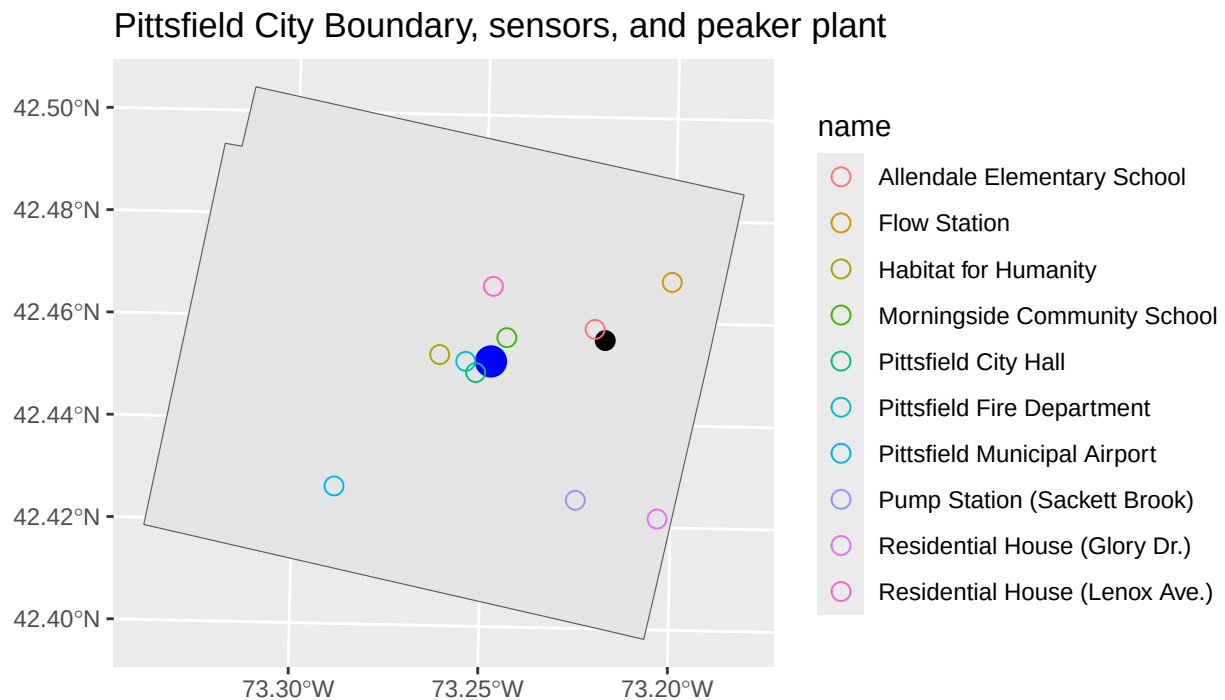
```

#get pittsfield boundaries and other shapefiles
pittsfield26986 <- sf::read_sf("data/input/pittsfield26986/pittsfield26986.shp")
peaker26986 <- sf::read_sf("data/input/peaker26986/peaker26986.shp")
sensors26986 <- sf::read_sf("data/input/sensors26986/sensors26986.shp")
bg26986 <- sf::read_sf("data/input/pittsfieldbg26986/pittsfieldbg26986.shp")
airport26986 <- sf::read_sf("data/input/airport26986/airport26986.shp")
majroads26986 <- sf::read_sf("data/input/majroads26986/majroads26986.shp")

#get population-weighted centroid
centroid4326 <- st_sf(
  geometry = st_sfc(st_point(c(-73.24816089, 42.45187495))),
  crs = st_crs(4326) # make sure CRS matches the raster
)
centroid26986 <- st_transform(centroid4326, st_crs(pittsfield26986))

#plot some points. COME BACK TO THIS
ggplot(pittsfield26986) + geom_sf(data = pittsfield26986) +
  geom_sf(data = peaker26986, color = "black", size = 3) +
  geom_sf(data = centroid26986, color = "blue", size = 5) +
  geom_sf(data = sensors26986, aes(color = name), shape = 21, size = 3) +
  labs(title = "Pittsfield City Boundary, sensors, and peaker plant")

```



```

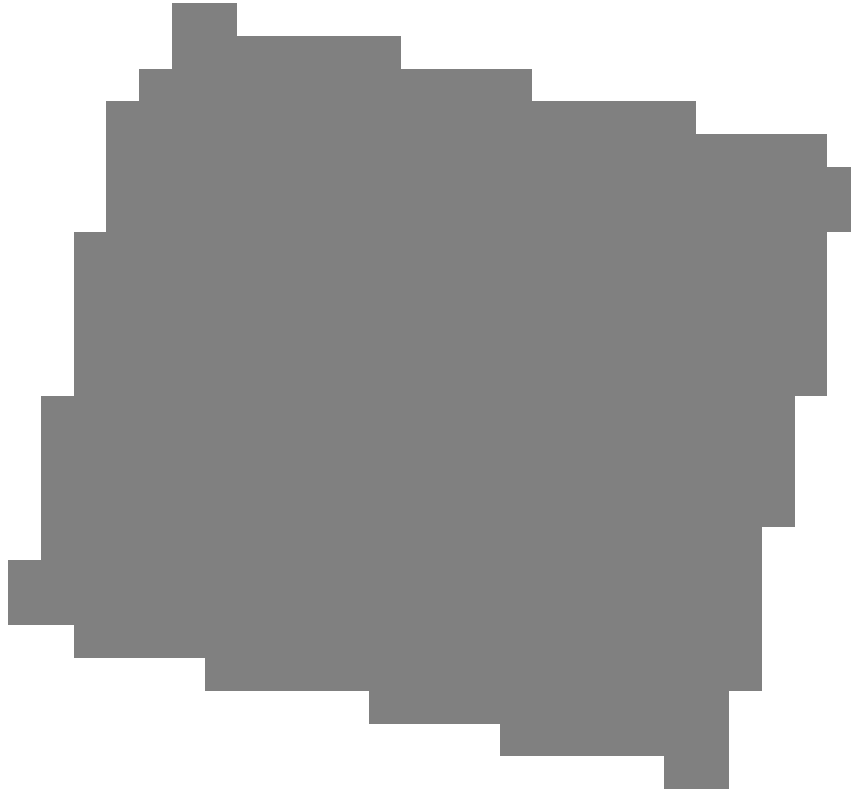
#create a 500m grid
grid500 <- sf::st_bbox(pittsfield26986) |> st_as_stars(dx=500) |>
  st_crop(pittsfield26986)

```

```
plot(grid500)
```

```
## Warning in plot.stars(grid500): breaks="quantile" leads to a single class;  
## maybe try breaks="equal" instead?
```

values



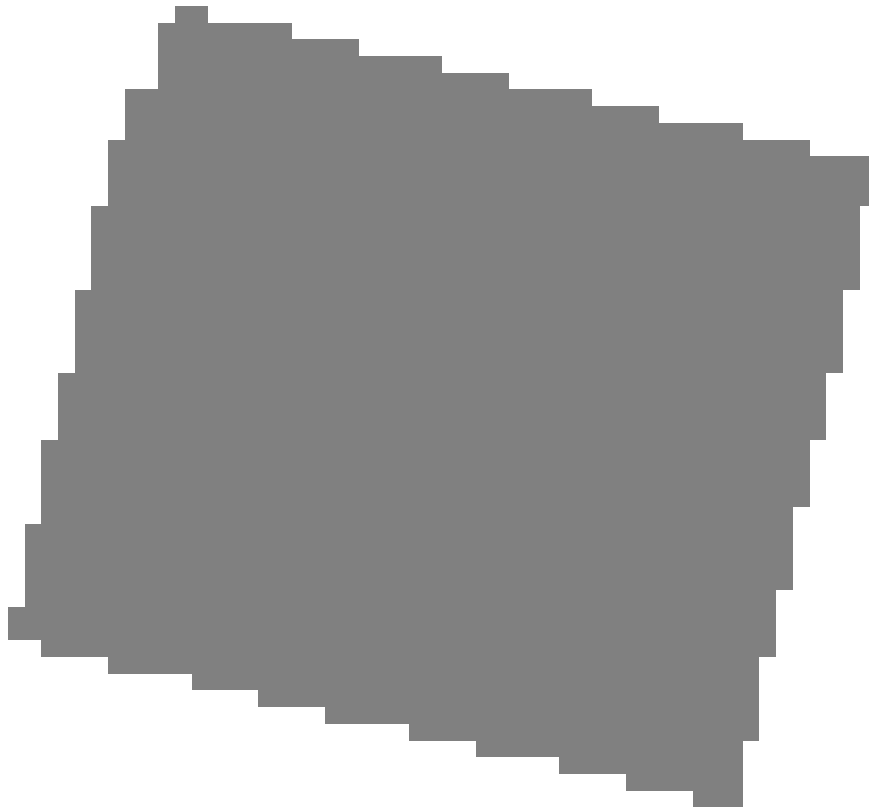
```
#create a 250m resolution grid
```

```
grid250 <- sf::st_bbox(pittsfield26986) |> st_as_stars(dx=250) |>  
  st_crop(pittsfield26986)
```

```
plot(grid250)
```

```
## Warning in plot.stars(grid250): breaks="quantile" leads to a single class;  
## maybe try breaks="equal" instead?
```

values



```
#test 2 using the terra package
#pittsfield_boundary <- pittsfield26986 |> st_cast("MULTILINESTRING")
#pittsfield_boundary_vect <- terra::vect(pittsfield_boundary)

#pittsfield_raster500 <- terra::rast(
  # ext(pittsfield_boundary_vect),
  #resolution = 500,
  #crs = st_crs(pittsfield_boundary)$proj4string
#)

#hasValues(pittsfield_raster500)

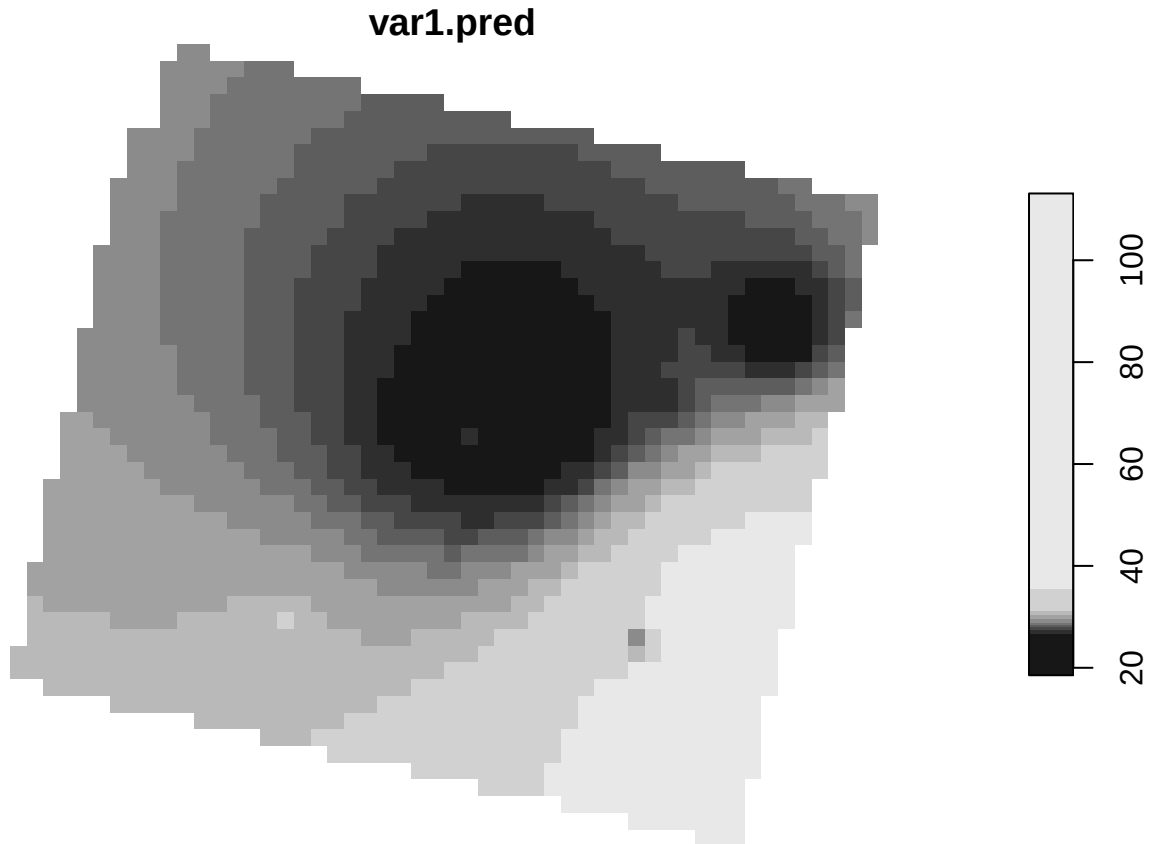
#raster500 <- as.points(pittsfield_raster500, values = FALSE)

##test interpolating a raster and extracting data at a point with dummy data
#test idw for no2, no filtering
#this is with dummy data so won't be scientifically accurate
idw_test1 <- idw(formula = no2mean ~1,
  locations = no2_26986 ,
  newdata = grid250,
  idp = 1)

## [inverse distance weighted interpolation]
names(idw_test1)

## [1] "var1.pred" "var1.var"
```

```
plot(idw_test1)["var1.pred"]
```



```
## NULL
```

```
summary(idw_test1$var1.pred)
```

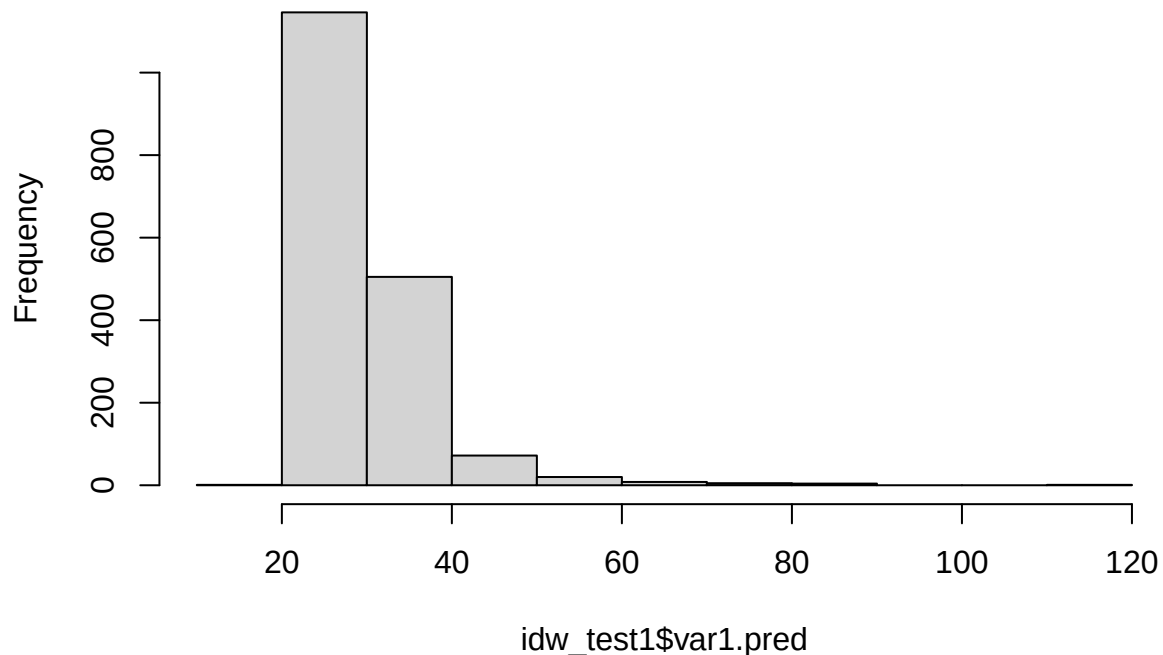
##	V1	V2	V3	V4
##	Min. :29.13	Min. :28.84	Min. :28.59	Min. :28.32
##	1st Qu.:29.14	1st Qu.:28.93	1st Qu.:28.72	1st Qu.:28.46
##	Median :29.16	Median :29.02	Median :28.86	Median :28.67
##	Mean :29.16	Mean :29.02	Mean :28.87	Mean :28.69
##	3rd Qu.:29.17	3rd Qu.:29.10	3rd Qu.:29.01	3rd Qu.:28.90
##	Max. :29.18	Max. :29.19	Max. :29.16	Max. :29.13
##	NA's :51	NA's :45	NA's :41	NA's :36
##	V5	V6	V7	V8
##	Min. :28.16	Min. :28.03	Min. :27.90	Min. :27.75
##	1st Qu.:28.25	1st Qu.:28.10	1st Qu.:27.98	1st Qu.:27.89
##	Median :28.49	Median :28.32	Median :28.19	Median :28.17
##	Mean :28.53	Mean :28.44	Mean :28.33	Mean :28.25
##	3rd Qu.:28.79	3rd Qu.:28.75	3rd Qu.:28.63	3rd Qu.:28.48
##	Max. :29.10	Max. :29.19	Max. :29.17	Max. :29.15
##	NA's :32	NA's :25	NA's :21	NA's :16
##	V9	V10	V11	V12
##	Min. :27.60	Min. :27.42	Min. :27.23	Min. :27.02
##	1st Qu.:27.79	1st Qu.:27.70	1st Qu.:27.56	1st Qu.:27.42
##	Median :28.15	Median :28.09	Median :27.96	Median :27.78
##	Mean :28.21	Mean :28.17	Mean :28.06	Mean :27.93

##	3rd Qu.:28.54	3rd Qu.:28.62	3rd Qu.:28.51	3rd Qu.:28.44
##	Max. :29.20	Max. :29.26	Max. :29.18	Max. :29.17
##	NA's :11	NA's :7	NA's :7	NA's :7
##	V13	V14	V15	V16
##	Min. :26.76	Min. :26.46	Min. :26.08	Min. :25.33
##	1st Qu.:27.27	1st Qu.:27.08	1st Qu.:26.74	1st Qu.:26.43
##	Median :27.61	Median :27.49	Median :27.39	Median :27.26
##	Mean :27.80	Mean :27.65	Mean :27.48	Mean :27.29
##	3rd Qu.:28.36	3rd Qu.:28.26	3rd Qu.:28.23	3rd Qu.:28.20
##	Max. :29.24	Max. :29.25	Max. :29.26	Max. :29.27
##	NA's :7	NA's :7	NA's :7	NA's :7
##	V17	V18	V19	V20
##	Min. :23.61	Min. :23.26	Min. :24.49	Min. :24.09
##	1st Qu.:26.25	1st Qu.:26.08	1st Qu.:26.25	1st Qu.:26.37
##	Median :27.19	Median :27.16	Median :27.32	Median :27.49
##	Mean :27.08	Mean :27.03	Mean :27.19	Mean :27.30
##	3rd Qu.:28.19	3rd Qu.:28.18	3rd Qu.:28.34	3rd Qu.:28.39
##	Max. :29.29	Max. :29.39	Max. :29.41	Max. :29.45
##	NA's :7	NA's :7	NA's :7	NA's :7
##	V21	V22	V23	V24
##	Min. :22.66	Min. :18.54	Min. :21.08	Min. :23.96
##	1st Qu.:26.30	1st Qu.:26.32	1st Qu.:26.32	1st Qu.:26.74
##	Median :27.82	Median :27.94	Median :28.14	Median :28.43
##	Mean :27.35	Mean :27.35	Mean :27.62	Mean :28.03
##	3rd Qu.:28.66	3rd Qu.:28.99	3rd Qu.:29.25	3rd Qu.:29.40
##	Max. :29.60	Max. :30.30	Max. :30.71	Max. :31.52
##	NA's :7	NA's :7	NA's :7	NA's :7
##	V25	V26	V27	V28
##	Min. :25.06	Min. :25.33	Min. :26.15	Min. :26.80
##	1st Qu.:26.74	1st Qu.:27.19	1st Qu.:27.52	1st Qu.:28.03
##	Median :28.71	Median :28.94	Median :29.12	Median :29.37
##	Mean :28.40	Mean :28.80	Mean :29.18	Mean :29.68
##	3rd Qu.:29.61	3rd Qu.:29.73	3rd Qu.:29.85	3rd Qu.:29.96
##	Max. :32.39	Max. :33.32	Max. :34.14	Max. :35.34
##	NA's :7	NA's :7	NA's :7	NA's :7
##	V29	V30	V31	V32
##	Min. :27.41	Min. :27.96	Min. :28.47	Min. :28.94
##	1st Qu.:28.39	1st Qu.:28.86	1st Qu.:29.27	1st Qu.:29.65
##	Median :29.60	Median :29.79	Median :29.93	Median :30.09
##	Mean :30.22	Mean :30.81	Mean :31.26	Mean :31.90
##	3rd Qu.:30.36	3rd Qu.:31.06	3rd Qu.:31.28	3rd Qu.:31.79
##	Max. :36.71	Max. :38.31	Max. :40.19	Max. :42.63
##	NA's :7	NA's :7	NA's :8	NA's :7
##	V33	V34	V35	V36
##	Min. :29.36	Min. :29.74	Min. :30.07	Min. :29.33
##	1st Qu.:30.10	1st Qu.:30.29	1st Qu.:30.39	1st Qu.:30.49
##	Median :30.27	Median :30.39	Median :30.61	Median :30.70
##	Mean :32.61	Mean :33.40	Mean :34.27	Mean :34.65
##	3rd Qu.:32.37	3rd Qu.:32.80	3rd Qu.:33.03	3rd Qu.:32.49
##	Max. :45.80	Max. :50.12	Max. :56.05	Max. :65.53
##	NA's :7	NA's :7	NA's :7	NA's :8
##	V37	V38	V39	V40
##	Min. :30.56	Min. : 30.64	Min. :30.72	Min. :30.80
##	1st Qu.:30.61	1st Qu.: 30.71	1st Qu.:30.82	1st Qu.:30.97

```
## Median :30.82 Median : 30.89 Median :31.24 Median :31.75
## Mean :35.88 Mean : 37.55 Mean :37.28 Mean :36.38
## 3rd Qu.:32.92 3rd Qu.: 34.28 3rd Qu.:35.24 3rd Qu.:36.11
## Max. :81.83 Max. :113.11 Max. :85.83 Max. :70.11
## NA's :7 NA's :7 NA's :9 NA's :14
## V41 V42 V43 V44
## Min. :30.95 Min. :31.16 Min. :31.54 Min. :32.45
## 1st Qu.:31.24 1st Qu.:31.74 1st Qu.:32.54 1st Qu.:33.98
## Median :32.70 Median :33.74 Median :35.02 Median :36.92
## Mean :36.76 Mean :37.15 Mean :37.68 Mean :38.66
## 3rd Qu.:37.98 3rd Qu.:39.70 3rd Qu.:41.20 3rd Qu.:42.66
## Max. :61.33 Max. :55.76 Max. :51.93 Max. :49.13
## NA's :19 NA's :23 NA's :27 NA's :32
## V45 V46 V47 V48 V49
## Min. :33.78 Min. :36.33 Min. :39.09 Min. :41.49 Min. : NA
## 1st Qu.:35.51 1st Qu.:38.12 1st Qu.:40.31 1st Qu.:41.78 1st Qu.: NA
## Median :38.23 Median :40.34 Median :41.54 Median :42.06 Median : NA
## Mean :39.06 Mean :40.47 Mean :41.45 Mean :42.03 Mean :NaN
## 3rd Qu.:42.29 3rd Qu.:42.81 3rd Qu.:42.67 3rd Qu.:42.30 3rd Qu.: NA
## Max. :46.41 Max. :44.84 Max. :43.58 Max. :42.53 Max. : NA
## NA's :37 NA's :42 NA's :46 NA's :50 NA's :53
```

```
hist(idw_test1$var1.pred)
```

Histogram of `idw_test1$var1.pred`



```
st_extract(idw_test1["var1.pred"], centroid26986)
```

```
## Simple feature collection with 1 feature and 1 field
```

```

## Geometry type: POINT
## Dimension:      XY
## Bounding box:   xmin: 56206.54 ymin: 912734.4 xmax: 56206.54 ymax: 912734.4
## Projected CRS:  NAD83 / Massachusetts Mainland
##   var1.pred      geometry
## 1  24.49184 POINT (56206.54 912734.4)

##create a function that runs IDW for one group and extracts the value at the population-weighted centroid
#create a for loop that interpolates a raster for when the power plant is ON and for a specific week
#prepare what I need for the loop

wks <-sort(unique(no2_26986$week))
operating_statuses <- sort(unique(no2_26986$operating))

idw_weekly_results <- list()

idw_rasters <- list()

row_id <- 1

for(week_id in wks){
  for(status in operating_statuses){

    #this filters data for unique combos of week ID and operation status
    no2group <- no2_26986 |> dplyr::filter(week == week_id,
                                          operating == status)

    #skip if there aren't any points since the power plant doesn't run every week
    if (nrow(no2group) < 3 || all(is.na(no2group$no2mean))) {
      next
    }

    #run the IDW formula, setting idp = 2
    idw_group <- gstat::idw(formula = no2mean~1,
                           locations = no2group,
                           newdata = grid250,
                           idp = 2
                           )

    idw_df <- as.data.frame(idw_group["var1.pred"], xy = TRUE)

    ggraster <- ggplot() + geom_raster(data = idw_df,
                                      aes(x = x, y = y, fill = var1.pred)) +
      geom_sf(data = pittsfield26986, fill = NA, color = "black", linewidth = 1) +
      geom_sf(data = no2group, aes(color= "Sensors", shape = "Sensors"), size = 2, stroke = 1) +
      geom_sf(data = centroid26986, aes(color = "Population-weighted centroid", shape = "Population-weighted centroid"), size = 4, stroke = 1) +
      geom_sf(data = airport26986, aes(color = "Airport", shape = "Airport"), size = 4, stroke = 2) +
      geom_sf(data = peaker26986, aes(color = "Peaker plant", shape = "Peaker plant"), size = 4, stroke = 1) +
      scale_fill_viridis_c(option = "turbo", na.value = NA, limits = c(0, 60)) +
      scale_color_manual(
        name = "Locations",
        values = c("Sensors" = "black", "Population-weighted centroid" = "blue", "Airport" = "gray", "Peaker plant" = "red")
      )
  }
}

```

```

scale_shape_manual(
  name = "Locations",
  values = c(
    "Sensors" = 21,
    "Population-weighted centroid" = 4,
    "Airport" = 24,
    "Peaker plant" = 24)) +
coord_sf() +
labs(title = paste0("Average hourly NO2 levels for Week ", week_id, ": ", status),
     fill = "NO2 (ppb)") +
theme_minimal()

print(ggraster)

# Clean file name (remove spaces)
status_clean <- gsub(" ", "_", status)

# Construct output file name
png_name <- paste0("plots/ggraster_week_", week_id, "_", status_clean, ".png")

ggsave(
  filename = png_name,
  plot      = ggraster,
  width     = 10,      # inches
  height    = 8,
  dpi       = 300      # high resolution
)

raster_name <- paste0("Week_", week_id, "_", status)
idw_rasters[[raster_name]] <- idw_group

#extract the value of the raster at the centroid point
centroid_no2mean <- st_extract(idw_group["var1.pred"], centroid26986)[[1]]

#assign the covariates to the extracted value for each week and operating status
centroidno2 <- no2group |>
  sf::st_drop_geometry() |>
  dplyr::summarise(
    week = first(week),
    operating = first(operating),
    fireworks2 = first(fireworks2),
    smoke2 = first(smoke2),
    temp = first(temp),
    humidity = first(humidity),
    windspeed = first(windspeed),
    year = first(year),
    .groups = "drop"
  )

#create a new column NO2 mean based on the extracted value of the centroid
centroidno2$no2mean <- centroid_no2mean

```

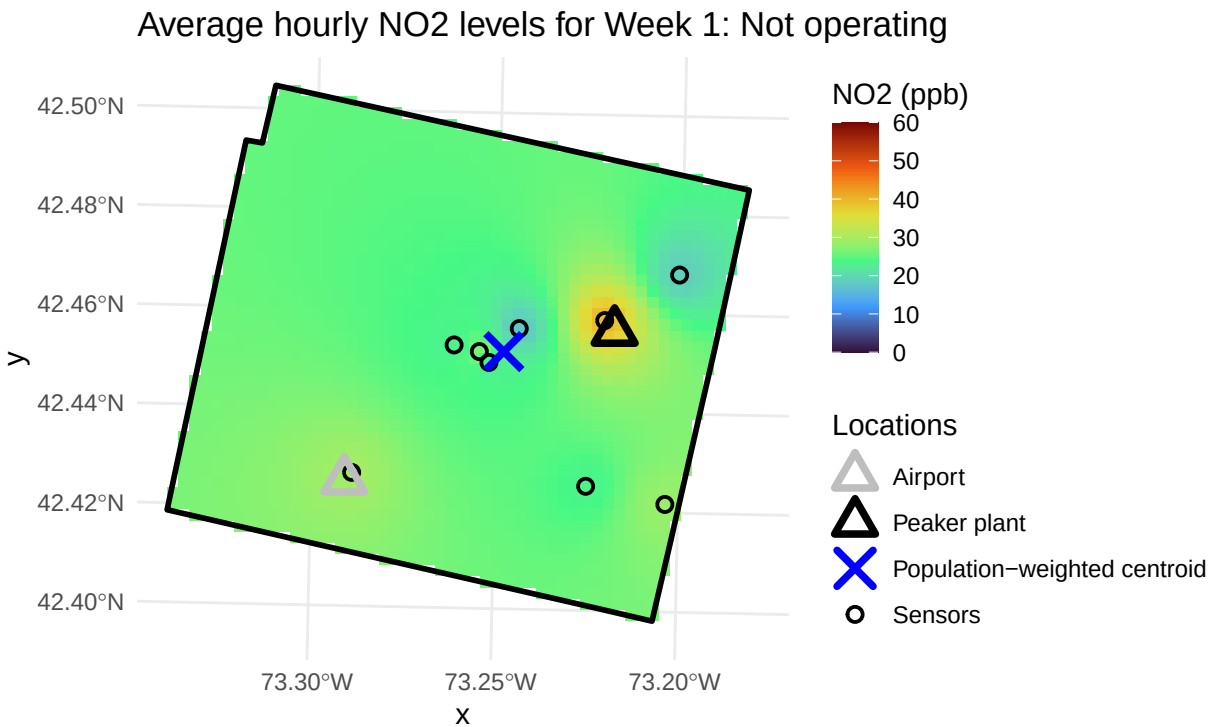
```

idw_weekly_results[[row_id]] <- centroidno2
row_id <- row_id + 1
}
}

## [inverse distance weighted interpolation]

## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).

```

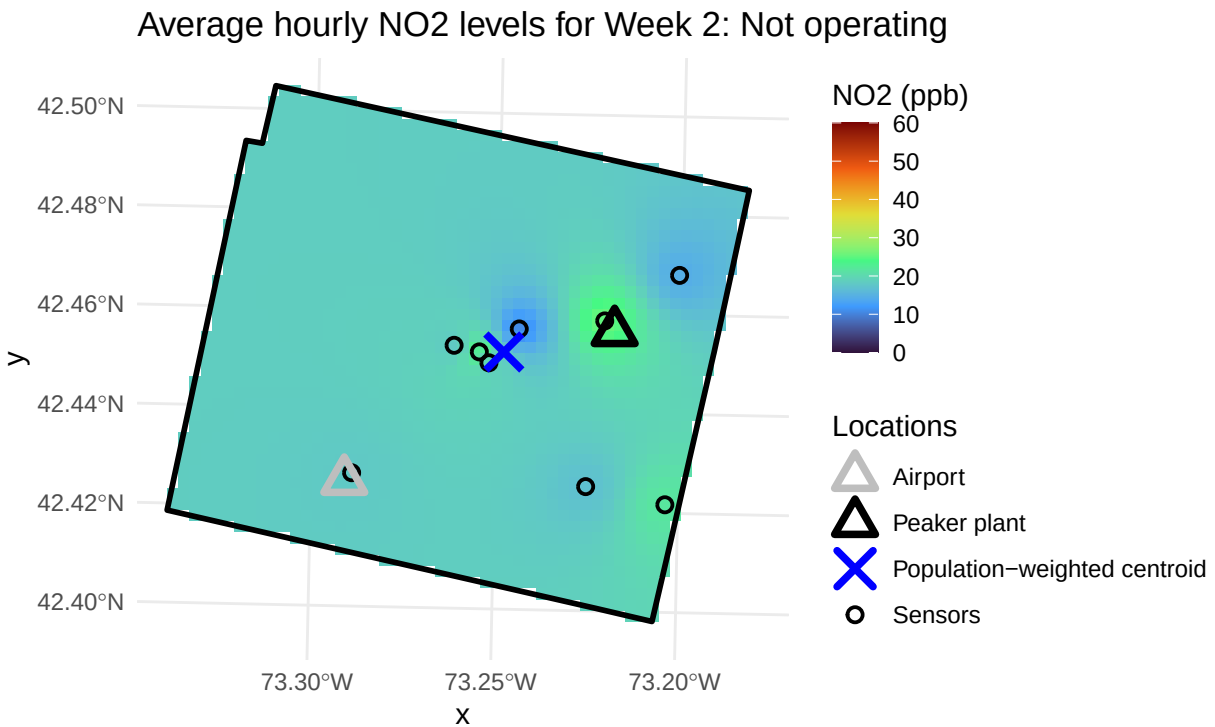


```

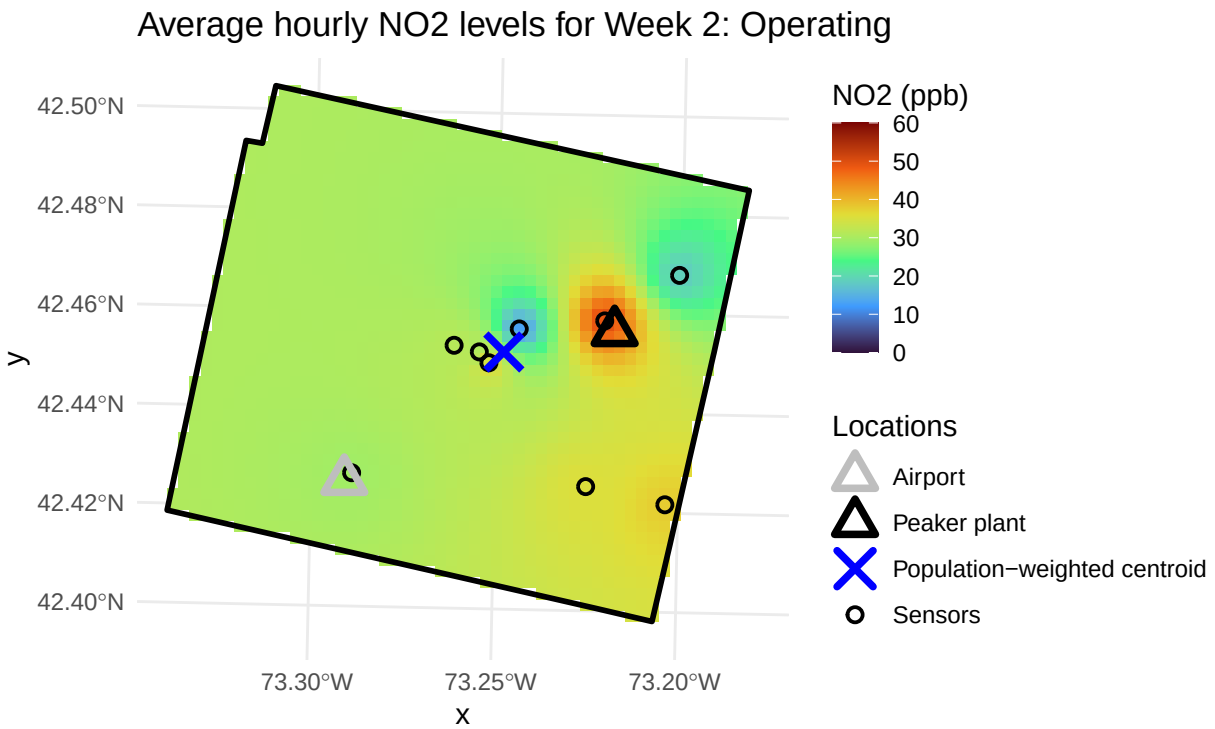
## [inverse distance weighted interpolation]

## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).

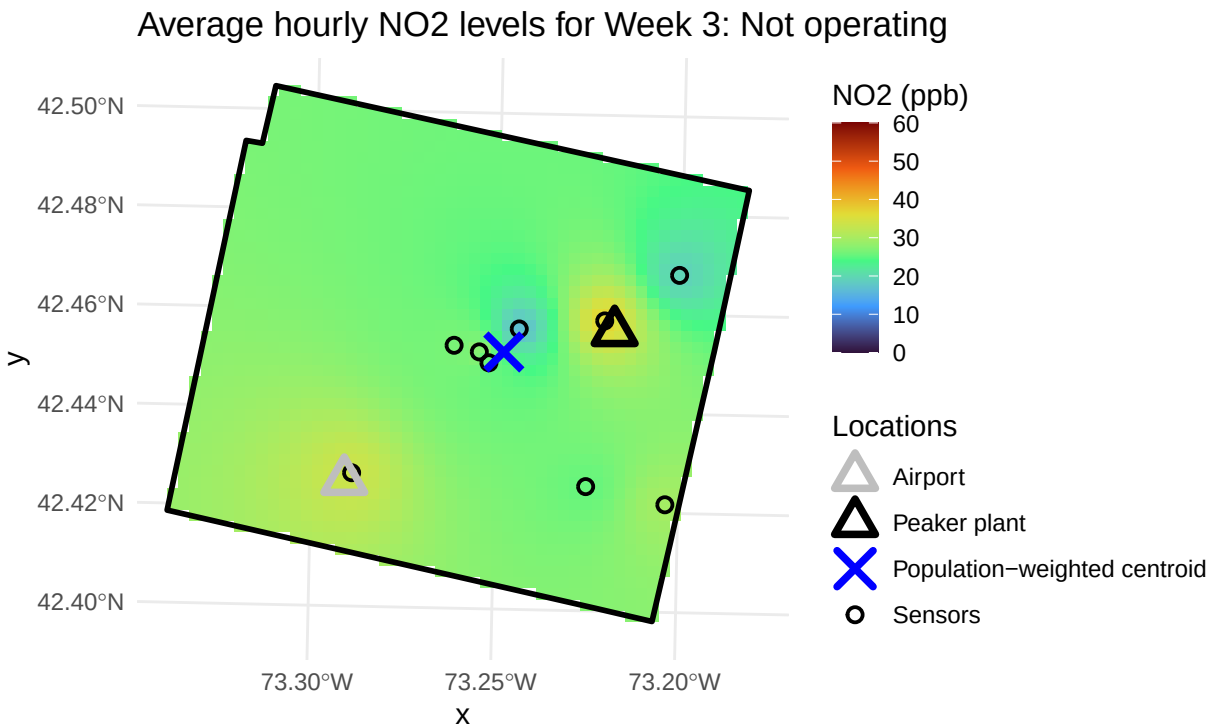
```



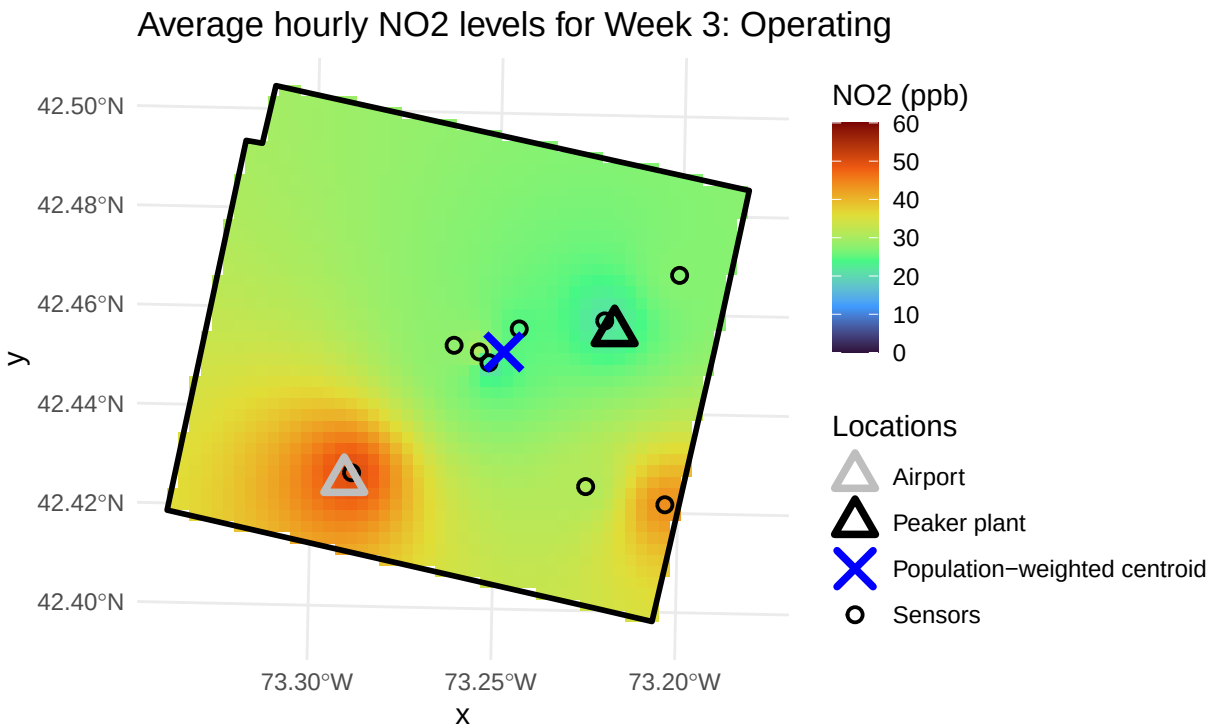
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



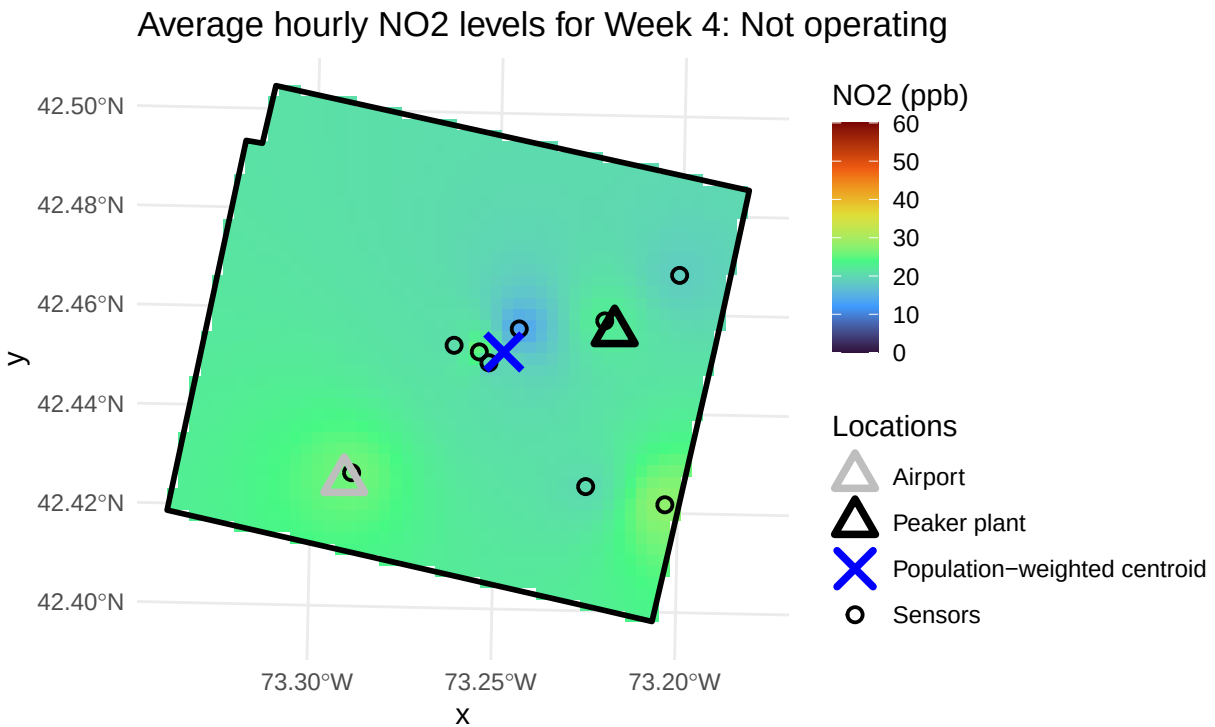
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



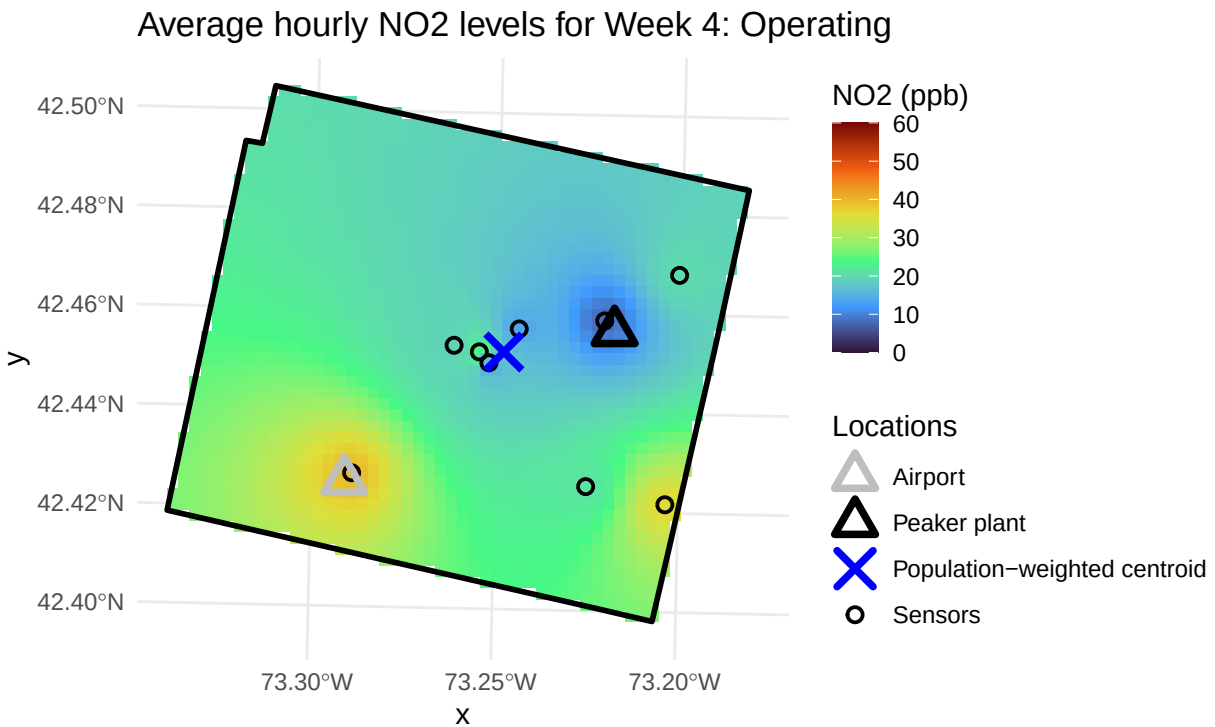
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



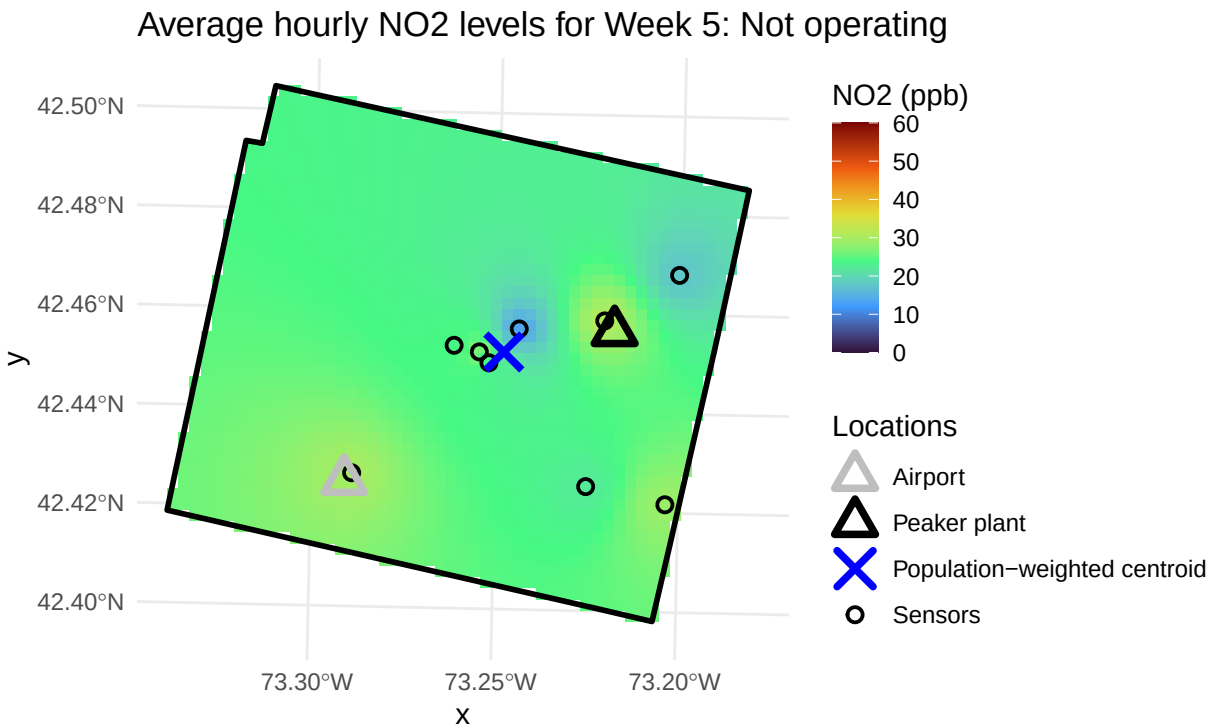
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



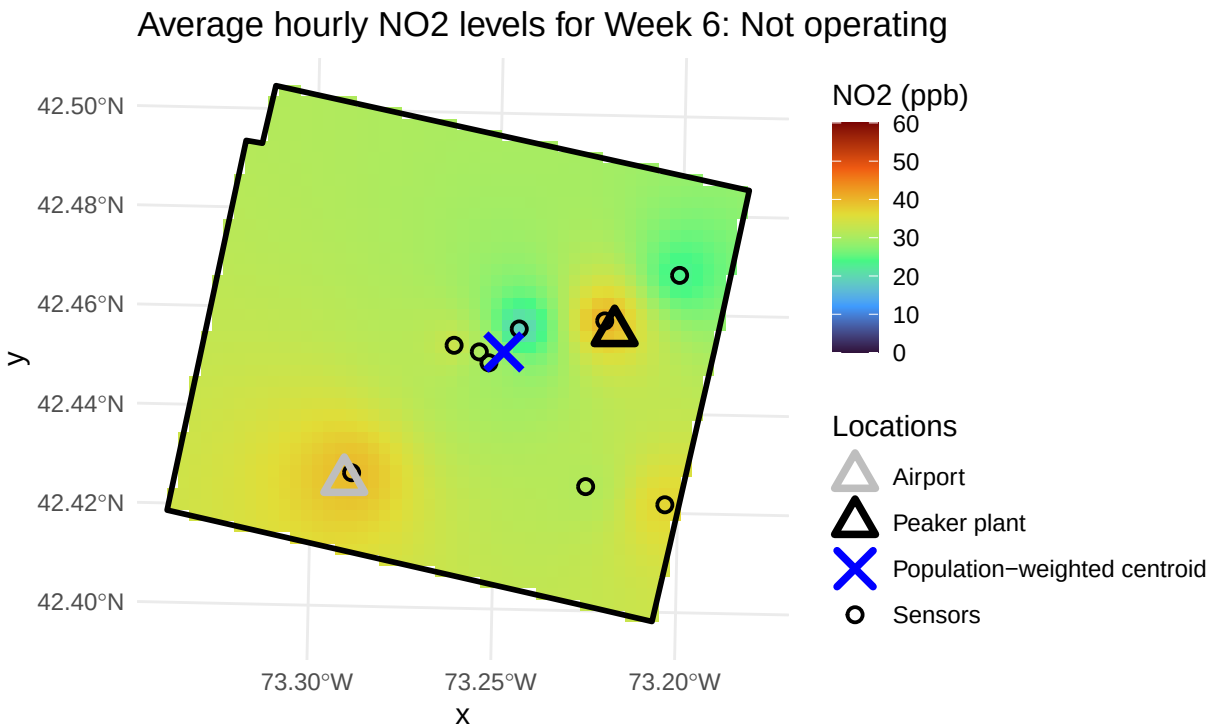
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



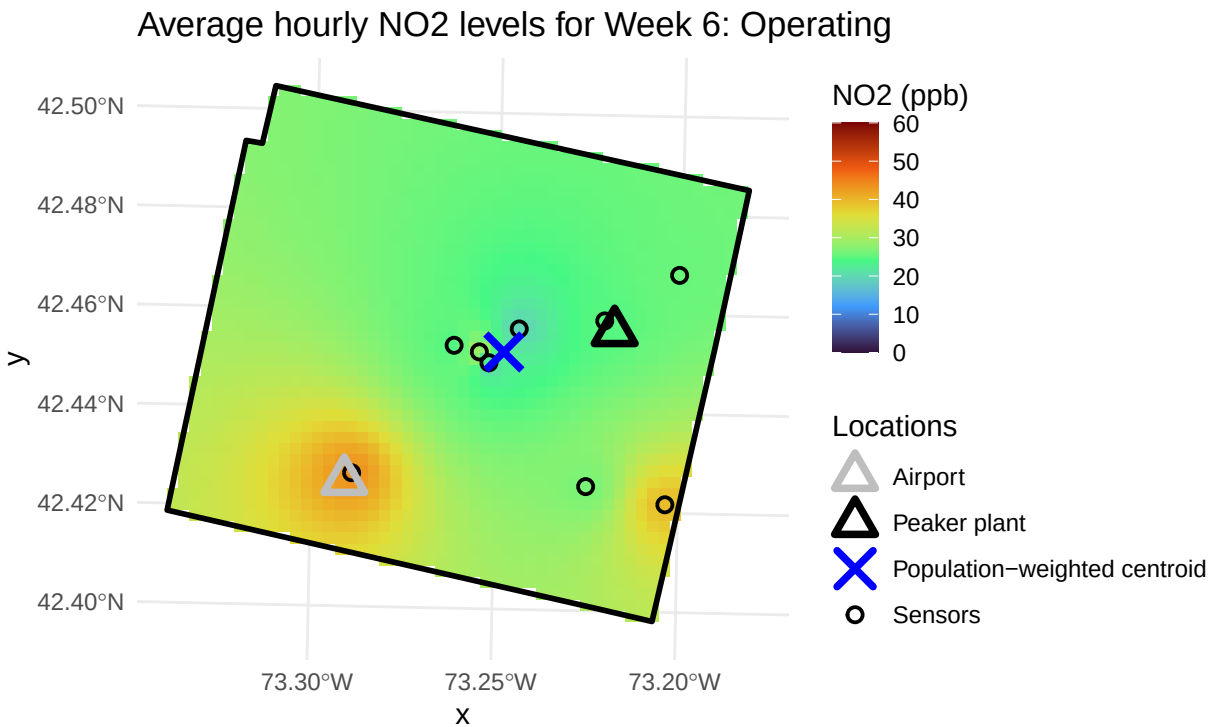
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



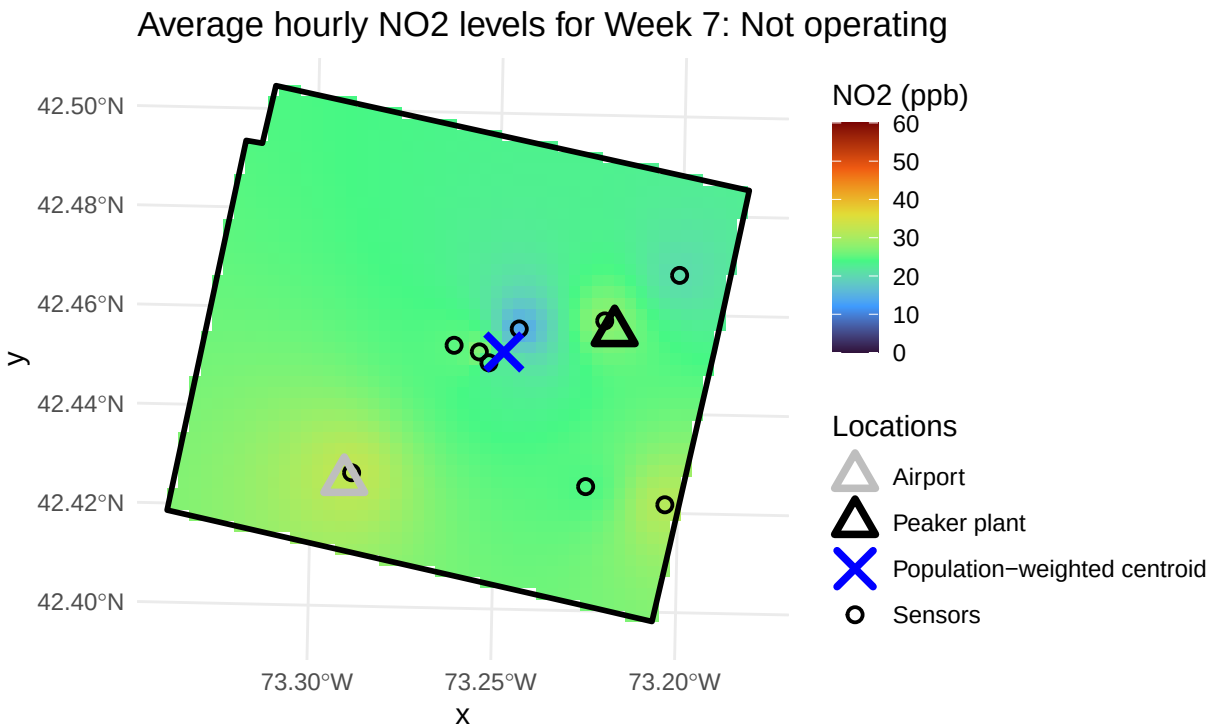
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



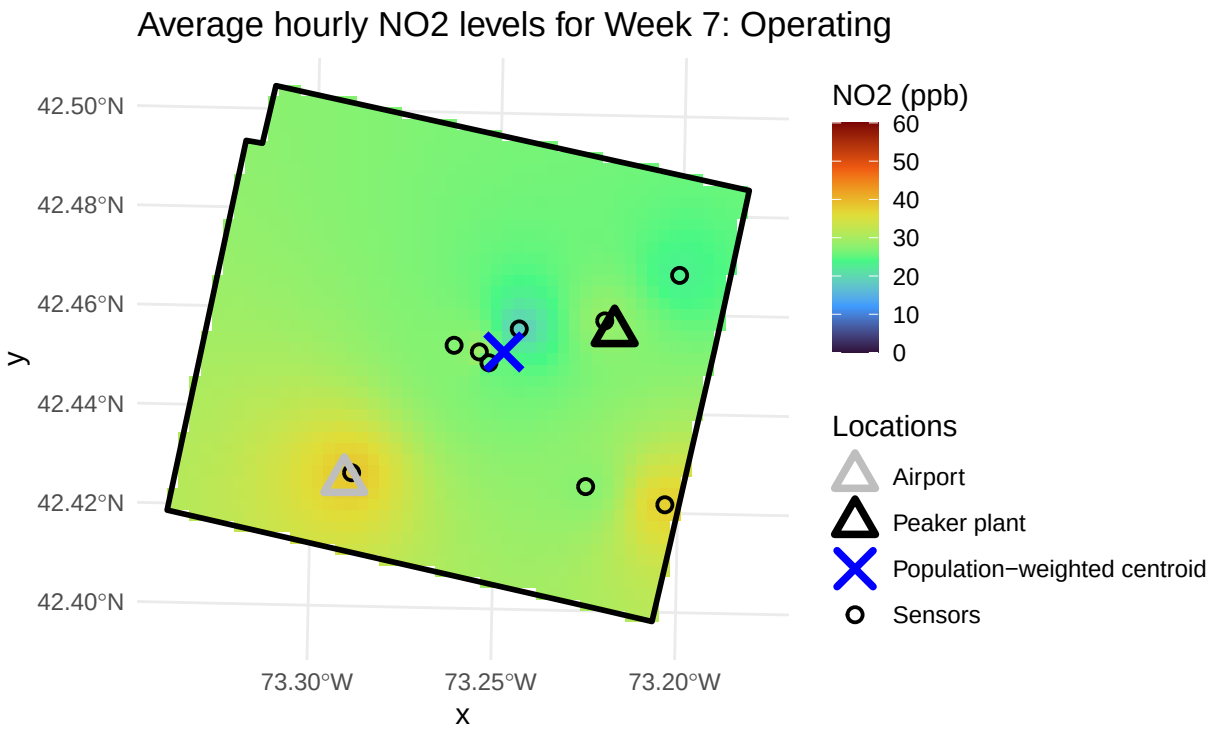
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



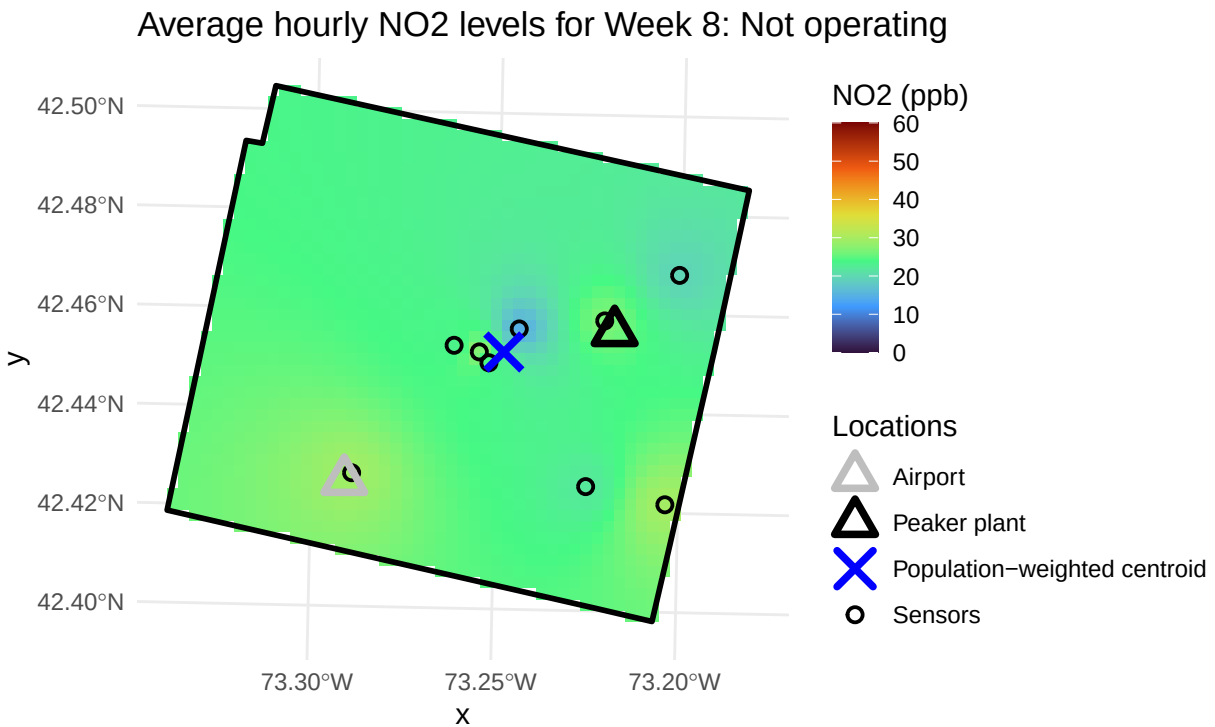
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



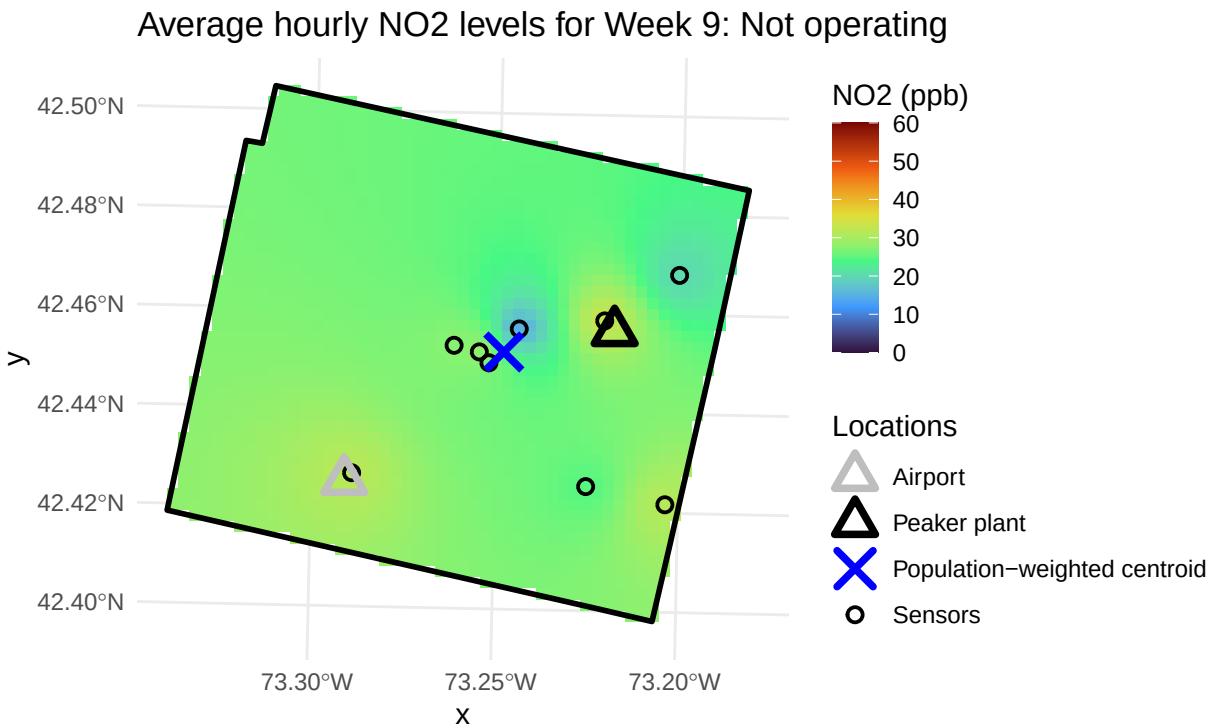
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



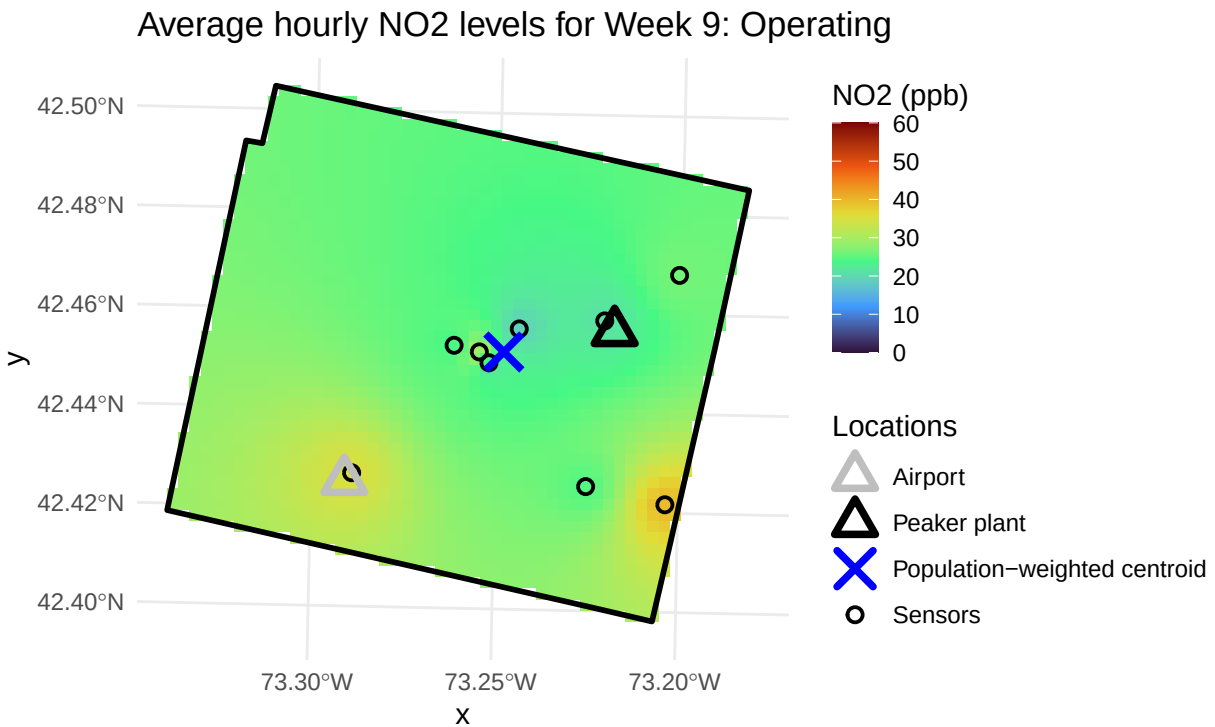
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



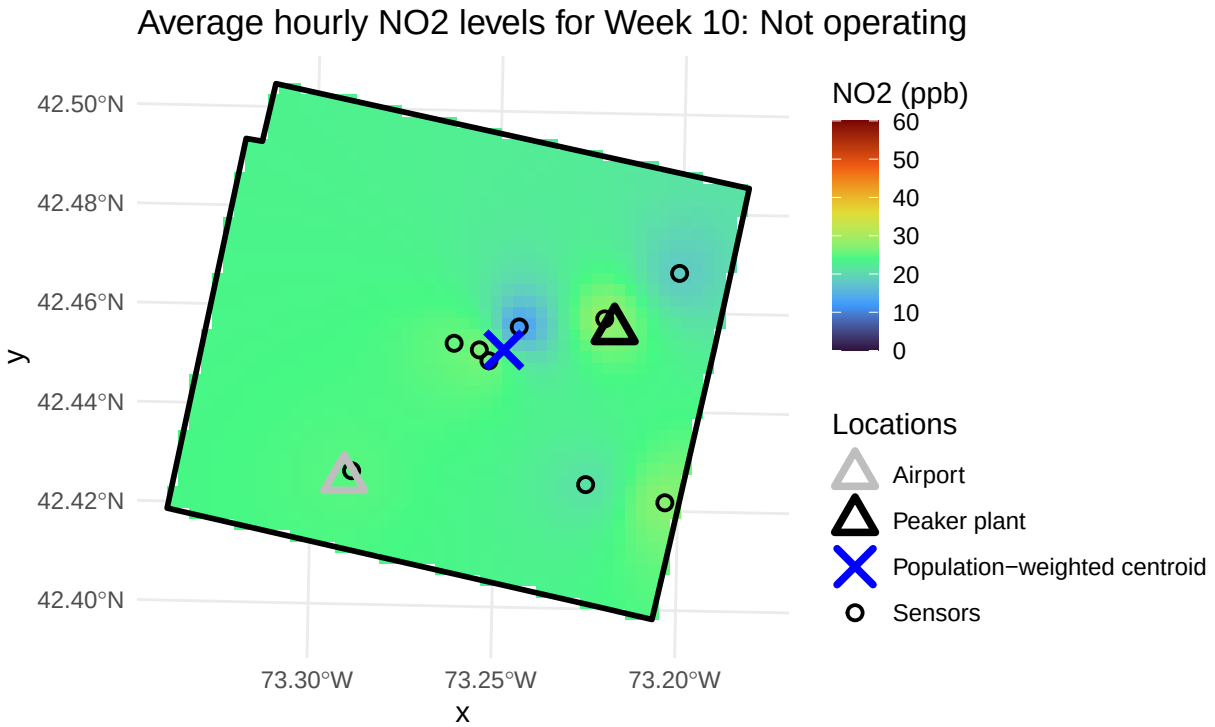
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



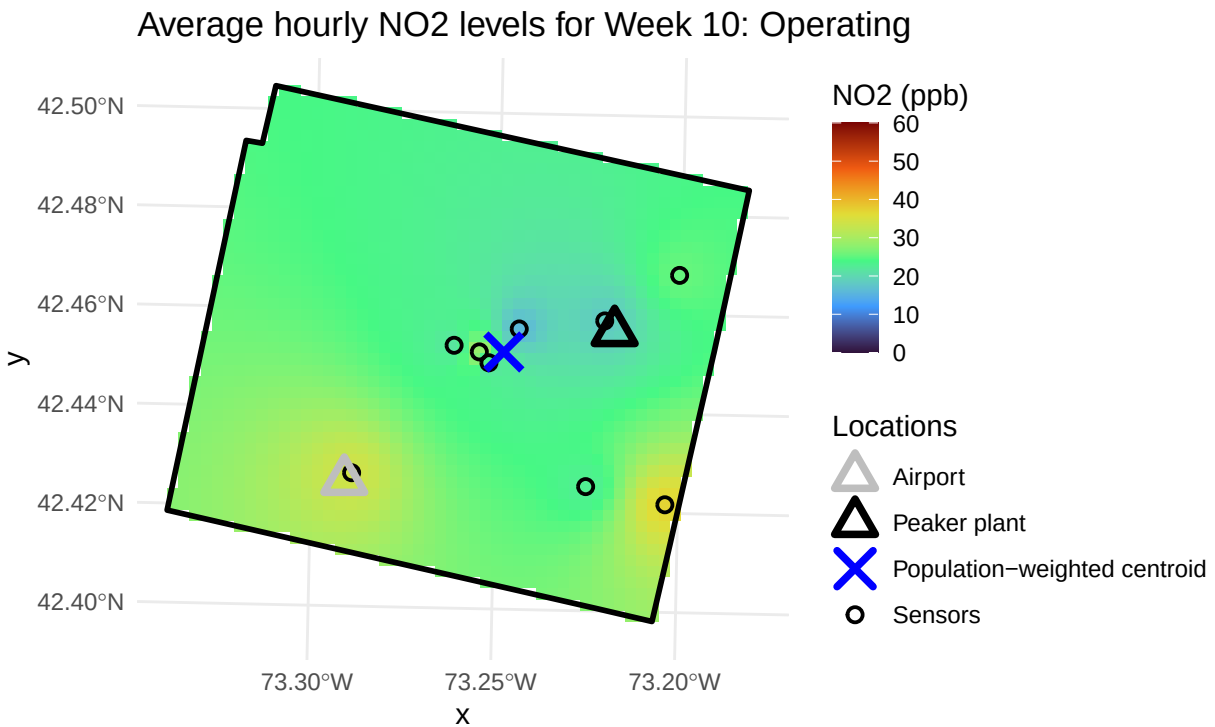
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



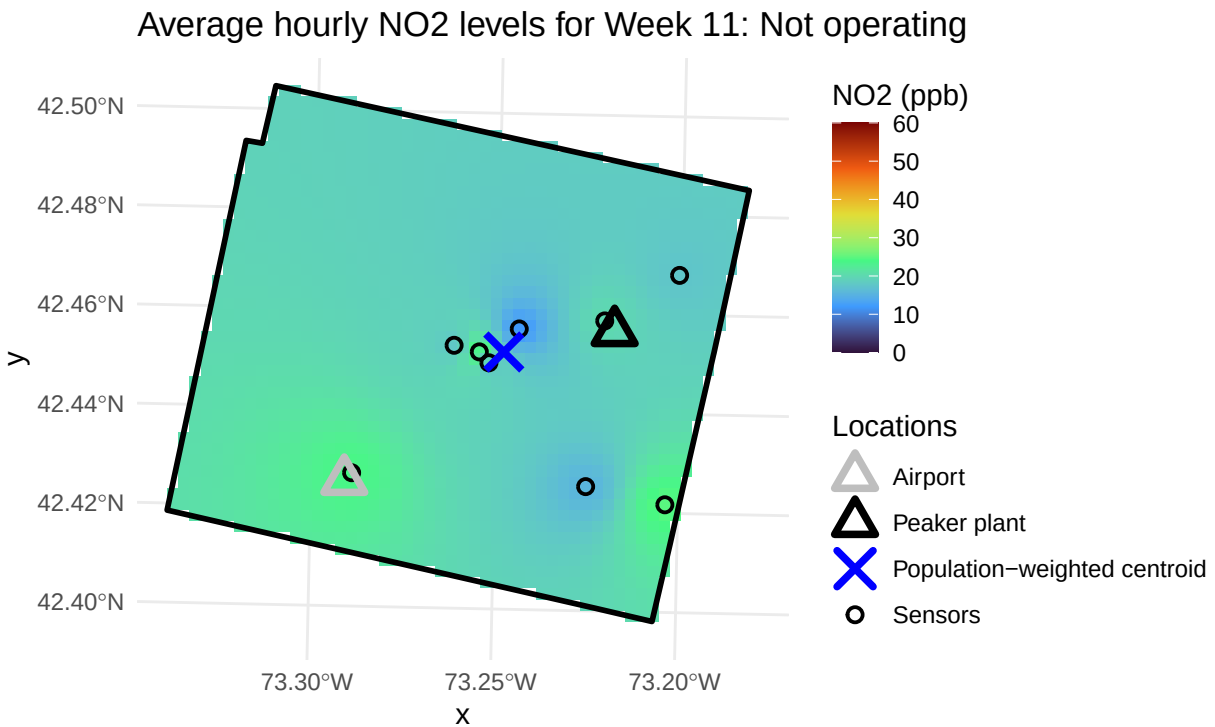
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



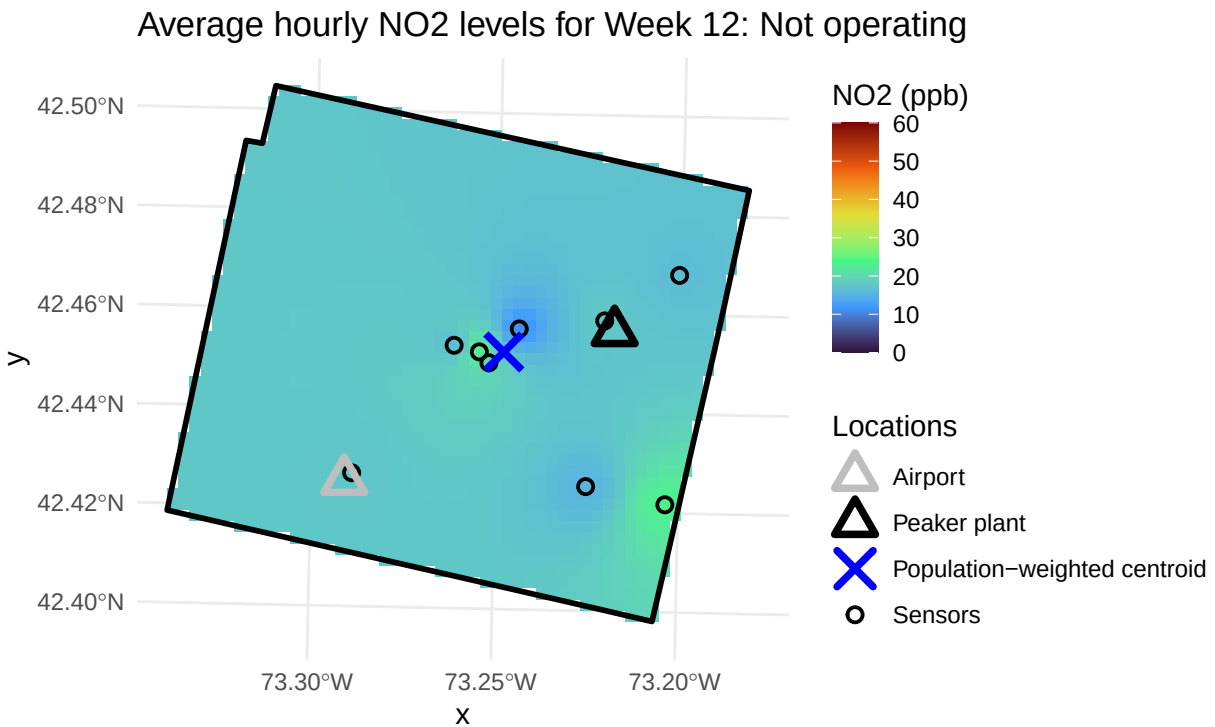
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



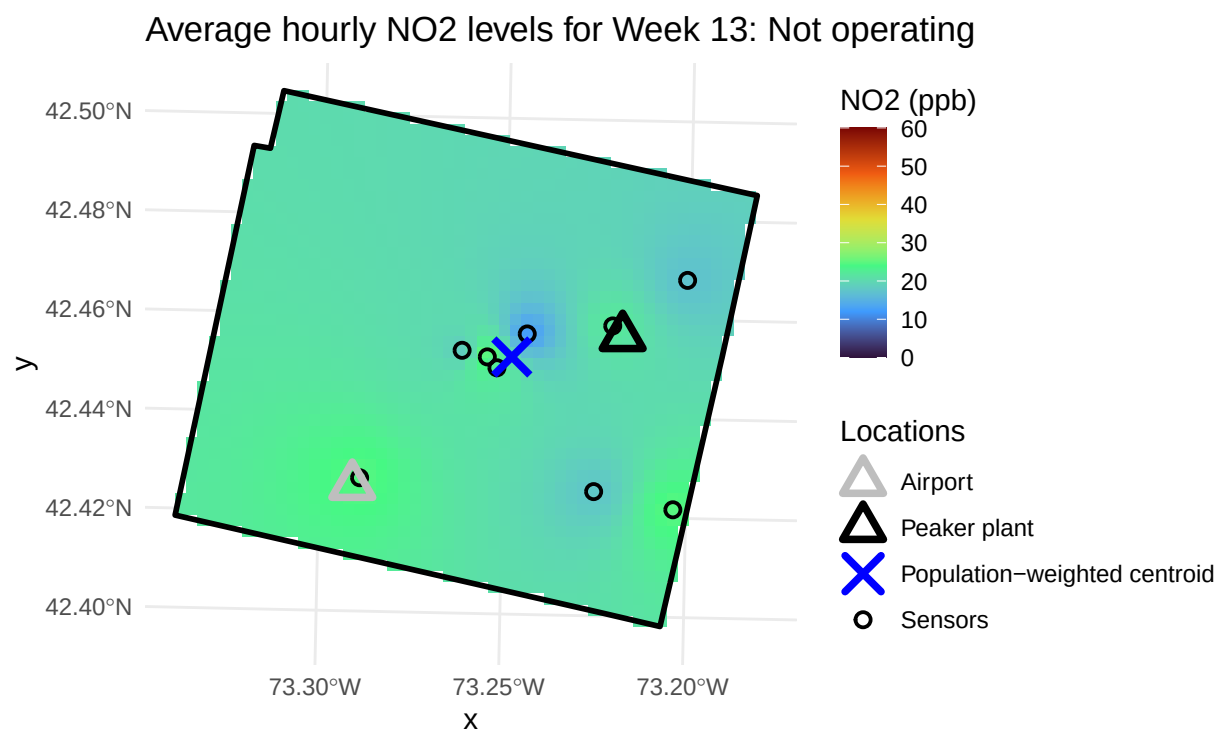
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



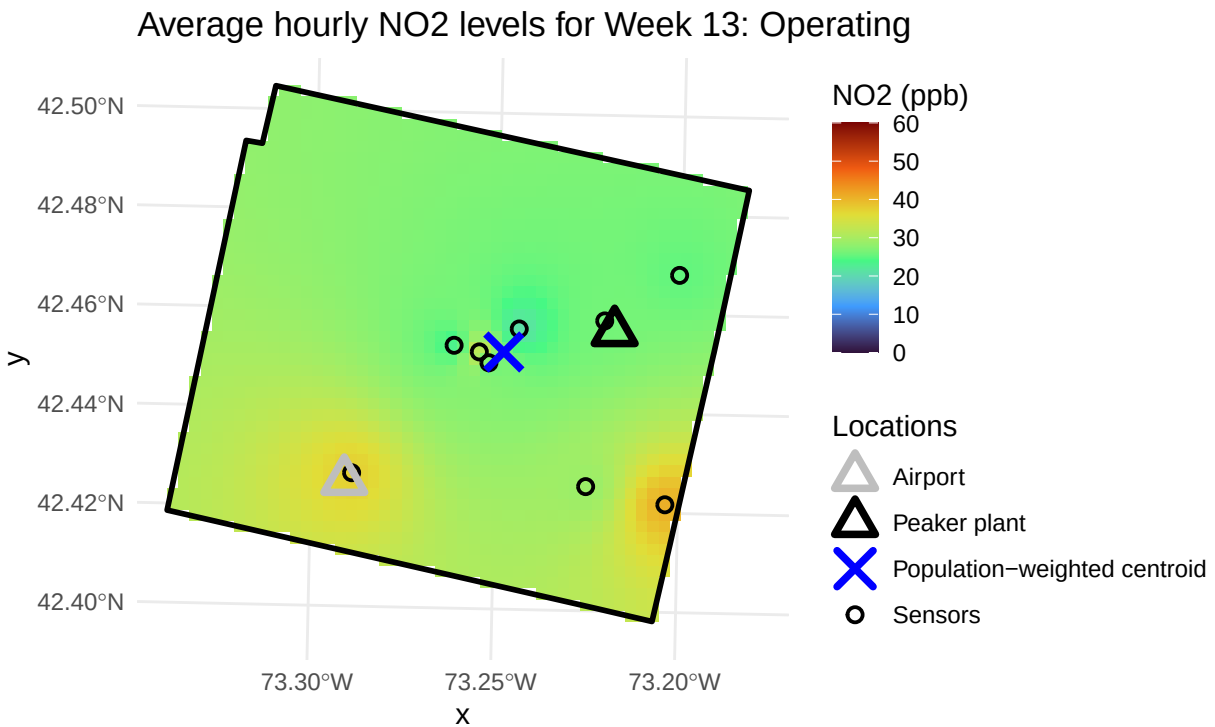
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```

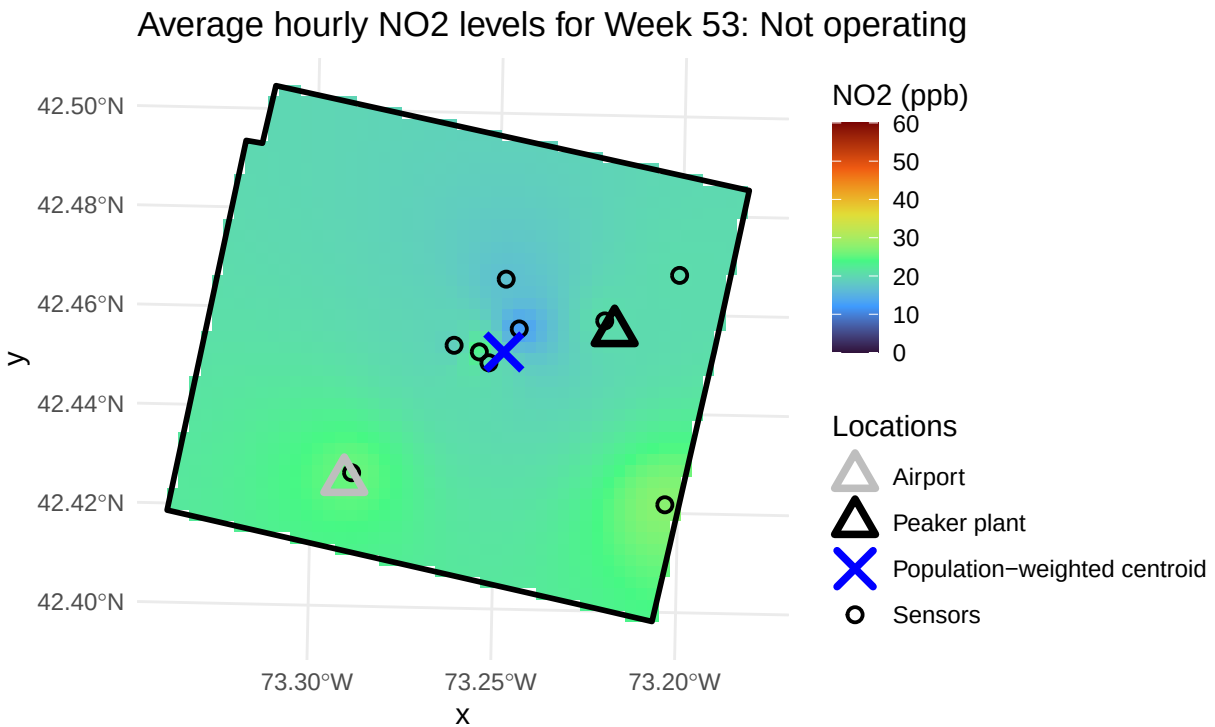


```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```

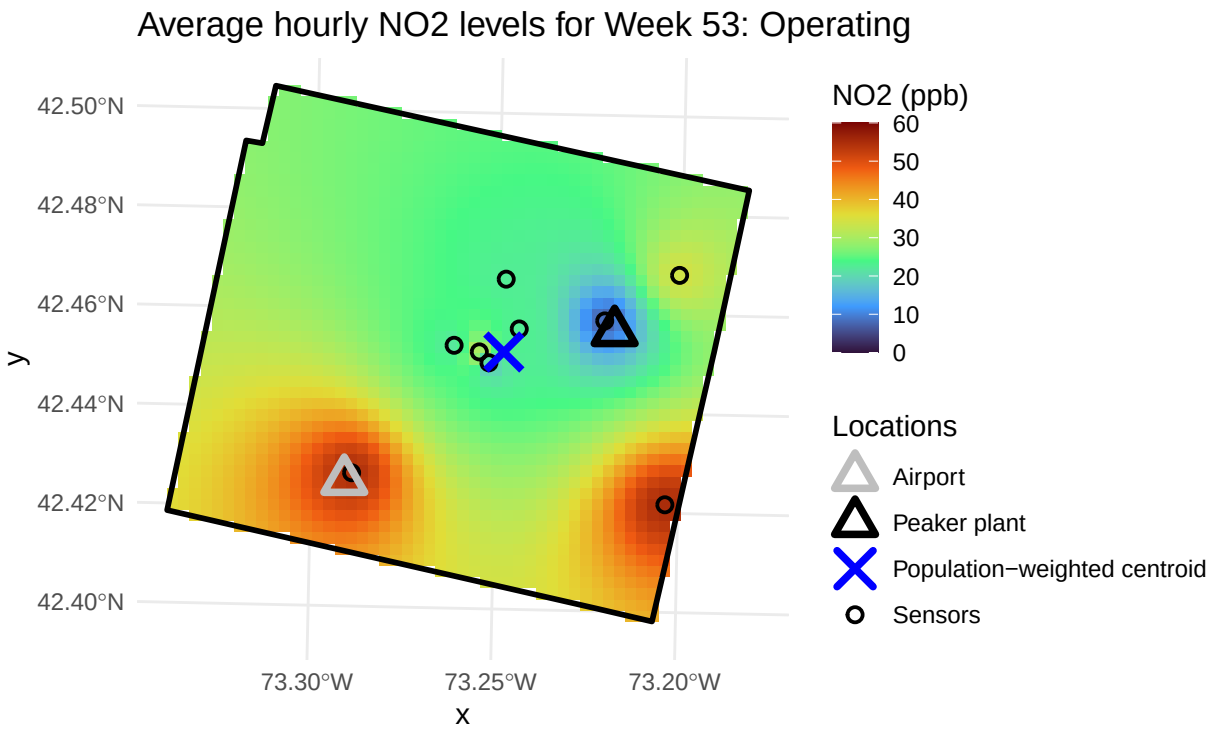


```
## [inverse distance weighted interpolation]

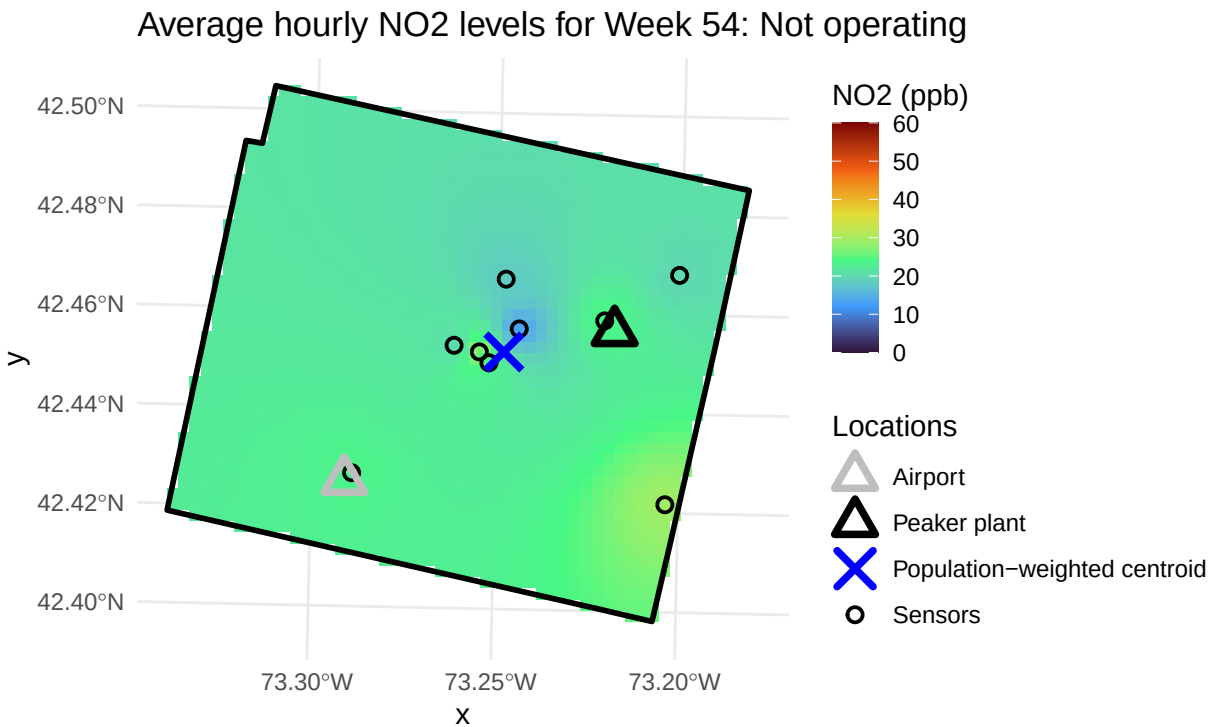
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



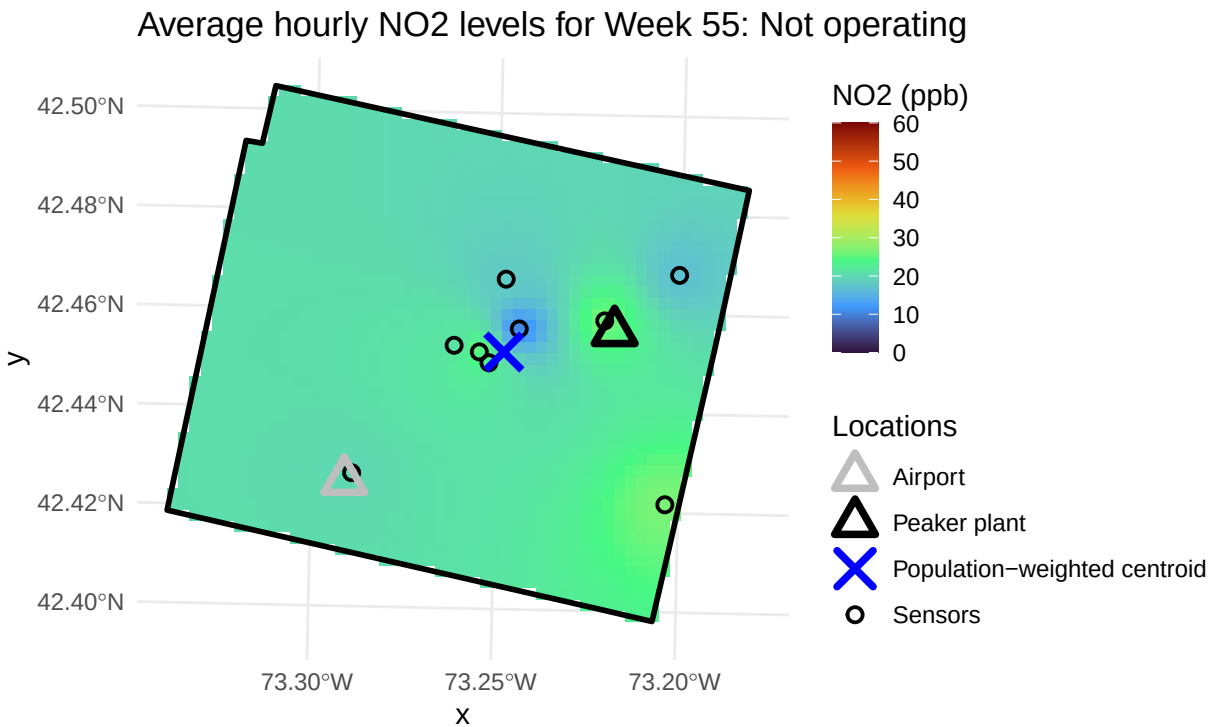
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



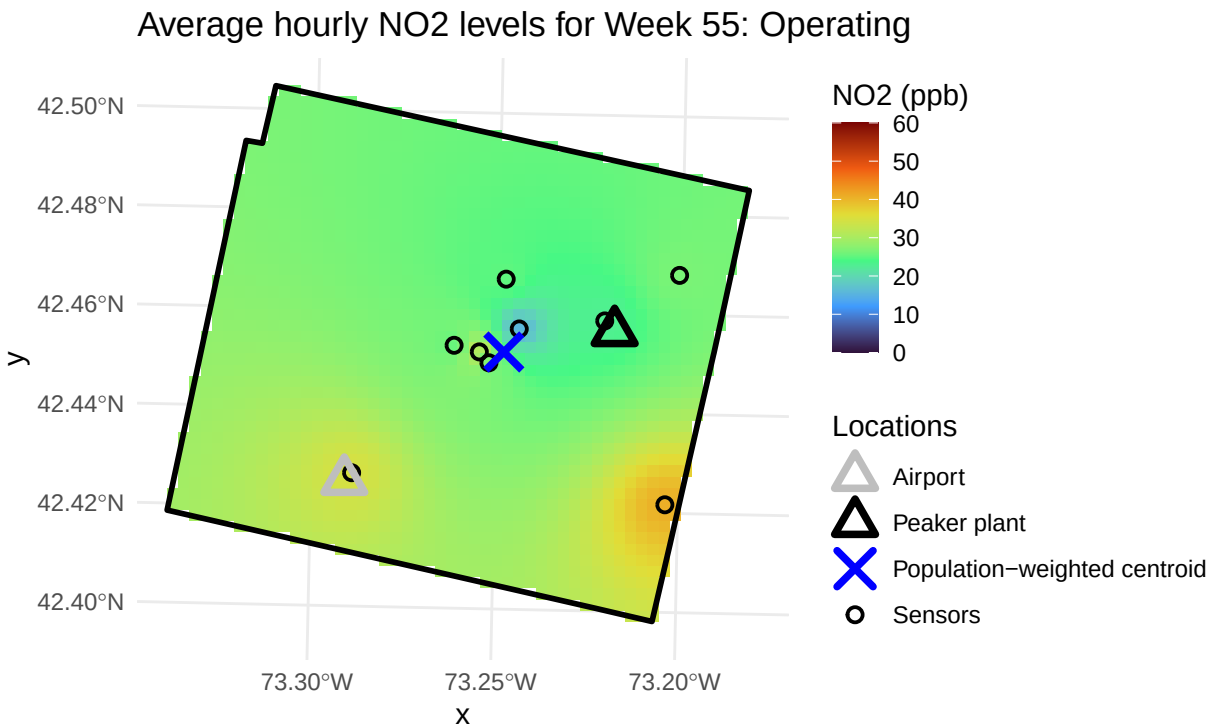
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



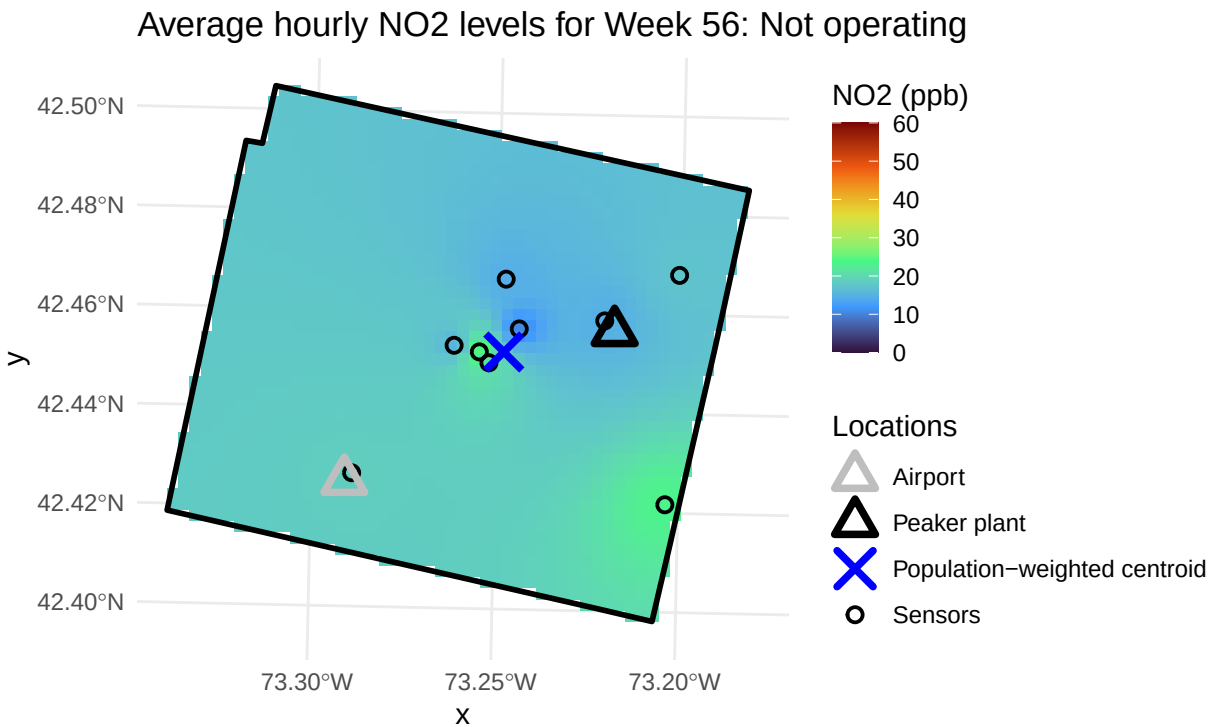
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



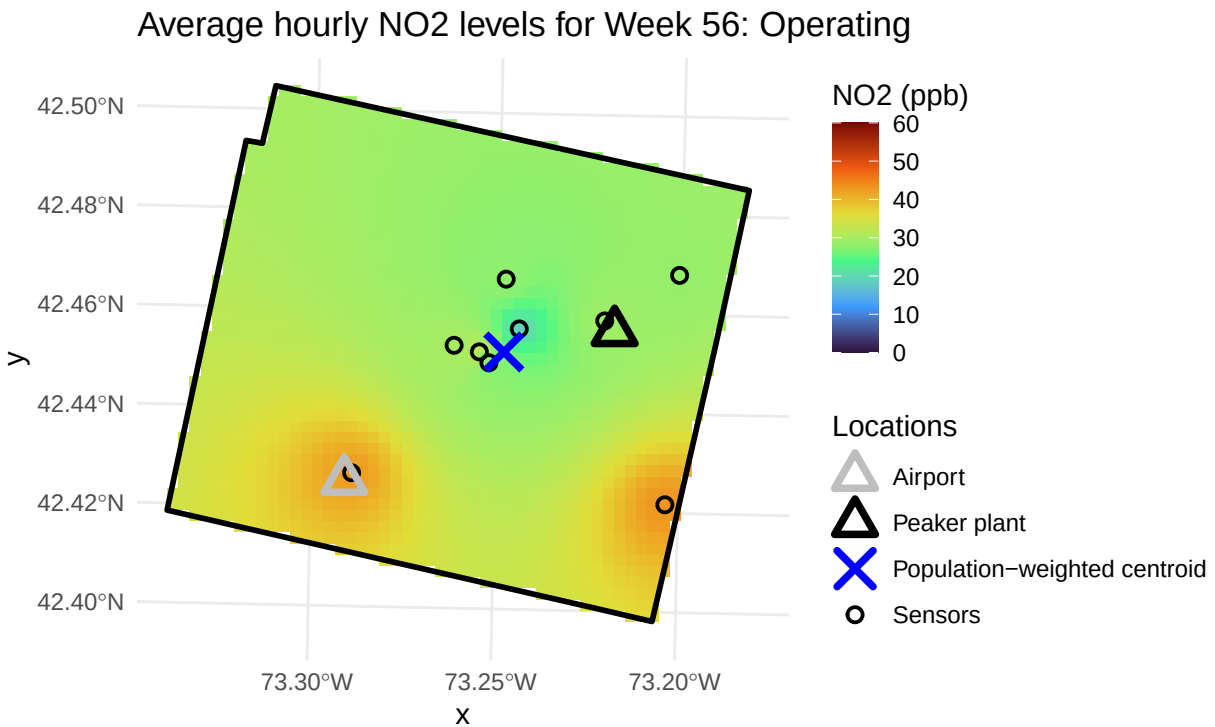
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



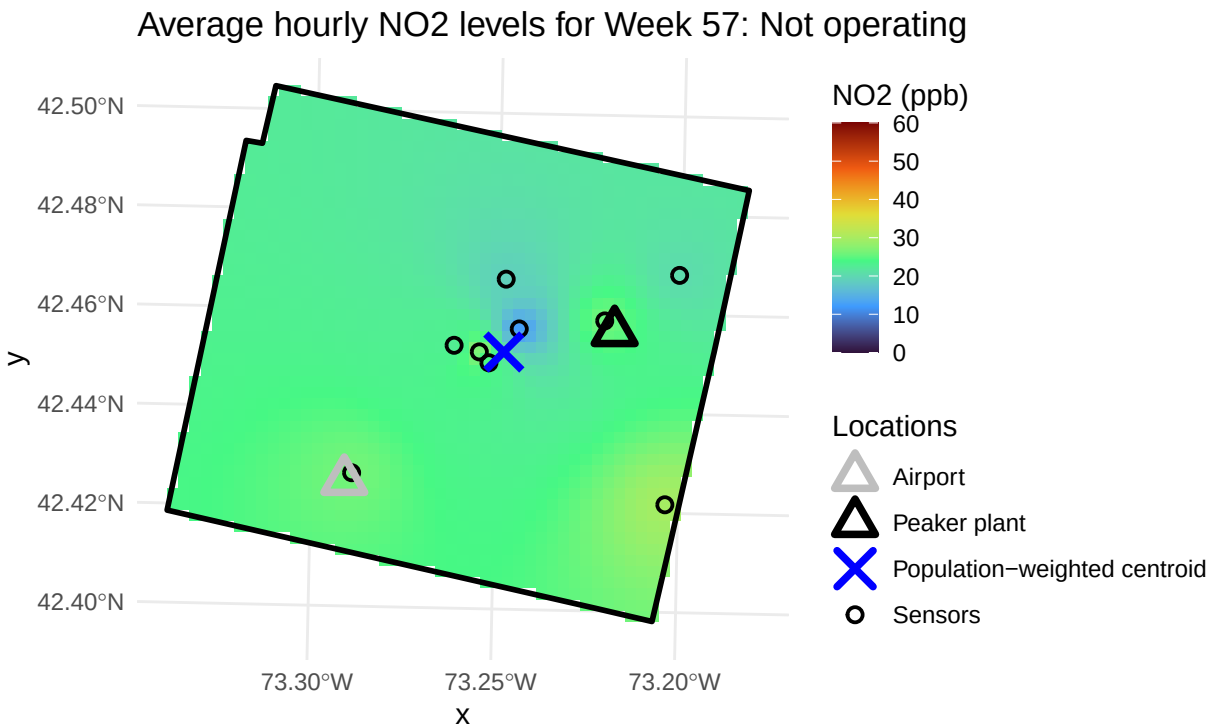
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



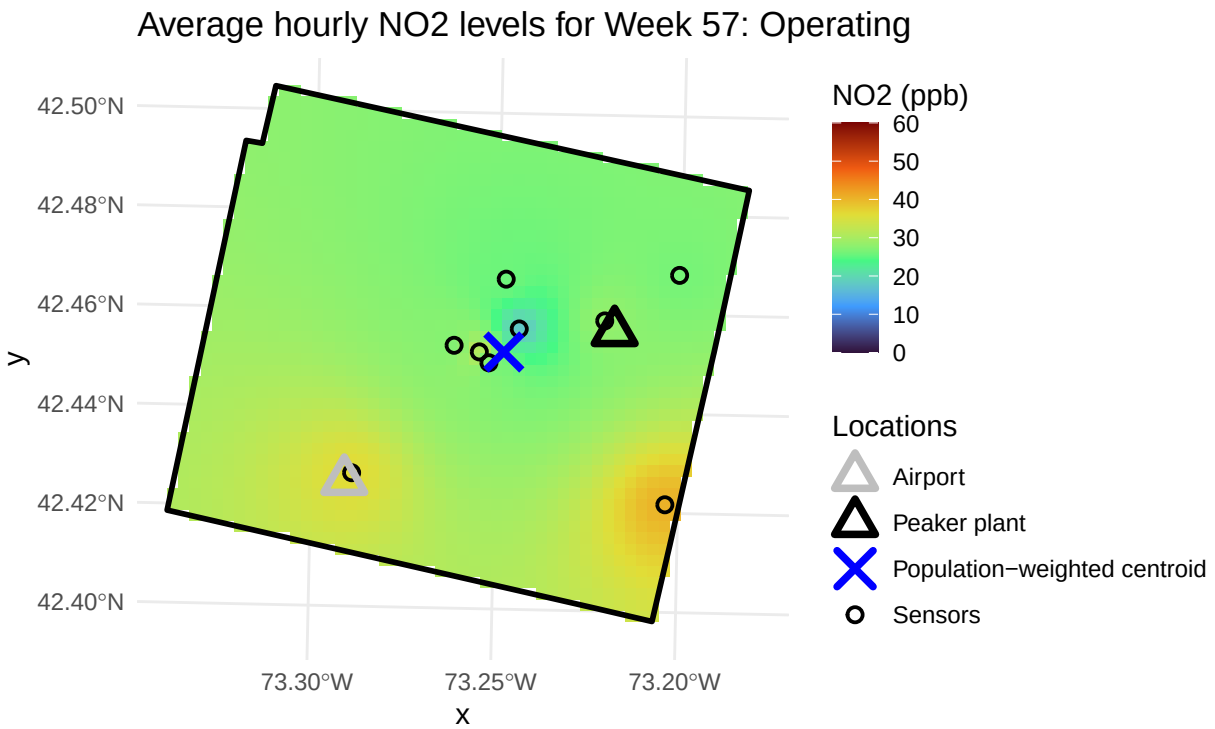
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```

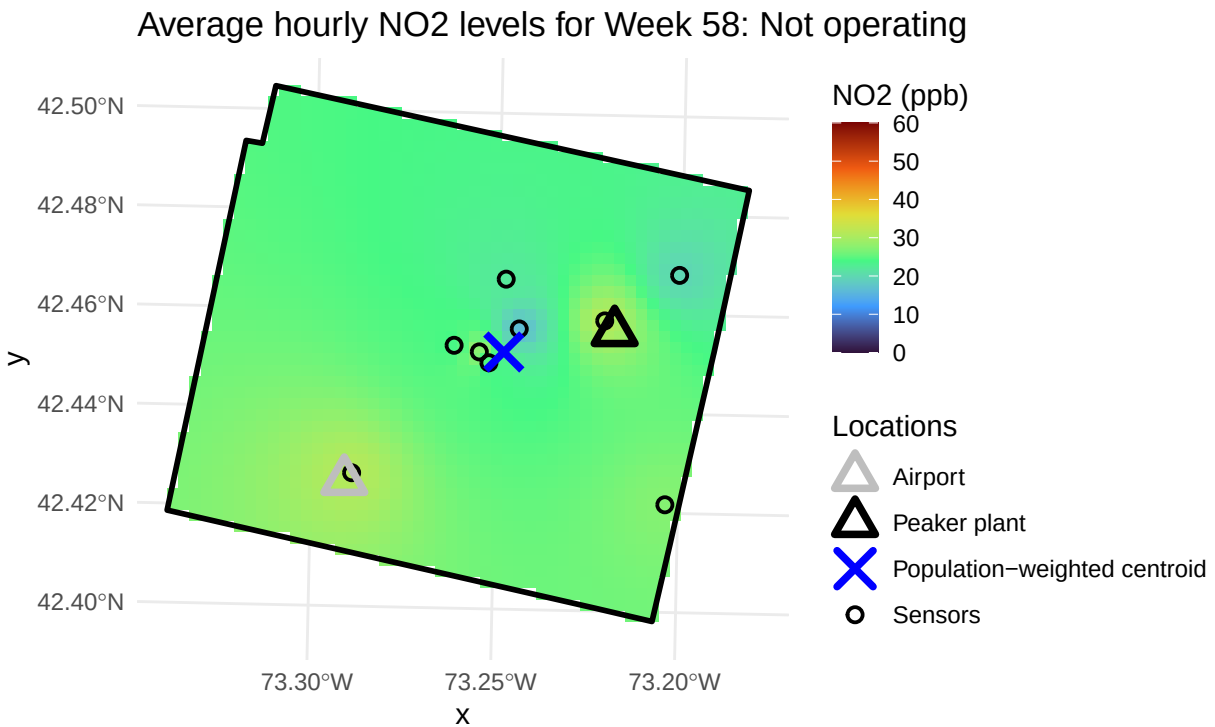


```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



```
## [inverse distance weighted interpolation]

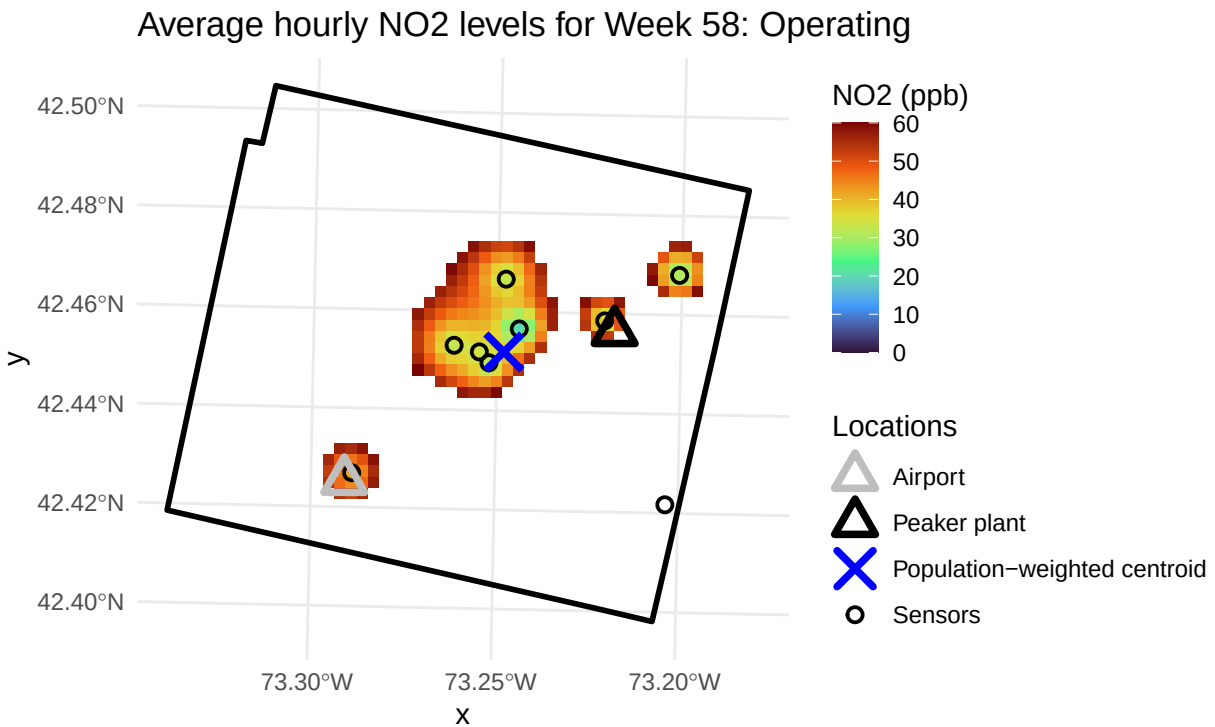
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



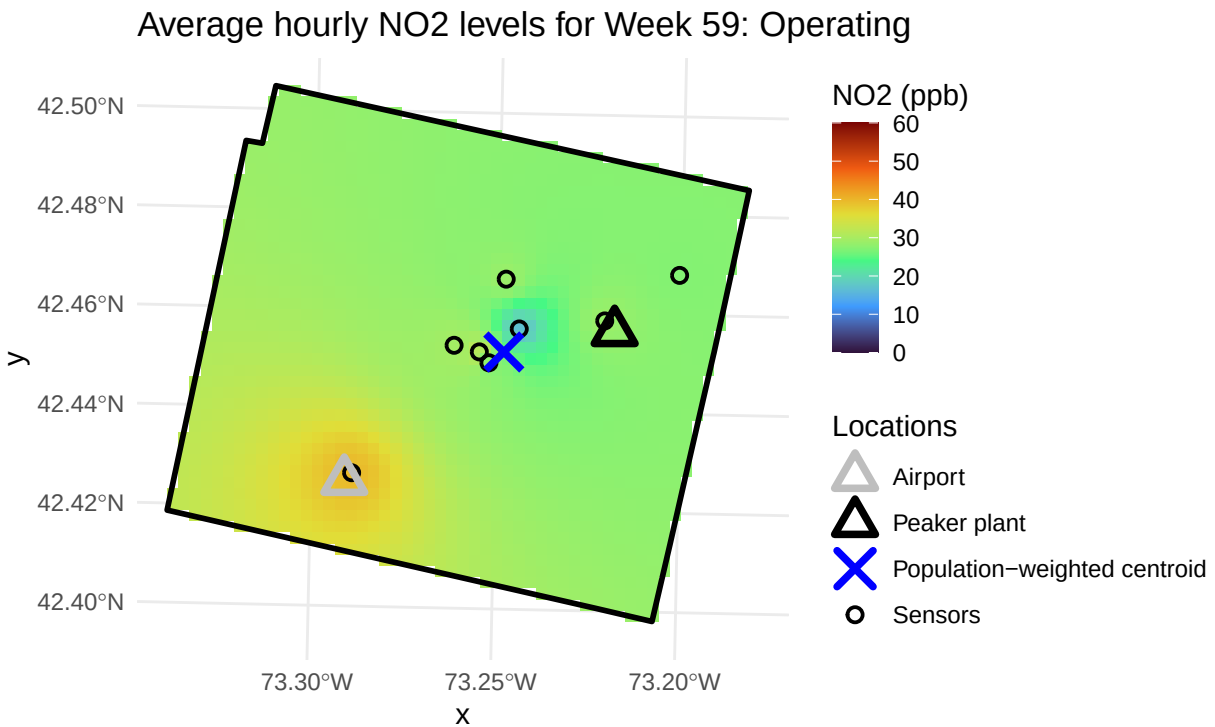
```
## [inverse distance weighted interpolation]
```

```
## Warning: Removed 2407 rows containing missing values or values outside the scale range  
## (`geom_raster()`).
```

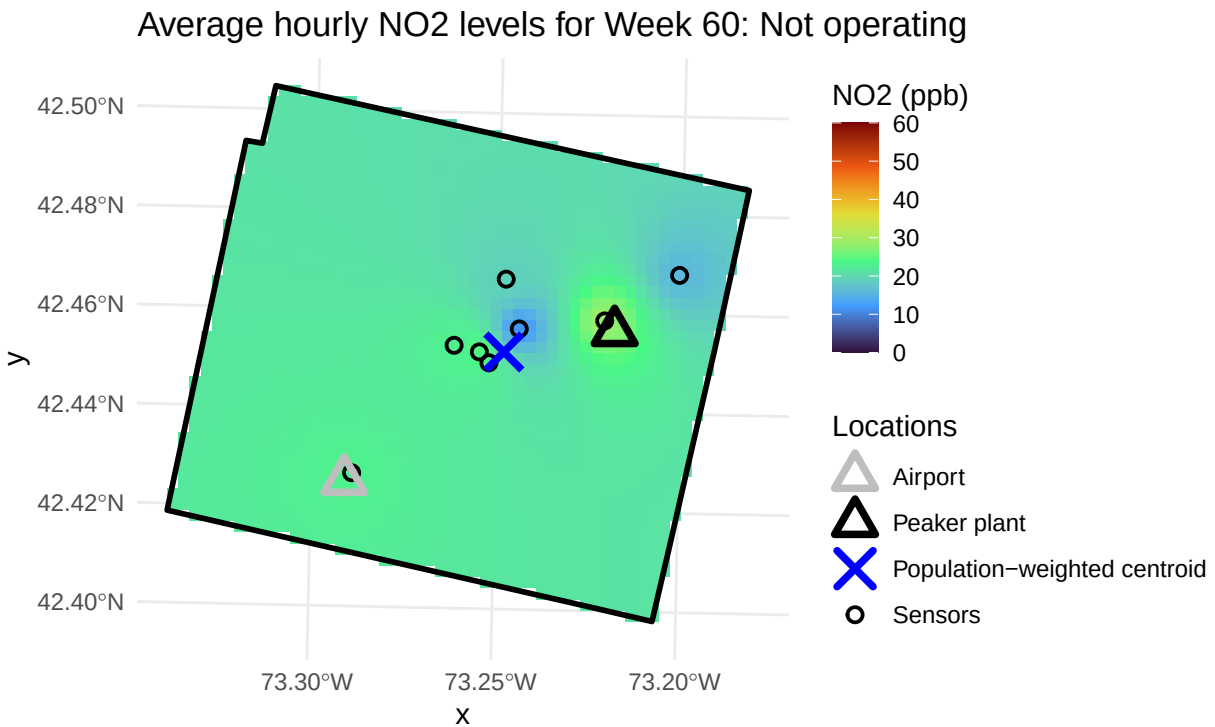
```
## Warning: Removed 2407 rows containing missing values or values outside the scale range  
## (`geom_raster()`).
```



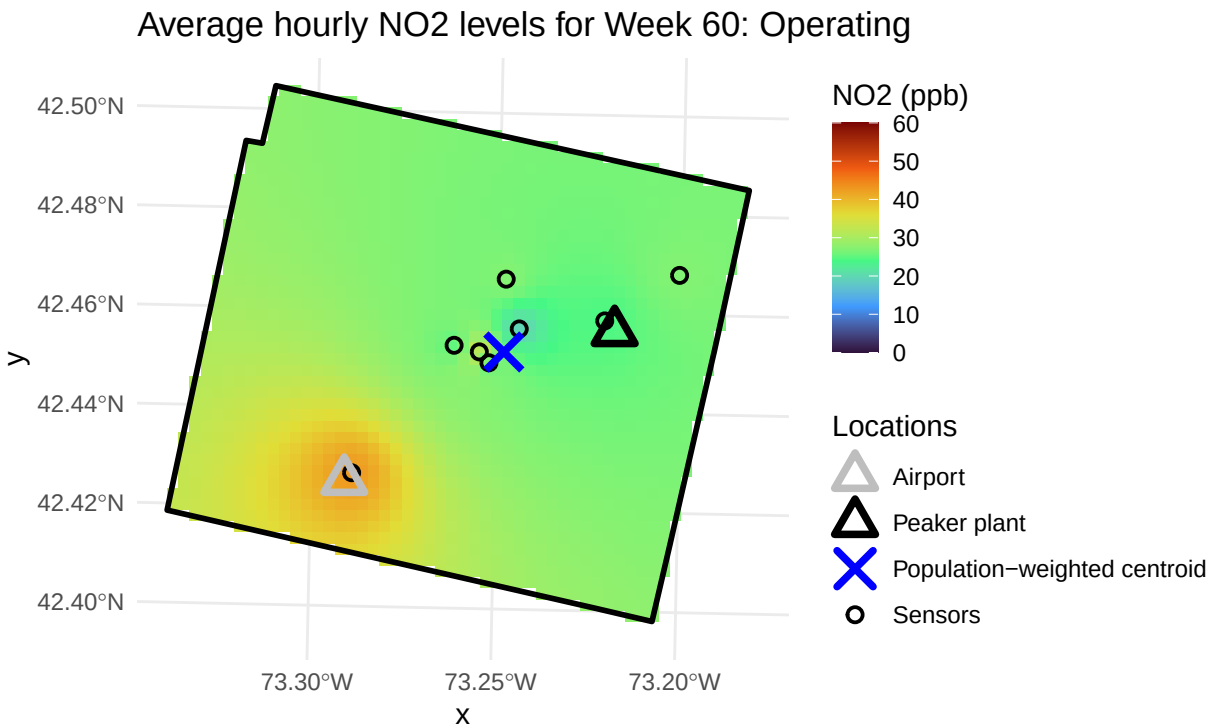
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



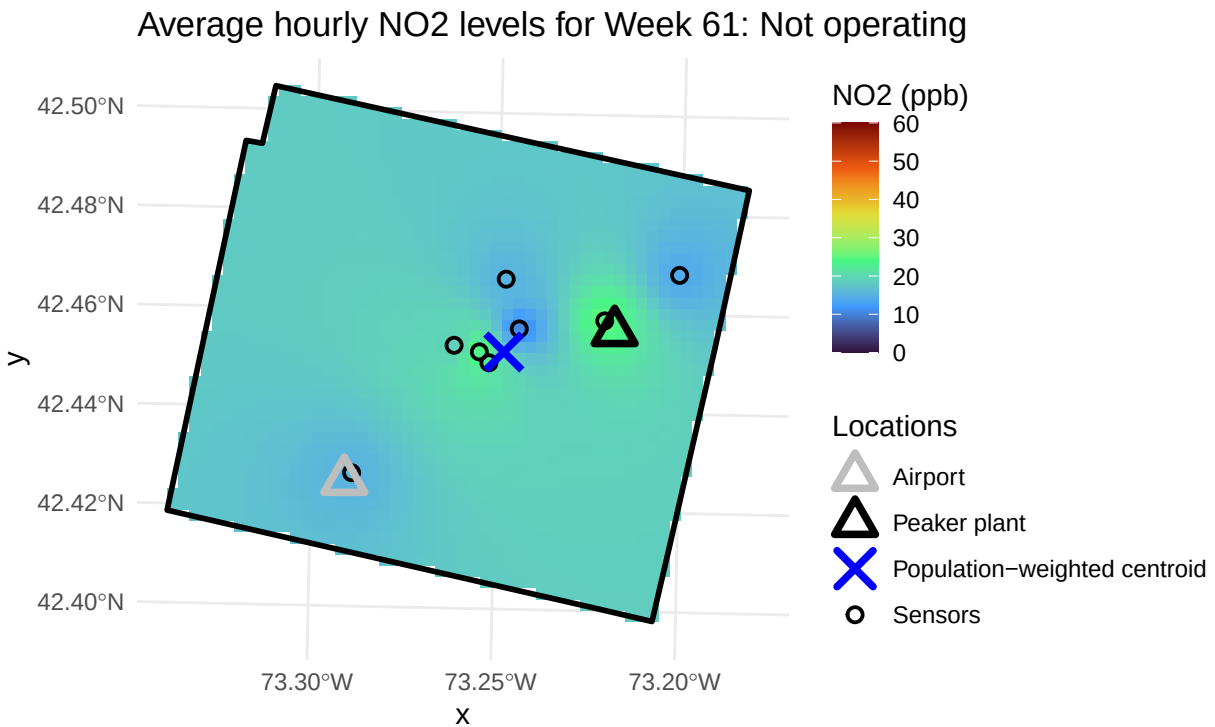
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



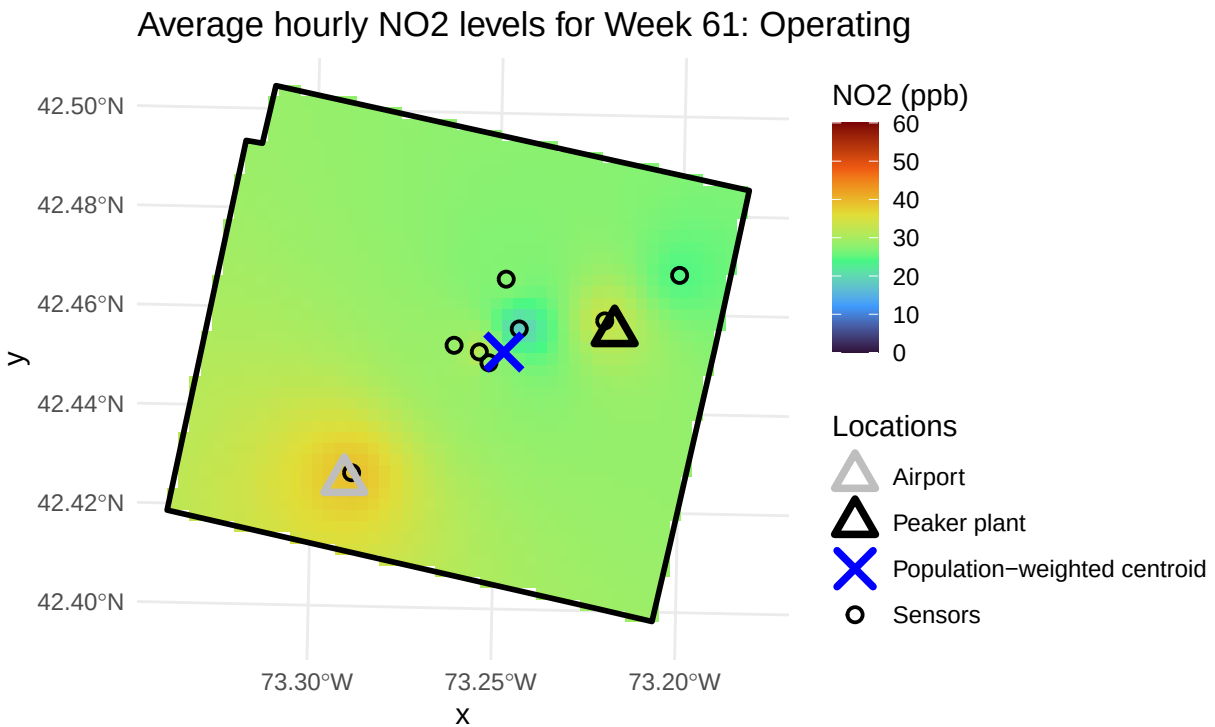
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



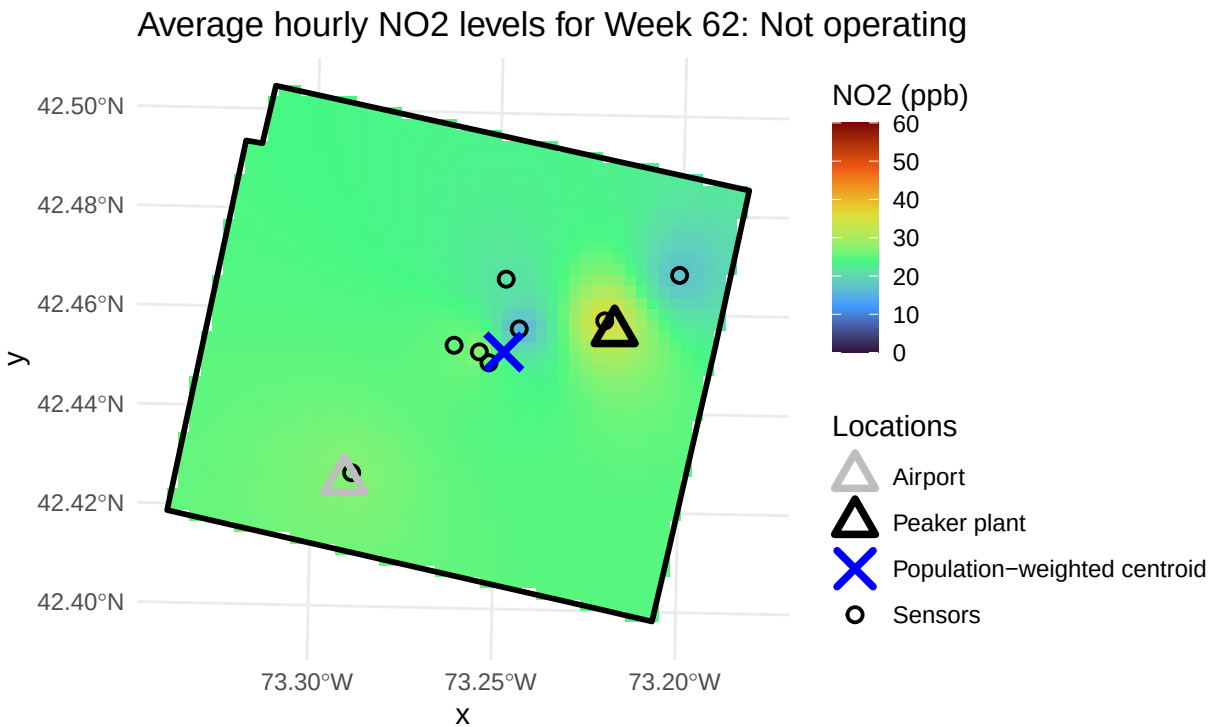
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



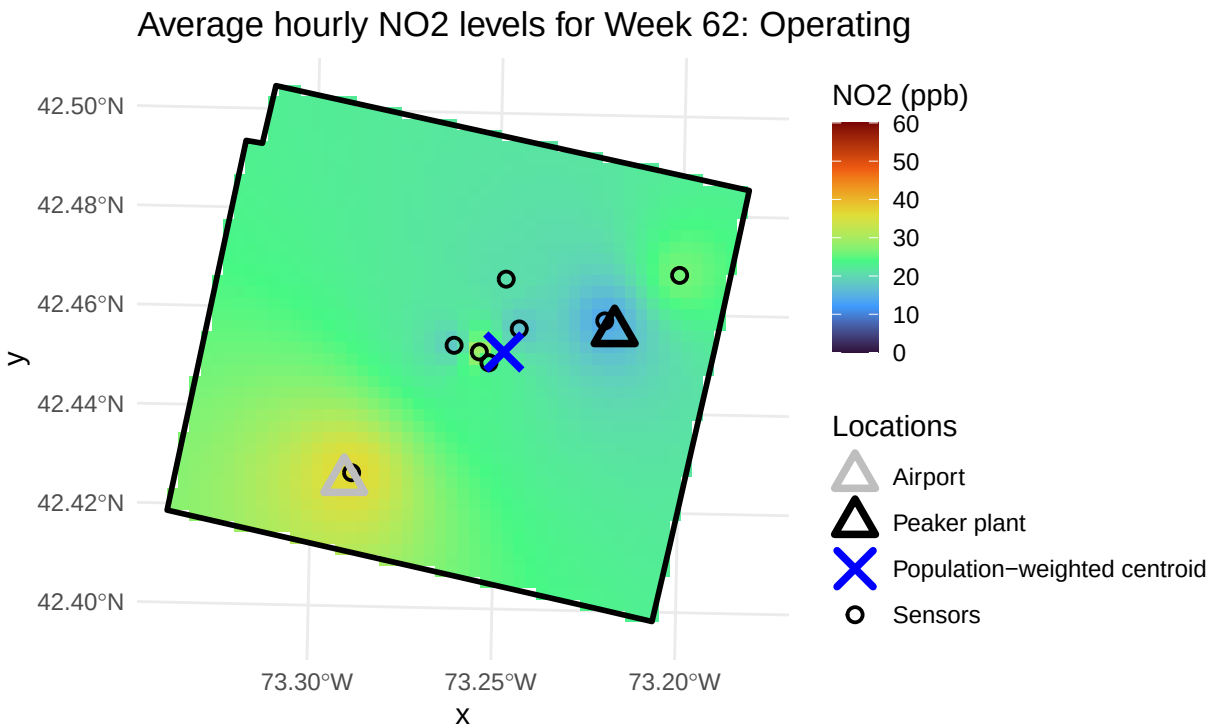
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



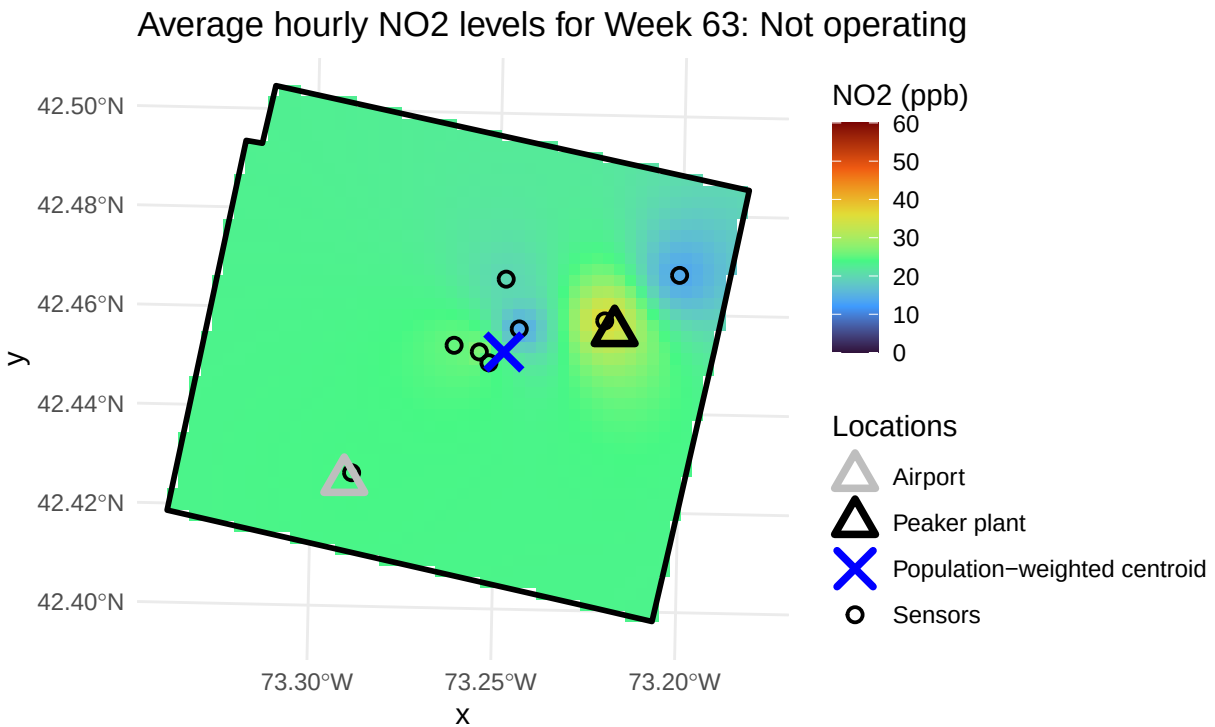
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



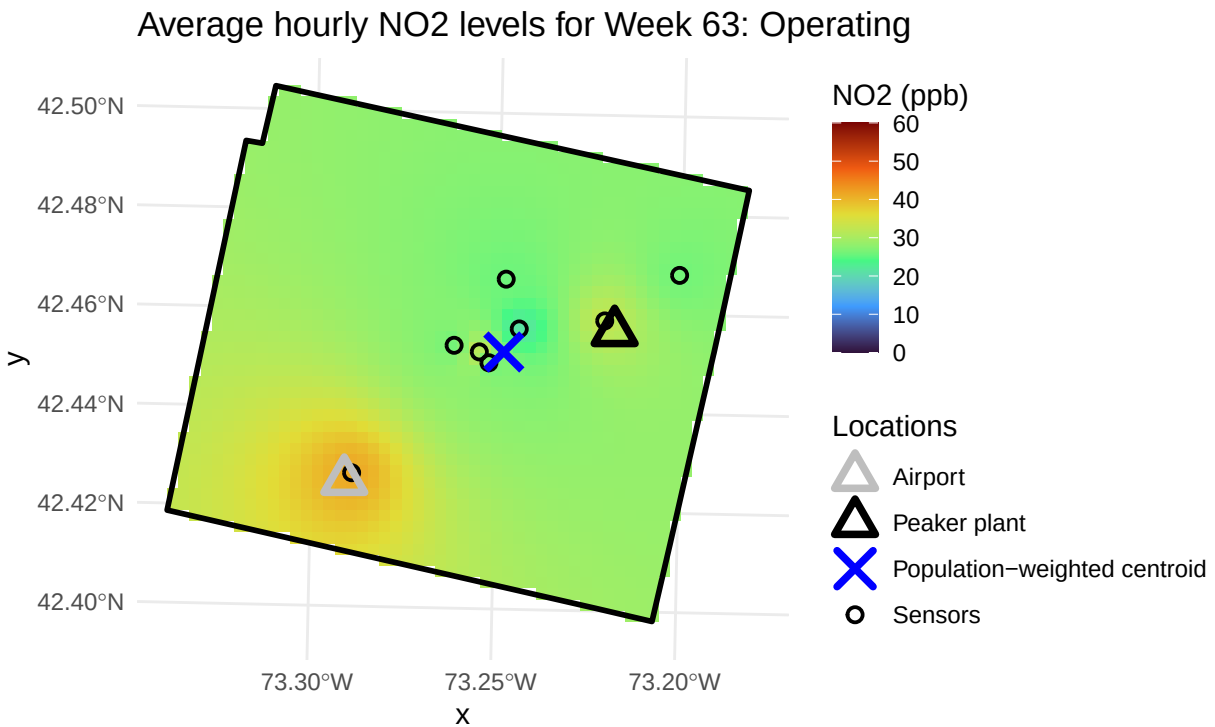
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



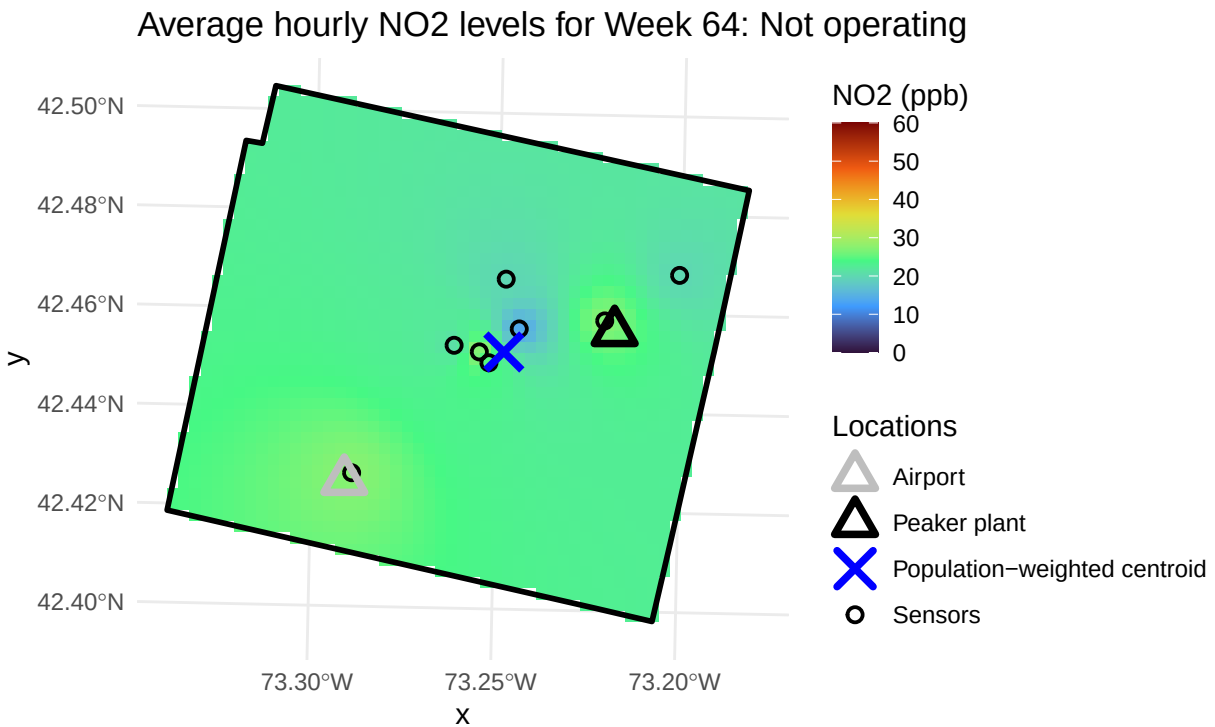
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



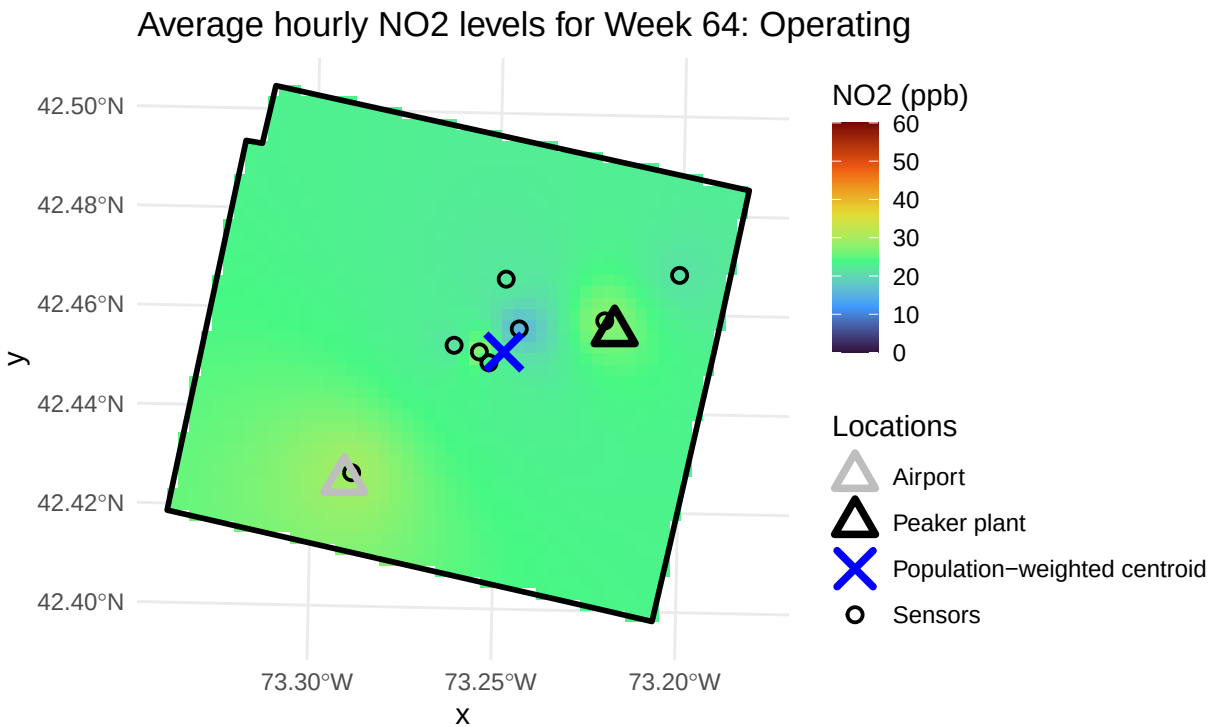
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



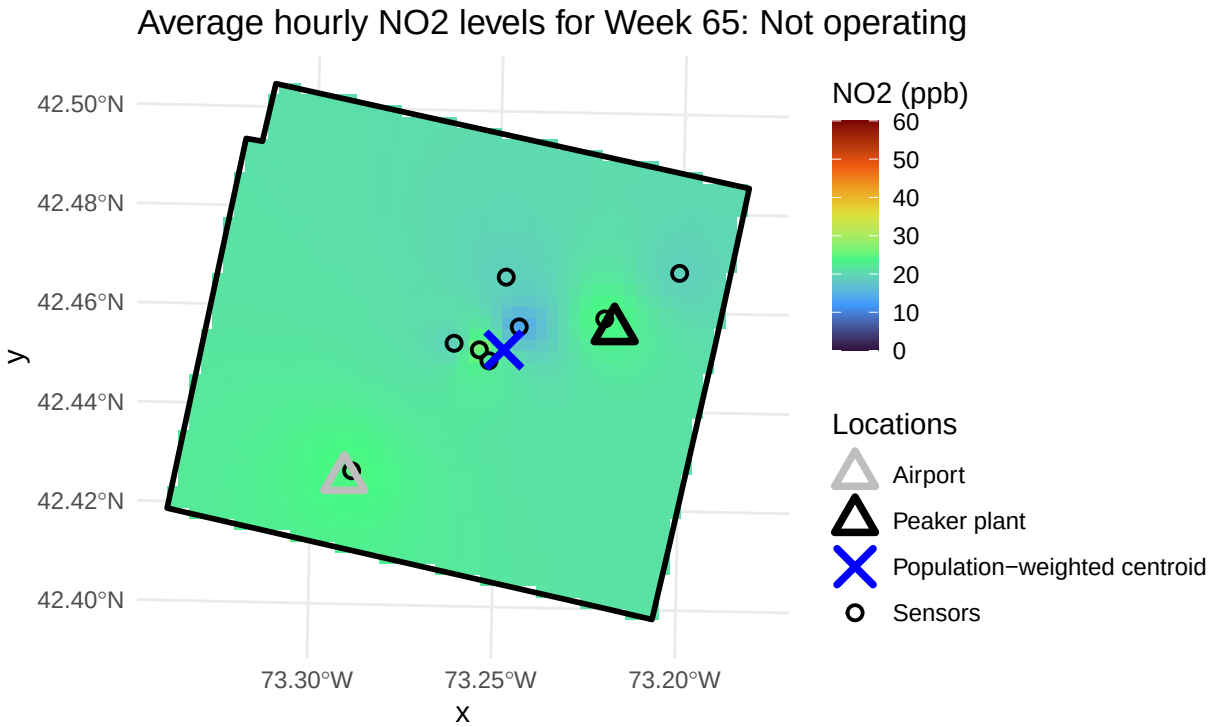
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



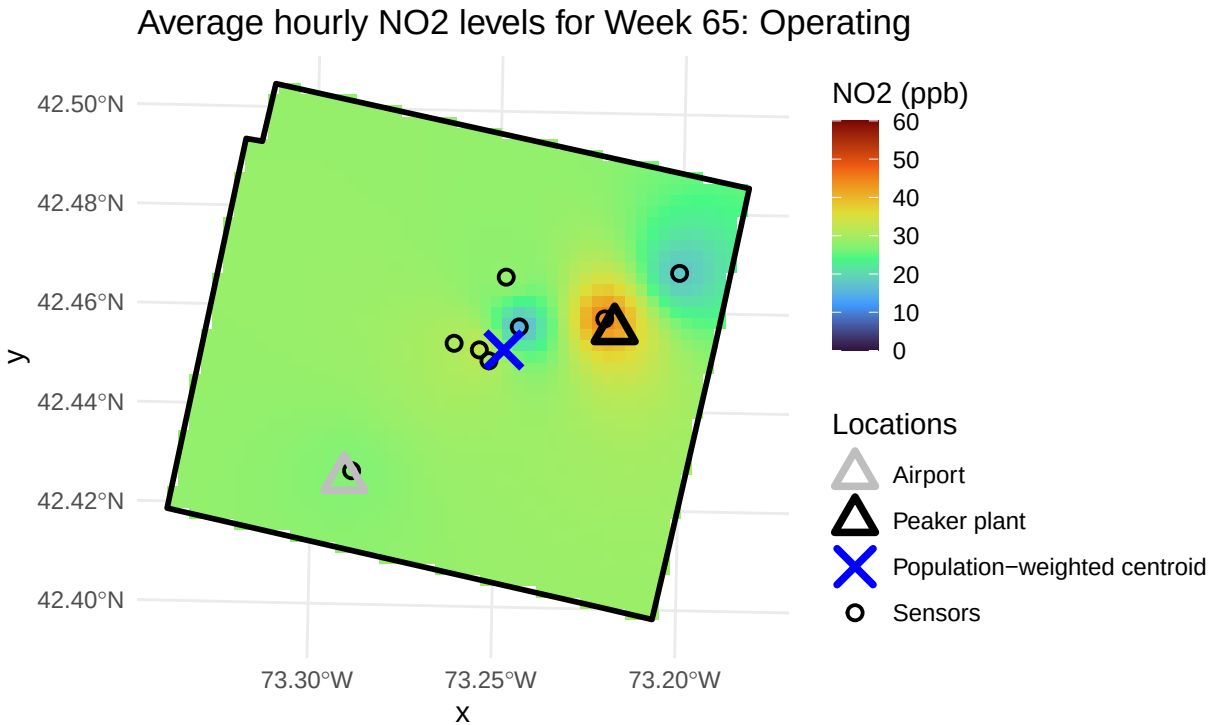
```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



```
## [inverse distance weighted interpolation]
## Warning: Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
## Removed 835 rows containing missing values or values outside the scale range
## (`geom_raster()`).
```



```
centroidno2_weekly <- dplyr::bind_rows(idw_weekly_results)
head(centroidno2_weekly)
```

```
## # A tibble: 6 x 9
##   week operating fireworks2 smoke2 temp humidity windspeed year no2mean
##   <fct> <chr>          <dbl> <dbl> <dbl> <dbl> <dbl> <chr> <dbl>
## 1 1 Not operating      0      0 67.0  67.1  NA    2024  22.1
## 2 2 Not operating      0      0 61.7  72.4  NA    2024  16.9
## 3 2 Operating          0      0 74    69    2.5   2024  26.2
## 4 3 Not operating      0      0 68    73.0  3.08  2024  22.9
## 5 3 Operating          0      0 82.4  64.2  NA    2024  24.9
## 6 4 Not operating      0      0 69.0  68.4  NA    2024  18.5
```

```
ggno2weeklybox <- ggplot(data = centroidno2_weekly, aes(x = factor(operating), y = no2mean)) +
  geom_boxplot(na.rm = TRUE) +
  labs(title = "Mean hourly N02 levels per week \nat population-weighted centroid",
       x = "Peaker plant operation status",
       y = "Concentration (ppb)") +
  theme_minimal() +
  theme(plot.title = element_text(size = 20),
        axis.title.x = element_text(size = 18),
        axis.title.y = element_text(size = 18),
        axis.text = element_text(size = 16))

ggsave("plots/ggno2weeklyboxplot.png", ggno2weeklybox, height = 6, width = 8, unit = "in")
```

```

write.csv(centroidno2_weekly, "data/output/centroidno2_weekly.csv")

#model to include weather data. excluded windspeed because
model2_lm1 <- lm(formula = no2mean ~ operating + week + smoke2 +
                  fireworks2 + humidity + temp, data = centroidno2_weekly)

summary(model2_lm1)

##
## Call:
## lm(formula = no2mean ~ operating + week + smoke2 + fireworks2 +
##      humidity + temp, data = centroidno2_weekly)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.700 -0.298  0.000  0.298  1.700
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -43.85192    10.75467  -4.077 0.002223 **
## operatingOperating    0.41032     1.30986   0.313 0.760524
## week2          -2.71358     1.79212  -1.514 0.160930
## week3          -3.62454     1.78237  -2.034 0.069391 .
## week4          -5.15709     1.71104  -3.014 0.013028 *
## week5          -4.50921     2.88451  -1.563 0.149058
## week6          -5.45944     2.93125  -1.862 0.092137 .
## week7          -6.23801     1.84718  -3.377 0.007038 **
## week8          -6.48505     2.10014  -3.088 0.011485 *
## week9          -7.59764     1.99952  -3.800 0.003487 **
## week10         -9.78839     2.24772  -4.355 0.001433 **
## week13         -5.61312     1.87619  -2.992 0.013533 *
## week53         -7.92857     2.62388  -3.022 0.012858 *
## week54         -3.83417     2.01231  -1.905 0.085861 .
## week55         -6.97285     2.22173  -3.138 0.010537 *
## week56         -2.79078     1.74530  -1.599 0.140897
## week57         -3.31825     2.17616  -1.525 0.158286
## week58         -1.81199     2.92253  -0.620 0.549120
## week60         -3.25747     1.80064  -1.809 0.100557
## week61         -8.96656     2.62674  -3.414 0.006618 **
## week62         -9.94742     3.01416  -3.300 0.008010 **
## week63         -2.58760     1.87208  -1.382 0.197000
## week64         -2.14682     2.03206  -1.056 0.315604
## week65          3.56702     2.06247   1.729 0.114410
## smoke2          5.69289     2.06887   2.752 0.020419 *
## fireworks2     -0.54870     2.06955  -0.265 0.796293
## humidity        0.41910     0.07405   5.659 0.000210 ***
## temp           0.56399     0.12198   4.624 0.000945 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.353 on 10 degrees of freedom
## (7 observations deleted due to missingness)
## Multiple R-squared:  0.9386, Adjusted R-squared:  0.7729
## F-statistic: 5.663 on 27 and 10 DF, p-value: 0.003321

```

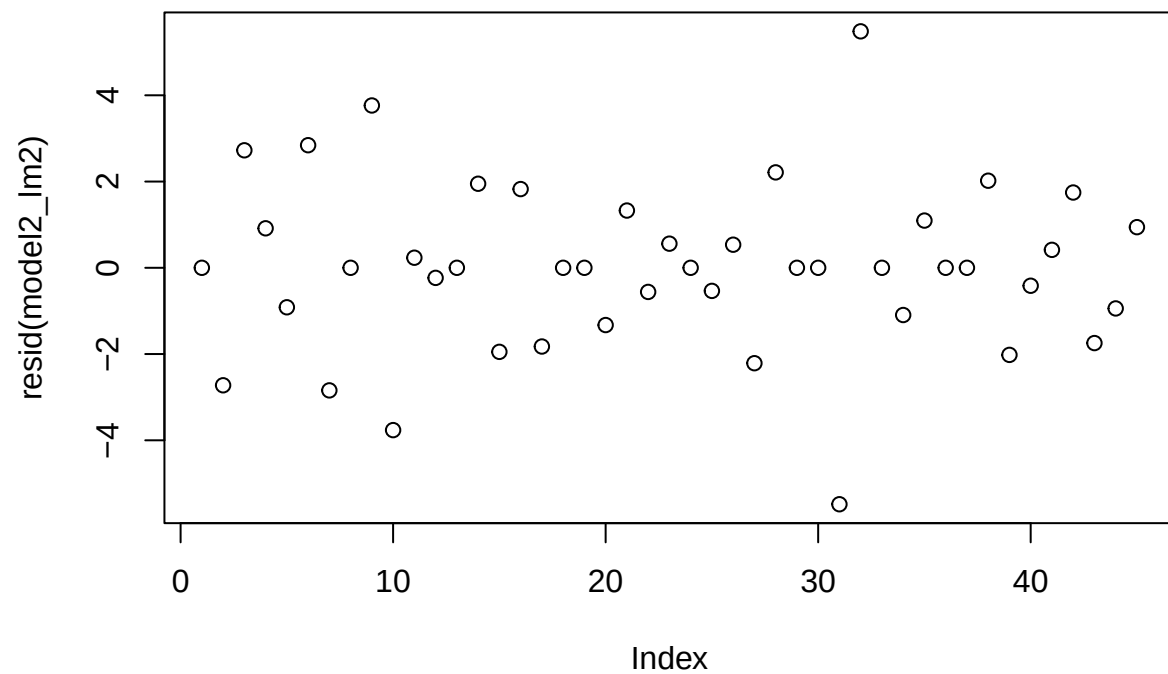
```

#without weather data.
model2_lm2 <- lm(formula = no2mean ~operating + week + smoke2 +
                  fireworks2, data = centroidno2_weekly)
summary(model2_lm2)

##
## Call:
## lm(formula = no2mean ~ operating + week + smoke2 + fireworks2,
##     data = centroidno2_weekly)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.4827 -0.9421  0.0000  0.9421  5.4827
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    22.0917     3.2480   6.802 4.25e-06 ***
## operating      3.8752     1.1141   3.478  0.0031 **
## week2        -2.4935     4.0168  -0.621  0.5435
## week3        -0.1451     4.0168  -0.036  0.9716
## week4        -6.4150     4.0168  -1.597  0.1298
## week5        -1.2423     6.5908  -0.188  0.8529
## week6         0.8021     6.3020   0.127  0.9003
## week7        -1.8068     4.0168  -0.450  0.6589
## week8        -0.9315     4.5934  -0.203  0.8419
## week9        -1.3487     4.0168  -0.336  0.7414
## week10       -2.8970     4.0168  -0.721  0.4812
## week11       -8.7213     6.5908  -1.323  0.2044
## week12       -5.2178     4.5934  -1.136  0.2727
## week13       -1.7591     4.0168  -0.438  0.6673
## week53       -7.2641     6.1019  -1.190  0.2512
## week54       -2.0596     4.5934  -0.448  0.6599
## week55       -3.0320     4.0168  -0.755  0.4613
## week56       -2.5748     4.0168  -0.641  0.5306
## week57       -1.4702     4.7265  -0.311  0.7598
## week58        5.8229     6.3020   0.924  0.3692
## week59       -1.2920     4.7265  -0.273  0.7881
## week60       -2.0456     4.0168  -0.509  0.6175
## week61       -4.2479     4.5934  -0.925  0.3688
## week62       -5.9065     6.1019  -0.968  0.3475
## week63       -0.5559     4.0168  -0.138  0.8916
## week64       -2.9451     4.0168  -0.733  0.4740
## week65       -1.3046     4.0168  -0.325  0.7496
## smoke2        3.8190     4.7265   0.808  0.4309
## fireworks2    -0.3505     4.7265  -0.074  0.9418
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.248 on 16 degrees of freedom
## Multiple R-squared:  0.712, Adjusted R-squared:  0.208
## F-statistic: 1.413 on 28 and 16 DF, p-value: 0.2369

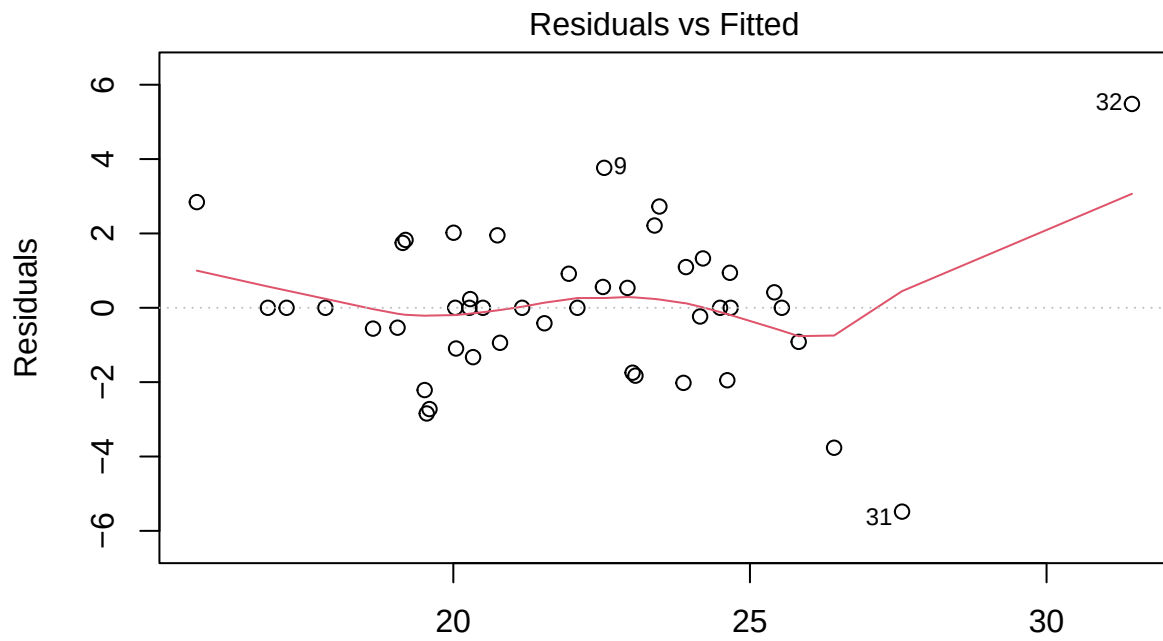
#check model residuals...
plot(resid(model2_lm2))

```

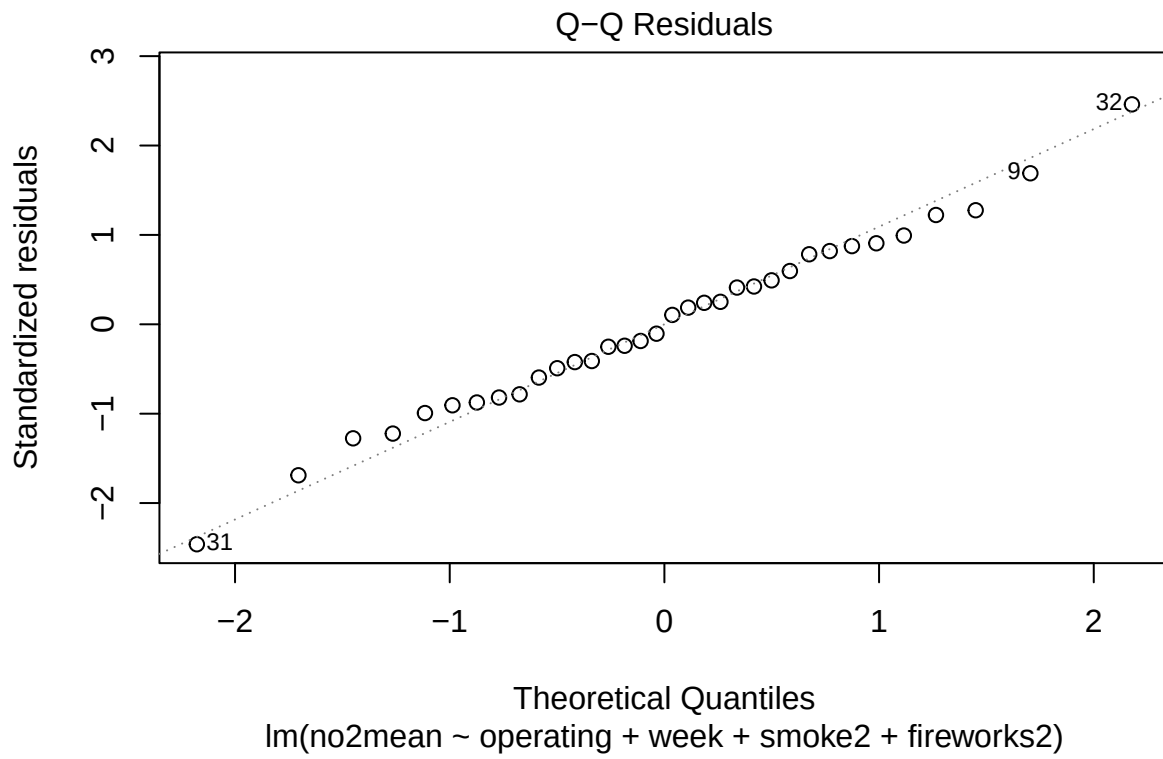


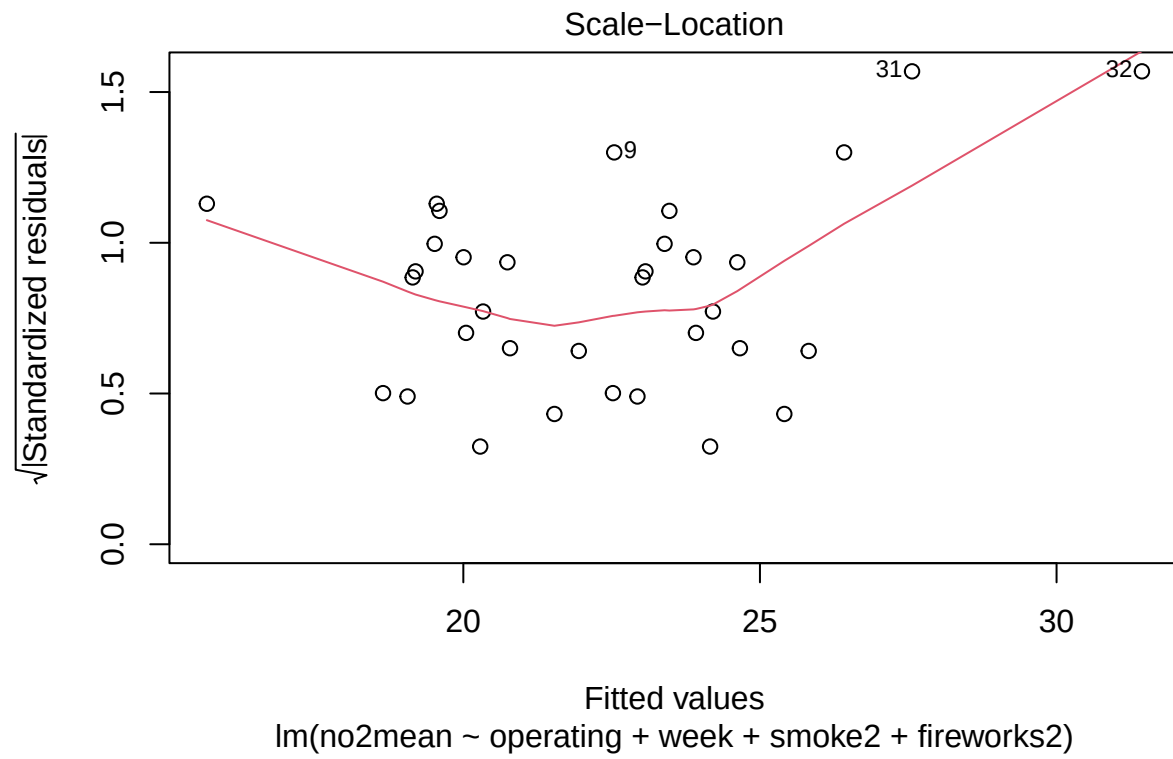
```
plot(model2_lm2)
```

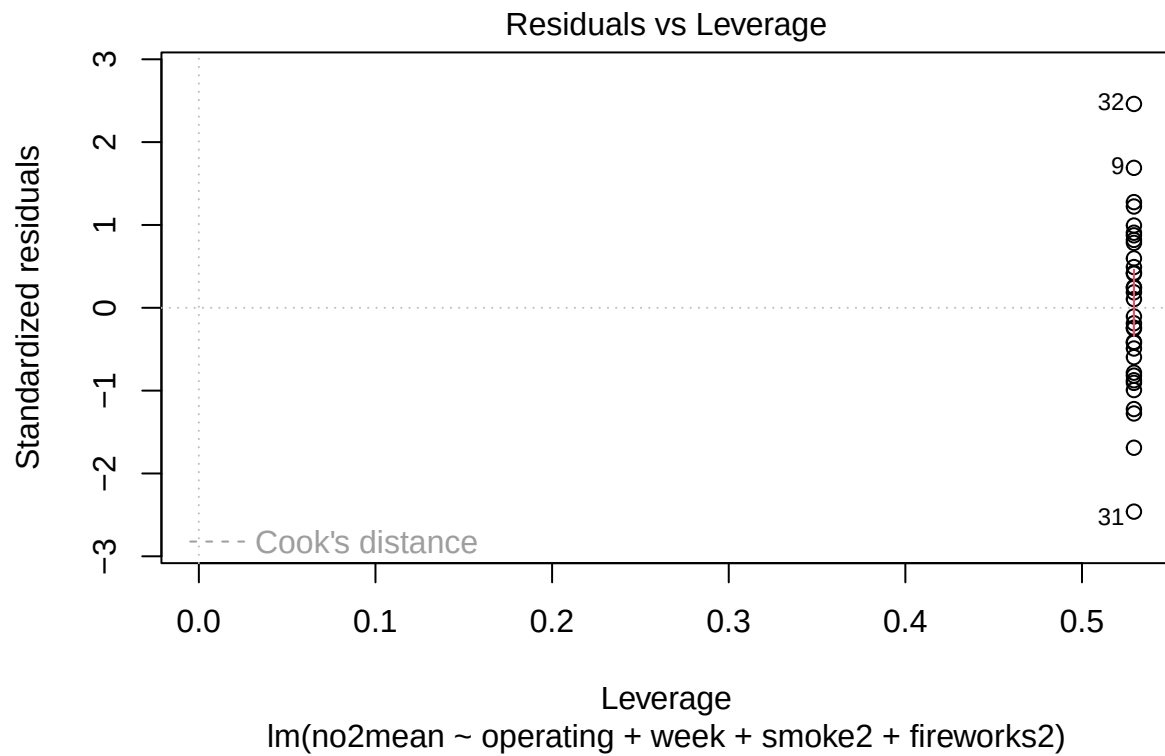
```
## Warning: not plotting observations with leverage one:  
## 1, 8, 13, 18, 19, 24, 29, 30, 33, 36, 37
```



Fitted values
 $\text{lm}(\text{no2mean} \sim \text{operating} + \text{week} + \text{smoke2} + \text{fireworks2})$

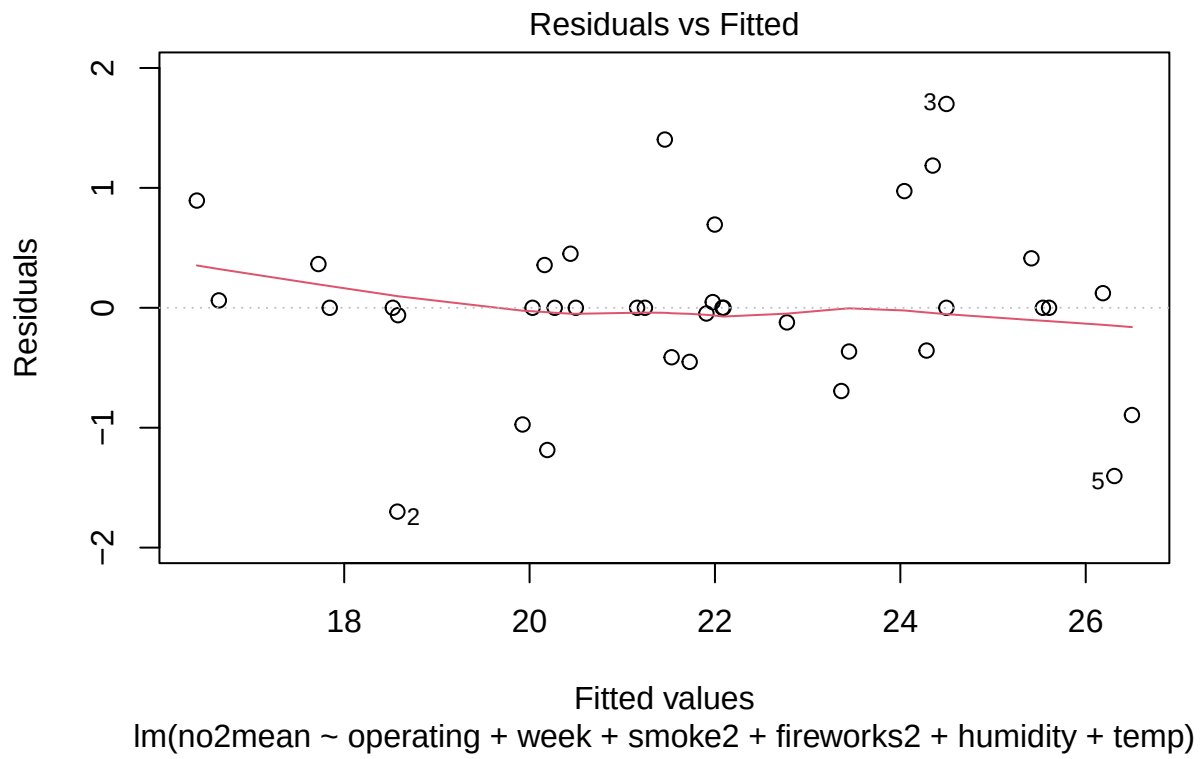


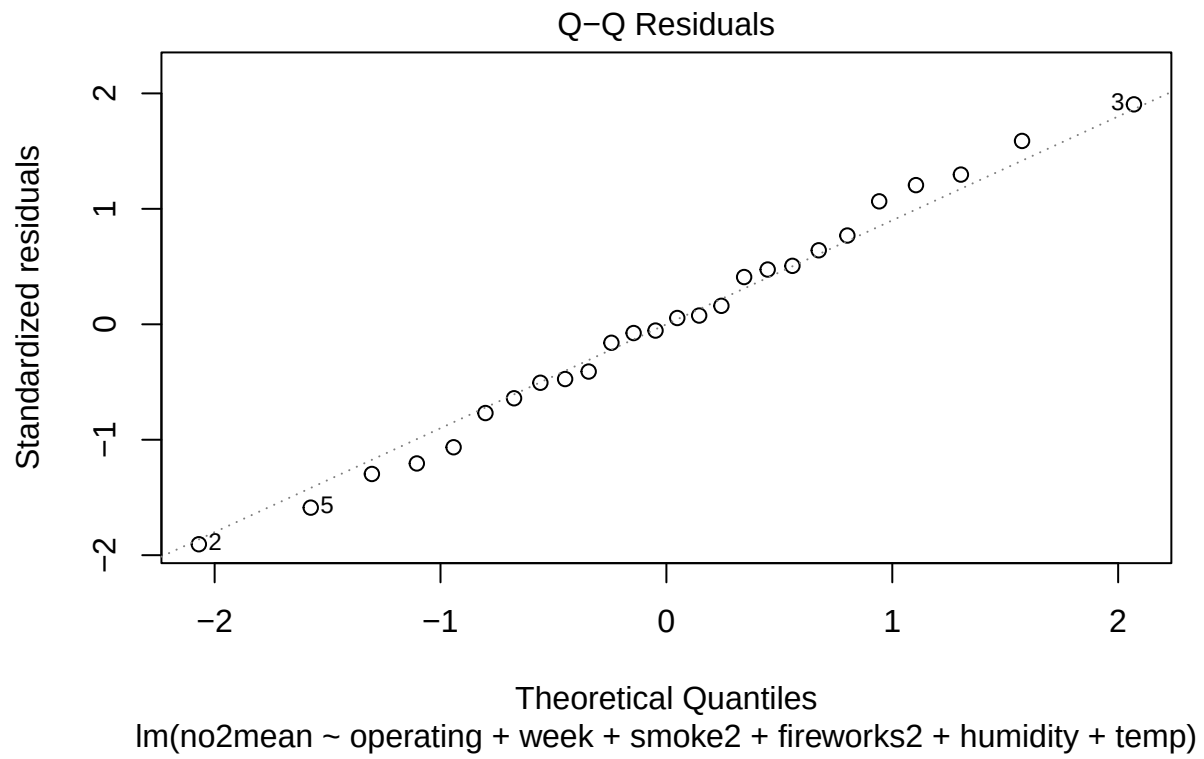


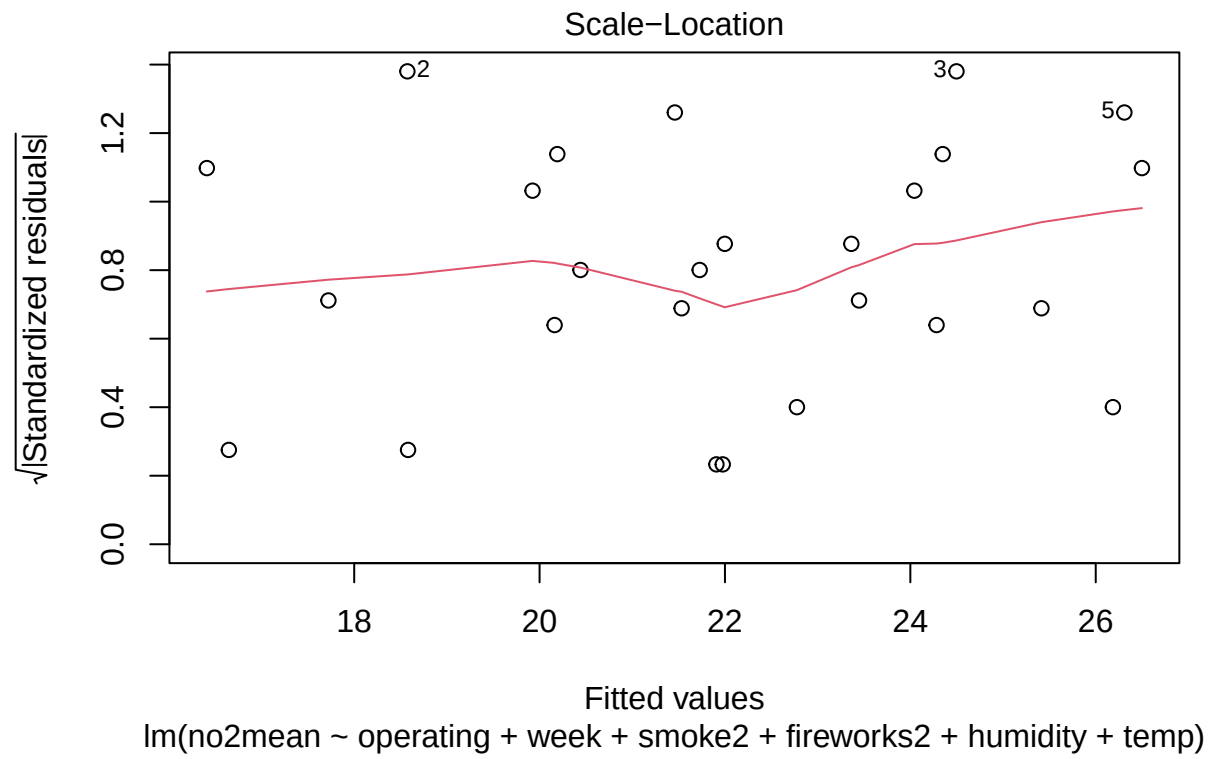


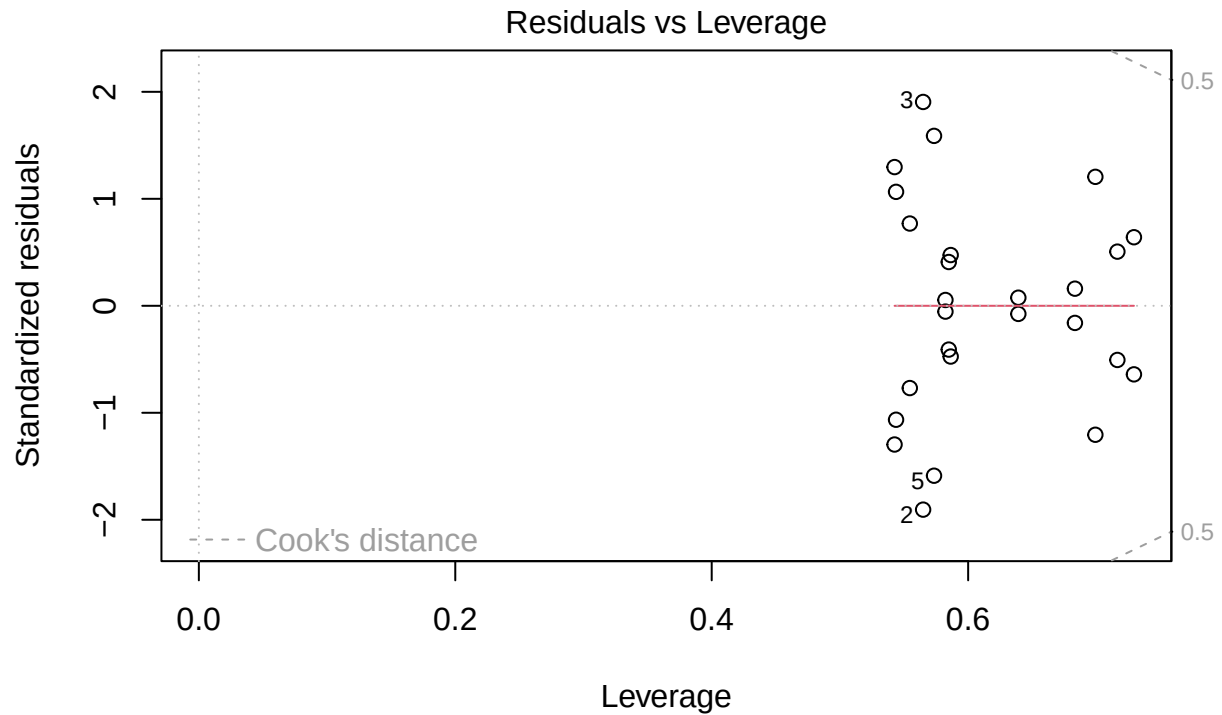
```
plot(model2_lm1)
```

```
## Warning: not plotting observations with leverage one:
## 1, 8, 13, 16, 21, 22, 25, 26, 27, 30, 31, 38
```









```
## Warning: package 'webshot2' was built under R version 4.4.3
```

```
## file:///C:/Users/anh8611/OneDrive - Harvard University/Documents/GitHub/eh204-pittsfield-aq/model-ai
```

Model 2 Regression Results		
Dependent variable:		
	no2mean	
	(1)	(2)
operatingOperating	3.875*** (1.114)	0.410 (1.310)
week2	-2.494 (4.017)	-2.714 (1.792)
week3	-0.145 (4.017)	-3.625* (1.782)
week4	-6.415 (4.017)	-5.157** (1.711)
week5	-1.242 (6.591)	-4.509 (2.885)
week6	0.802 (6.302)	-5.459* (2.931)
week7	-1.807 (4.017)	-6.238*** (1.847)
week8	-0.932 (4.593)	-6.485** (2.100)
week9	-1.349 (4.017)	-7.598*** (2.000)
week10	-2.897 (4.017)	-9.788*** (2.248)
week11	-8.721 (6.591)	
week12	-5.218 (4.593)	
week13	-1.759 (4.017)	-5.613** (1.876)
week53	-7.264 (6.102)	-7.929** (2.624)
week54	-2.060 (4.593)	-3.834* (2.012)
week55	-3.032 (4.017)	-6.973** (2.222)
week56	-2.575 (4.017)	-2.791 (1.745)
week57	-1.470 (4.727)	-3.318 (2.176)
week58	5.823 (6.302)	-1.812 (2.923)
week59	-1.292 (4.727)	
week60	-2.046 (4.017)	-3.257 (1.801)
week61	-4.248 (4.593)	-8.967*** (2.627)
week62	-5.907 (6.102)	-9.947*** (3.014)
week63	-0.556 (4.017)	-2.588 (1.872)
week64	-2.945 (4.017)	-2.147 (2.032)
week65	-1.305 (4.017)	3.567 (2.062)
smoke2	3.819 (4.727)	5.693** (2.069)
fireworks2	-0.350 (4.727)	-0.549 (2.070)
humidity		0.419*** (0.074)
temp		0.564*** (0.122)
Constant	22.092*** (3.248)	-43.852*** (10.755)
Observations	45	38
R ²	0.712	0.939
Adjusted R ²	0.208	0.773
Residual Std. Error	3.248 (df = 16)	1.353 (df = 10)
F Statistic	1.413 (df = 28; 16)	5.663*** (df = 27; 10)
Note: *p<0.1; **p<0.05; ***p<0.01		
Model 2 Regression Results		
Average hourly NO2 (micrograms/m3) per week		

```

#st_write(centroid26986, "data/output/centroid26986.shp")

#sometimes you get an error that centroid26986.shp already exists, which it does
#so I commented that code out

#statistics for weekly interpolated no2 levels at the centroid
mean(centroidno2_weekly$no2mean, na.rm = TRUE)

## [1] 21.99565
median(centroidno2_weekly$no2mean, na.rm = TRUE)

## [1] 21.86015
quantile(centroidno2_weekly$no2mean, na.rm = TRUE)

##      0%      25%      50%      75%     100%
## 16.70919 19.84507 21.86015 24.49671 36.92203
sd(centroidno2_weekly$no2mean, na.rm = TRUE)

## [1] 3.649701
centroidno2on <- centroidno2_weekly |> filter(operating == "Operating")
mean(centroidno2on$no2mean, na.rm = TRUE)

## [1] 24.36077
median(centroidno2on$no2mean, na.rm = TRUE)

## [1] 24.58581
quantile(centroidno2on$no2mean, na.rm = TRUE)

##      0%      25%      50%      75%     100%
## 16.70919 22.66560 24.58581 25.55463 36.92203
sd(centroidno2on$no2mean, na.rm = TRUE)

## [1] 3.716457
centroidno2off <- centroidno2_weekly |> filter(operating == "Not operating")
mean(centroidno2off$no2mean, na.rm = TRUE)

## [1] 20.10356
median(centroidno2off$no2mean, na.rm = TRUE)

## [1] 20.27104
quantile(centroidno2off$no2mean, na.rm = TRUE)

##      0%      25%      50%      75%     100%
## 16.87386 18.51937 20.27104 21.16021 26.30725
sd(centroidno2off$no2mean, na.rm = TRUE)

## [1] 2.257294
#check distance from allendale to the peaker plant
allendale26986 <- sensors26986 |> dplyr::filter(name == "Allendale Elementary School")
st_distance(allendale26986, airport26986)

```

```
## Units: [m]
##          [,1]
## [1,] 6843.304
```

```
#check distance from allendale to centroid
st_distance(centroid26986, airport26986)
```

```
## Units: [m]
##          [,1]
## [1,] 4590.585
```

```
#st_write(allendale26986, "data/output/allendale26986.shp")
```

```
#sometimes you get an error that allendale26986.shp already exists, which it does
#so I commented the code out
```